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Technical Report

3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; Study on Architecture support of Ambient power-enabled Internet of Things (Release 19)





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Contents

Foreword 9
1 Scope 11
2 References 11
3 Definitions of terms and abbreviations 12
3.1 Terms 12
3.2 Abbreviations 12
4 Architectural Assumptions and Requirements 12
4.1 Architectural Assumptions 12
4.2 Architectural Requirements 13
5 Key Issues 13
5.1 Key Issue #1: Architecture support of Ambient IoT Devices 13
5.1.1 Description 13
5.2 Key Issue #2: Identification, Subscription, Registration and Connection management 14
5.2.1 Description 14
5.3 Key Issue #3: Support of Ambient IoT Services 14
5.3.1 Description 14
6 Solutions 15
6.0 Mapping of Solutions to Key Issues 15
6.1 Solution #1: AIoT Temporary Identifier Control 15
6.1.1 Description 15
6.1.2 Procedures 16
6.1.3 Impacts on services, entities and interfaces 16
6.2 Solution #2: AIoT Device ID with home network, owner and instance identification 17
6.2.1 Description 17
6.2.2 Procedures 18
6.2.3 Impacts on services, entities and interfaces 18
6.3 Solution #3: Lightweight Ambient IoT system 19
6.3.1 Description 19
6.3.1.1 Introduction 19
6.3.1.2 Definitions 19
6.3.1.3 Assumptions 19
6.3.1.4 Reference architecture 19
6.3.1.5 Network function description 20
6.3.1.5.1 Reader 20
6.3.1.5.2 AIoT Controller 20
6.3.1.5.3 AF 21
6.3.1.6 Protocol Stacks 21
6.3.1.6.1 Protocol Stack between Reader and AIoT Controller 21
6.3.1.6.2 Protocol Stack between AIoT Device and Application Function 23
6.3.2 Procedures 24
6.3.2.1 Inventory procedure 24
6.3.2.2 Command procedure 26
6.3.3 Impacts on services, entities and interfaces 27
6.4 Solution #4: Simplified system for AIoT 27
6.4.1 Description 27
6.4.1.1 General 27
6.4.1.2 Abbreviations 28
6.4.1.3 Terms 28
6.4.1.4 Reference Architecture and NFs 29
6.4.1.4.1 Pure Control Plane 29
6.4.1.4.2 User Plane of UE reader 30
6.4.1.5 Protocol Stack 30
6.4.2 Procedures 32
6.4.2.1 Purely via control plane: Inventory procedure 32

6.4.2.2 Purely via control plane: Command procedure 33
6.4.2.3 Discover UE reader via UE reader control plane and AIoT device related data delivery via UE reader user plane 34
6.4.2.4 UE reader activation 35
6.4.3 Impacts on services, entities and interfaces 35
6.5 Solution #5: NAS-based information transfer for Ambient IoT Services 36
6.5.1 Description 36
6.5.2 Procedures 37
6.5.3 Impacts on services, entities and interfaces 38
6.6 Solution #6: AIoT device authentication, ID validation and AIoT communication 39
6.6.1 Key Issue mapping 39
6.6.2 Functional Description 39
6.6.3 Procedures 41
6.6.4 Impacts on existing services, entities and interfaces 43
6.7 Solution #7: Bulk Introduction of Devices to the Network 43
6.7.1 Description 43
6.7.2 Procedures 44
6.7.3 Impacts on services, entities and interfaces 44
6.8 Solution #8: Inventory for AIoT Devices using AIOTF 45
6.8.1 Description 45
6.8.2 Procedures 47
6.8.2.1 Application Inventory Procedure 47
6.8.2.2 Periodic Inventory Procedure 50
6.8.3 Impacts on services, entities and interfaces 51
6.9 Solution #9: Information Transfer without a PDU Session 51
6.9.1 Description 51
6.9.2 Procedures 52
6.9.3 Impacts on services, entities and interfaces 53
6.10 Solution #10: Registration procedure for Ambient IoT Devices 53
6.10.1 Description 53
6.10.2 Procedures 54
6.10.2.1 Procedures for AF triggered Registration for Topology 1 54
6.10.2.2 Procedures for AF triggered Registration for Topology 2 57
6.10.3 Impacts on services, entities and interfaces 59
6.11 Solution #11: 5GC support for AF to communicate with AIoT devices 60
6.11.1 Description 60
6.11.2 Procedures 61
6.11.2.1 Communication with AIoT devices 61
6.11.2.2 Selection of intermediate nodes 63
6.11.3 Impacts on services, entities and interfaces 68
6.12 Solution #12: UE Function Delegation in Intermediate Node for AIoT Device 69
6.12.1 Description 69
6.12.2 Procedures 70
6.12.2.1 UFDM instantiation and provisioning 70
6.12.2.2 UFDM Registration 71
6.12.3 Impacts on services, entities and interfaces 72
6.13 Solution #13: Multi-level Device ID management 72
6.13.1 Description 72
6.13.2 Procedures 72
6.13.3 Impacts on services, entities and interfaces 73
6.14 Solution #14: AIoT Device ID with consideration on credential holder, service provider and mobile network 74
6.14.1 Description 74
6.14.2 Procedures 75
6.14.3 Impacts on services, entities and interfaces 77
6.15 Solution #15: Ambient IoT Device States 77
6.15.1 Description 77
6.15.1.1 Overview 77
6.15.1.2 New AIoT Device States 77
6.15.2 Procedure for State Toggling in AIoT Devices 78
6.15.3 Impacts on services, entities and interfaces 80
6.16 Solution #16: AIoT Registration/Onboarding 80
6.16.1 Description 80

6.16.2 Procedures 81
6.16.3 Impacts on services, entities and interfaces 82
6.17 Solution #17: Enable Ambient IoT Management within 5GC to support AIoT services 83
6.17.1 Description 83
6.17.1.1 General 83
6.17.1.2 Architecture Principles 83
6.17.1.3 Function Principles 84
6.17.2 Procedures 86
6.17.2.1 General 86
6.17.2.2 AIoT Service High-Level Information Flow 87
6.17.2.3 NEF AIoT Service 90
6.17.3 Impacts on services, entities and interfaces 90
6.18 Solution #18: Solution for support of Ambient IoT services 91
6.18.1 Description 91
6.18.2 Procedures 92
6.18.2.1 Ambient IoT service - Network initiated Ambient IoT service procedure 92
6.18.3 Impacts on services, entities, and interfaces 94
6.19 Solution #19: AIoT communication for connectivity topology 1 95
6.19.1 Key Issue mapping 95
6.19.2 Functional Description 95
6.19.3 Procedures 96
6.19.3.0 General 96
6.19.3.1 N2' association establishment 96
6.19.3.2 AIoT session establishment between the AIoTF and the AIoT AF/AS and AIoTF and the BS AIoT Reader 98
6.19.4 Impacts on existing services, entities and interfaces 100
6.20 Solution #20: E2E handling for interaction of AIoT Device related information 100
6.20.1 Description 100
6.20.2 Procedures 103
6.20.2.1 The interaction for AIoT Device among AIoT device, reader and AIoT Function 103
6.20.2.2 The interaction for AIoT Device among AIoT device, reader, AIoT Function and AF 105
6.20.3 Impacts on services, entities and interfaces 106
6.21 Solution #21: Architecture enhancement to support AIoT services for topology 2 106
6.21.1 Description 106
6.21.2 Procedures 107
6.21.3 Impacts on services, entities and interfaces 108
6.22 Solution #22: AF Based Solution for Topology 2 109
6.22.1 Description 109
6.22.1.0 General 109
6.22.1.1 Reference Architecture 109
6.22.1.2 Protocol Stack 110
6.22.2 Procedures 111
6.22.2.1 AIoT Service Authorization for Intermediate UE 111
6.22.2.2 Inventory Procedure 111
6.22.2.3 Command Procedure 112
6.22.3 Impacts on services, entities and interfaces 114
6.23 Solution #23: UE reader selection for inventory procedure 114
6.23.1 Description 114
6.23.1.1 Definition 114
6.23.1.2 General 114
6.23.1.3 Architecture and Principles 115
6.23.2 Procedures 115
6.23.3 Impacts on services, entities and interfaces 116
6.24 Solution #24: Availability Reporting without Ambient IoT Device Timers 117
6.24.1 Functional Description 117
6.24.2 Procedures 117
6.24.3 Impacts on existing services, entities and interfaces 119
6.25 Solution #25: Subscription Management for Ambient IoTs 119
6.25.1 Description 119
6.25.2 Procedures 121
6.25.3 Impacts on services, entities and interfaces 123
6.26 Solution #26: Unified API for Topology 1 and Topology 2 for an AF 123
6.26.1 Description 123

6.26.1.1 General 123
6.26.1.2 Example Target Sets 124
6.26.1.2 NEF IoT Services 126
6.26.1.2.0 Overview 126
6.26.1.2.1 Inventory service operation 126
6.26.2 Procedures 127
6.26.3 Impacts on services, entities and interfaces 127
6.27 Solution #27: AIoT service operation with privacy and authentication protection. 128
6.27.1 Description 128
6.27.2 Procedures 129
6.27.2.1 Inventory procedure 129
6.27.2.2 Command procedure 130
6.27.3 Impacts on services, entities and interfaces 131
6.28 Solution #28: Device-Reader Binding Information 131
6.28.1 Description 131
6.28.2 Procedures 132
6.28.2.1 Device-Reader Binding Information Storage 132
6.28.2.2 Device-Reader Binding Information Usage 133
6.28.3 Impacts on Existing Nodes and Functionality 135
6.29 Solution #29: Periodic triggered Ambient IoT Devices for registration update in Topology-1 135
6.29.1 Description 135
6.29.2 Procedures 135
6.29.2.1 Periodic triggered AIoT devices to perform Registration Update 135
6.29.2.2 Periodic triggered AIoT devices to perform Registration Update (Continued) 137
6.29.2.3 BS relocation according the update of the latest registration area of AIoT device 138
6.29.2.4 Deregistration 138
6.29.3 Impacts on services, entities and interfaces 139
6.30 Solution #30: Support AIoT Devices using AIOTF for Topology 2 140
6.30.1 Description 140
6.30.1.0 Introduction 140
6.30.1.1 Reference Architecture 140
6.30.1.2 Protocol Stack 141
6.30.2 Procedures 141
6.30.2.1 AIoT Service Authorization for Intermediate UE 141
6.30.2.2 AIOTF Keep Track of Intermediate UE 142
6.30.2.3 Application Inventory Procedure 142
6.30.2.4 Periodic Inventory 144
6.30.2.5 Command Procedure 144
6.30.3 Impacts on services, entities and interfaces 147
6.31 Solution #31: fixed type of UE reader solution for Ambient IoT services 147
6.31.1 Description 147
6.31.2 Procedures 148
6.31.2.1 Reader UE registration procedure with UE reader type info, Location, and AF ID 148
6.31.2.2 Ambient IoT service - Network initiated Ambient IoT service procedure with fixed UE reader 150
6.31.3 Impacts on services, entities, and interfaces 152
6.32 Solution #32: AIoT communication for connectivity topology 2 and AIoT data transmission in the UP of PDU Session 152
6.32.1 Key Issue mapping 152
6.32.2 Functional Description 152
6.32.2.1 General 152
6.32.2.2 Protocol Stack 153
6.32.3 Procedures 154
6.32.4 Impacts on existing services, entities and interfaces 156
6.33 Solution #33: Solution for different Ambient IoT Device Types 157
6.33.1 Description 157
6.33.1.1 Introduction 157
6.33.1.2 Functional Description 157
6.33.1.3 Reference Architecture 158
6.33.1.4 Protocol Stack 158
6.33.2 Procedures 159
6.33.2.1 AIoT Service Operation Procedure for Type A AIoT Device 159
6.33.2.2 AIoT Service Operation Procedure for Type B AIoT Device 160

6.33.3 Impacts on services, entities and interfaces 160
6.34 Solution #34: UE assisted paging for AIoT Devices 161
6.34.1 Functional Description 161
6.34.2 Procedures 161
6.34.3 Impacts on existing services, entities, and interfaces 164
6.35 Solution #35: An compatible format of AIoT device 164
6.35.1 Description 164
6.35.2 EPC format 164
6.35.3 Design AIoT device ID 166
6.35.4 Procedure 166
6.35.5 Impacts on services, entities and interfaces 166
6.36 Solution #36: Device ID(s) for AIoT Devices 167
6.36.1 Description 167
6.36.2 Procedures 168
6.36.3 Impacts on services, entities and interfaces 168
6.37 Solution #37: Multiple Temporary Reusable Identities for AIoT Devices 169
6.37.1 Description 169
6.37.2 Procedures 170
6.37.2.1 Temporary Reusable ID Allocation 170
6.37.2.2 Temporary Reusable ID Revocation and Re-Allocation 171
6.37.2.3 Temporary Reusable ID Conflict Detection and Resolution 171
6.37.3 Impacts on services, entities and interfaces 171
6.38 Solution #38: AIoT device tracking for inventory 172
6.38.1 Description 172
6.38.2 Procedures 173
6.38.3 Impacts on services, entities and interfaces 174
6.39 Solution #39: Configuration of Ambient IoT device frequency 174
6.39.1 Description 174
6.39.2 Procedures 174
6.39.2.1 Enable Ambient IoT device(s) communication in operator network 175
6.39.2.2 Disable Ambient IoT device(s) communication in operator network 177
6.39.3 Impacts on services, entities and interfaces 178
6.40 Solution #40: Reader dependent Inventory Response 178
6.40.1 Description 178
6.40.2 Procedures 179
6.40.3 Impacts on services, entities and interfaces 180
6.41 Solution #41: Simple AIoT device and AIoT solution in the certain deployment 180
6.41.1 Description 180
6.41.2 Procedures 181
6.41.3 Impacts on services, entities and interfaces 182
6.42 Solution #42: AIoT data forwarding over UE reader in topology 2 182
6.42.1 Description 182
6.42.2 Procedures 184
6.42.3 Impacts on services, entities and interfaces 185
6.43 Solution #43: Harmonized RAN-CN solution using AIOTF for Topology 2 185
6.43.1 Description 185
6.43.1.1 Reference Architecture 185
6.43.1.2 Protocol Stack 186
6.43.2 Procedures 187
6.43.2.1 NG Setup 187
6.43.2.2 AIoT Service Authorization for Intermediate UE 187
6.43.2.3 AIOTF Keep Track of Intermediate UE 187
6.43.2.4 Application Inventory 188
6.43.2.5 Periodic Inventory Procedure 190
6.43.2.6 Command Procedure 190
6.43.3 Impacts on services, entities and interfaces 193

7 Overall Evaluation 194

8 Conclusions 194
8.1 Conclusion on Key Issue #1 194
8.1.1 General 194
8.1.2 Architecture to Support Topology 1 196
8.1.2.1 General 196

8.1.2.2 AIOT RAN and the AIOTF communicate directly 196
8.1.2.3 AIOT RAN and the AIOTF communicate indirectly via an AMF 198
8.1.3 Architecture to Support Topology 2 200
8.1.3.1 General 200
8.1.3.2 User-plane option 201
8.1.3.3 RRC option 202
8.1.3.4 Interim Conclusion for authorization 202
8.2 Conclusion on Key Issue #2 202
8.2.1 Identifier and Identification Management 202
8.2.2 Subscription Management 203
8.3 Conclusion on Key Issue #3 204
8.3.1 General 204
8.3.2 AIoT services supported by the 5GC 204
8.3.3 NEF exposure 205
8.3.4 Principles on the procedures to support AIoT services 205
8.3.5 Information provided by the AF 206

Annex A: Overview of the Electronic Product Code 207

Annex B: Change history 209

Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

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z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

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should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

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can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document studies the architecture support of Ambient IoT Devices, based on the services requirements defined in TS 22.369 [2] applicable to the Device types, traffic types, use cases and connectivity topologies defined in TR 38.769 [8].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 22.369: "Service requirements for Ambient power-enabled IoT".

[3] 3GPP RP-234058: "New SID: Study on solutions for Ambient IoT (Internet of Things) in NR".

[4] 3GPP TS 23.501: "System Architecture for the 5G System (5GS); Stage 2".

[5] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[6] 3GPP TS 23.503: "Policies and Charging control framework for the 5G System; Stage 2".

[7] 3GPP TR 38.848: "Technical Specification Group Radio Access Network; Study on Ambient IoT (Internet of Things) in RAN".

[8] 3GPP TR 38.769: "Study on solutions for Ambient IoT (Internet of Things) in NR".

[9] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[10] GS1 TDS Release 2.1: "EPC Tag Data Standard".

[11] GS1 Organisation: "The Global Language of Business". Available at: https://www.gs1.org/.

[12] GS1: "Standards in the Healthcare Supply Chain". Available at: https://www.gs1.ch/de/media/1117.

[13] 3GPP TS 23.288: "Architecture enhancements for 5G System (5GS) to support network data analytics services".

[14] 3GPP TS 38.401: "NG-RAN; Architecture description".

[15] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[16] BS ISO-IEC 15963-1-2020: Information technology - Radio frequency identification for item management - Part 1: Unique identification for RF tags numbering systems.

3 Definitions of terms and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

Ambient IoT Device: An Ambient IoT device is an IoT device powered by energy harvesting, with limited energy storage capability. The other characteristics of the Ambient IoT device are defined in TR 38.769 [8].

NOTE 1: The final decision on the term name is to be determined in TR conclusion or normative phase.

Ambient IoT Services: The functionalities and procedures to support Ambient IoT use cases.

NOTE 2: the functionalities and procedures for Ambient IoT Services are left to outcome of the study. The Ambient IoT use case(s) can be referred to TR 38.848 [7] and TS 22.369 [2].

NOTE 3: The final definition on the term is to be determined in TR conclusion or normative phase.

Device-originated - device-terminated triggered (DO-DTT): The device originated traffic is triggered by the device terminated traffic or signalling.

Device-terminated (DT): The traffic is terminated at the AIoT device.

Electronic Product Code: A universal identifier that provides a unique identity for any physical object as defined by GS1 in the EPC Tag Data Standard (GS1 TDS Release 2.1 [10]) and used for identification needs of various business domains.

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

AIoT	Ambient IoT
DO-A	Device-originated - autonomous
DO-DTT	Device-originated - device-terminated triggered
DT	Device-terminated
EPC	Electronic Product Code
RFID	Radio-Frequency IDentification

4 Architectural Assumptions and Requirements

4.1 Architectural Assumptions

- The following traffic types for Ambient IoT Device are to be studied:
  - DT: Device-terminated; and
  - DO-DTT: Device-originated - device-terminated triggered.

NOTE 1: The DO-DTT additionally includes traffic from AIoT Devices, which is triggered by RAN/UE as reader, without CN sending traffic towards the AIoT Devices.

NOTE 2: The final decision for including DO-A (Device-originated - autonomous) in the study depends on RAN decision.

- The following two connectivity topologies as defined in TR 38.848 [7] are to be studied:
  - Topology 1: BS <--> Ambient IoT Device;

  - Topology 2: BS <--> intermediate node <--> Ambient IoT Device: Only a UE can act as an intermediate node which is under the network control.

- The communication spectrum is assumed to be licensed.
- Handover is not supported.
- RRC states are not supported by AIoT Devices (see TR 38.769 [8])
- No mobility (i.e. at least no cell selection/re-selection-like function) supported by AIoT Devices (see TR 38.769 [8])

Editor's note: The meaning of no mobility is to be clarified by RAN in TR 38.769 [8].

NOTE 3: Coordination with RAN is required to determine the Ambient IoT Device capabilities in relation to system level of functionality (considering e.g. traffic scenarios, connectivity topologies etc.).

NOTE 4: The security aspects for Ambient IoT requires coordination with SA WG3.

NOTE 5: The charging aspects for Ambient IoT will be studied by SA WG5.

NOTE 6: The NAS based Congestion control is not in the scope of this study.

NOTE 7: A dedicated Ambient IoT core network can be deployed by the operator to handle the traffic produced by ambient IoT devices.

4.2 Architectural Requirements

The following architectural requirements are applicable to this study:

- Support for AIoT Services needs to adhere to the nature of the AIoT Devices (e.g. ultra-low complexity, power, cost and resource-constrained).
- Support of the security aspects needs to consider the nature of the AIoT Devices (e.g. ultra-low complexity power, cost and resource-constrained) while addressing e.g. confidentiality, integrity, etc.

5 Key Issues

5.1 Key Issue #1: Architecture support of Ambient IoT Devices

5.1.1 Description

This key issue will address the system architecture to support Ambient IoT Devices, especially on the following aspects:

- System architecture identified along with the solutions for KI#2 and KI#3.
- Authentication and authorization for the Ambient IoT Device;
- Validation of the Ambient IoT Device identifier;

NOTE 1: Format of the Ambient IoT Device identifier is addressed in KI#2.

- Whether and how to secure device operations and services for an Ambient IoT Device or a group of Ambient IoT Devices;

NOTE 2: This key issue will take into account the outcome of RAN study in TR 38.769 [8].

NOTE 3: The security aspects related to this key issue, including the enable/disable device operation, requires coordination with SA WG3.

5.2 Key Issue #2: Identification, Subscription, Registration and Connection management

5.2.1 Description

This Key Issue pertains to the authorization and management of Ambient IoT Devices to support Ambient IoT services.

Considering that Ambient IoT Devices are a new type of reduced capabilities devices, the existing subscription model may not be suitable. Specifically, there is the need to study the device identification method to support Ambient IoT devices which are under operator control.

Based on the above consideration, the aspects to be studied in this key issue include:

- Study whether subscription management, registration management and/or connection management are necessary for an Ambient IoT Device or a group of Ambient IoT Devices, and if so identify the necessary state machine(s), procedures and functionality considering the Ambient IoT Devices capability and characteristics.
- Study whether and how reachability and paging apply to Ambient IoT Device(s) considering the Ambient IoT devices capability and characteristics, and if so, what are the impacts.
- Study how to identify Ambient IoT Device or group of devices and how to format the identifier.

NOTE: NAS based Congestion control are not in the scope of this study.

5.3 Key Issue #3: Support of Ambient IoT Services

5.3.1 Description

This Key Issue pertains to the AIoT services. Considering that AIoT Devices are a new type of devices with reduced capabilities, the following need to be supported:

- Inventory.
- Command.

NOTE 1: Further detailing of the inventory and commands will be addressed by solutions.

The key issue will study the following aspects:

- Study how to support information transfer for Ambient IoT services and related system functionality, including the information transfer for an Ambient IoT device and for a group of Ambient IoT Devices.

NOTE 2: The above aspect includes studying whether there is a need to support session based transfer between Ambient IoT Device and the network considering the device types and capabilities.

- Study which of the enabled Ambient IoT services are exposed to AF and how, e.g. for the case AF requests Ambient IoT service for an Ambient IoT Device and for a group of Ambient IoT Devices.

6 Solutions

6.0 Mapping of Solutions to Key Issues

Table 6.0-1: Mapping of Solutions to Key Issues

	Key Issues		
Solutions	**Key Issue #1**	**Key Issue #2**	**Key Issue #3**
#1		X	
#2		X	
#3	X	X	X
#4	X	X	X
#5	X	X	X
#6	X	X	X
#7	X	X	
#8	X	X	
#9	X	X	X
#10	X	X	
#11			X
#12	X	X	X
#13		X	
#14		X	
#15		X	X
#16		X	
#17	X	X	X
#18			X
#19	X		X
#20	X	X	X
#21	X	X	X
#22	X	X	X
#23			X
#24		X	
#25		X	
#26			X
#27			X
#28			X
#29	X	X	
#30	X	X	X
#31		X	X
#32	X		X
#33	X	X	X
#34		X	
#35		X	
#36		X	
#37		X	
#38			X
#39		X	
#40			X
#41	X	X	X
#42			X
#43	X	X	X

6.1 Solution #1: AIoT Temporary Identifier Control

6.1.1 Description

This solution addresses KI#2. The basic principle in 5GS is that the permanent subscriber identifier shall never be sent in clear text over Uu interface and a UE temporary identifier shall only be sent over Uu interface in clear text once e.g. when the UE is paged. This solution propose that same principle shall apply also for Ambient IoT.

The solution is based on the following assumptions:

- The AIoT device has higher complexity than a RFID tag that only reflects a preconfigured device ID when exited by RF power, but significantly lower complexity than a 3GPP CIoT device.
- The AIoT device has a non-volatile storage capability.
- The Temp ID generation algorithm is light weight but enough to avoid unauthorized AIoT device tracking.

6.1.2 Procedures

As the available power in an AIoT device is very limited, the message exchange between the device and the network must be minimized. The solution is based on the following principle:

- The initial temporary identifier (TempID) is known by both the CN NF and the AIoT device. After the AIoT device has been onboarded to the network the CN NF provision the AIoT device with the initial TempID and/or parameters to derive the initial Temp ID i.e. parameters for the Temp ID generation algorithm.

NOTE 1: Onboarding procedure will be studied under KI#2 and related conclusions needs to be considered when concluding the AIoT device Identifier control. It is assumed that during the onboarding procedure the CN NF can retrieve information from another NF or AF to onboard the AIoT device. The initial message from the UE during onboarding could e.g. include a URL or FQDN to establishing IP connection to a AF that holds additional onboarding information needed.

- Every time the TempID has been sent over the radio interface and a response from the AIoT device is sent, both CN NF and AIoT device locally generate a new TempID.

Editor's note: Details on the algorithm that locally generates a new TempID needs to be defined by SA WG3.

- If the CN NF and the AIoT device TempID out of sync is detected, the CN NF and the AIoT re-synchronize the TempID.

Editor's note: How the out-of-sync detection and re-synchronization are performed is FFS, e.g. it may be a counter value not matching the expected value. Details depends on the Temp ID generation algorithm.

- AIoT device considers and responds to the DT message if the AIoT device can match the TempID used in the DT message.
- AIoT device responds with DO-DTT message if the AIoT device can match the TempID used in the trigger message.

NOTE 2: The CN NF that manages the TempID will be defined together with the system architecture design to support the Ambient IoT in 5GC.

6.1.3 Impacts on services, entities and interfaces

Impacts on existing entities:

CN NF:

- Support the AIoT TempID handling including the generation of initial TempID, new TempID and re-synchronization between AIoT device and CN NF.

AIoT device:

- Recieving a initial TempID from the network.
- Generation of new TempID locally, when TempID used over Uu interface.
- re-synchronization between AIoT device and CN NF.

6.2 Solution #2: AIoT Device ID with home network, owner and instance identification

6.2.1 Description

This solution addresses the KI#2 and the aspect about how to identify and format the identifier of Ambient IoT Device in order to identify specific Ambient IoT Device or a group of Ambient IoT Devices.

In this solution, it is assumed that the Ambient IoT Device is configured with 3GPP-defined identifier and optionally be configured with 3rd Party-defined identifier in some specific scenarios.

According to TS 22.369 [2], the 5G system shall provide suitable mechanisms to support communication between an authorized 3rd party and an Ambient IoT device or group of Ambient devices. In addition, subject to user consent, operator's policy and 3rd party request, the 5G system shall provide information about an Ambient IoT device or a group of Ambient IoT devices (e.g. position) to the 3rd party via the 5G network. Based on these service requirements, the relationship among the Ambient IoT Device, Mobile Network Operator and the 3rd party is illustrated below.

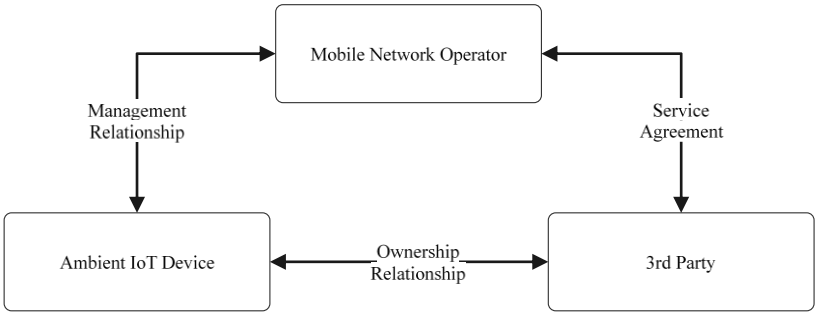


Figure 6.2.1-1: Relationship Among Ambient IoT Device, Mobile Network Operator and 3rd Party

It is assumed an Ambient IoT Device is owned by a third party who has a service agreement with a Mobile Network Operator to enable Ambient IoT service in 3GPP system. The MNO manages the Ambient IoT Device (e.g. holds the credentials of Ambient IoT Device, etc) in order to support communication between the 3rd party and the Ambient IoT Device via the 5G network. Therefore, based on this model, in the 3GPP system, different MNOs may need to manage different Ambient IoT Devices owned by different third parties. If uniqueness of Device ID cannot be guaranteed, operation of Ambient IoT services provided by 3GPP system may be impacted. Hence, considering of this, the Device ID used by an Ambient IoT Device shall enable the identification of the MNO it is managed by, the identification of the 3rd party it belongs and the identification of the Ambient IoT itself.

Based on the above consideration, following components are considered necessary to compose the Ambient IoT Device ID, which is defined by 3GPP:

- **Home Network Identifier:** an identifier used to identify the home MNO;
- **Owner Identifier:** an identifier used to identify a 3rd party who sends service requests to trigger 5GC to perform Ambient IoT service operation;

NOTE 1: The Owner Identifier is allocated by the home MNO corresponding to the Home Network Identifier.

Editor's note: The use of the Owner Identifier is FFS and may depend on the solution for Inventory, etc.

- **Instance Identifier:** an identifier used to identify a specific Ambient IoT device owned by the 3rd party.

NOTE 2: The Instance Identifier is allocated by the home MNO which may coordinate with the 3rd party.

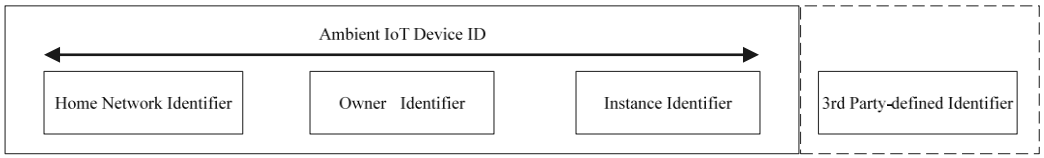


Figure 6.2.1-2: Structure of Ambient IoT Device ID and optional 3rd Party-defined Identifier

The device can be configured with either:

- Only the 3GPP-defined identifier (i.e. Ambient IoT Device ID); or
- 3GPP-defined identifier (i.e. Ambient IoT Device ID) and 3rd Party-defined identifier.

The Ambient IoT Device ID is a permanent identifier used by the MNO to derive subscription-like data related to Ambient IoT Devices. In terms of each component of the Ambient IoT Device ID, the Owner Identifier has to be unique within the MNO identified by the Home Network Identifier. The Instance Identifier has to be unique within the Owner Identifier (which in turn is unique within a Home Network Identifier). The means that the Ambient IoT Device ID is unique globally. In this way, the length of the Ambient IoT Device ID can be shortened.

In the case of only a 3GPP-defined identifier, the 3rd party relies on the Ambient IoT Device ID allocated by the operator to perform Ambient IoT services.

Additionally, an Ambient IoT Device may be assigned an 3rd party-controlled identifier which, for example, can be used by the 3rd party to group Ambient IoT Devices together. As this identifier will not be used by the 3GPP system to uniquely identify the Ambient IoT Device, its allocation can be fully under the 3rd party's control and the MNO can retain control of the allocation and management of the Device ID used to locate the subscription-like data for an Ambient IoT Device.

The 3rd party can use the 3rd Party-defined identifier component to perform Ambient IoT services on specific Ambient IoT devices.

For example, when a 3rd party (e.g. AF) sends a service request (e.g. inventory or command) to 5GC, the 5GC can trigger the reader(s) to inventory a group of Ambient IoT Device by broadcasting a partial/full Ambient IoT Device ID or partial/full 3rd Party-defined Identifier or both. The Ambient IoT Devices matching the broadcasted message will perform random access responding to the broadcast message. For example, when the partial value is the Home Network Identifier, Owner Identifier, the Ambient IoT Devices matching that the partial value (i.e. belonging to a specific 3rd party), or if the partial value is (part of) the 3rd Party-defined Identifier (i.e. matching the 3rd-party defined identifier), will respond to the broadcast message to perform random access and report their Device ID to the network.

With such format, the network can enable different group of Ambient IoT Device to respond the broadcast message for inventory.

Editor's note: It is FFS whether it can be assumed that the device and the CN can be pre-provisioned with Ambient IoT Device ID and the optional 3rd Party-defined identifier.

6.2.2 Procedures

How to utilize such format of Ambient IoT Device ID when network performs service operations will be specified in the call flows in other solutions.

6.2.3 Impacts on services, entities and interfaces

Editor's note: This clause captures impacts on existing services, entities and interfaces.

6.3 Solution #3: Lightweight Ambient IoT system

6.3.1 Description

6.3.1.1 Introduction

This solution proposes a lightweight Ambient IoT system.

6.3.1.2 Definitions

- **Command:** Refers to an instruction sent by an AF to an AIoT Device. The following instructions may be supported:
  - Read: Reading data from an AIoT Device;
  - Write: Writing data to an AIoT Device;
  - Disable: Disable an AIoT Device temporarily or permanently.
- **Command Response:** Refers to the message sent by an AIoT Device in response to a Command. This may include an acknowledgement and optionally data (e.g. in case of the Read operation).
- **Enrichment Data**: Additional information that a Reader may include when providing Inventory information or a Command Response to the AIoT Controller. Enrichment Data may include information about the signal strength for each detected Device ID and the location of the Reader (if known).
- **Filter Criteria**: Criteria to limit an Inventory or Command to AIoT Devices that match certain criteria. Filter Criteria may consist of the following:
  - an EPC as defined by GS1 in the EPC Tag Data Standard (GS1 TDS Release 2.1 [10]) and optionally a bitmask;
  - a Closest indication.

NOTE 1: The Closest indication only applies to the Inventory procedure.

NOTE 2: The term Filter Criteria can be revisited to avoid clashes with existing terminology.

- **Inventory:** Refers to determining the identity of all or a subset of AIoT Devices in range of a reader.

6.3.1.3 Assumptions

This solution makes the following assumptions:

- Commands and Command Responses are end-to-end protected between AF and AIoT Device.

NOTE 1: Details of the end-to-end protection of Commands and Command Results are assumed to be addressed by SA WG3.

- Commands and Command Responses are transparent to the AIoT Controller and the Reader, i.e. the AIoT Controller and Reader are not aware of the contents of Commands and Command Responses.
- AIoT Devices are assumed to be pre-provisioned with a Device ID and the security material for the end-to-end protection of Commands and Command Results. The EPC as defined by GS1 in the EPC Tag Data Standard (GS1 TDS Release 2.1 [10]) is used as the Device ID.

NOTE 2: Whether also dynamic (re-)provisioning of Ambient IoT Devices can be supported is up to SA WG3.

- The radio configuration (frequency bands, etc.) of the Reader is assumed to be configured through an OAM system, which is beyond the scope of SA WG2.

6.3.1.4 Reference architecture

This solution proposes the reference architecture depicted in Figure 6.3.1.4-1 and Figure 6.3.1.4-2.

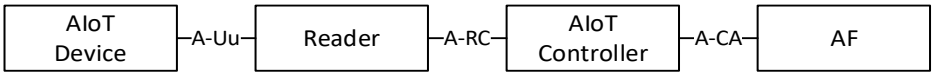


Figure 6.3.1.4-1: Ambient IoT system architecture for Topology 1

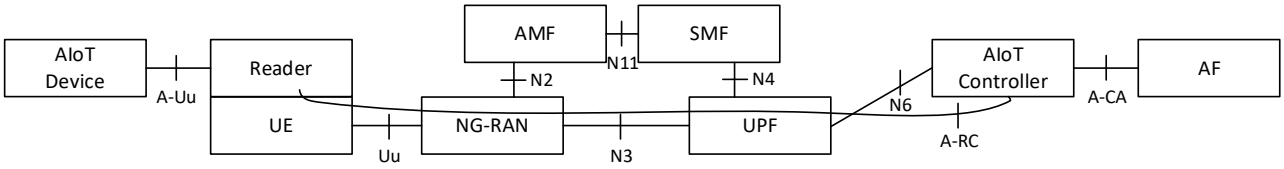


Figure 6.3.1.4-2: Ambient IoT system architecture for Topology 2

The Ambient IoT system can support different deployment options. For example, the Reader may be co-located with a 3GPP UE so that the communication between Reader and AIoT Controller uses a PDU Session to enable Topology 2 (enabling Topology 2 as depicted in Figure 6.3.1.4-2). Alternatively, Readers may be deployed independent of 3GPP UEs, i.e. as stand-alone base stations to enable Topology 1 (Figure 6.3.1.4-1).

The AIoT controller is assumed to be a 3GPP core network entity that enables Ambient IoT scenarios in the context of 5G. The 5GC network functions shown in Figure 6.3.1.4-2 (AMF, SMF, UPF) are assumed to be in the PLMN domain.

NOTE: For Topology 2, the UE is assumed to be configured with the DNN/S-NSSAI for the PDU Session to be used for Reader to AIoT Controller communication and also with an FQDN to discover the AIoT Controller.

Editor's note: Whether there is a need to validate the AIoT Device ID by a serving network and if so, how to achieve this is FFS.

6.3.1.5 Network function description

6.3.1.5.1 Reader

The Reader supports the following functionality:

- Supports the A-Uu air interface towards Ambient IoT Devices.
- Registers with an AIoT Controller.
- Supports the following functionality based on requests from an AIoT Controller:
  - Perform one-time or periodic Inventory, deliver Inventory result to AIoT Controller.
  - Delivers Commands from an AIoT controller to an AIoT Device.
  - Delivers Command Responses received from an AIoT Device to an AIoT Controller.

6.3.1.5.2 AIoT Controller

The AIoT Controller supports the following functionality:

- Register Readers.
- Authenticate and authorize AFs.
- Based on requests from an AF:
  - Verify whether an AF is entitled to issue a specific Inventory Request.
  - Select Readers to fulfil Inventory or Command request by AFs.
  - Forward Inventory request to Readers and deliver Inventory result to AF.
  - Forward Command to Readers and Command Responses to AF.

- Optionally collect usage data per AF, e.g. for charging purposes.
- Store last known Reader information for AIoT Devices.

This solution does not use the NEF. Instead, to authenticate and authorize AFs and to protect the AIoT Controller - AF communication, the security mechanisms defined for the Network Exposure Function in TS 33.501 [15] (TLS for mutual authentication, integrity protection, replay protection and confidentiality protection, OAuth-based authorization) are assumed to also be used for the AIoT Controller. Details can be specified by SA WG3.

NOTE: The details of usage data collection per AF are in the scope of SA WG5. It is up to SA WG5 to decide whether CHF functionality can be reused to achieve that.

The solution assumes that the CAPIF framework can also be used to register and discover APIs of the AIoT Controller.

6.3.1.5.3 AF

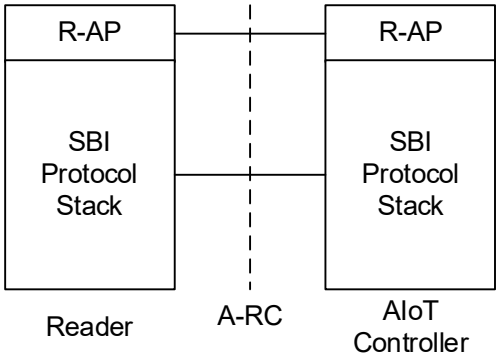
The AF is assumed to support the following functionality:

- Authenticate towards the AIoT Controller.
- Send Inventory and Command Requests.
- Receive Inventory Responses and Command Responses.

NOTE: The AF is assumed to be preconfigured with the IP address or FQDN of the AIoT Controller.

6.3.1.6 Protocol Stacks

6.3.1.6.1 Protocol Stack between Reader and AIoT Controller



Legend:
- **Reader Application Protocol (R-AP):** Application Layer Protocol between the Reader and the AIoT Controller.
- **Service-based Interface (SBI) Protocol Stack:** The protocol stack for service-based interfaces as defined in TS 29.500 [9].

Figure 6.3.1.6.1-1: Control Plane between Reader and AIoT Controller for Topology 1

NOTE 1: Whether an SBI protocol stack (consisting of IP/TCP/HTTP2/JSON) as defined in TS 29.500 [9] or an SCTP-based stack will be used will be decided in coordination with RAN wg3.

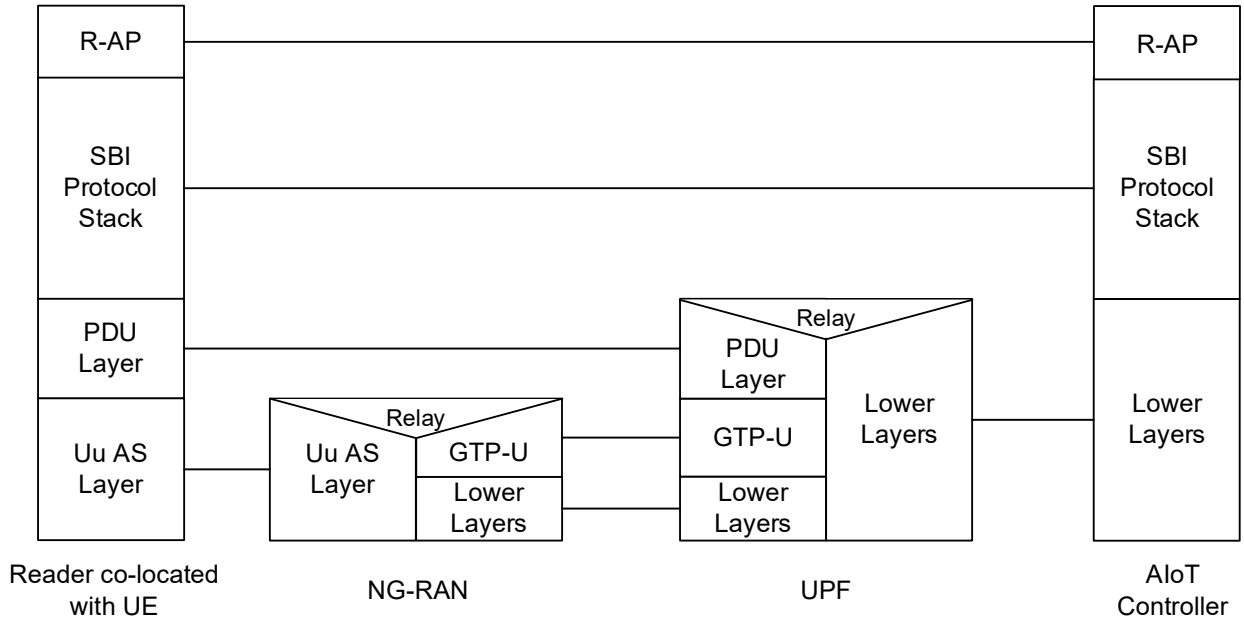


Figure 6.3.1.6.1-2: Control Plane between Reader and AIoT Controller for Topology 2 using a PDU Session between UE and UPF

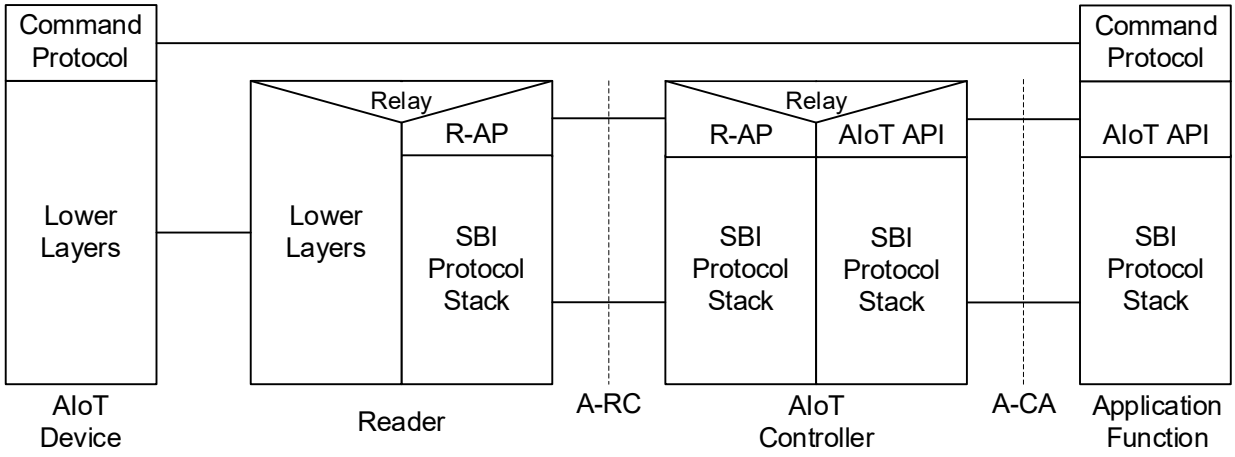
NOTE 2: Whether an SBI protocol stack (consisting of IP/TCP/HTTP2/JSON) as defined in TS 29.500 [9] or an SCTP-based stack will be used will be decided in coordination with CT WG1.

NOTE 3: The R-AP protocol is assumed to be defined by RAN WG3 in coordination with SA WG2.

NOTE 4: The motivation to propose the R-AP protocol instead of reusing NGAP is that (a) NGAP terminates on the AMF, while AMF is not assumed to be used by this solution and (b) that most of the underlying concepts of NGAP (existence of UE contexts at RAN nodes, PDU Sessions, support of UE mobility, etc.) do not apply to AIoT in this solution.

NOTE 5: Whether instead of a new R-AP protocol a simplified version of NGAP can be defined to support AIoT scenarios will be decided in coordination with RAN WG3.

6.3.1.6.2 Protocol Stack between AIoT Device and Application Function



Legend:

- **Reader Application Protocol (R-AP):** Application Layer Protocol between the Reader and the AIoT Controller.
- **Service-based Interface (SBI) Protocol Stack:** The protocol for service-based interfaces as defined in TS 29.500 [9].
- **AIoT API:** The API between AIoT Controller and Application Function to support Inventory and Command Procedures.
- **Command Protocol:** Application Layer Protocol between AIoT Device and Application Function to support Commands and Command Responses.

Figure 6.3.1.6.2-1: Protocol stack between AIoT Device and Application Function for Topology 1

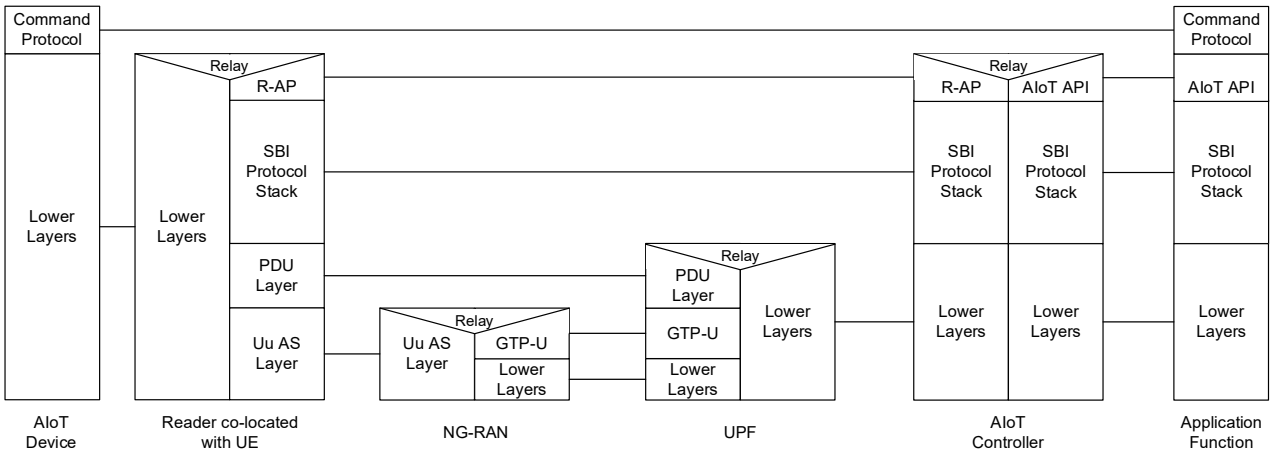


Figure 6.3.1.6.2-2: Protocol stack between AIoT Device and Application Function for Topology 2

NOTE 1: The AIoT API is assumed to be defined by SA WG2 (Stage 2 aspects) and CT WG3 (Stage 3 aspects).

NOTE 2: The Command Protocol is assumed to be defined by SA WG2 (Stage 2 aspects) and CT WG1 (Stage 3 aspects).

6.3.2 Procedures

6.3.2.1 Inventory procedure

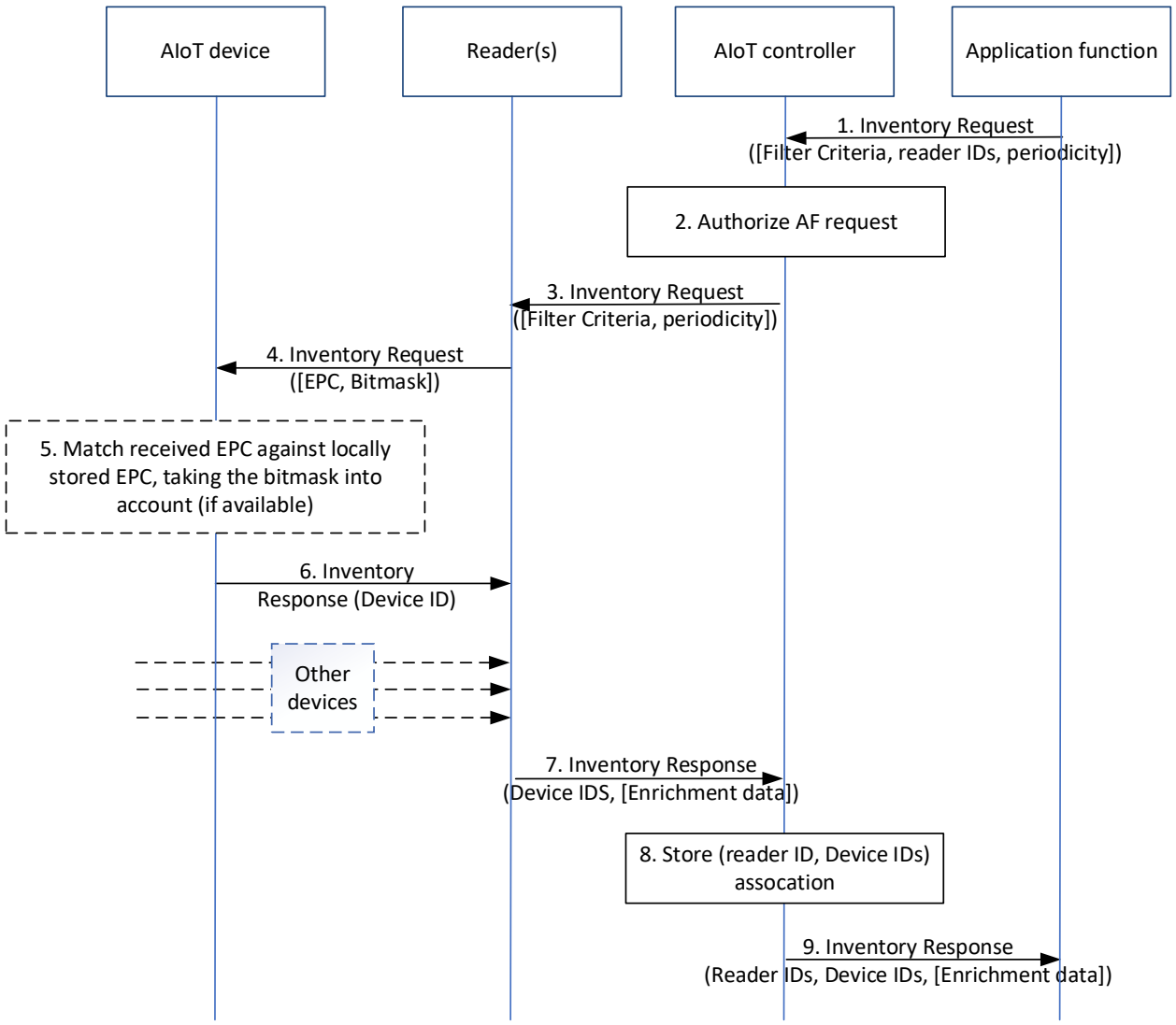


Figure 6.3.2.1: Inventory procedure

1. The AF sends an Inventory Request to the AIoT controller. The AF may optionally include the following information:

   - Filter Criteria to limit the inventory to AIoT Devices matching those criteria;

NOTE 1: If the AF does not provide Filter Criteria, then a full inventory of all AIoT devices reachable by the Reader(s) will be performed.

   - a list of reader IDs to limit the inventory to specific Readers;
   - a periodicity value to request the inventory to be performed periodically.

   As part of the Filter Criteria an AF may provide an EPC and optionally a bitmask. This enables the AF to trigger a targeted inventory:

   - For example, an AF may want to inventory complete shipments only (identified by AIoT devices attached to the pallets that carry the shipment), without having to unnecessarily inventory also all AIoT devices which may be located on the pallets. This can be achieved by limiting the inventory to AIoT devices that carry an EPC based on the Serial Shipping Container Code EPC scheme. To do so, the AF may provide a bitmask that only matches on the Header value of the EPC and an EPC value that contains the

standardized Header value for the Serial Shipping Container Code EPC scheme. (See Annex X for further details on EPC schemes and EPC Header values.).

- Another example is an AF that may only want to inventory individual items from a specific company. This can be achieved by limiting the inventory to AIoT devices that carry an EPC that is based on the Serialised Global Trade Item Number (SGTIN) EPC scheme and where the GS1 Company Prefix contained in the SGTIN matches the company prefix of the specific company that the AF is interested in. To do so, the AF may provide a bitmask matching on the Header value of the EPC and the bits of the GS1 Company Prefix contained in the Serialised Global Trade Item Number (SGTIN) EPC scheme and may provide an EPC that contains the standardized Header value for the Serialised Global Trade Item Number (SGTIN) and the Company Prefix value of the company that the AF is interested in.

  As part of the Filter Criteria an AF may provide a Closest indication. The AF may only include the Closest indication if the inventory is limited to a single reader (i.e. if the AF includes a single reader ID in the Inventory Request). The Closest indication in the Filter Criteria enables the AF to limit the Inventory Response to the AIoT device closest to the Reader.

Editor's note: Whether this requires the AF to have knowledge about how the Reader has determined closeness is FFS. Other solutions are not excluded.

NOTE 2: This assumes that the AF is aware of the Reader ID of the reader for which to trigger the inventory (same as for other solutions that propose a Reader ID for triggering an Inventory on specific Readers).

2. The AIoT controller verifies whether the AF is entitled to make the received Inventory Request, e.g. the AIoT controller verifies whether the AF is allowed to issue an Inventory Request with the specified Filter Criteria or without any Filter Criteria, without providing reader IDs, etc.

3. The AIoT controller sends the Inventory Request to the Readers identified by the Reader IDs or to all readers, includes the Filter Criteria and periodicity information, if provided by the AF.

4. Each Reader that received the Inventory Request from the AIoT Controller performs the Inventory procedure according to the Filter Criteria, if provided. If the Filter Criteria contain an EPC and optionally a bitmask, then the Reader sends the EPC and the bitmask as part of the inventory request. The Readers either perform a one-time inventory or perform the inventory periodically according to the received periodicity.

5. If the Inventory Request contains an EPC, then the AIoT Device compares the received EPC with its Device ID, i.e. the locally stored EPC. If the Inventory Request also contains a bitmask, then the AIoT Device only compares the relevant bits according to the bitmask. If the Inventory Request does not contain an EPC, then the AIoT Device skips this step.

6. If the Inventory Request contains an EPC and optionally a bitmask, and the received EPC matches the locally stored EPC (or parts of the EPC in case a bitmask was provided) as described in the previous step, then the AIoT Device responds to the Reader.

   If the Inventory Request does not contain an EPC, then the AIoT Device responds to the Reader.

NOTE 3: The AIoT Device responds to Inventory Requests that do not contain an EPC to enable the case that an AF wants to perform a full inventory.

   The AIoT Device includes its Device ID in the Inventory Response.

7. The Readers collects the received Device IDs and provides the Device IDs to the AIoT Controller. The Readers may optionally include Enrichment Data. If the Inventory Request from the AIoT Controller contain Filter Criteria including a Closest Indication, then the Reader only provides the Device ID of the AIoT Device closest to the Reader to the AIoT Controller. If the Reader cannot determine a single AIoT Device as the closest one, then the Reader returns an error cause to the AIoT Controller.

NOTE 4: How the Reader determines which of the AIoT Devices is closest to the Reader is assumed to be up to Reader implementation, for example signal strength can be used to achieve this. Whether the Closest Indication can be supported depends on RAN WGs.

8. For each reported Device ID, the AIoT Controller stores the reader ID that reported the Device ID together with a timestamp. The timestamp enables the AIoT controller to purge outdated last known Reader information; the details of this are up to AIoT Controller implementation.

9. The AIoT Controller provides the Device IDs and the Reader ID that reported the Device ID, respectively, and optionally the Enrichment Data to the AF.

NOTE 5: The AIoT Controller provides the Device IDs as received to the AF, i.e. the AIoT Controller is not assumed to verify Device IDs.

6.3.2.2 Command procedure

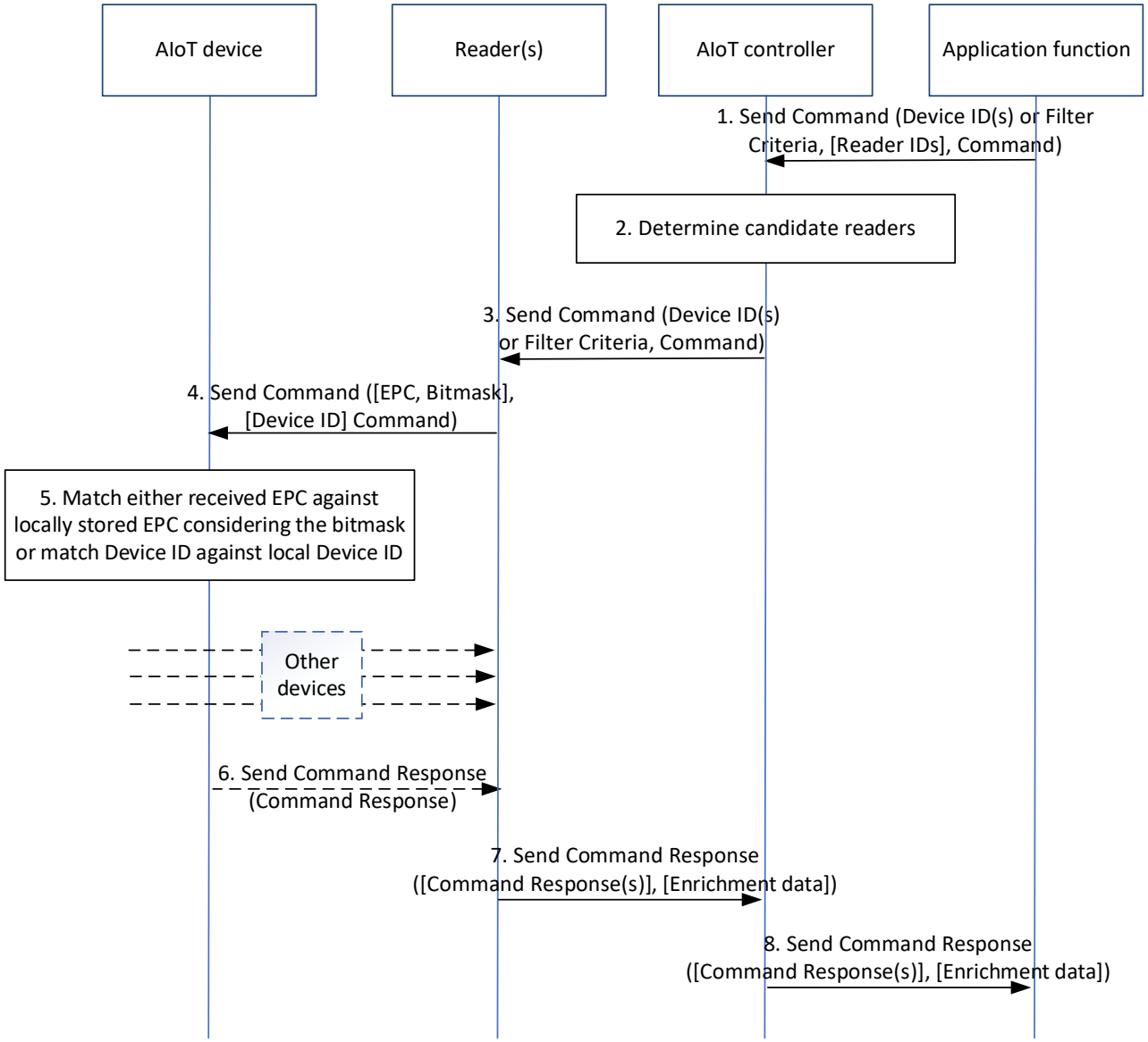


Figure 6.3.2.2: Command procedure

1. The AF issues a Send Command request, which includes the Command to be sent and either:

   - a list of Device IDs that the Command is destined to; or
   - Filter Criteria that identify the AIoT Devices that are supposed to act upon the Command.

   In addition, if the AF provides Filter Criteria, the AF may additionally include a list of reader IDs to use for sending the Command.

NOTE 1: Including the list of reader IDs enables the AF to limit sending of the Command to a specific area.

2. The AIoT Controller determines the list of Readers to use for sending the Command taking the Reader IDs (if provided by the AF) and any stored information about the last known Reader for specific Device IDs into account. The details of determining candidate Readers are up to AIoT Controller implementation.

3. The AIoT Controller provides the Command and the Device IDs or Filter Criteria (whichever has been provided by the AF) to the Readers.

NOTE 2: Whether the Reader needs to perform an Inventory before sending the command to the AIoT devices is a RAN WG2 decision.

4. Each Reader that receives the Send Command request from the AIoT Controller, sends the Command to AIoT Devices as follows:

   - If the Send Command request contains one or more Device IDs, then the Reader sends the Command message to the AIoT Devices identified by the Device IDs.

   - If the Send Command request contains Filter Criteria consisting of an EPC and a bitmask, then the Reader sends the EPC and the bitmask as part of the Command message to AIoT Devices.

5. If the Command Message contains an EPC, then the AIoT Device compares the received EPC with its Device ID, i.e. the locally stored EPC. The AIoT Device only compares the relevant bits according to the bitmask. If the Command Message contains not contain a Device ID, then the AIoT Device skips compares the received Device ID with its local Device ID.

6. If there was a match in the previous step, then the AIoT Device executes the Command and provides the Command Response to the Reader.

7. The Readers send the received Command Response(s) to the AIoT Controller. The Readers may optionally include Enrichment Data. If a Reader has not received any Command Response, then the Reader informs the AIoT Controller accordingly.

8. The AIoT Controller sends the received Command Response(s) and optionally Enrichment Data to the AF. If all Readers that the AIoT Controller sent the Command to in step 3 indicate to the AIoT Controller that no Command Responses have been received, then the AIoT Controller may perform any of the following actions:

   - Attempt to deliver the Command using additional Readers, i.e. continue the procedure from step 2 and select additional Readers;

   - Buffer the Command and attempt sending the Command again later (details of this are up to AIoT Controller implementation);

   - Inform the AF that the Command could not be delivered.

NOTE 3: Support of sending the same Command to multiple AIoT Devices requires support of group security for the Command (e.g. group keys for protecting the Command), which depends on SA WG3.

6.3.3 Impacts on services, entities and interfaces

New network entities and interfaces are proposed.

6.4 Solution #4: Simplified system for AIoT

6.4.1 Description

6.4.1.1 General

The solution addresses KI#1, #2 and KI#3.

It provides an e2e solution to support AIoT services with regard to topology1 and 2.

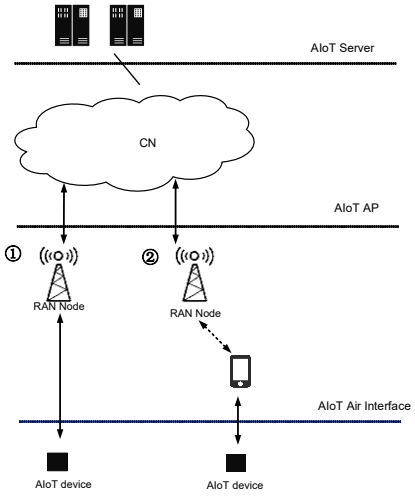


Figure 6.4.1.1-1: Topology 1&2

The following assumptions are taken into account for the solutions

- No matter what Topologies are applied (including more topologies in future release), The Topologies types are transparent to the AIoT devices and the AIoT devices are common designed.
- In topology2, when the UE out of Uu coverage in some blind area but the AIoT air coverage goes well, the AIoT operation at the UE reader will not stop and continue.
- The AIoT Air interface is assumed to be a new air interface and layers above the AS layer is in the scope of SA WG2 study.

6.4.1.2 Abbreviations

6.4.1.3 Terms

The following terms are used for the solution:

- **AIoT operation:** the operation with communicating to AIoT devices, e.g. inventory and command.
  - **Inventory**: filter and/or discovery one or multiple Ambient IoT device(s).
  - **Command**: e.g. read**,** write, control, enable or disable one or multiple Ambient IoT device(s).
- **AIoT function:** the NF providing management and control for AIoT services and AIoT operation.
- **AIoT API:** the service-based API to provide AIoT Services which can be invoked by the AF.
- **AIoT Air interface:** the air interface between reader and AIoT devices.
- **Reader:** the device which supports to communicate one or multiple Ambient IoT device(s). It can operate as a UE and be called UE reader, or can operate as a RAN node and be called RAN reader.

6.4.1.4 Reference Architecture and NFs

6.4.1.4.1 Pure Control Plane

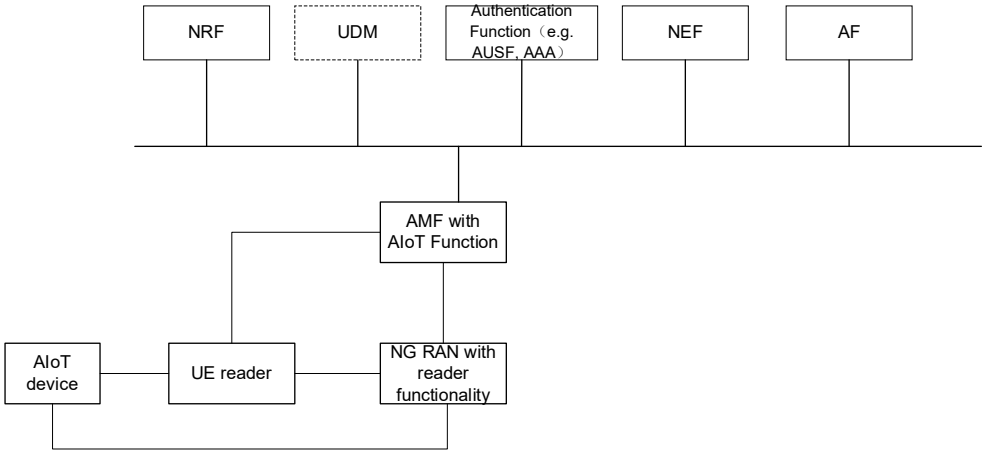


Figure 6.4.1.4.1-1: The example of reference architecture which supporting topology 1 & 2

Up to different deployment, the above NF may be collocated or standalone. The above reference architecture can be separate into the following alternatives:

- Alternative1: a standalone network supporting AIoT functionality only.
- Alternative2: a network supporting both legacy 5GS functionality and additional AIoT functionality.

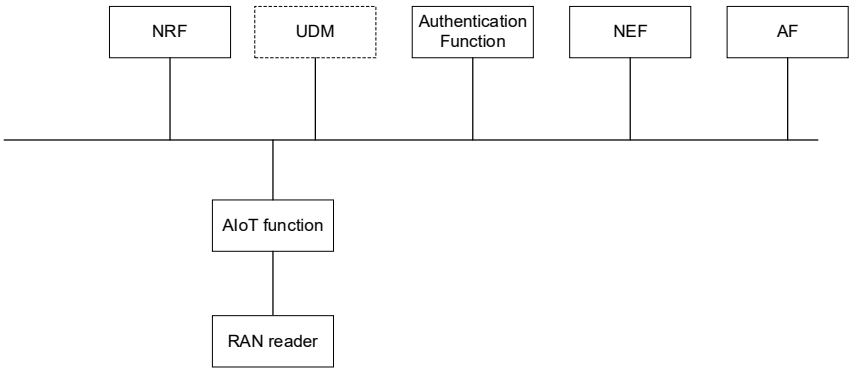


Figure 6.4.1.4.1-2: The example of reference architecture for alternative 1

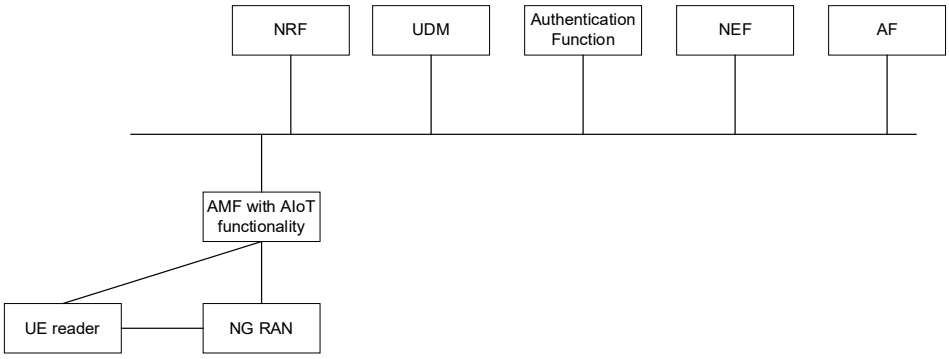


Figure 6.4.1.4.1-3: The example of reference architecture for alternative 2

The functionalities may include the following NFs and/or devices:

- The reader (i.e. UE reader and/or RAN reader).

NOTE: a 5GS UE may additionally support UE reader functionality. a NG RAN may additionally support RAN reader functionality.

- The AIoT Function: provides the AIoT control, which may be collocated with AMF or a standalone NF.
- In some scenario (e.g. operator owned AIoT devices), there might be ta UDM which stores the data of AIoT device.

Editor's note: It is FFS whether it can be assumed for all scenarios that the device and the CN can be pre-provisioned with CN level per device information (e.g. network layer AIoT device ID, security material).

- The NRF is responsible for NF discovery as legacy and support AIoT function discovery.
- The Authentication Function (e.g. AUSF/AAA) provide authentication for AIoT devices. The Authentication Function may be located in the serving network or in the 3rd party.
- The NEF is responsible to authorize the AF request as legacy and support the new scenario of AIoT, i.e. authorize the AIoT operation request from the AF.

6.4.1.4.2 User Plane of UE reader

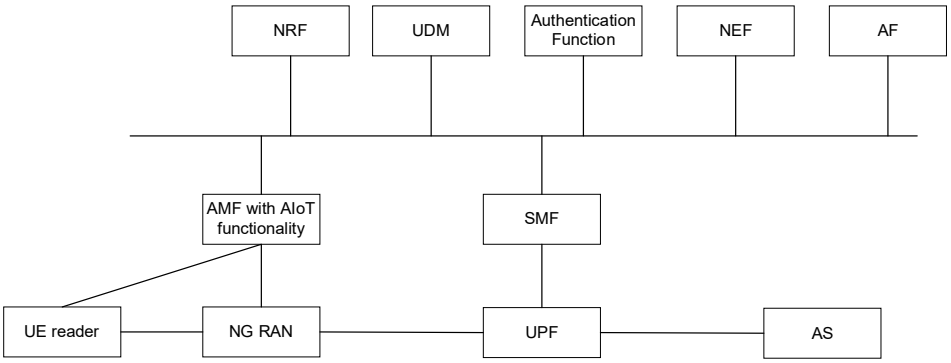


Figure 6.4.1.4.2-1 An example of reference architecture for discovering UE reader via control plane and AIoT device related data delivery via UE reader user plane

6.4.1.5 Protocol Stack

Both the App layer and the AIoT layer are beyond the AIoT AS layer and belongs to non-access layer.

- The App layer are used to transmit the information between the AIoT device and the Server.
- The AIoT layer is used to transmit the information between the AIoT device and UE reader / CN.

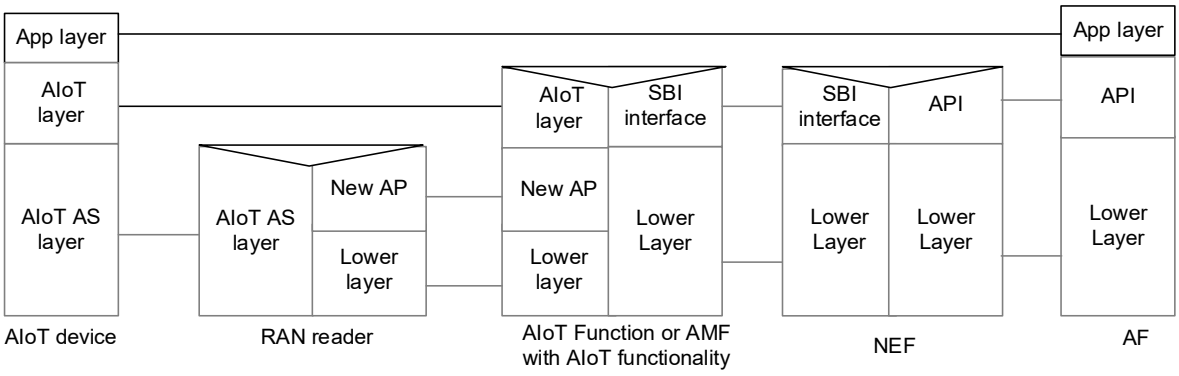


Figure 6.4.1.5-1: protocol stack example for Topology1

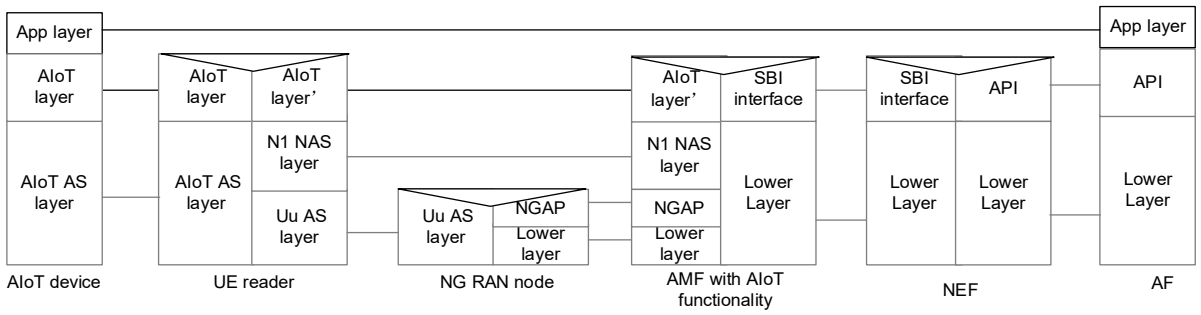


Figure 6.4.1.5-2: protocol stack example-A for Topology2

Both the App layer and the AIoT layer are beyond the AIoT AS layer and belongs to non-access layer.

- The App layer are used to transmit the information between the AIoT device and the Server.
- The AIoT layer is used to transmit the information between the AIoT device and UE reader / CN.

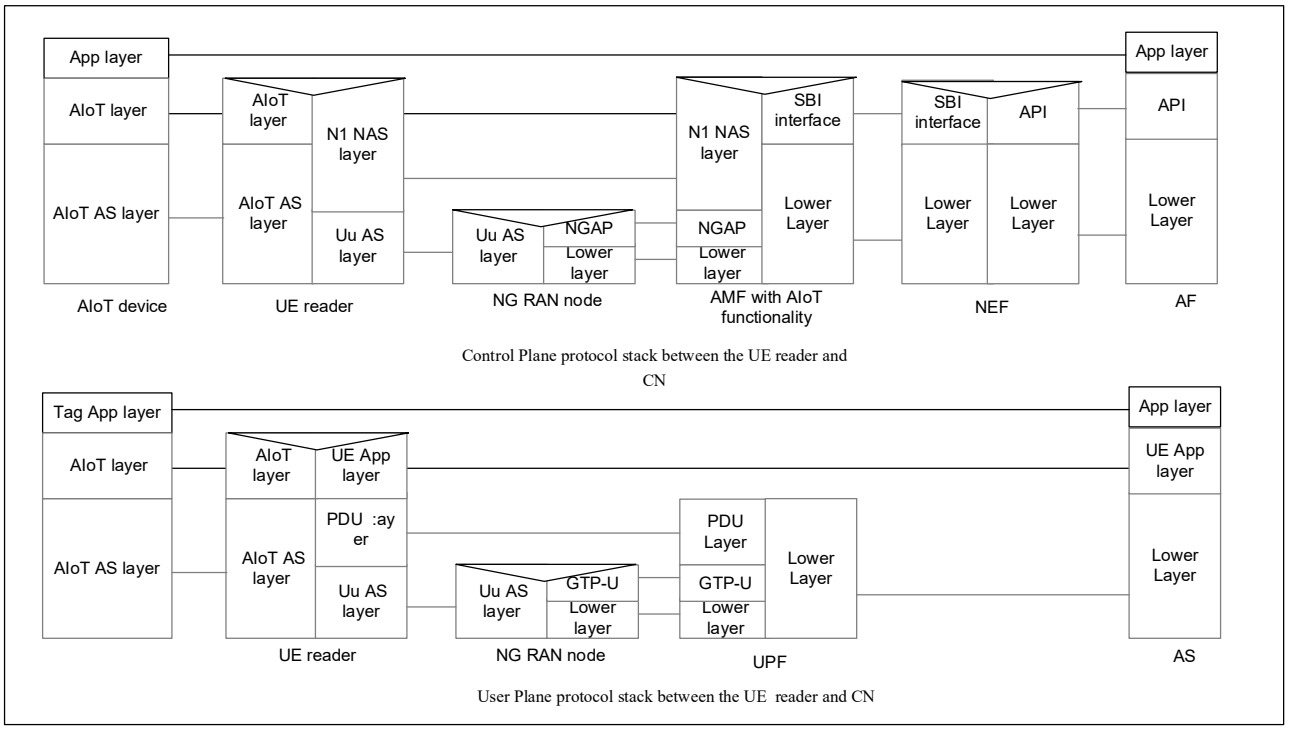


Figure 6.4.1.5-2: Protocol stack example-B for Topology2

In protocol stack example-B for Topology2

- The AIoT layer is used to transmit the information between the AIoT device and UE reader.
- There is no user plane for AIoT devices, but for a UE reader, there are control plane and user plane:
  - In control plane: the AF can request to CN to discover and select reader based on e.g. the target area or reader ID.
  - In user plane: the UE reader can communicate with AS to report the AIoT operation result and interact the AIoT device related information, i.e. the information target to the AIoT device or the information from the AIoT device.

6.4.2 Procedures

6.4.2.1 Purely via control plane: Inventory procedure

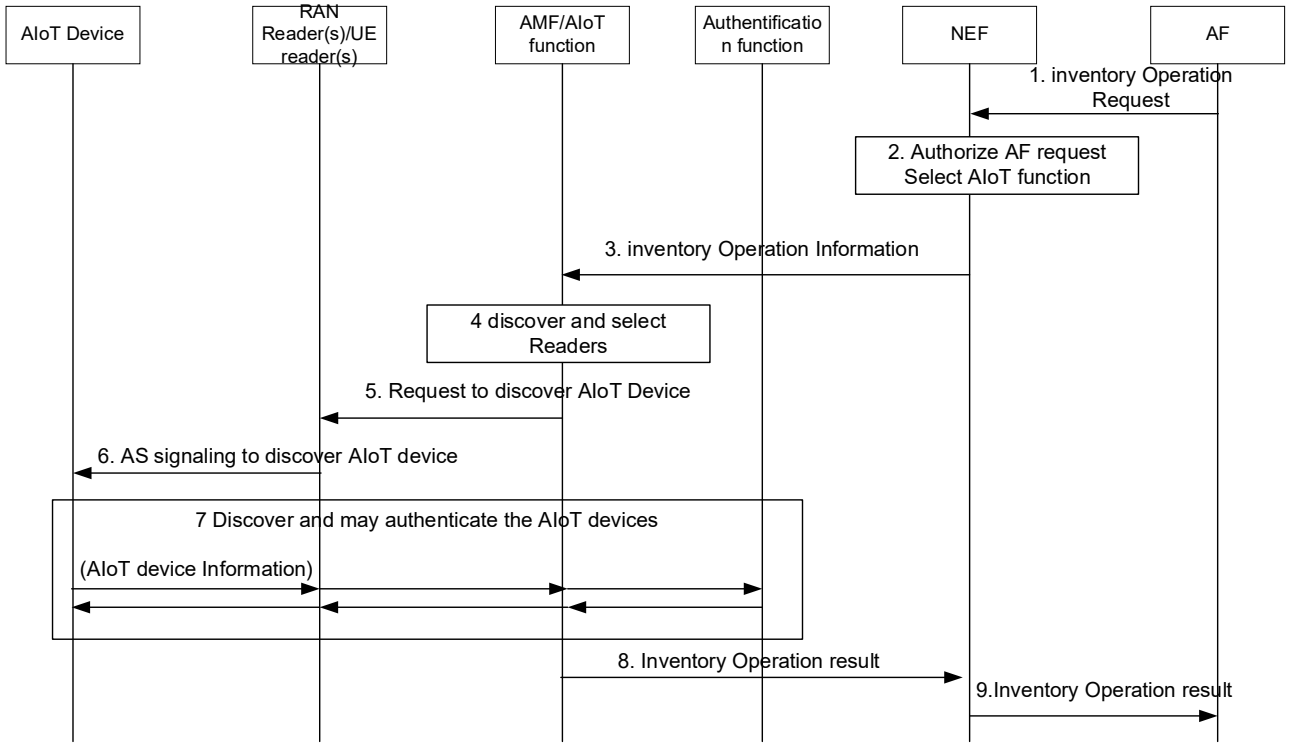


Figure 6.4.2.1-1: Inventory Service Flow

1. AF sends Inventory Operation Request with the following information: the number of AIoT devices, target area for the operation, client which requests the operation, match information which is used to filter and discover the target AIoT devices for the operation. The number of AIoT devices is the exact number or an approximate number range.

NOTE 1: The number of AIoT devices is to assist the resource allocation for inventory at the reader. Generally, resource for different device number are different.

2. The NEF authorizes the AF request. If the AF request is authorized, The NEF discovers the AIoT function using the information in Inventory Operation Request, e.g. using the target area for the operation to discovery the AIoT function from NRF. If the target area for the operation matches the AIoT service area of the AIoT Function, the NRF returns the information for the AIoT function to the NEF.

3. The NEF forwards the Inventory Operation information to each the selected AIoT function.

4. AIoT function discovers and selects reader (i.e. UE reader(s) and/or RAN reader(s)) to perform Inventory Operation according to the Inventory Operation information. e.g. using the target area for the operation to discovery the readers. If the target area for the operation matches the AIoT service area supported by the readers, those readers can be selected for execute the inventory operation.

Editor's note: It is FFS how to discover and select UE readers.

5. For each of the selected reader, the AIoT Function sends Inventory with the Inventory Operation information.

6. The select reader executes the inventory operation towards the target AIoT Devices.

7. The AIoT device which matches to the match information will responses the AIoT device information (e.g. the device ID) to the reader and the AIoT device authentication procedure may be triggered. The step7 can be repeated for multiple AIoT devices.

NOTE 2: How the AIoT device authentication performed is left to SA WG3. How to avoid the leakage of target AIoT devices number caused by forged as inventory from faked ran reader is to be addressed by SA WG3.

8-9. AIoT Function report the successfully discovered AIoT device(s) to the AF vie NEF.

6.4.2.2 Purely via control plane: Command procedure

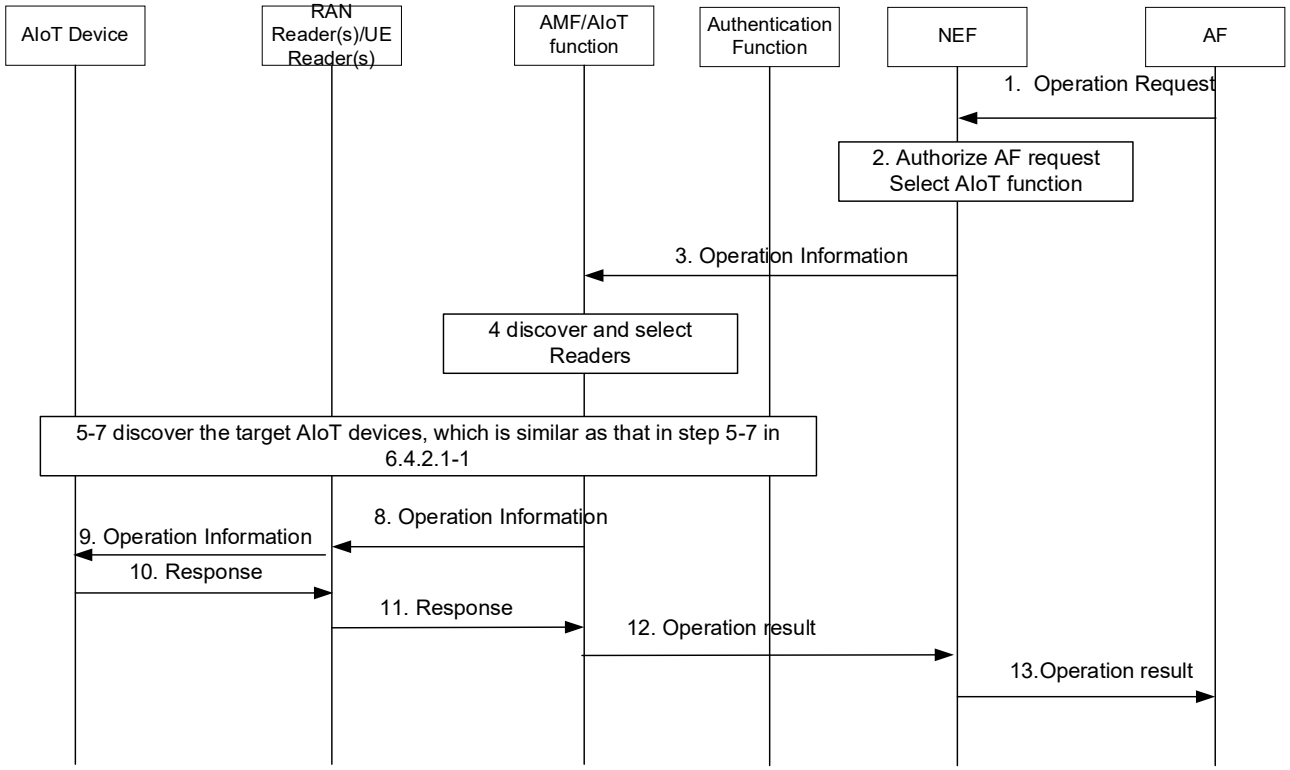


Figure 6.4.2.2-1: Command Service Flow

1. AF sends a command request along with the following information: the number of AIoT devices (if the AF have it), target area for the operation, client which requests the command operation, match information which is used to filter and discover the target AIoT devices for the operation container. The number of AIoT devices is the exact number or an approximate number range. The match information may include the type(s) of the acquired information. The AIoT devices matching the match information need to response.

NOTE 1: The number of AIoT devices is to assist the resource allocation for operation. Generally, resource for different device number are different. If the number reader for AIoT operation is only one, it is useful. If the number of readers for AIoT operation is multiple, it is also useful as a reference to represent the approximate maximum number, which is better than no number is provided.

Editor's note: It is FFS whether the AF provided number of AIoT devices is still useful in case there are multiple readers to perform AIoT operations because the AF provided number of AIoT devices may be more than the number of AIoT devices responding to one reader.

NOTE 2: The reader might require information (e.g. read operation) from the AIoT devices, however, not all of the target AIoT devices have the acquired information. Let's assume there are 1000 devices matching the filtering information of a paging like signalling and trigger response the reader, but only 50 devices with the acquired information, then random access competition under the control of the reader for the leftover 950 device are useless. Add the type(s) of the acquired information in the filtering information can save the useless response from the device without the type(s) of the acquired information.

NOTE 3: The command procedure in this solution assumes the command content is transparent to the CN and can be carried by some container between the AIoT device and AF.

2. The NEF authorizes the AF request. If the AF request is authorized, The NEF discovers the AIoT function using the information in the Operation Request. How AIoT function is discovered and select is as described in step 2 in clause 6.4.2.1.

3. The NEF forwards the Operation information to each the selected AIoT function.

4. AIoT function discovers and selects reader (i.e. UE reader(s) and/or RAN reader(s)) to perform the Operation according to the Operation information. How reader is discovered and selected is as described in step 4 in clause 6.4.2.1.

5-7. Discover the target AIoT devices. How to discover and authenticate the AIoT devices is similar as that within the inventory operation procedure. The AIoT device may also authenticate the reader during the procedure.

NOTE 4: How the reader is authenticated by the AIoT device and how AIoT device is authenticated is left to SA WG3.

8-9. The reader sends the Operation container to each of the target AIoT devices.

10-11. AIoT Function report the operation result to the AF vie NEF/ASAF.

6.4.2.3 Discover UE reader via UE reader control plane and AIoT device related data delivery via UE reader user plane

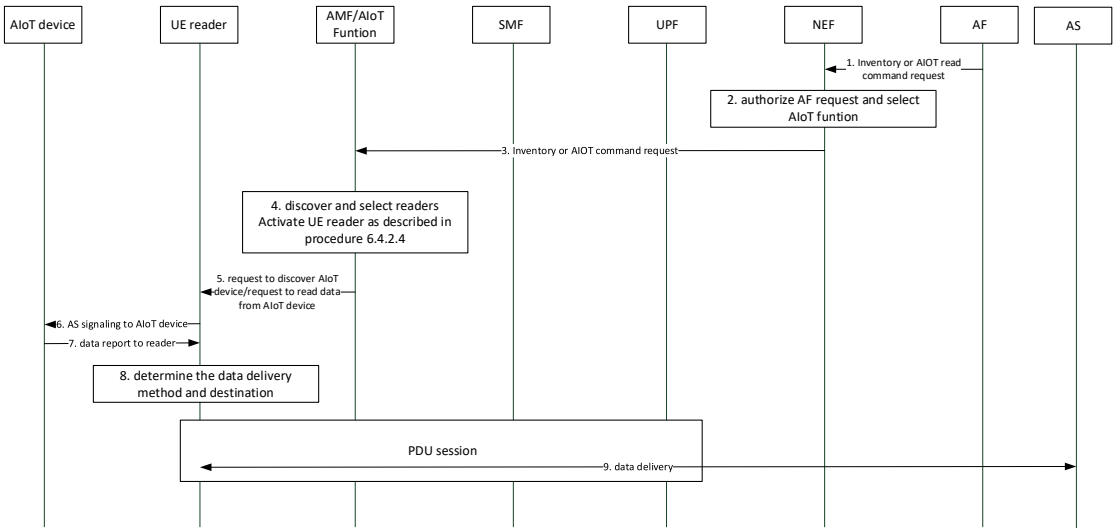


Figure 6.4.2.3-1: Procedure flow of user plane data delivery for multiple AIoT devices

NOTE 1: In this procedure, it is assumed serving network is responsible to provide resource for reader, but the authentication happens among AIoT device, UE reader and Application Server.

Editor's note: Whether there is a need to validate the AIoT Device ID by a serving network for this procedure and if so, how to achieve this is FFS.

1. AF sends AIoT operation request (e.g.: inventory or command) with the following information: the number of AIoT devices, target area, UE reader's ID/IP address, destination IP address of application server, the information used to indicate or filter the AIoT device. The match information may include "there is buffered data" or the type(s) of the acquired information.

2. The NEF authorizes the AF request as legacy. If the AF request is authorized, The NEF discovers the AIoT function using the information in AIoT operation request.

3. The NEF forwards the AIoT operation information to AIoT function.

4. AIoT function discovers and selects reader to perform AIoT Operation according to the AIoT operation information.

5. For each of the selected reader, the AIoT Function sends AIoT operation with the AIoT Operation information. If a UE is select as UE reader, the AMF/AIoT function authorize the UE as reader and let the RAN to allocate the time frequency resource for the UE reader as described in clause 6.4.2.4.

6. The selected reader executes the AIoT operation towards the target AIoT Devices.

7. The AIoT device which matches to the information will responses the AIoT device information (e.g. the device ID) to the reader.

8. According destination IP address of application server, the UE reader interacts the AIoT device related information with application server via PDU session.

NOTE 2: The PDU Session should be able reach to the DN where application server is.

6.4.2.4 UE reader activation

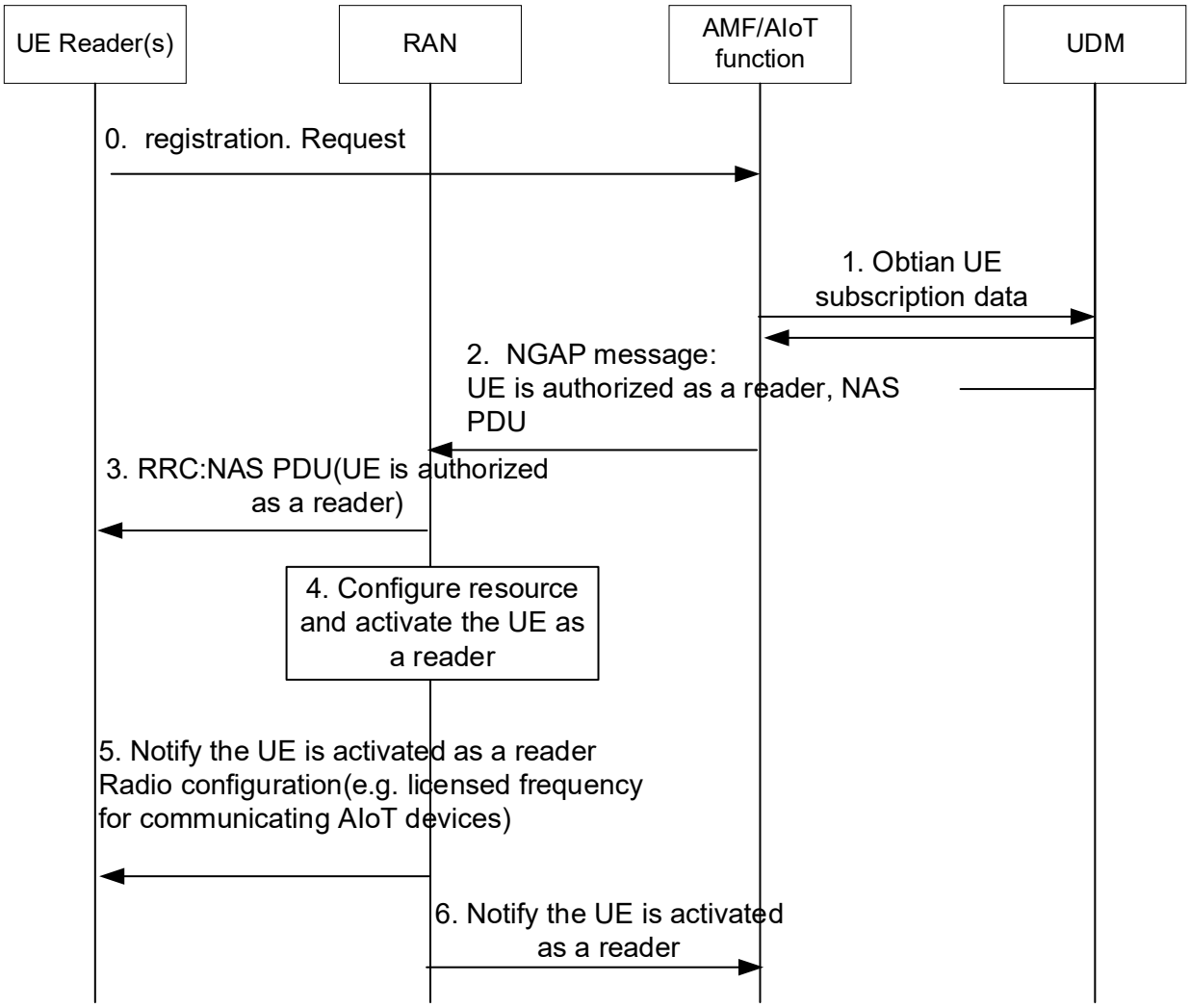


Figure 6.4.2.4-1: Service Flow

0. During Registration Request, the UE reader provide its capability as UE reader.

1. The AMF obtains the UE subscription data during the interaction with UDM.

2-3. Based on the UE subscription, the AMF authorizes the UE as a reader and notifies the authorization information to the RAN node and UE.

4. The RAN node activates the UE as its intermedia node if the UE is authorized as a reader.

5. The RAN node notifies the UE that the UE is activated as a reader and provides the radio configuration (e.g. licensed frequency for communicating AIoT devices).

6. The RAN node notifies the AMF that the UE is activated as a reader.

6.4.3 Impacts on services, entities and interfaces

Editor's note: This clause captures impacts on existing services, entities and interfaces.

6.5 Solution #5: NAS-based information transfer for Ambient IoT Services

6.5.1 Description

The solution applies to the Key Issue#1 "Architecture support of Ambient IoT Devices", the Key Issue #2 "Identification, Subscription, Registration and Connection management" and the Key Issue #3 "Support of Ambient IoT Services". This solution applies to Topology 1 as defined in TR 38.848 [7].

The Figure 6.5.1-1 illustrates the system architecture to support AIoT service for Topology 1.

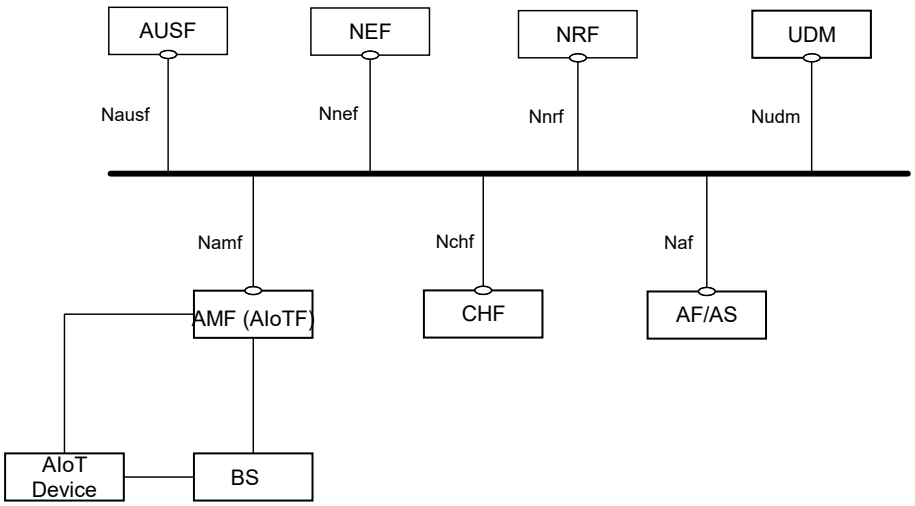


Figure 6.5.1-1: System Architecture to support AIoT Services for Topology 1

The system architecture and functions as defined in TS 23.501 [4] are used as basis to support the Ambient IoT services (e.g. Inventory, Command) with the following clarifications:

- AMF is enhanced to support AIoT Functions, including transfer of service request from AF (via NEF) to AIoT Device (via BS), and transfer of service response from AIoT Device (via BS) to AF (via NEF).

NOTE 1: The AIoT Function could be integrated into AMF or be a standalone entity, and for the latter new interfaces need to be defined.

- NEF is enhanced to support transfer of service request from AF to AMF (AIoTF), and transfer of service response from AMF (AIoTF) to AF.

- UDM is enhanced to support authorize AIoT service request from AF, and AIoT service response from AIoT Device, and stores the mapping between Device ID and served AMF/AIoTF address.

The Figure 6.5.1-2 illustrates the protocol stack to support AIoT services for Topology 1.

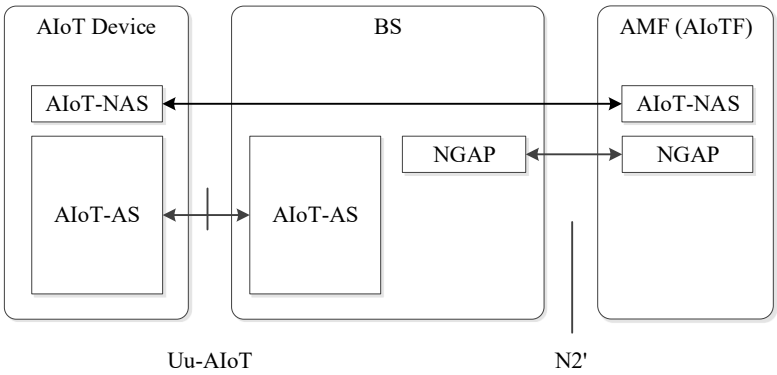


Figure 6.5.1-2: Protocol stack to support AIoT Services for Topology 1

As shown in the Figure 6.5.1-2, it is assumed AIoT Device communicates with BS via a new interface Uu-AIoT. Both Access Stratum layer and Non-Access Stratum layer are supported for AIoT Device.

NOTE 2: The Access Stratum layer functionality is to be defined by RAN WGs.

Considering the nature of the AIoT Devices (e.g. ultra-low complexity power, cost and resource-constrained), the PDU Session/QoS Flow based data transfer is not suitable for such AIoT Devices. This solution proposes to use NAS-based message for information transfer for Ambient IoT Services. The AF requests service operation for an AIoT Device or a group of AIoT Devices via the NEF, and the service operation is forwarded to the AMF/AIoTF (new function introduced to support AIoT services) which then triggers the N2 like procedure and AS procedure to the AIoT Devices. After receiving AS message, the AIoT Devices includes Device ID and AIoT data requested by service operation in the NAS message and send it to the AMF/AIoTF, and then the AMF/AIoTF sends the Device ID and AIoT data to the AF via the NEF.

6.5.2 Procedures

Depicted in Figure 6.5.2-1 is the procedure for NAS-based information transfer for Ambient IoT Services.

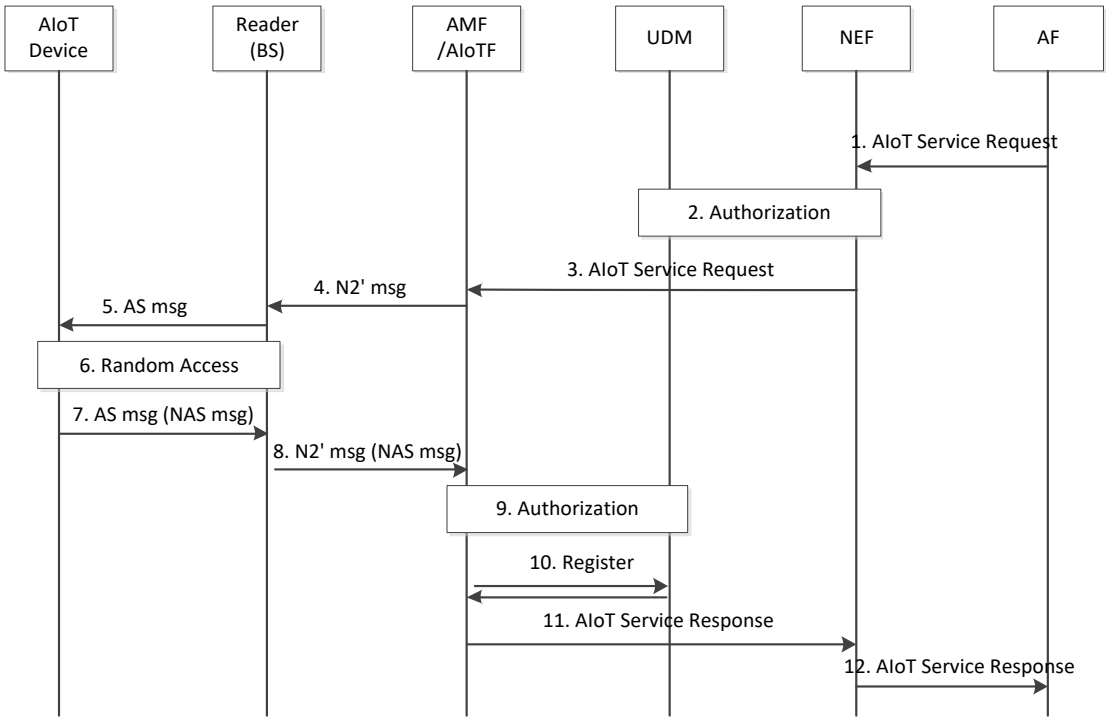


Figure 6.5.2-1: Procedure for NAS-based information transfer for Ambient IoT Services

1. AF sends an AIoT Service Request (AF Identifier, Device ID, requested service operation, area information) message to the NEF. The requested service operation indicates the service (e.g. Inventory, Command) the AF requested for the Ambient IoT Device(s). Device ID is used to identify an Ambient IoT Device or a group/all of

Ambient IoT Devices. The area information is used to indicate the area the service operation will be applied, and the area information could be geographic area.

The service operation could be an end-to-end parameter between AF and Ambient IoT Device which means it is transparent to CN and BS, it also could be a parameter that can be understood by CN. The following steps further clarify how these two options work.

2. The NEF checks if the AF is authorized to request the AIoT service and requests for the served AMF/AIoTF information. The NEF provides the Device ID to the UDM, and the UDM may return the AMF/AIoTF information if it has served the AIoT Device before.

3. The NEF selects the AMF/AIoTF supporting AIoT service either using the served AMF/AIoTF information retrieved from UDM in step 3 or mapping the area information to the corresponding AMF/AIoTF, and forwards the requested service operation to the AMF/AIoTF using AIoT Service Request (Device ID, requested service operation) message.

Editor's note: The details on Reader (BS) selection are FFS.

4. The AMF/AIoTF may derive the service operation based on the requested service operation and include the service operation in an IoT-NAS message, or the AMF/AIoTF may include the requested service operation as a container in an IoT--NAS message. The AMF/AIoTF sends a N2 like message (NAS message (Device ID, service operation)) to Reader (BS).

NOTE 1: It is assumed an IoT-NAS message is used between AMF/AIoT Function and AIoT Device, and the IoT-NAS message is based on connection-less transmission.

5. The Reader (BS) performs AS procedure (e.g paging-like procedure) with Ambient IoT Devices. The IoT- NAS message (Device ID, service operation) is included in the AS message to Ambient IoT Devices.

6. For each Ambient IoT Device, if Ambient IoT Device receives AS message and is matched with the Device ID, the Ambient IoT Device initiates Random Access like procedure.

NOTE 2: The steps 5 and 6 are to be defined by RAN WGs.

7. The Ambient IoT Device sends the IoT-NAS message (Device ID, AIoT data) over AS message. If the service operation is Inventory, the Device ID is included in the IoT- NAS message. If the service operation is Command (e.g. Read), the AIoT data along with the Device ID are included in the IoT-NAS message if requested by the service operation.

8. The IoT- NAS message is forwarded by Reader (BS) to the AMF/AIoTF.

Editor's note: The security protection of IoT-NAS message including group security is FFS.

9. The AMF/AIoTF checks if the Ambient IoT Device is authorized to transfer the AIoT data.

Editor's note: It is FFS whether it can be assumed for all scenarios that the device and the CN can be pre-provisioned with CN level per device information (e.g. network layer AIoT device ID, security material).

10. The AMF/AIoTF register its address information along with the AIoT Device ID to the UDM.

11. The AMF/AIoTF responds to the NEF using AIoT Service Response (Device ID, AIoT data) message.

12. The NEF responds to the AF using AIoT Service Response (Device ID, AIoT data) message.

6.5.3 Impacts on services, entities and interfaces

NEF:

- The NEF authorizes AF request, retrieves the served AMF/AIoTF information or maps the area information to the corresponding AMF/AIoTF, and forwards service request/response between AF and AMF/AIoTF.

AMF/AIoTF:

- The AMF/AIoTF selects BS and delivers service request to the AIoT Device via BS, and receives service response from the AIoT Device and performs validation.

AIoT Device:

- The AIoT Device receives service request from the AMF/AIoTF and responds with service response to the AMF/AIoTF.

6.6 Solution #6: AIoT device authentication, ID validation and AIoT communication

6.6.1 Key Issue mapping

This solution addresses the following requirements:

- from Key Issue #1 the requirement:
  - *System architecture identified along with the solutions for KI#2 and KI#3.*
  - *Validation of the Ambient IoT Device identifier.*
  - *Whether and how to secure device operations and services for an Ambient IoT Device or a group of Ambient IoT Devices.*
- from Key Issue #2 the requirements:
  - *Study whether subscription management, registration management and/or connection management are necessary for an Ambient IoT Device or a group of Ambient IoT Devices, and if so identify the necessary state machine(s), procedures and functionality considering the Ambient IoT Devices capability and characteristics.*
  - *Study whether and how reachability and paging apply to Ambient IoT Device(s) considering the Ambient IoT devices capability and characteristics, and if so, what are the impacts.*
- from Key Issue #3 the requirements:
  - *Study how to support information transfer for Ambient IoT services and related system functionality, including the information transfer for an Ambient IoT device and for a group of Ambient IoT Devices.*

  NOTE: Including whether there is a need to support session based transfer between Ambient IoT Device and the network considering the device types and capabilities.
  - *Study which of the enabled Ambient IoT services are exposed to AF and how, e.g. for the case AF requests Ambient IoT service for an Ambient IoT Device and for a group of Ambient IoT Devices.*

6.6.2 Functional Description

The security protection of AIoT data can be categorized in the following way:

- Application layer security between the AIoT device and AIoT server. The application layer security is outside the scope of 3GPP.
- Network layer security between the AIoT device and network (NG-RAN and 5GC). The AIoT data/signalling transmitted to/from the AIoT devices may need to be security protected, i.e. encrypted and/or integrity protected.

This solution proposes to apply network layer security protection between the AIoT device and a new NF called Ambient AIoT gateway (AIoT-GW) which is introduced in the 5GC control plane. The AIoT-GW also handles the AIoT information transfer from the 5GC to the AIoT devices.

The assumed architecture for AIoT device authentication and communication is shown in Figure 6.6.2-1. The architecture covers both topology 1 and topology 2 as introduced in Architectural Assumptions in clause 4.1. The AIoT reader includes either a UE as intermediate node or base station (BS) node.

It is assumed that the AIoT reader and the AIoT-GW establish an association for AIoT communication as follows:

- The BS acting as AIoT reader establishes a N2' association with the AIoT-GW where the protocol used on the N2' association may be a different from NGAP or simplified NGAP.

- The UE acting as AIoT reader is registered with the AMF via the N1 interface. The communication between the UE and AIoT-GW is not in the scope of this solution, i.e. will be described in solutions for topology 2 (e.g. the UE may exchange with the AIoT-GW via a PDU Session or the AMF may register the UE with the AIoT-GW like registering with the SMSF, etc.).

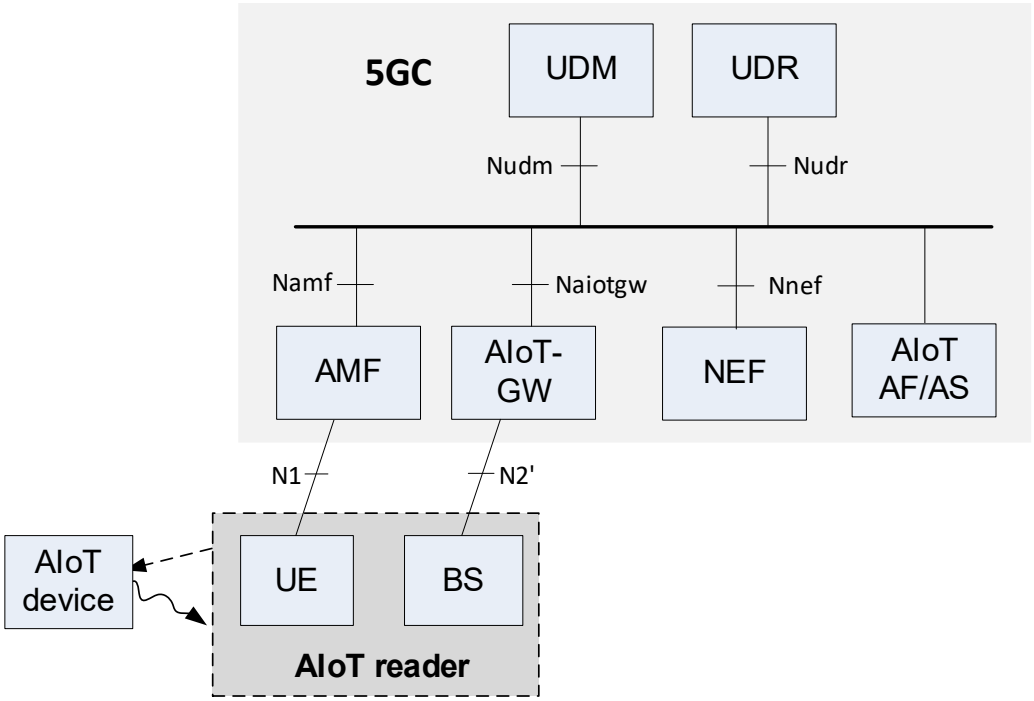


Figure 6.6.2-1: Assumed architecture for AIoT device authentication and AIoT communication

In particular, this solution applies for inventory AIoT service or other services where the destination of the AIoT data/signalling is either a group of AIoT devices or all devices subscribed for the AIoT service The main principles of the solution are as follows:

- The security protection of the DL AIoT data/signalling is created in the AIoT-GW by applying either 1) the group/service credential(s) corresponding to the destination group/service ID or 2) the credentials corresponding to the destination individual AIoT device ID. The AIoT device uses the preconfigured credential(s) corresponding to the destination ID (i.e. either group ID or individual ID) to verify the received DL AIoT data/signalling.

- The security protection of the UL AIoT data/signalling is created in the AIoT device by applying individual credential corresponding to the individual device ID. The individual device ID is transmitted together with the UL AIoT data. The AIoT-GW uses the device individual credentials from the AIoT device subscription data to verify the device individual ID and the UL AIoT data/signalling.

It is further proposed that the AIoT devices do not implement USIM module, do not register with the 5GS and no CM state is maintained in the 5GS. The UDM/UDR is provisioned with AIoT device subscription data which may contain (in addition to the identifiers and credentials) a status and location for the device. The status and location of the AIoT device can be updated in the UDM by the AIoT-GW when the AIoT device sends AIoT data/signalling.

Editor's note: The storage of AIoT device subscription in the of UDM/UDR is intended for the use case of network-controlled authentication/verification of the AIoT device. This assumption of network-layer security for the AIoT device will be evaluated during the conclusion phase.

Editor's note: This solution assumes 1) the transmission of the IDs over the radio interface without encryption and 2) the use of group credentials for security protection of a group of devices (e.g. configured in the UDM/UDR and AIoT devices). Coordination with the SA WG3 WG is needed to verify the feasibility.

6.6.3 Procedures

The Figure 6.6.3-1 describes the signalling flow for the authentication and ID validation of AIoT devices and secure AIoT communication.

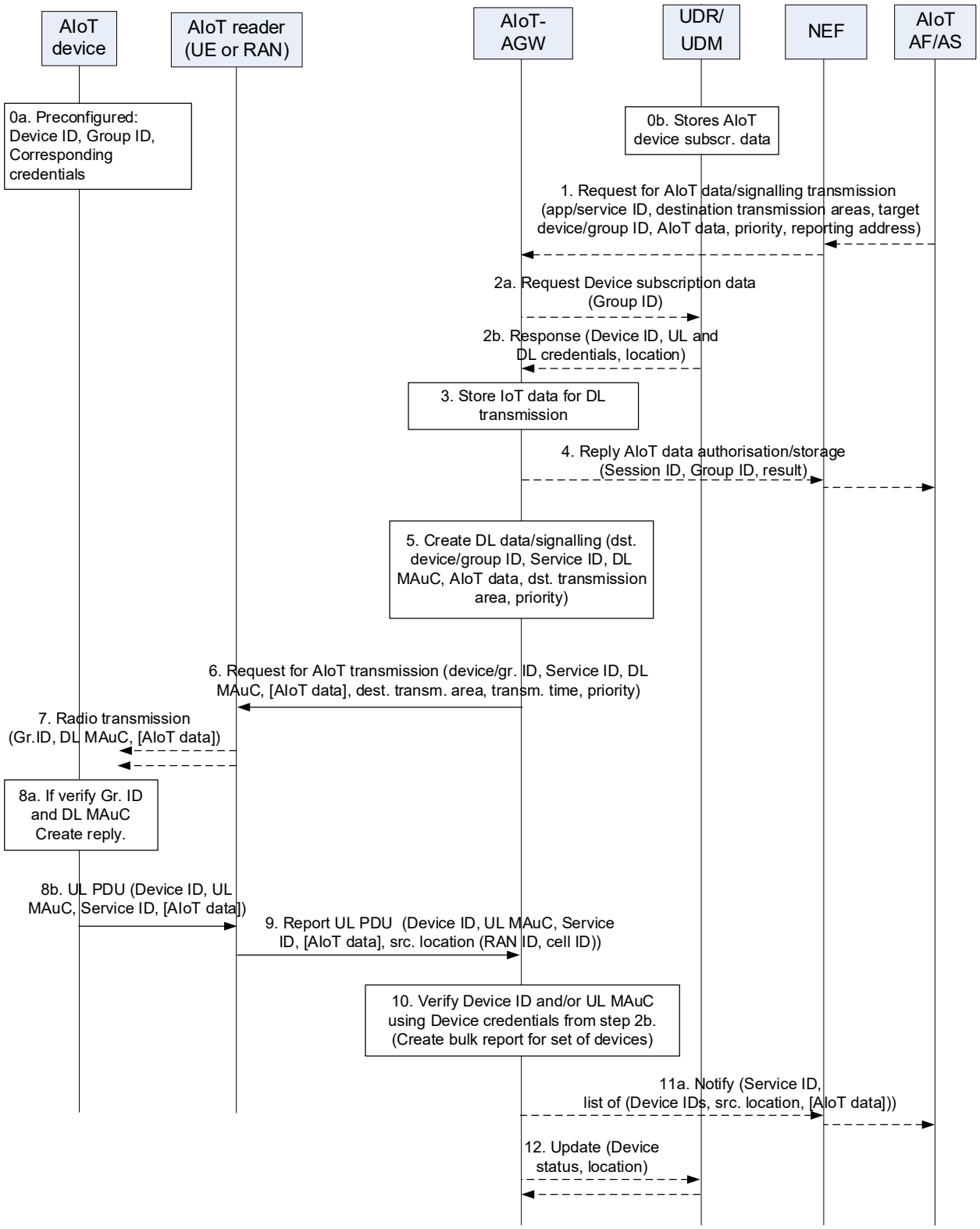


Figure 6.6.3.1-1: Signalling flow for the authentication and ID validation of AIoT devices and secure AIoT communication

The detailed description of the steps is provided as follows:

0a. The AIoT device is (pre-)configured with a device individual identifier and corresponding credentials, as well as with group/service identifier and corresponding credentials.

- Device individual ID and corresponding credentials are used for the security protection of the individual ID and the AIoT data destined to the individual ID. The device ID may be in the format of <hardwareID@MNO-ID> or <hardwareID@manufacturer-ID>.

- Group ID and corresponding credentials are used for the security protection of the group ID and AIoT data destined to the group ID. The group of devices can be organized by sharing the same service, so the group IDs may be in the format of <applicaitonID@SP-ID> wherein the SP-ID is the service provide ID or AIoT AF/AS ID.

0b. The UDM/UDR is provisioned with subscription data for each AIoT device, for which the AIoT service is enabled in the communication system. The AIoT device subscription data can include the following parameters:

- Individual device ID and corresponding credentials for security protection.

- a list of AIoT services (or applications) for which the device is subscribed, including the group/service IDs and corresponding credentials for security protection.

- a status of the individual ID validation, e.g. the status being success or failure. This status may be updated by the AIoT-GW when the AIoT-GW performs verification of the individual device ID, e.g. when the AIoT device sends UL AIoT data.

- enabled or disabled state for the AIoT device. It means whether the AIoT device is allowed or not allowed to transmit AIoT data or to be on service.

- location information of the AIoT device, e.g. last AIoT reader node ID or AIoT cell ID.

1. The AIoT-GW is triggers DL data/signalling to the AIoT device or a group of AIoT devices. This may be triggered by data received from the AIoT AF/AS. The request from the AIoT AF/AS can include: AIoT Session ID, list of destination transmission areas, destination Group ID or list of Device IDs, AIoT data (e.g. containing a "command" to the device) which may include the AIoT service type, AIOT service policy (priority for the AIOT data transmission, service duration, periodic transmission frequency if it’s periodic transmission, aggregation waiting time for group transmission if the communication is for a group of AioT devices), reporting AIoT AS address.

2a. The AIoT-GW may retrieve from the UDM either the AIoT device subscription data (if device individual ID is included instep 1), or service/application subscription data or both.

2b. The response from the UDM may include the information described in step 0b.

3. The AIoT-GW stores the IoT data/signalling for DL transmission. In this sense, the AIoT -GW applies the store-and-forward functionality for DL data.

4. The AIoT-GW can send a response to the AIoT AF to acknowledge (or indicate failure) that the AIoT data/signalling from step 1 has been authorised and stored for DL transmission.

5. The AIoT-GW creates DL AIoT signalling for transmission or store the AIoT data received in step 1. The AIoT signalling is meant to be processed by the AIoT reader node which performs the AIoT radio transmission. The DL AIoT data/signalling may contain:

- destination identifier, e.g. individual device ID or group ID.

- service/application identifier, e.g. AIoT service ID, which is meant to identify the AIoT application at the AIoT device, since there may be multiple AIoT applications installed or running at the AIoT device. This identifier may further include a port ID of the AIoT application.

- DL Message Authentication Code (MAuC) corresponding to the destination identifier.

- DL AIoT signalling or DL AIoT data. The difference is that the AIoT signalling is used for inventory type of service where no data/command is written in the AIoT device, whereas the AIoT data carries a data/command to be written or stored in the AIoT device.

- AIOT service policy includes destination transmission area, priority, service duration (e.g. start and stop time), periodic transmission frequency during the service duration if it’s periodic transmission, etc.

The AIoT-GW also selects an AIoT reader node (e.g. BS or UE as intermediate node) to which to forward the request for AIoT data/signalling transmission.

6. The AIoT-GW sends the AIoT data/signalling request message to the selected AIoT reader. The AIoT data request message may the parameters listed in step 5.

7. When the AIoT service duration time starts if those parameters are presented in step 5, The AIoT reader transmits the received AIoT data or AIoT signalling over the radio interface to the AIoT device.

8a. The AIoT device verifies whether the authentication code (e.g. MAuC) and the destination identifier (device ID or group ID) are correct by using the credential as described in step 0a.

8b. The AIoT device creates an UL AIoT data/signalling for transmission (e.g. for backscattering). The UL message or PDU is transmitted using the individual device ID, optional group ID if group ID is included in the DL AIoT data/signalling and security protected by UL MAuC created with the credentials corresponding to the individual device ID.

9. The AIoT reader can create and sends an AIoT report to the AIoT-GW. The AIoT report includes the device ID, optional group ID if group ID is included in the DL AIoT data/signalling, UL MAuC, AIoT service ID, AIoT data/signalling, AIoT reader ID (e.g. including the location identifier).

10. The AIoT-GW verifies the Device ID, or Group ID, and/or UL MAuC using the credentials for the individual device ID (e.g. received in step 2b) or group ID.

11. The AIoT-GW creates and transmits a notification message to the AIoT AF/AS which includes AIoT Serving ID, AIoT signalling result (Device ID, status (e.g. type of good to which the device is attached), ID verification failed/succeeded), AIoT device location. If the AIoT report includes group ID, AIoT-GW performs aggregation and buffering the UL data of the devices with the same group ID during the aggregation waiting time provided by the AF/AS. When the aggregation waiting time expires, the AIOT-GW sends the UL reports of those AIoT devices belonging to the same group.

12. The AIoT-GW can update to the UDM for the AIoT device ID status (e.g. verified ID) or device location.

6.6.4 Impacts on existing services, entities and interfaces

The following NFs and interfaces are impacted:

- AIoT-GW: a new NF with the functionality described in clause 6.6.2. AIoT-GW can support the functionality to aggregate the UL data of the devices with the same group and provide the aggregated data to the AF.

- AIoT reader (BS or UE as intermediate node): supporting 1) AIoT capabilities to communication with the AIoT devices and 2) capabilities to receive and store AIoT signalling/data from the an AIoT-GW and to transmit AIoT report to the AIoT-GW.

- UDM/UDR: store AIoT device subscription data and AIoT application subscription data.

6.7 Solution #7: Bulk Introduction of Devices to the Network

6.7.1 Description

This solution is targeting KI#2 Identification, Subscription, Registration and Connection management. It is related with KI#1 as well.

Regarding the Registration, the following should be considered.

- AIoT Devices have ultra-low complexity/power/cost and are resource-constrained.

- The network needs to support a huge number of AIoT Devices.

Considering the above, registration from each of the AIoT Device to the network is difficult both for the AIoT Device and for the network.

This solution proposes a Bulk Introduction of Devices to the Network performed by AF based on pre-Registration of Ambient IoT Devices (Stores the credentials of Ambient IoTs per a service). The pre-Registration of Ambient IoT Devices may be done during provisioning phase, even before Ambient IoT Device deployment.

It is assumed that UDM is provisioned with the Ambient IoT Device data before bulk introduction procedure.

After this Bulk Introduction of Devices to the Network, AMF has context for the Ambient IoT devices, and so the messages from the devices need not undergo additional authentication/authorization checks.

6.7.2 Procedures

Bulk Introduction of Devices to the Network procedure for Ambient IoT ID is depicted in Figure 6.7.2-1.

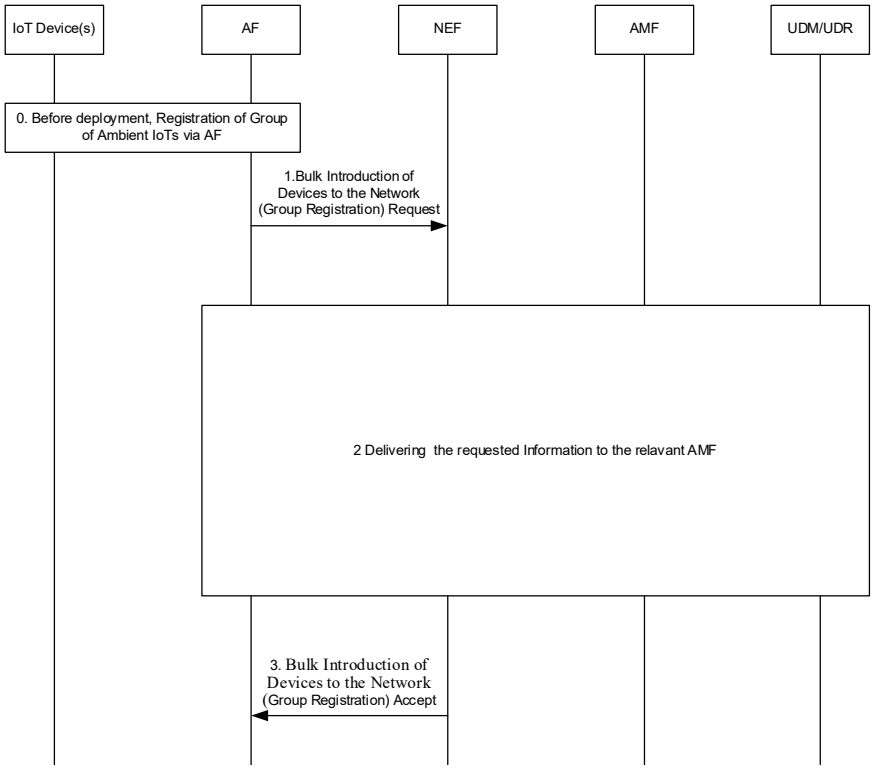


Figure 6.7.2-1: Bulk Introduction of Devices to the Network procedure for Ambient IoT

0. Before deployment, Registration of Group of Ambient IoTs via AF is done. AF stores all the credentials of Ambient IoTs per a service

1. Bulk Introduction of Devices to the Network Request sent by AF to NEF. Bulk Introduction of Devices to the Network Request includes AF ID, list of device IDs, Registration Type. The Registration type can be Group Initial Registration.

2. Delivering the request information to the relevant AMF.

Editor's note: How to find relevant AMF is FFS, For example, AMF can be selected by pre-configuration.

Editor's note: In this step, whether and how the procedure for Authentication for Group of Ambient IoT devices is FFS.

3. NEF sends Bulk Introduction of Devices to the Network Accept to AF.

6.7.3 Impacts on services, entities and interfaces

AF:

- Performs Bulk Introduction of Devices to the Network with the PLMN. This Registration of devices can be done before deployment.

- Stores all the credentials of Ambient IoTs per a service.

NEF:

- Supports signalling for Bulk Introduction of Devices to the Network between AF and AMF.

AMF:

- Supports Bulk Introduction of Devices to the Network of Ambient IoT Devices.
- Retrieves Access and Mobility Subscription data for group of devices from UDM. Subscription data for group of devices can be stored in AF.
- Creates UE context of group of devices. UE context of group of devices is identified by AF ID.

UDM:

- Supports retrieval of subscription data for group of devices from UDM.
- May authenticate and authorize whole devices at a time.

6.8 Solution #8: Inventory for AIoT Devices using AIOTF

6.8.1 Description

This solution addresses subscription, registration and connection management aspect of Key Issue #2, which is based on the following system architecture.

The functional entities defined in TS 23.501 [4] are reused with the exception for the following additions:

- Ambient IoT Function (AIOTF): AIOTF is introduced to support AIoT services, with some AMF's functionalities integrated, which includes:
  - A-RAN (Ambient IoT RAN) connectivity.
  - Inventory handling and device context management.
  - Authentication and authorization for the access, which triggers interaction with AUSF/UDM.
  - Collect charging data and interact with CHF for charging.
  - Routing the request from AF (via NEF) to A-RAN, for DO-DTT/DT traffic types.
  - Routing the response from A-RAN to AF (via NEF) for DO-DTT traffic type.

Editor's note: It is FFS whether AIOTF needs to support further functionalities.

- UDM: UDM is enhanced to store and manage the AIoT device information. The device information contains the device ID, device status information (e.g. enabled/disabled/permanently disabled), as well as CN related information (e.g. serving NF).
- NEF: NEF is enhanced to expose AIoT specific services towards AF.
- CHF: CHF is enhanced for the charging for AIoT services.

NOTE 1: The charging aspects is to be studied by SA WG5.

- NRF: NRF is enhanced to support the new NF type AIOTF and the corresponding NF profile.
- AUSF: AUSF is enhanced for the authentication for the access from AIoT devices.

NOTE 2: The security aspects are to be studied by SA WG3.

Editor's note: The functions of the NFs need to be updated to align with the solutions for KI#2 and KI#3, and to be aligned with A-RAN WGs, SA WG3 (for security), SA WG5 (for charging).

Editor's note: It is FFS whether it can be assumed for all scenarios that the device and the CN can be pre-provisioned with CN level per device information (e.g. network layer AIoT device ID, security material).

The Figure 6.8.1-1 illustrates the enhanced architecture to support AIoT devices using AIOTF for inventory.

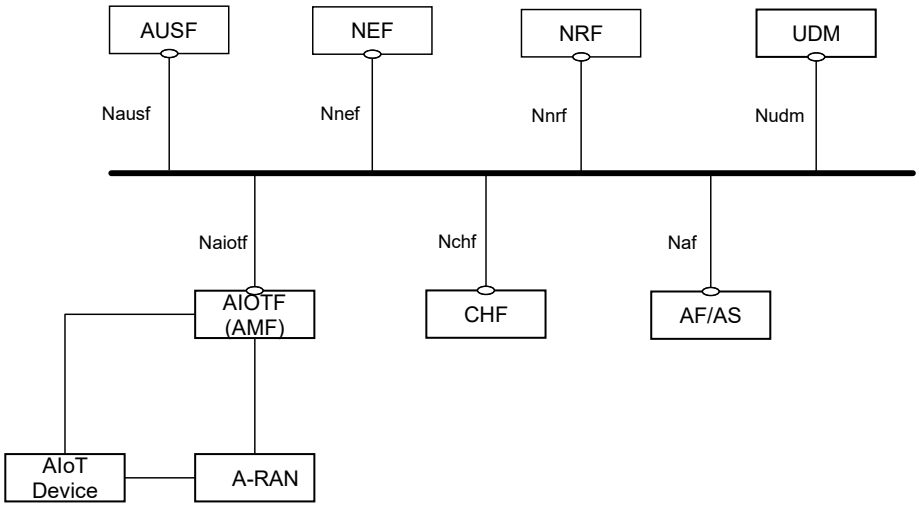


Figure 6.8.1-1: System Architecture to support AIoT Devices using AIOTF for Inventory

Editor's note: It is FFS whether and how to address other services/use cases than inventory with this architecture.

Editor's note: It is FFS whether and how much of NGAP is re-used for interface between A-RAN and AIOTF.

This solution focuses on Topology 1.

Editor's note: It is FFS whether and how the solution can be evolved for Topology 2.

The AIoT device information can be stored in UDM, which is similar as the subscription data. The device information contains the device ID, device status information (e.g. enabled/disabled/permanently disabled), as well as CN related information (e.g. serving NF).

For enabling registration management, the some AIoT devices are assumed not to be able to initiate the registration on their own. However, such passive AIoT devices can respond to messages from the network, which make them discoverable by the network. The procedure that is used to make the AIoT devices discoverable can be called an inventory procedure.

Editor's note: It is FFS whether none of the AIoT device types can initiate registration on their own.

The inventory procedure can be triggered by an AF sending an inventory request towards CN, and CN sends a request to A-RAN. The AIoT devices respond to the inventory request from A-RAN and send their device IDs. Such inventory procedure triggered by an AF can be called application inventory procedure.

The AF may further provide the inventory strategy information (e.g. inventory frequency, inventory period) to enable CN or A-RAN to perform periodic inventory to allow the newly coming AIoT devices to be discovered, without further explicit requests from the AF.

For the application inventory or periodic inventory, depends on device capabilities, the authentication and authorization may be performed, and the CN allocated device ID which is similar as 5G-GUTI may be passed to the device. For those devices, the device contexts are stored in AIOTF, including the security contexts.

When application inventory or periodic inventory procedure is triggered, the network may indicate whether all targeted devices need to respond the inventory, or only those devices which haven't been "inventoried" towards this A-RAN node should respond.

When AIoT devices move to a new A-RAN node, by responding the inventory, the network can also keep track of the AIoT devices location e.g. on a serving A-RAN node granularity, so that the network can route the request effectively which is sent from the AF to one or more specific AIoT devices.

NOTE 3: When AIoT devices move, the AIoT devices does not perform cell re-selection like logic.

For connection management, when in CM-CONNECTED state, the device is known by the network and DL data can be delivered to the service A-RAN node directly. When in CM-IDLE state, the detailed location of the device is unknown, so that before delivering DL data, CN needs to look up the device.

Editor's note: It is FFS whether there are needs to enable functionality enabling similar functionality as CM-IDLE and CM-CONNECTED states enables, and whether CN (i.e. AIOTF) needs to perform different actions accordingly, when DL data need to be delivered.

6.8.2 Procedures

NOTE: The message names in the procedures below are descriptive. It is assumed that the names are updated with corresponding SBI based names where applicable during the normative phase.

6.8.2.1 Application Inventory Procedure

The application inventory procedure is initiated by the AF to discover one or more AIoT devices in a specific area.

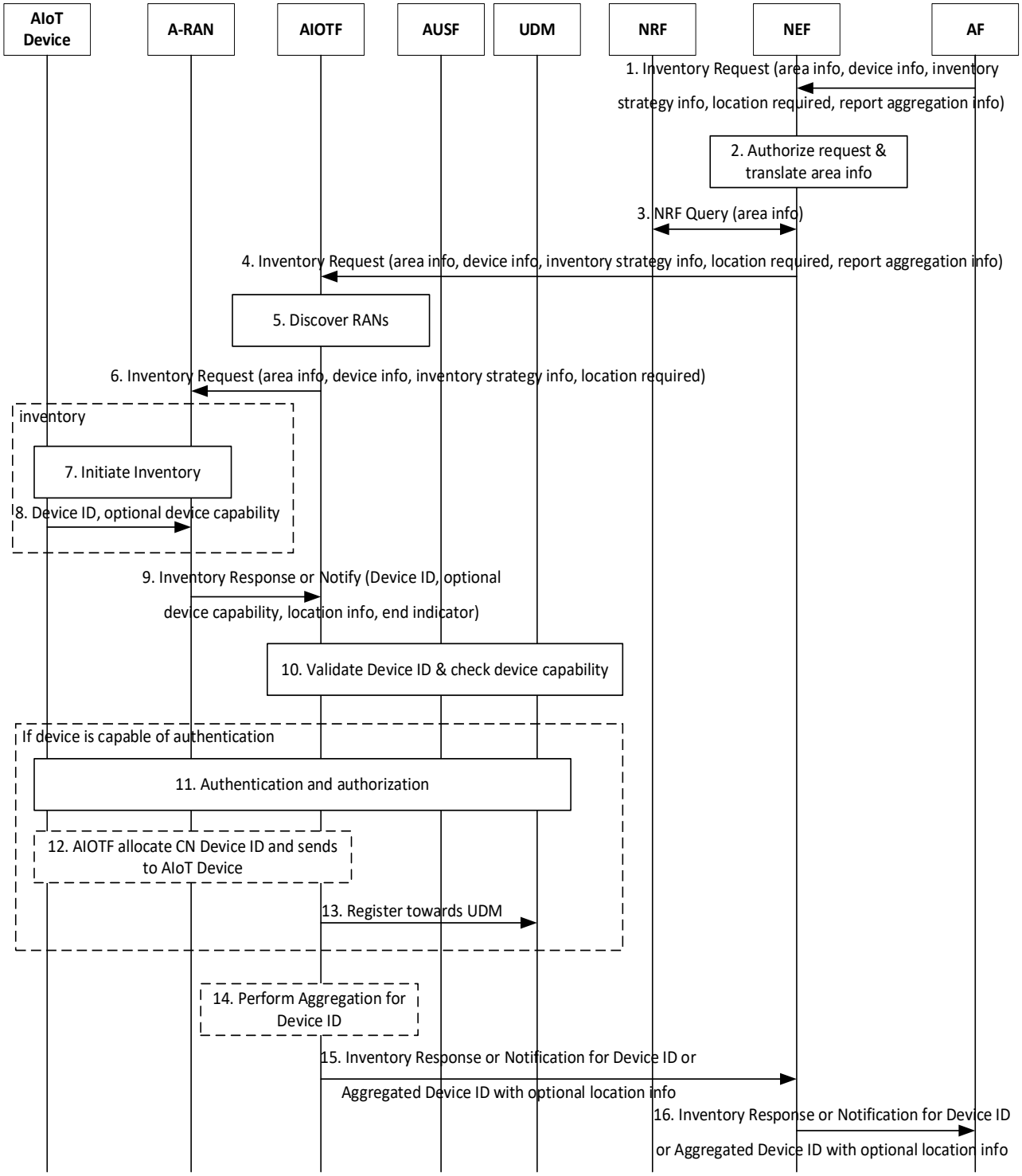


Figure 6.8.2.1-1: Inventory Procedure

1. The AF sends Inventory Message Request to the NEF, containing the area information, device information, optional inventory strategy information, and optional report aggregation info.

   - The area information could be the external geographical area information.

   - The device information could be device ID, device group ID, and/or device type.

Editor's note: Details of device type is FFS.

   - The inventory strategy information contains, e.g. the inventory frequency and inventory period to guide the readers to perform the inventory periodically. It also indicates whether all the targeted devices need to respond (full inventory), or only those who haven't performed the inventory procedure (delta inventory) should respond.

- The location required indicates whether the AF requests the location information of the AIoT devices provided.
- The report aggregation info indicates whether the reports need to be aggregated or not for a specific aggregation period, and whether the reports are needed after the aggregation period.

2. The NEF authorizes the request from the AF and perform the area translation to translate external area information to the internal area information. Within the authorization, the NEF further check whether the AF is authorized to get the location information of the device.
3. The NEF sends NRF query with internal area information to query AIOTFs serving the area.
4. The NEF sends the Inventory Request to the AIOTFs with the internal area information, device information and optionally inventory strategy information, optional location required information.
5. The AIOTF discovers A-RANs based on internal area information.
6. The AIOTF sends an NGAP message (Inventory Request) to the A-RANs with the internal area information, device information and inventory strategy information, and optional location required information.
7. The A-RAN (reader) initiates inventory based on device information as well as the inventory strategy information provided by the AF. The A-RAN may provide reader identity information (e.g. A-RAN ID) to enable the AIoT devices to understand they are read by which A-RAN node.
8. The AIoT Device reports the device ID, and optional device capability information. If the Inventory procedure indicates only who haven't performed the inventory procedure should respond, and if the AIoT Device has performed the inventory procedure towards this A-RAN node, it should skip the reporting.
9. The A-RAN sends an NGAP message (Inventory Response or Inventory Notify) to the AIOTF, containing the device ID and the optional device capability information provided by the device. The A-RAN may further provide location information (ULI) of the device, if requested from the AIOTF and allowed by local policy. The A-RAN may further provide an end indicator to inform the AIOTF whether it is the last inventory response for the inventory round.

Editor's note: Details of what location information A-RAN provides is FFS.

10. The AIOTF validates the device ID via interacting with AUSF and UDM. The AIOTF may further check the device capability information from the device subscription data stored in the UDM for the device capability information.

Editor's note: The check of the device capability information and device ID is FFS.

Editor's note: It is FFS which entity defines the AIoT device info and how to store such info in the UDM.

Based on the device capability info from the device and/or device information data in UDM, if the AIoT device is capable of handling authentication and authorization, step 11 - step 13 are executed:

Editor's note: The use of the step 11-13 is FFS.

11. The AIOTF together with AUSF and UDM, triggers authentication and authorization procedures towards the AIoT device.
12. The AIOTF may further allocates CN device ID and sends to the AIoT device.
13. The AIOTF registers with the UDM (UECM) for the device access.
14. The AIOTF may perform aggregation for the device ID, based on the report aggregation information provided by the AF. Within the aggregation period, the AIOTF will buffer the device IDs reported from the AIoT devices. The AIOTF may stop buffering and send report immediately, if it receives end indicator from A-RAN in step 9. When the aggregation period expires, the AIOTF sends the report. For those device ID report after the aggregation period, if it is needed by the AF, the AIOTF sends the report. Otherwise, it will be dropped.
15. The AIOTF sends Inventory Response or Notification Request towards the NEF for the device ID or the aggregated device ID information.
16. The NEF sends Inventory Response or Notification Request towards the AF for the device ID or the aggregated device ID information.

6.8.2.2 Periodic Inventory Procedure

The periodic inventory procedure is initiated by the CN or A-RAN, which follows the instructions from the AF,

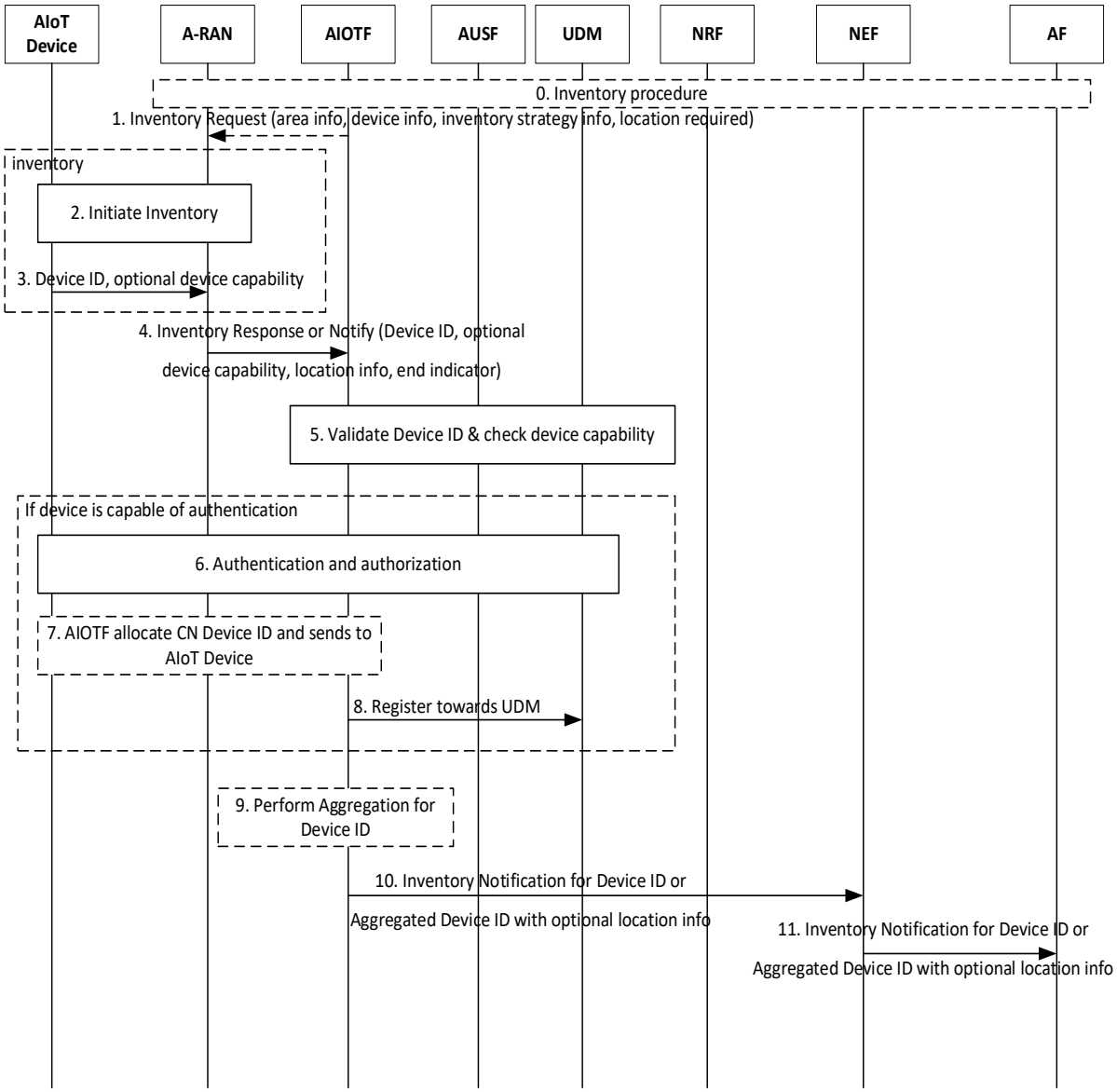


Figure 6.8.2.2-1: Periodic Inventory Procedure

The AIOTF or the A-RAN may initiate the periodic inventory procedure based on the inventory strategy information provided by the AF, to enable the AIoT devices to be discovered when they newly enter the coverage area.

0. An Inventory Procedure may have taken place resulted in inventory strategy information stored in the AIOTF. e.g. inventory frequency.

The step 1 - step 11 are similar as step 6 - step 16 in clause 6.8.2.1, with the following additions:

1. It will be performed only when it is AIOTF to initiate the periodic inventory. The inventory strategy should be set to delta inventory.

2. It can be performed without step 1, if it is A-RAN who initiates the periodic inventory. The inventory strategy should be set to delta inventory.

3. The AIOT device responds if it hasn't been "inventoried". Using the reader identity information (e.g. A-RAN ID) over the air, AIOT device determines if it is being read by a new A-RAN node and responds the inventory. The device may provide CN allocated device ID if it received before.

6. Can be skipped if the AIOTF finds AIoT device has performed the authentication and authorization with the network, based on CN allocated device ID.

7. Can be skipped if the AIOTF decides not to allocate a new CN allocated device ID by local policy.

8. Can be skipped if the AIOTF has registered towards UDM.

9. The AIOTF may use the aggregation period provided by the AF, or a locally configured value or even skip the aggregation, based on local policy. If the AIoT device responds the inventory procedure due to the A-RAN is a new reader, the device ID may not be sent to AF via NEF, unless AF requires the location information.

10. The AIOTF sends Inventory Notify Request towards the NEF for the device ID or the aggregated device ID information.

11. The NEF sends Inventory Notify Request towards the AF for the device ID or the aggregated device ID information.

6.8.3 Impacts on services, entities and interfaces

Editor's note: The services, entities and interfaces are FFS.

6.9 Solution #9: Information Transfer without a PDU Session

6.9.1 Description

This solution is to address the KI#1, KI#2 and KI#3. This solution is to support the DO-DTT and DT traffic type and an enhancement to the 5GS to support the AIoT Device. The main points are as following:

- The AIoT Device still performs the initial registration procedure, and after the initial registration procedure, the AMF can set up the transmission tunnel towards the AIoT NF; No PDU Session has been established at the AIoT device and the CN. No mobility registration update registration procedure is performed. During the registration procedure, a temporary ID is allocated to the AIoT Device for the security reason.

- There is no CM state at the AIoT Device or the AMF. After the initial registration procedure, there is no active NAS connection maintained between the AIoT Device and the AMF; however the AIoT Device and the AMF require to maintain MM context.

- For the DT traffic (e.g command from the AF) over the air interface, the RAN can perform paging (or paging like);

- For the DO-DTT traffic (e.g feedback or the DO-DTT data reporting to the AF), it can be sent over the NAS;

- If the AF is in the 3rd party domain, the NEF function can be combined into the AIoT NF.

Figure 6.9.1-1 shows the 5GS enhancement to support AIoT Device.

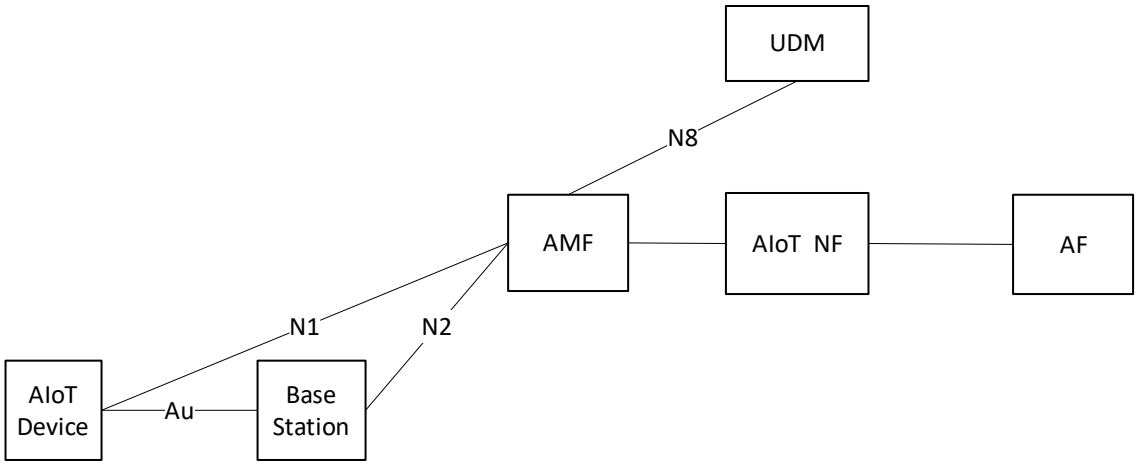


Figure 6.9.1-1 5GS enhancement to support AIoT Devices

6.9.2 Procedures

Figure 6.9.2-1 shows the signalling flow for the DT traffic (e.g command) delivery and DO-DTT data reporting.

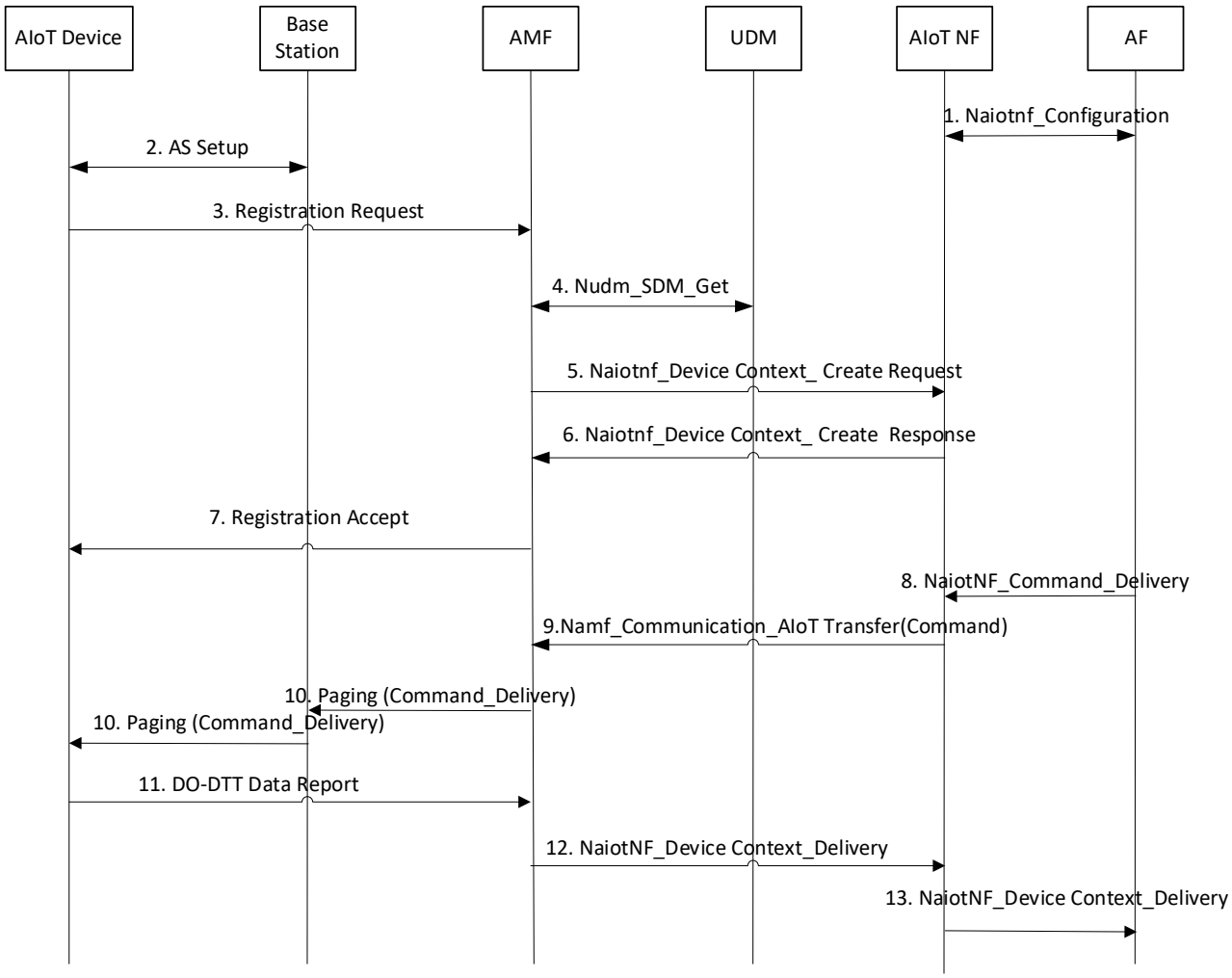


Figure 6.9.2-1 Signalling flow for Information Transfer

1. AF performs the service configuration. This service configuration can provide the AIoT Device ID(s) information to the AIoT NF. The AIoT NF can further provide the AIoT Device ID information to the UDR.

2. The AIoT Device performs the AS procedure.

Editor's note: Whether and how to trigger the AS procedure is to be studied in RAN WGs and to be defined in TR 38.769 [8].

3. The AIoT Device performs the Initial registration procedure. In this step, the AIoT Device ID is included in the Registration Request.

Editor's note: whether to reuse the initial registration procedure defined in TS 23.502 [5] or to define a new registration procedure is FFS.

Editor's note: The format of AIoT Device ID is FFS and how to protect the Device ID is FFS.

4. The AMF performs the Device ID validation by retrieving the subscription data from the UDM. The subscription data can also include the AIoT NF information.

Editor's note: The validation of AIoT Device requires coordination with SA WG3.

Editor's note: Whether the UDM has the subscription data per Device ID or group of Devices or other granularities is FFS.

Editor's note: It is FFS whether it can be assumed that the device and the CN can be pre-provisioned with CN level per device information (e.g. network layer AIoT device ID, security material).

5-6. The AMF sets up a transmission tunnel towards the AIoT NF. And the AIoT NF acks this setting up procedure. Multiple AIoT devices can share this transmission tunnel.

Editor's note: The concept of transmission tunnel, how to set up the transmission tunnel and what is the granularity of the transmission tunnel is FFS.

7. The AMF accepts the registration procedure if the AIoT Device has been validated. The AMF also allocates a temporary id to the AIoT Device in the Registration Accept message.

8. AF delivers the command to the AIoT NF including the Device ID(s) and detailed "Command" information.

9. AIoT NF forwards the AIoT Device ID information and detailed "Command" information to the AMF.

10. Since the AMF does not maintain the CM state for the AIoT Device, the AMF utilizes the RAN information where the AMF receiving the N2 message for the AIoT Device for paging. The paging (or paging-like) includes the temporary ID and detailed "Command" information. The AMF sends paging to the RAN and RAN performs the paging over the air interface.

Editor's note: How to protect the detailed "Command" information is FFS.

Editor's note: The paging (or paging-like) requires coordination with RAN WGs.

11. The AIoT Device checks the temprory ID in the paging message and if the tempory ID is matched, the AIoT Device then handles "Command". If feedback or DO-DTT (e.g. sensor data reporting) is required, the AIoT Device includes DO-DTT traffic over the NAS.

12. The AMF forwards the DO-DTT traffic to the AIoT NF.

13. The AIoT NF forwards the DO-DTT traffic to the AF.

6.9.3 Impacts on services, entities and interfaces

Editor's note: The impacts on services, entities and interfaces are FFS.

6.10 Solution #10: Registration procedure for Ambient IoT Devices

6.10.1 Description

This solution is for Key Issue #2 "Identification, Subscription, Registration and Connection management".

As depicted in Architecture Requirements, the DT and DO-DTT traffic types will be studied in this stage. The Ambient IoT devices could be driven by the network for Topology1 or UE for topology 2 before registering to the network. This proposal proposes one potential mechanism for identification, subscription, registration management, and registration procedures as well.

As we know, 5GS uses SUPI with PLMN info, PEI, and GPSI as identifications for device ID based access control, identification, authentication, routing steering, and service purposes. However, due to cost and power consumption, most ambient IoT devices may not have these kinds of IDs. Referencing the RFID solution, most likely, the ambient IoT device has the following IDs:

1) TID (See GS1 TDS Release 2.1 [10]): The AIoT manufacturer configures the tag ID in advance. It is only readable. It can be treated as PEI in the ambient IoT solution.

2) EPC (See GS1 TDS Release 2.1 [10]): Electrical Product Code. The ID can be used to flexibly define one or a group of ambient IoT devices in the ambient IoT service procedure. It can be treated as GPSI in the ambient IoT solution.

Unfortunately, neither TID nor EPC has the operator ID information. Therefore, they are not suitable for device ID based access control, identification, and routing steering purposes. The solution introduces a new device ID. It can be treated as SUPI. The device ID is either allocated by 5GC or the third party. This ID need to be kept internally in 5GS and ambient IoT devices for access control, identification, and route steering purposes. The format of the device ID is Operator ID + Group ID + Instance ID.

Editor's note: Detailed information carried in Operator ID and Group ID is FFS.

The solution has a default device ID concept. The Instance ID is set to zero in the default device ID during the pre-configuration phase. Once the AF-triggered registration procedure is performed successfully, the 5GC will generate a new device ID with a unique and non-zero instance ID or will obtain the new device ID from the third party to replace the default device ID in the ambient IoT device.

In order to keep a competitive solution compared to the RFID solution, it is possible to trigger not only one or a few ambient IoT devices defined by the TID list but also numerous unregistered ambient IoT devices in the whole serving zone in the AF-triggered registration procedure. Then, it is possible to perform an automatic AF-triggered registration procedure for unregistered ambient IoT devices in the whole zone during the middle of the night without human resources involved. However, ambient IoT devices are not allowed to keep dedicated status info. How is the ambient IoT device aware of the registered or unregistered status in this case? In the solution, if the instance ID in the device ID is set to zero, such as the default device ID mentioned above, it will be implicit to show an unregistered status.

Besides, the credential holder can be flexibly placed in the serving operator, roaming operator, enterprise, or third AF based on Operator ID and Group ID derived from the default device ID to realize different networking architectures.

The principles/assumptions are given below:

- A new network function named Ambient IoT NF may be adopted to manage Ambient IoT devices and procedures. If not, this relevant function can be supported by AMF.
- In 5GC, each Ambient IoT device has a unique device ID that consists of the Operator ID, group ID, and Instance ID.
- The device ID is kept internally in 5GS and ambient IoT devices for access control, identification, and route steering purposes.
- The ambient IoT device and Credential holder are pre-configured with the default ambient IoT device ID, TID, and default credentials.
- The solution is based on an operator-controlled Ambient IoT device.

6.10.2 Procedures

6.10.2.1 Procedures for AF triggered Registration for Topology 1

The following figure presents a procedure of AF triggered registration for Topology 1.

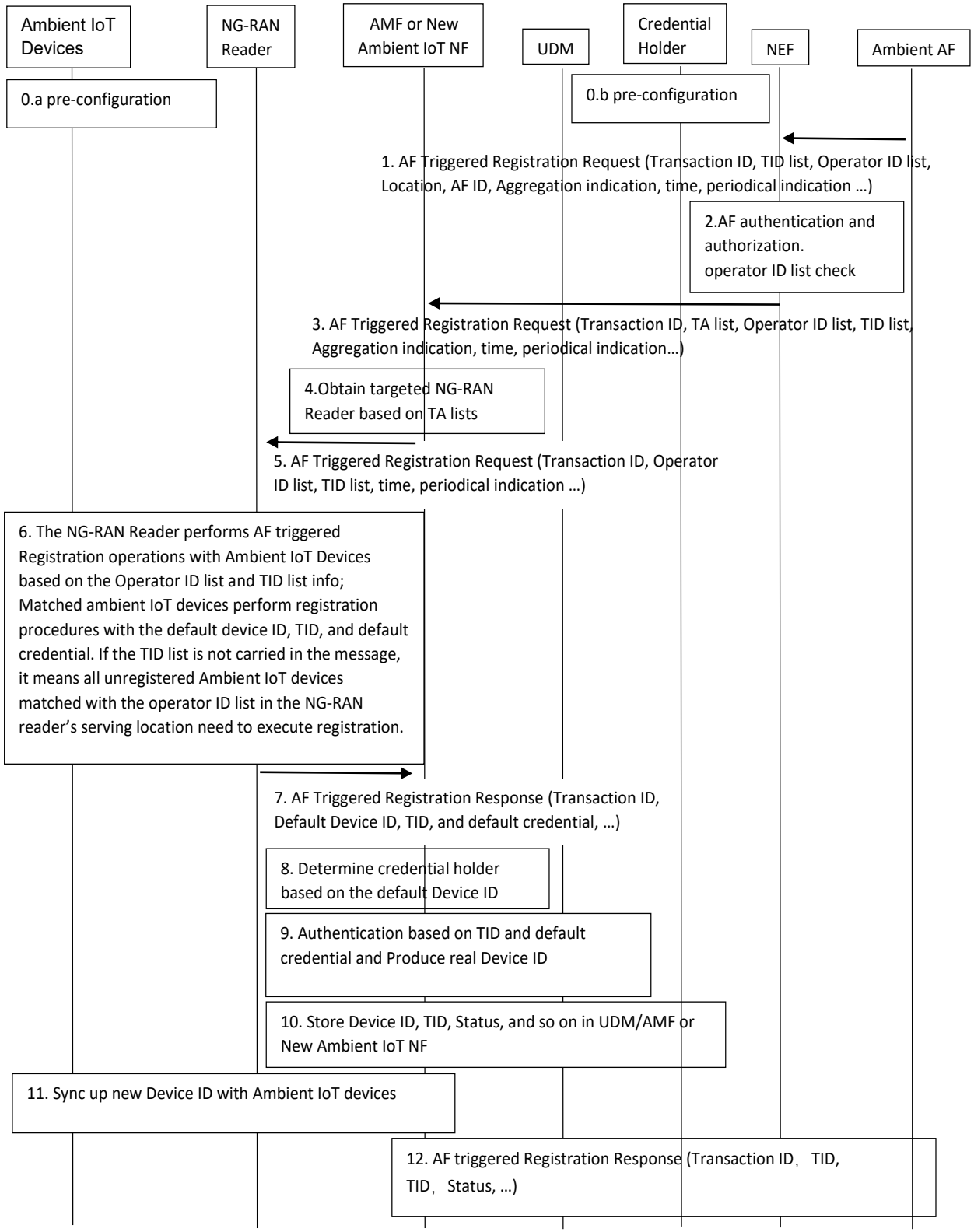


Figure 6.10.2-1: AF triggered Registration Procedure for Topology 1

0.a. The Ambient IoT devices are pre-configured with default AIoT device ID and default credentials. TID (See GS1 TDS Release 2.1 [10]) is configured by AIoT manufacturers in advance. The instance ID value in the default AIoT device ID is set to zero. The ambient IoT device will implicitly know its active or inactive status based on the instance value in the solution.

NOTE 1: How information is configured into ambient IoT device is out of scope of 3GPP.

0.b Meanwhile, the 5GC or the third credential holder is also pre-configured with the TID, default credentials, status (inactive), and so on.

1. AF sends an AF Triggered Registration Request to NEF with the following parameters: Transaction ID, TID list, Operator ID list, location, AF ID, Aggregation indication information, time, periodical indication, and so on.

   - Operator ID list: This information is used to define the targeted ambient IoT devices from the operator's perspective, supporting multiple operator ambient IoT device access. It could be one specific operator or a list of operators.

   - TID list: The information can be used to define one or a few targeted ambient IoT devices. If the TID list is carried in the message, it means AF wants to trigger one or a few ambient IoT devices defined by the TID list to perform the registration procedure. If the TID list is not carried in the message, it means AF wants to trigger all unregistered Ambient IoT devices in this location to perform the registration procedure. The ambient IoT device will implicitly know its registered or unregistered status based on the instance value in the device ID. If the instance has all zero values in the device ID, then the ambient IoT device has an unregistered status.

   - Aggregation indication: It is used to tell the 5GS how to handle the response messages - aggregation response or not.

   - Time: when the service will be carried out exactly. This parameter is only carried in the AF-triggered whole unregistered ambient IoT devices performing registration procedure scenarios.

   - Periodical indication: whether the service operation will be executed periodically. This parameter is only carried in the AF-triggered whole unregistered ambient IoT devices performing registration procedure scenarios.

2. NEF will perform the below actions towards the third Ambient AF

   - Perform authentication to the third Ambient AF to decide whether it is allowed to access 5GS or not.

   - Check the authorization to determine whether the third Ambient AF is allowed to perform an AF triggered registration operation or not.

   - Check the authorization to determine whether operators in the operator list ID are allowed or not.

   - Converting location information to TA List information.

     The NEF obtains serving AMF or Ambient IoT NF based on the TA list.

3. The NEF sends AF Triggered Registration Request to the AMF/Ambient IoT NF, including the TID list ID, Transaction ID, Operator ID list, TA list, aggregation indication, time, periodical indication, and so on.

4. The AMF/Ambient IoT NF selects the NG-RAN reader based on the TA list.

5. The AMF/Ambient IoT NF forwards AF triggered the registration request to the NG-RAN reader, including the TID list, Operator ID list, time, periodical indication, and so on.

6. NG-RAN activates one or a few targeted ambient IoT devices that match both the TID list and Operator ID list to perform the registration procedure with the default device ID, TID, and default credentials. If the TID list is not carried by the NG-RAN reader, all unregistered ambient IoT devices that match the operator ID list have to execute the registration procedure . The ambient IoT device will implicitly know its registered or unregistered status based on the instance value in the device ID. If the instance has all zero values in the device ID, then the ambient IoT device has an unregistered status. A receiving limit time may be configured on NG-RAN. Once timeout, the message received after this time will be discarded by NG-RAN.

7. The NG-RAN reader forwards the registration messages with the default device ID, TID, and default credentials to AMF.

NOTE 2: Considering the NG-RAN reader sequence scanning behaviour, don't suggest that NG-RAN perform a response aggregation operation during the interactive authentication procedure between ambient IoT devices and 5GC. Aggregation operations may only be performed in 5GC in this scenario.

8. Upon receiving the registration message from NG-RAN, AMF/New AIoT NF can determine the targeted credential holder based on local configurations related to operator ID and/or group ID derived from the default device ID.

9. Perform an authentication operation based on TID as username and default credentials between AMF/New AIoT NF and the credential holder. Once authentication is successful, a Real device ID with a unique and non-zero instance ID will be produced internally by 5GC or the third party..

10. 5GC stores the new device ID, TID, status (active), and so on in UDM, AMF/New AIoT NF, 5GC sets the device status as active. The status information will be used in the Ambient IoT service procedures to verify the Ambient IoT device.

11. AMF/New AIoT NF will sync up new device IDs with relative ambient IoT devices.

12. The AMF/New AIoT NF returns AF Triggered Registration Response to the AF via NEF.

NOTE 3: Security related mechanisms will be in the remit of SA WG3.

6.10.2.2 Procedures for AF triggered Registration for Topology 2

The registration may be triggered by AF via the UE reader. The UE reader interacts with AMF /New Ambient IoT NF via NG-RAN, which could be regarded as an AF triggered registration procedure for topology 2.

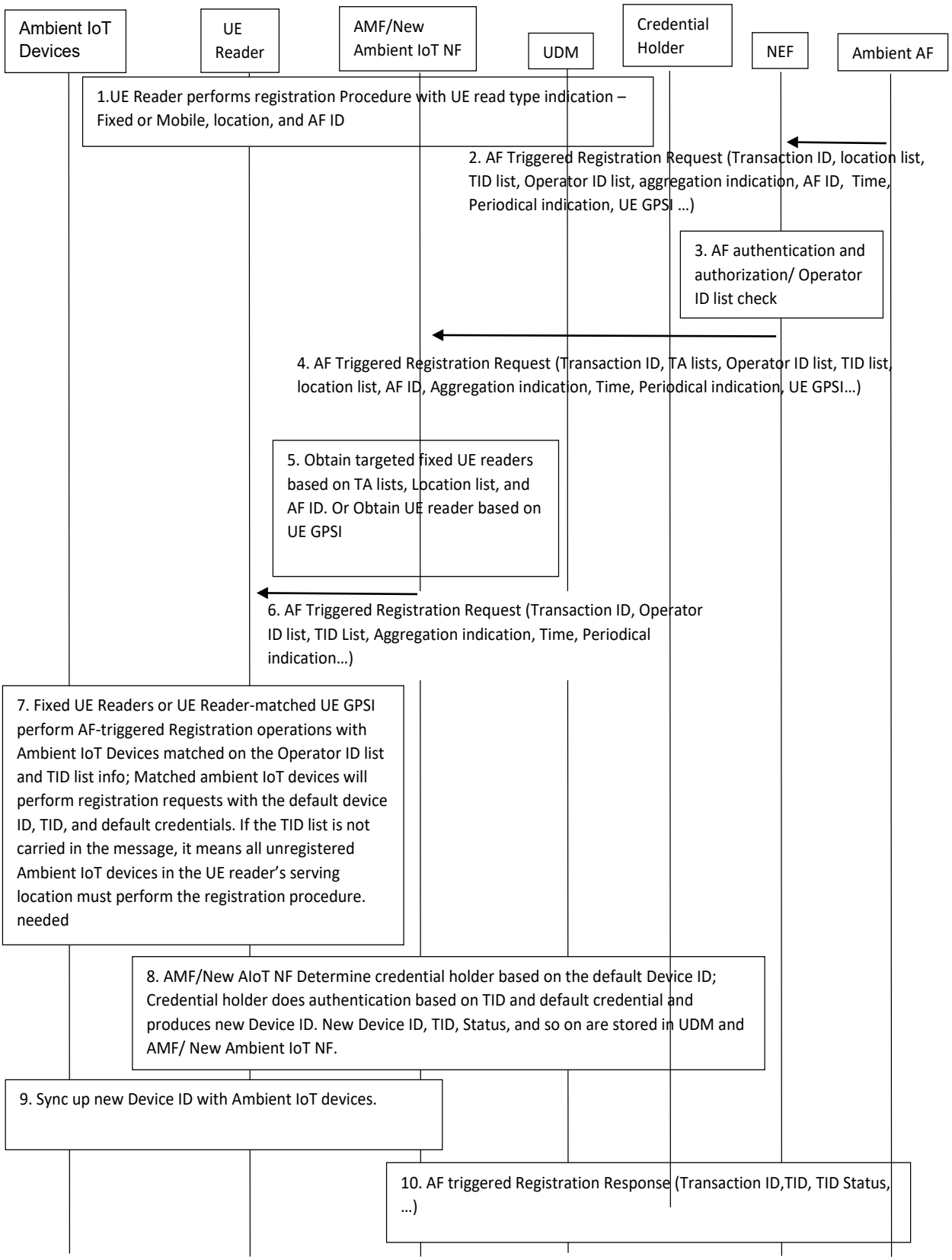


Figure 6.10.2-2: AF triggered Registration Procedure for Topology 2

1. The UE Reader performs the registration procedure, including the fixed or mobile type of UE reader, location, and AF ID.

2. AF sends an AF Triggered Registration Request to NEF with the following parameters: Transaction ID, TID list, Operator ID list, location, AF ID, Aggregation indication information, time, periodical indication, UE GPSI, and so on.

   If UE GPSI is included in the message, the third AF wants to perform an AF-triggered registration procedure with a designated UE reader defined by UE GPSI. In addition, time and periodical indication parameters are not included in the message.

   If UE GPSI isn't included in the message, the third AF wants to perform an AF-triggered registration procedure without designated UE readers. 5GS determines the involved UE readers based on input info from the third AF, such as location, AF ID, and so on. Considering UE reader posture, such as UE reader antenna pointing direction, only fixed-type UE readers will be involved in the procedure.

3. NEF will perform the below actions towards the third Ambient AF:

   - Perform authentication to the third Ambient AF to decide whether it is allowed to access 5GS or not.
   - Check the authorization to determine whether the third Ambient AF is allowed to perform an AF-triggered registration operation.
   - Check the authorization to determine whether operators in the operator list ID are allowed or not.
   - Converting location information to TA List information.

   The NEF obtains serving AMF or Ambient IoT NF based on the TA list.

4. The NEF sends AF Triggered Registration Request to the AMF/Ambient IoT NF, including the Transaction ID, Operator ID list, TID list ID, TA list, AF ID, aggregation indication, time, periodical indication, UE GPSI and so on.

5. If UE GPSI isn't included in the message, the AMF/Ambient IoT NF selects the fixed-type UE readers based on the TA, Location list, and AF ID. Alternatively, if UE GPSI is carried in the message, the AMF/Ambient IoT NF selects the designated UE reader based on UE GPSI.

6. First, The AMF/Ambient IoT NF sets up the connection with targeted UE readers. Then, it forwards the AF-triggered registration request to the fixed-type UE readers or the designated GPSI UE reader, including the TID list, Operator ID list, time, periodical indication, and so on.

7. the fixed-type UE readers or the designated GPSI UE reader activate one or a few targeted ambient IoT devices that match both the TID list and Operator ID list to perform the registration procedure with the default device ID, TID, and default credentials. If the TID list is not carried in the message, all unregistered ambient IoT devices in the fixed-type UE readers' or the designated GPSI UE reader's serving area that match the operator ID list have to execute the registration procedure. The ambient IoT device will implicitly know its registered or unregistered status based on the instance value in the device ID. If the instance has all zero values in the device ID, then the ambient IoT device has an unregistered status. A receiving limit time may be configured on NG-RAN. Once timeout, the message received after this time will be discarded by NG-RAN.

NOTE: Considering the latency requirement and reader scanning capability, don't suggest that UE Reader performs a response aggregation operation during the interactive authentication procedure between ambient IoT devices and 5GC. Whether and when the UE Reader should engage the response aggregation should be decided on the service requirements and service scenario, e.g. when the latency exceeds the scanning speed or when the Topology 1 and Topology 2 are in one Registration request.

8. AMF/New AIoT NF Determines the credential holder based on the default Device ID; the credential holder authenticates the ambient IoT device based on the TID and default credential. If authentication is successful, 5GS produces a new Device ID, 5GC sets the device status as active. At the same time, new Device ID, TID, Status (active), and so on are stored in UDM and AMF/ New Ambient IoT NF. The status information will be used in the Ambient IoT service procedures to verify the Ambient IoT device

9. AMF/New AIoT NF will sync up new device IDs with relative ambient IoT devices.

10. The AMF/New AIoT NF returns AF Triggered Registration Response to the AF via NEF.

6.10.3 Impacts on services, entities and interfaces

NEF:

- Capability to authentication and authorization to the third ambient IoT AF.
- Capability to support verification on operator ID list.

AMF/Ambient IoT NF:

- The AMF/Ambient IoT NF selects the NG-RAN reader and/or UE reader based on the TA list, Location, and AF ID.
- Capability to aggregate the Ambient IoT devices' response.
- Capability to be aware of the Topology types.

UDM/Credential holder:

- The UDM stores the profile information of Ambient IoT devices.
- Authenticating the ambient IoT device and producing new device IDs.

NG-RAN Reader:

- The NG-RAN reader performs paging and receives responses from Ambient IoT devices in Topology 1.

UE Reader:

- The UE Reader performs paging and receives responses from Ambient IoT devices in Topology 2.

Ambient IoT device:

- The ambient IoT device stores the default AIoT device ID or the and default credentials and updates the device ID if the registration procedure is performed successfully.
- Performing registration procedure once it is matched during the paging procedure.

6.11 Solution #11: 5GC support for AF to communicate with AIoT devices

6.11.1 Description

This solution address aspects of key issue #3 on Support of Ambient IoT Services.

This solution enables the AF to communicate with the target AIoT devices for DO-DTT and DT traffic types via 5GC, dedicated for only Topology 2.

This solution considers scenarios in which an application service provider has prior information about intermediate nodes (abbreviated as I-node) expected to be located in specific places (e.g. intermediate nodes being used only in particular warehouses). Additionally, scenarios where the provider knows candidate locations for the target AIoT devices (e.g. the AIoT devices attached to goods are expected to be in particular warehouses or retail markets) are also considered.

The assumption and high-level procedures of this solution are as follows:

- The AIoT device ID is defined by the external application and provided by the AF to the 5GC when requesting the AIoT-related services. The uniqueness of the AIoT device ID per application is assumed to be guaranteed by the application itself, such as by adhering to standards like EPC (Electronic Product Code) and utilizing EPCglobal allocation, which manages a global registry for unique prefixes among different companies.
- AIoT devices are not registered with the 5GC.
- AIoT devices do not have NAS layer.
- Application layer security between AIoT devices and AS is assumed.
- AIoT Devices are provisioned with AIoT Device ID and associated security materials.
- A UE, additionally capable of directly communicating with the AIoT devices, is acting as the intermediate node.

- A UE acting as an intermediate node is registered with 5GC using the existing mechanism, with some enhancements to indicate its capability of acting as an intermediate node, and is authorized as an intermediate node (UE) during the registration procedure.
- The AF possesses information about the candidate location(s) of the target AIoT devices or about the preferred intermediate node (i.e. external UE ID) and provides them when requesting the AIoT related services to 5GC.
- If the AF is in the 3rd party domain, the AF can communicate with the 5GC via NEF that can be combined with the AIoT NF.
- When information about the preferred intermediate node is provided while it is not operational, or when the location(s) of the target AIoT devices are given, the 5GC selects the intermediate node based on the network information. The process of selecting such an intermediate node is not within the scope of this solution. This means that this solution supports both static and dynamic binding with the intermediate node.
- How to support security between each entity (e.g. between AIoT device and I-node, and AIoT device and CN) involved in the procedures (e.g. relying on application security or providing network security based on the information provided by the application) is assumed to be addressed by SA WG3.
- The Uu interface between the intermediate node and the gNB is assumed to be used, while possibly a new protocol stack, defined by RAN, is assumed to be used between the AIoT device and the intermediate node.
- For DO-DTT traffic (e.g. response from the AIoT device to the network), it can be sent over the NAS of the intermediate node.

Figure 6.11.1-1 shows the 5GS enhancement to support AIoT Device.

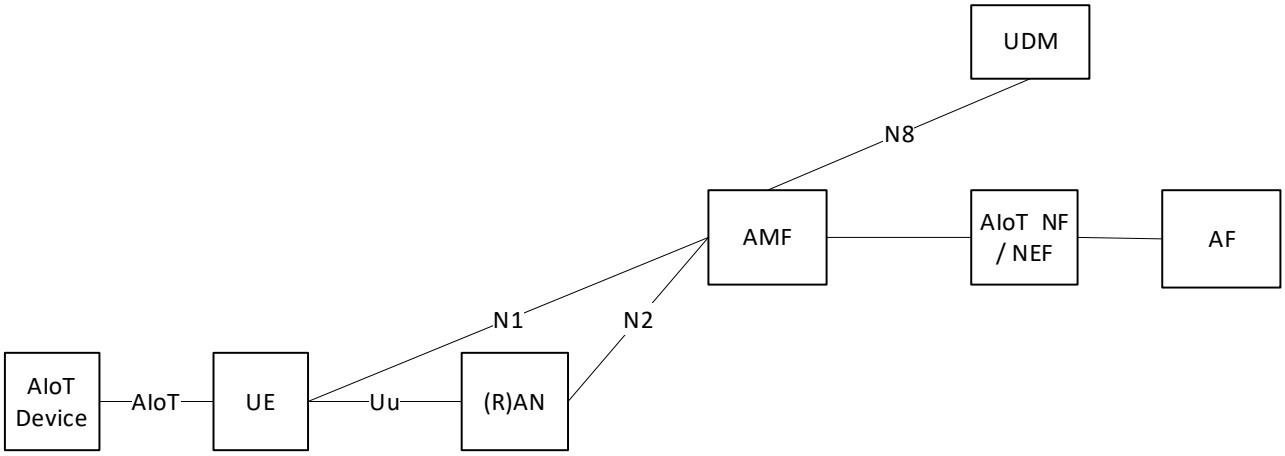


Figure 6.11.1-1 5GS enhancement to support AIoT Devices for Topology 2

6.11.2 Procedures

6.11.2.1 Communication with AIoT devices

The procedure for communicating with AIoT devices based on AF request is depicted in figure 6.11.2.1-1.

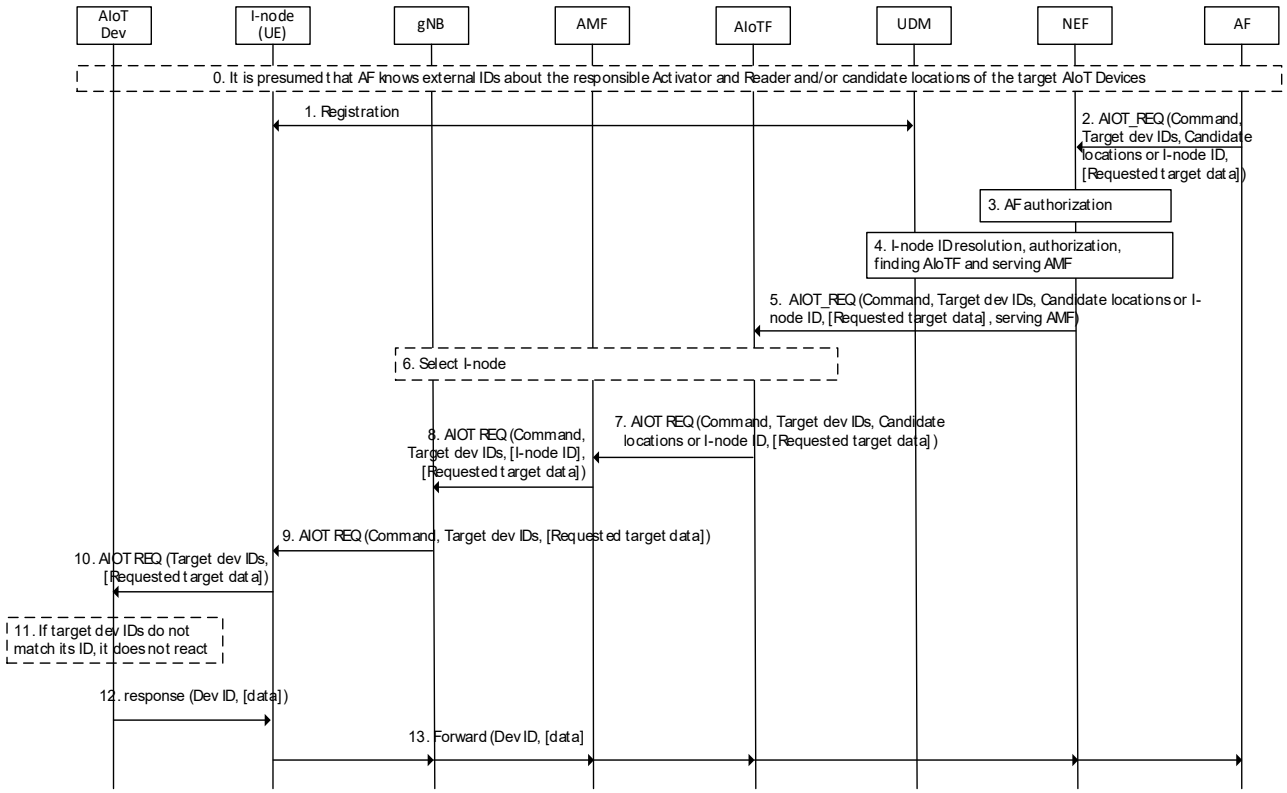


Figure 6.11.2.1-1: Information Flow for AIoT data retrieval

To facilitate communication between an external AF with AIoT devices for data retrieval, this solution proposes certain measures to be taken:

0. It is assumed that the AF requesting 5GC to communicate with specific AIoT devices possesses their AIoT device IDs, either defined by the device manufacturer or the AF itself, along with candidate locations of the AIoT devices or information (e.g. external ID) of the intermediate node covering the target AIoT devices.

1. The intermediate node is a UE with the capability to directly communicate with the AIoT devices. The intermediate UE performs the registration procedure as specified in clause 4.2.2.2 of TS 23.502 [5] with the following enhancement:

   - The UE indicates AIoT support in the MM capability.

   - The subscription data of the UE is also enhanced to indicate whether the UE is allowed to perform the AIoT operation.

   - The Registration Accept message also indicates whether the UE has been authorized to perform the AIoT operation. Validity information may also be included for the AIoT operation. Validity information can be time validity information (defined by start and end times) to indicate when the UE is allowed to perform the AIoT operation or location validity information (defined by TAI or cell ID) to indicate where the UE is allowed to performed the AIoT operation.

2. The AF sends a request to 5GC to communicate with the target AIoT devices for either DT or DO-DTT traffic type. The AF's request includes following parameters: the target AIoT device IDs defined by the external application and installed in the AIoT devices, the candidate locations of the target AIoT devices or the external ID (i.e. external UE ID) of the intermediate node that can directly communicate with the target AIoT devices, "command" information (e.g. read/write), optionally specific target data to be read by the AF.

3. Upon receiving the AF request, the NEF authorizes the AF request based on SLA.

4. The NEF translates the external ID (e.g. GPSI) of the intermediate node, if provided, to the SUPI via UDM and verifies whether the UE possessing the translated SUPIs is authorized to function as intermediate node through UDM. The NEF translates the location, if provided by the AF, into 3GPP based location information (Tracking areas TA(s), cell ID(s), etc). The NEF identifies the serving AMF(s) via UDM by utilizing the SUPI of the intermediate node, if applicable, or by considering the candidate locations of the AIoT devices. The NEF selects

AIoTF that can be a standalone NF or collocated with the NEF. If multiple locations are provided, the overall procedures can be performed for each location.

5. An AIOT service request message is sent to the AIoTF, including the following parameters: AIoT session ID, the target AIoT devices, candidate locations or the SUPI of the intermediate node, Serving AMF(s) as derived in step 4, "command" information (e.g. read/write), optional Requested target data.

6. The AIoTF may select the intermediate node based on the information provided in step 5 to communicate with the target AIoT devices. The detailed procedures for selecting the intermediate node are described in clause 6.11.2.2. The selected UEs can be configured with information necessary for AIoT services, using the UE configuration update procedure.

7. The AIoTF requests the serving AMF to initiate the communication, which includes AIoT session ID, the target AIoT device IDs, candidate locations or SUPI of the selected intermediate node, "command" information (e.g. read/write) and optional Requested target data.

8. The AMF identifies the serving gNB by utilizing the SUPI of the intermediate node, if applicable, or by considering the candidate locations (i.e. a list of TAIs and cell IDs) of the AIoT devices. The AMF then requests the gNB to send a AIoT service request, which contains the target AIoT device IDs, candidate locations or UE NGAP ID of the selected intermediate node, "command" information (e.g. read/write), and optional Requested target data, to the selected I-node. The paging request message, if the selected I-node is in CM-IDLE state, or N2-AP Downlink NAS Transport message, if the selected I-node is in CM-CONNECTED state, can be used with some extensions to convey the AMF request, for example.

9. The gNB requests the selected intermediate node to communicate with the AIoT data, providing the target AIoT device IDs, "command" information (e.g. read/write), and the Requested target data (if applicable). RRC paging message can be used by the gNB to send the request to the intermediate node if the selected I-node is in CM-IDLE or RRC-INACTIVE states. Alternatively, a dedicated RRC message can be used if the selected I-node is in RRC-CONNECTED state, for example. When RRC paging is used, the gNB may restrict the paging area only to the identified cells for optimization, if available.

10. The selected intermediate node sends out a request which includes the target AIoT device IDs and the Requested target data (if applicable).

NOTE: The radio resources required for communicating with AIoT devices are to be defined by RAN WGs.

11. The AIoT devices check whether their device IDs match any of the target AIoT device IDs included in the received request. If there's no match, the AIoT devices do not react.

12. If there's a match, the AIoT devices check the presence of Requested target data within the request as well as "command" information to check whether DT or DO-DTT traffic type is requested. In case of DO-DTT type requested, if the Requested target data is absent, the AIoT devices send a response message incorporating only their device IDs. If the Requested target data is present, the AIoT devices send a response message including their device IDs and only the data requested to the intermediate node.

13. The intermediate node receives the response message from the AIoT devices. If the AIoT device ID within the received message corresponds to any of the target AIoT device IDs obtained in step 10, the intermediate node assesses which AIoT devices, from the list of expected AIoT devices, have provided a response. The intermediate node may aggregate the response messages from the AIoT devices according to the configuration information if it was provided by the CN in advance. The intermediate node then forwards this information, encompassing AIoT device data (if applicable) and the AIoT device ID(s) over NAS signalling, to the AF via gNB, AMF, AIoTF and NEF. The AMF or AIoTF may aggregate response messages from some or all of the targeted AIoT devices during the forwarding procedures to AF.

6.11.2.2 Selection of intermediate nodes

The selection of the intermediate node can be performed based on the geographical location of the target AIoT devices or the identity of the intermediate node from AF. The procedure for selection of intermediate nodes based on geographical location is depicted in figure 6.11.2.2-1.

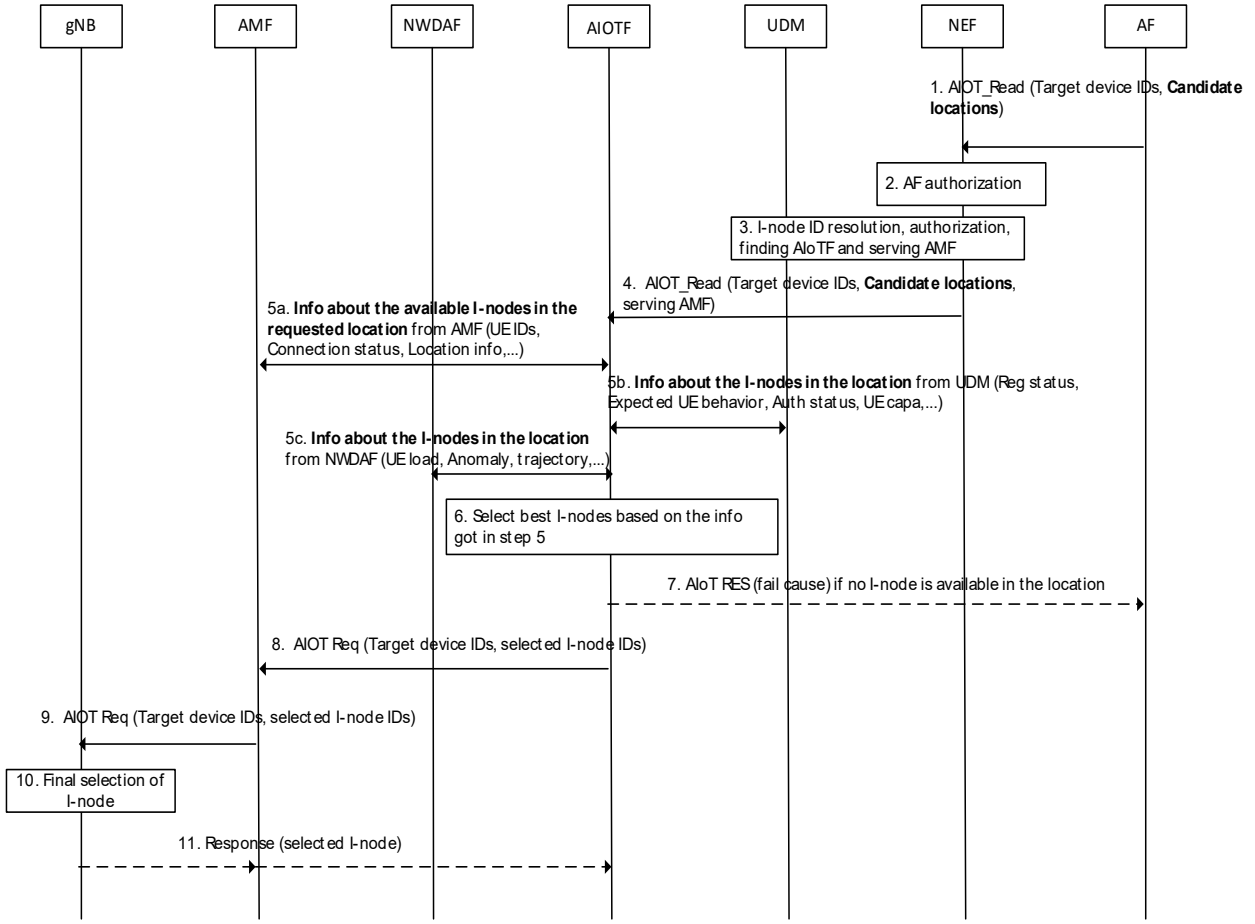


Figure 6.11.2.2-1: Information Flow for selecting I-node based on location of target devices from AF

1-4. Step 2 to 5 in clause 6.11.2.1 are performed.

5. The AIOTF collects necessary information from relevant NFs, for I-node selection, based on the requested location info.

   a. The AIOTF first discovers the UEs present in the requested location and may also retrieve the UE related information from AMF, including connection status and reachability.

   b. The AIOTF may retrieve information related to the UEs discovered in step 5a from UDM. This information enables checking, for example, whether the UEs are registered, authenticated, authorized to act as I-nodes. Additionally, the information from UDM can include the Expected UE behaviour, as described in clause 4.15.6.3 of TS 23.502 [5], that can be used to check the battery indication, stationary indication among other parameters.

   c. The AIOTF may retrieve information related to the UEs discovered in step 5a from NWDAF. This information enables checking, for example, if the discovered UEs are not too much loaded to serve the devices, function well without anomalies, or will not move out of the coverage for a significant period based on the UE related analytics as specified in clause 6.7 of TS 23.288 [13].

6. The AIOTF, based on the information collected in step 5, selects a list(s) of potential I-nodes. The number of I-nodes selected can be based on factors such as the number of device IDs. If there are many available candidate I-nodes, the AIoTF may prioritize I-nodes in CM-CONNECTED state to optimize signalling. For I-nodes in CM-IDLE or RRC-Inactive states, the AIoTF may further refine the selection, for example, by checking if the last visited cell matches the location specified in step 4.

7. In case no I-nodes could be selected at step 6, AIOTF sends AIOT Response with the failure reason to the AF, and the subsequent steps are skipped.

8. The AIoTF requests the serving AMF to initiate the communication with the AIoT devices, which includes the target AIoT device IDs, and the selected I-node IDs.

9. The AMF forwards the AIOT read request to gNB along with the target AIOT device IDs and the selected I-node IDs.

10. The gNB performs additional validation to assess if the received I-nodes are optimal for serving the AIoT devices, considering its local conditions such as response time, etc. Subsequently, the gNB makes the final selection on the best I-node. It is also possible that the gNB does not select any of the requested I-nodes if local conditions are not satisfied.

11. The gNB may respond to the AIOTF via AMF with the final list of selected I-nodes. In case the gNB does not selection any I-node, the response may indicate there is no available UE.

As an alternative to figure 6.11.2.2-1, the following procedure as depicted in figure 6.11.2.2-2 can be supported.

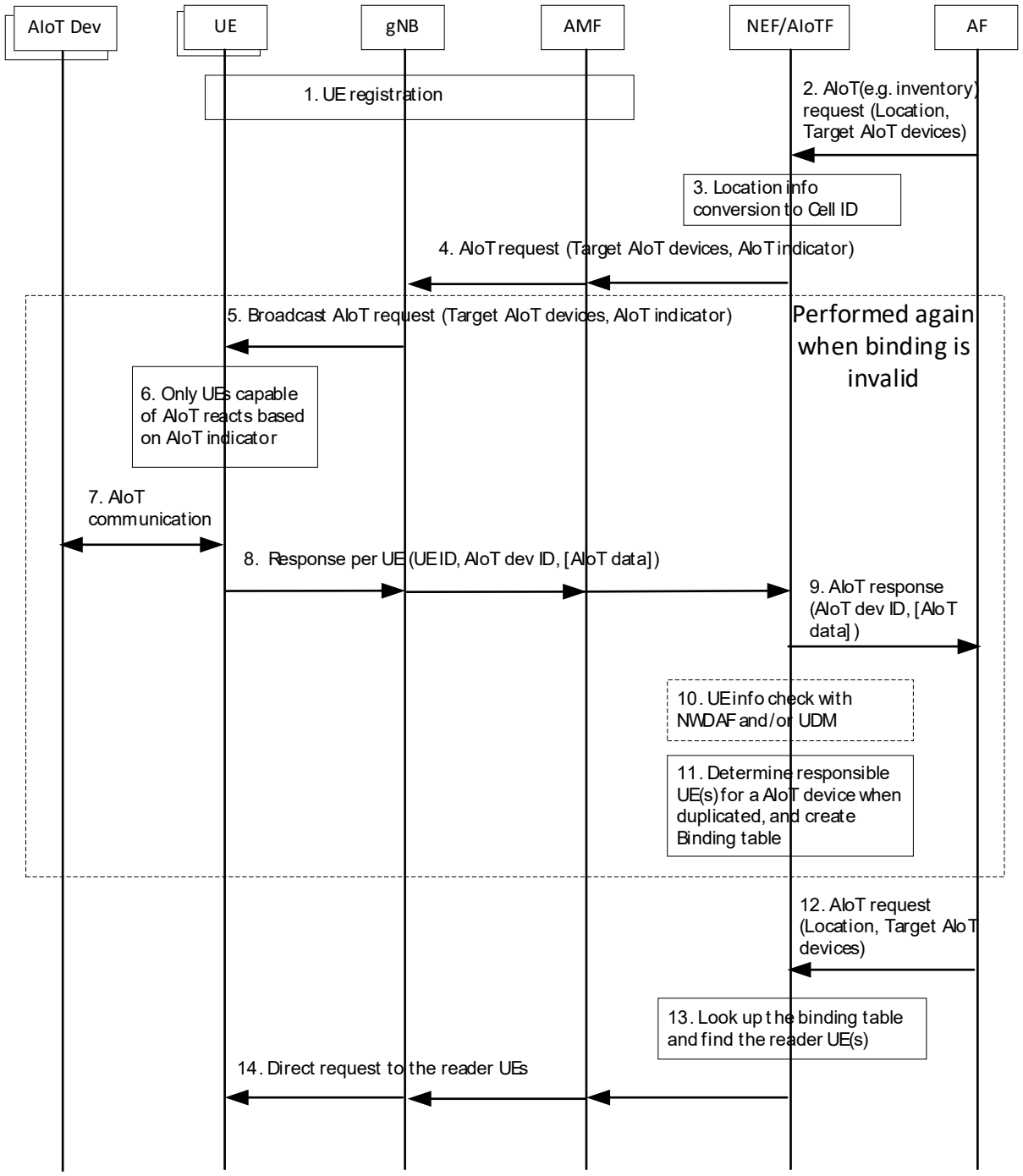


Figure 6.11.2.2-2

1. UE Registration.

The User Equipment (UE) registers with the 5G network. This involves the UE connecting to the gNodeB (gNB) and the network's core. The registration process ensures that the UE is authenticated and authorized for AIoT services.

2. AIoT Request.

   A request is initiated for an AIoT (Ambient IoT) service, such as inventory management. This request includes details such as the estimated location of the target AIoT devices and the target AIoT device IDs that need to be managed or interacted with. If the target AIoT device IDs are missing, it can be considered that the AF wants to inventory all the available AIoT devices within the interested location.

3. Location Info Conversion to Cell ID.

   The location information provided in the AIoT request is converted into a Cell ID managed within 3GPP. The Cell ID is a unique identifier for the cell (coverage area) that the base station (gNB) is serving.

4. 5GC AIoT request initiation.

   The network initiates an AIoT service request process to inventory the target AIoT devices and to locate the available UE readers within the specified cell. This request includes an AIoT service indicator that can be used by the UEs to decide whether to react or not.

5. Broadcast the AIoT request within the Cell.

   The discovery information is broadcast within the cell. This broadcast includes an AIoT indicator and the target AIoT device IDs.

NOTE 1: How a BS broadcasts the AIoT request (e.g. using MBS or SIBs) is assumed to be addressed by RAN WGs.

NOTE 2: It is assumed that the BS sending the broadcast message is installed indoors, so the cell coverage for the UE reader is presumed to be equivalent to the AIoT service area.

6. UE decision.

   All UEs in the specified cell receive the broadcast message as in step 5, but only UEs that are capable of AIoT communication will react to the AIoT indicator broadcast in step 5 and perform the following steps. This ensures that only relevant devices participate in the AIoT communication process.

7. AIoT Communication.

Communication is performed between the UE reader(s) and the AIoT devices. This step involves the actual data exchange, such as sending commands to the AIoT devices and receiving their responses.

8. Response per UE.

   The network collects responses from each UE involved in the AIoT communication. These responses include the UE ID, AIoT device ID, and optionally any relevant AIoT data that was exchanged.

9. AIoT Response.

   The AIoT response is consolidated and sent back to the AF. This response includes the AIoT device ID and the data collected from the AIoT devices, if any.

10. UE Info Check with NWDAF/UDM.

    Optionally, the UE information is checked against the Network Data Analytics Function (NWDAF) and/or User Data Management (UDM) systems. This step can provide additional information about the UE readers, allowing the 5GC to determine which UE to use as the reader for the AIoT devices when there are duplications. For instance, if two UEs are delivering data from the same AIoT device, the 5GC can decide to use only one of them by taking the additional information into consideration. If one of the UEs is expected to leave in 5 minutes, it would be safer to choose the other one as the reader for further communication.

11. Determine Responsible UE(s).

    The network determines which UEs are responsible for specific AIoT devices, especially in cases where multiple UEs may have interacted with the same device. This decision can be made based on the operator policy and/or

the additional information retrieved in step 10, if available. A binding table is created in the AIoTF to map these relationships.

12. Additional AIoT Request.

    Another AIoT request can be initiated for the same target AIoT devices. This step might be part of an iterative process to refine the inventory and interaction with AIoT devices.

13. Look Up the Binding Table.

    The core network (e.g. AIoTF) looks up the binding table created in step 11 to find the UEs that are responsible for the specific AIoT devices. This ensures that requests are directed to the appropriate UEs. If the target AIoT devices are missing, the network can broadcast the request to all the UEs within the location.

14. Direct Request to Reader UEs.

    Requests can be sent directly to the UEs identified as readers for the AIoT devices in the binding table. This step ensures efficient and accurate communication with the AIoT devices.

    When communication with some of the target AIoT devices through the UE reader selected from the binding table fails, it may indicate that the target AIoT devices are out of coverage or broken. In this case, the network can remove the binding information from the table and perform steps 5 to 10 to maintain the latest binding status.

The procedure for selecting intermediate nodes when I-node ID is provided in the AF request is depicted in figure 6.11.2.2-3.

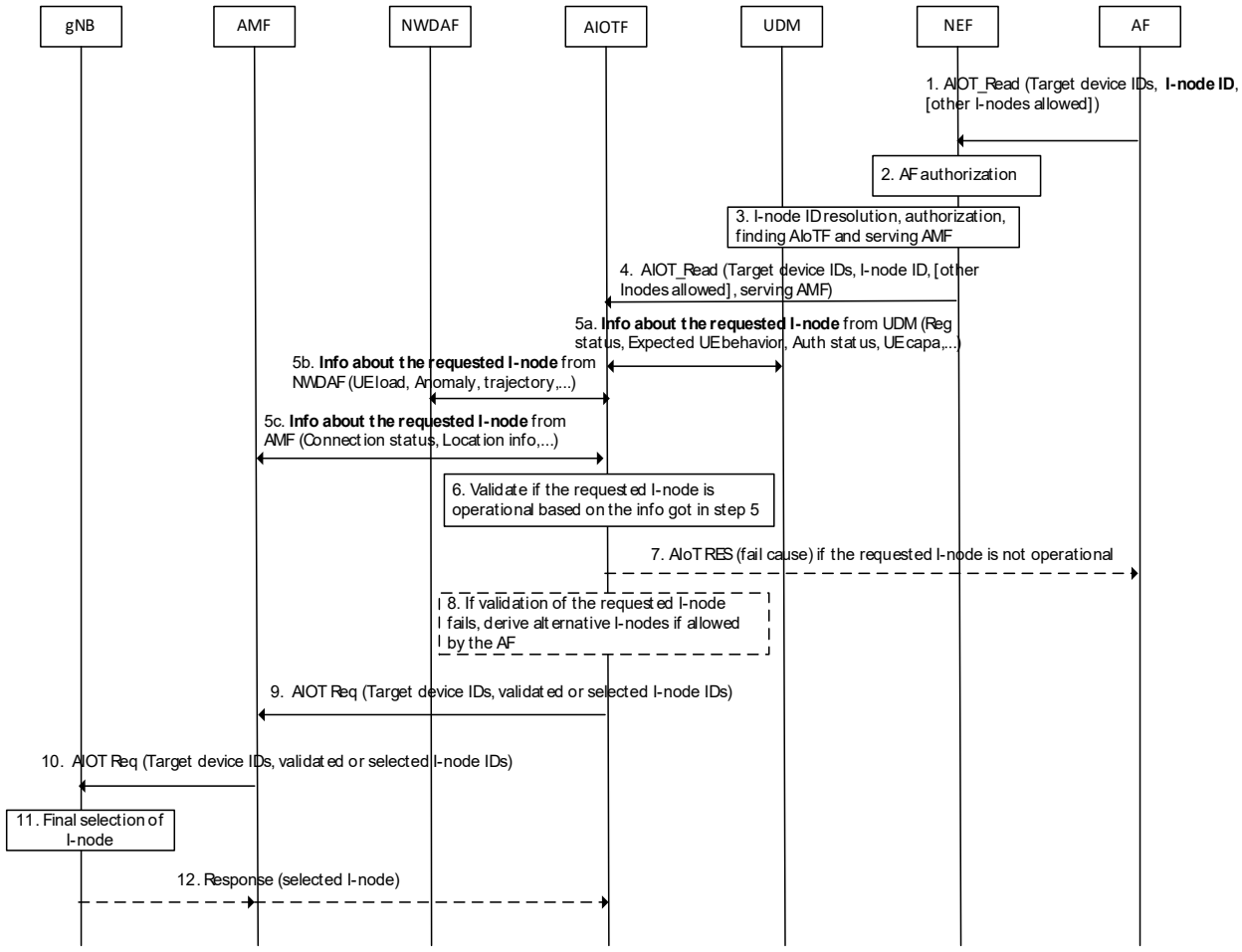


Figure 6.11.2.2-3: Information Flow for selection of I-nodes based on list of I-nodes from AF

1-4. Step 2 to 5 in clause 6.11.2.1 are performed.

5. The AIOTF collects necessary network information related to the requested UEs from relevant NFs.

a. The AIOTF may retrieve information related to the requested UE from UDM. This information enables checking, for example, whether the UEs are registered, authenticated, authorized to act as I-nodes. Additionally, the information from UDM can include the Expected UE behaviour, as described in clause 4.15.6.3 of TS 23.502 [5], that can be used to check the battery indication, stationary indication among other parameters.

b. The AIOTF may retrieve information related to the requested UE from NWDAF. This information enables checking, for example, if the discovered UEs are not too much loaded to serve the devices, function well without anomalies, or will not move out of the coverage for a significant period based on the UE related analytics as specified in clause 6.7 of TS 23.288 [13].

c. The AIOTF may retrieve information related to the requested UE from AMF, including its location, connection status and reachability.

6. The AIOTF, based on the information collected in step 5, validates whether the requested UE ID provided in the AF request can serve as an intermediate node.

7. If the requested UE is not operational and other "I-nodes allowed" flag was not set in the AF request, AIOTF sends AIOT Response with the failure reason to the AF, and the subsequent steps are skipped.

8. If the validation of the requested I-node fails and the 'other I-nodes allowed' flag was set in the AF request, the AIOTF may select a different list of UEs for the requested operation than the UE requested by the AF. This can be achieved by performing the step 5 in the figure 6.11.2.2-1 with the location set to the location information of the requested UE retrieved from AMF in step 5c.

9. The AIoTF requests the serving AMF to initiate the communication with the AIoT devices, which includes the target AIoT device IDs, and the selected I-node IDs.

10. The AMF forwards the AIOT read request to gNB along with the target AIOT device IDs and the selected I-node IDs.

11. The gNB performs additional validation to assess if the received I-nodes are optimal for serving the AIoT devices, considering its local conditions such as response time, received signal strength, etc. Subsequently, the gNB makes the final selection on the best I-node. It is also possible that the gNB does not select any of the requested I-nodes if local conditions are not satisfied.

12. The gNB may respond to the AIOTF via AMF with the final list of selected I-nodes.

6.11.3 Impacts on services, entities and interfaces

Impacts on existing entities:

NEF:

- Supports exposing a new API to AF for AIoT services.

UDM:

- Supports validating if a UE is authorized to act as an I-node.

AMF:

- Supports forwarding the AIOTF requests to gNB.

gNB:

- Supports handling new AIoT related services.
- Supports making the final selection of the I-node.

UE (I-node):

- Supports indicating its capability to support AIoT services during UE registration.
- Supports communicating with AIoT devices.

New NF:

- AIOTF:
- This can be a standalone NF or combined with NEF.
- Supports necessary network information for specific UEs or specific location from multiple NFs (e.g. AMF, UDM, NWDAF).
- Supports selecting or validating I-nodes.
- Supports forwarding the AF requests to AMF.

6.12 Solution #12: UE Function Delegation in Intermediate Node for AIoT Device

6.12.1 Description

This solution applies to those AIoT devices that are not capable of direct communication with a 5G network, and instead, may communicate with the network through an Intermediate Node (IN), or "IN-UE". The AIoT device and the IN-UE may communicate with each other using backscattering communication or other sidelink technologies. It is assumed that the AIoT device is able to identify itself, e.g. using its Device Identifier, over the "AIoT device - UE" interface.

NOTE 1: How the AIoT device communicate with the IN-UE is not in the scope of this solution.

The general principles of this solution are:

For a AIoT device or a group of AIoT devices that the IN-UE acts as Intermediate Node, the IN-UE instantiates a "UE Function Delegation Module" (UFDM) which represents the AIoT device and appears as a "UE" towards the 5GC. It is also referred to as "UFDM-UE" in this solution. The instantiation may be initiated after the IN-UE has been authorized as the IN. The AIoT device(s) for which the IN-UE acts as the IN may be preconfigured in the IN-UE or obtained during the IN authorization procedure.

NOTE 2: The IN authorization procedure is not covered by this solution.

After a new UFDM is instantiated, the IN-UE initiates an "AIoT device provisioning" procedure with the 5GC and through this procedure the UFDM is allocated a temporary UE identifier, e.g. a temporary SUPI, associated credentials, and other necessary configuration and policies (e.g. URSP policy). A UFDM maintains its own NAS connection and registration state with the 5GC. The UFDM-UE's NAS connection and IN-UE's NAS connection share the IN UE's RRC connection and the UFDM's NAS messages may be sent as payload of IN UE's NAS messages.

Though from the perspective of the network, a UFDM functions as a separate UE, the 5GC maintains the association between the intermediate node UE and other "UE(s)" represented by the UFDM(s) residing inside the intermediate node UE, so the network can optimize some procedures, e.g. paging procedure.

The IN UE maintains the association between the UFDM and the AIoT device and manage the lifecyle of the UFDM. For example, when the IN-UE detects that the AIoT device is not responding to the Command or activation signal, the corresponding UFDM instance may be deactivated and the 5GC may also be informed. In that case, the previously provisioned UE identifiers and credentials for the UFDM-UE is invalidated.

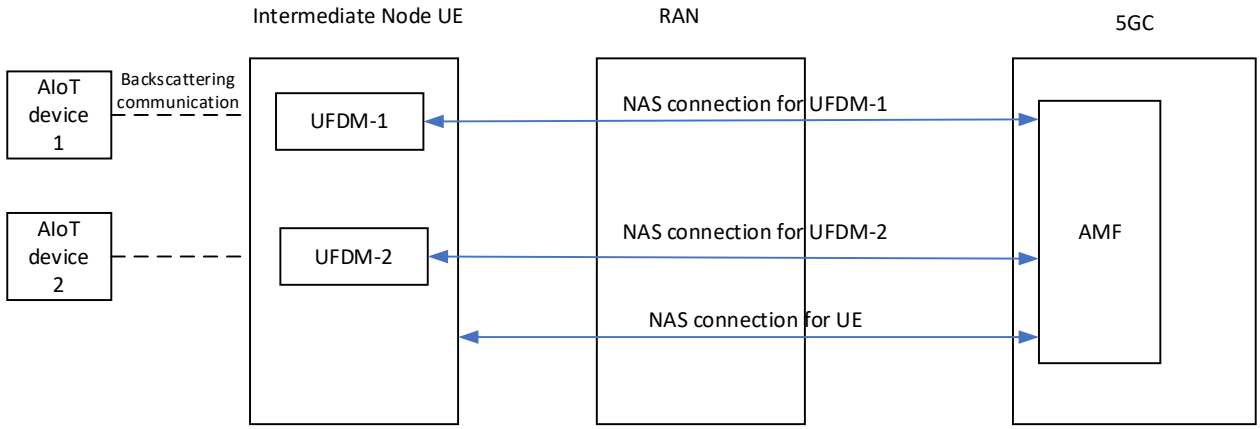


Figure 6.12.1-1: UFDM in AIoT Capable UE for AIoT devices

The UFDM handles the data forwarding between its associated AIoT device and the network. For example, the UFDM may use Control Plane 5GS Optimization for CIoT mechanism to forward AIoT device data to the application server or vice versa.

6.12.2 Procedures

6.12.2.1 UFDM instantiation and provisioning

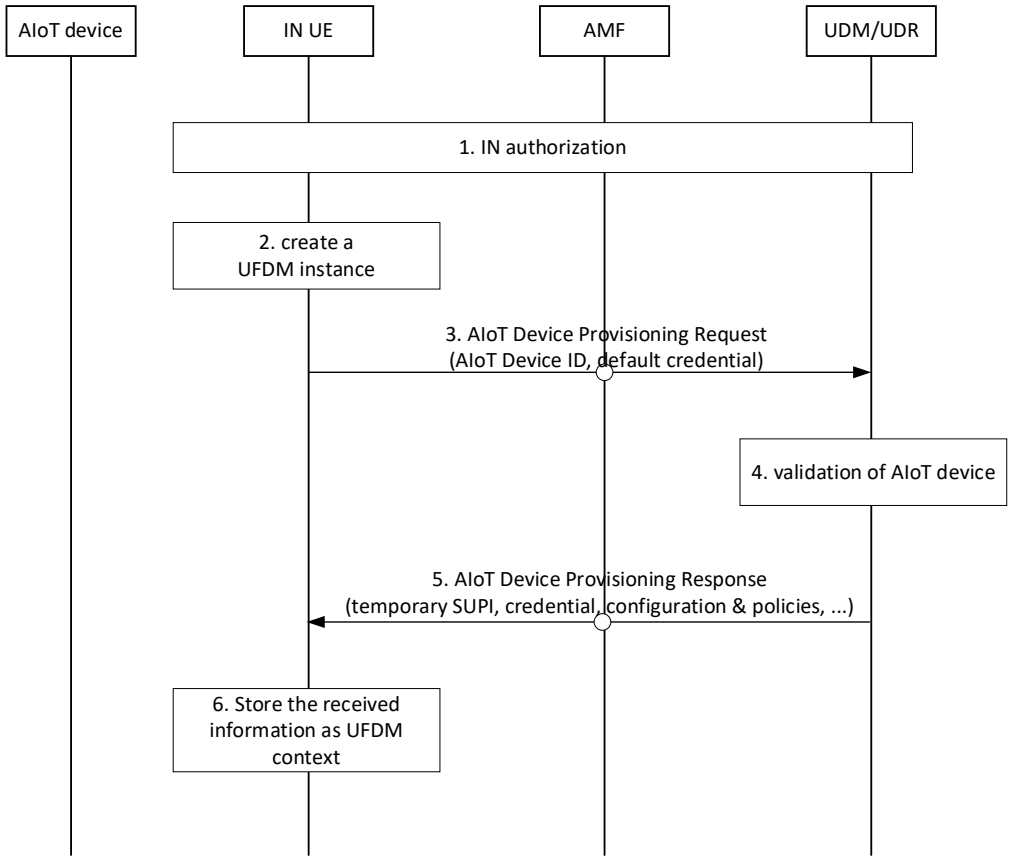


Figure 6.12.2.1-1: UFDM instantiation and provisioning

1. The IN-UE is authorized as a IN. This may happen during the IN-UE Registration based on IN-UE capabilities and subscription information. The AIoT devices that the IN-UE can act as IN may be preconfigured in the UE and provided by the network during the authorization procedure.

2. The UFDM instantiation in the IN UE is triggered by step 1. For each device that the UE can act as IN, The IN UE creates a UFDM instance, allocates a UFDM instance ID and associate it with the AIoT device.

3. The IN UE initiates AIoT Device Provisioning Request towards the 5GC UDM and includes the AIoT device identifier in the request. If the IN UE is preconfigured with default credentials for the AIoT device, it may also include the default credentials in the request.

4. The UDM validates the AIoT device ID and default credentials (if available). If necessary, the UDM may interact with external servers via NEF to complete the validation process.

NOTE: How the device ID and credential is validated is not covered in this solution.

Editor's note: The details of the provisioning procedure (Step 3 and 4) are FFS and the security aspects may need SA WG3 study.

5. If the AIoT device is valid, the UDM allocates a temporary SUPI, associated credentials (e.g. AKA keys) and provide them to the IN UE in the response message. Other necessary configuration and policies, e.g. URSP rules, default QoS rule, etc. may also be provided.

6. The IN UE stores the received information as the UFDM context.

6.12.2.2 UFDM Registration

UFDM acts as a separate "UE" towards the network so it performs Registration procedures similar to what a normal UE would do. However, to avoid unnecessary signalling, some of the UFDM's Registration procedures (e.g. periodical Registration update or mobility Registration update) may be combined with the IN UE's own Registration procedures.

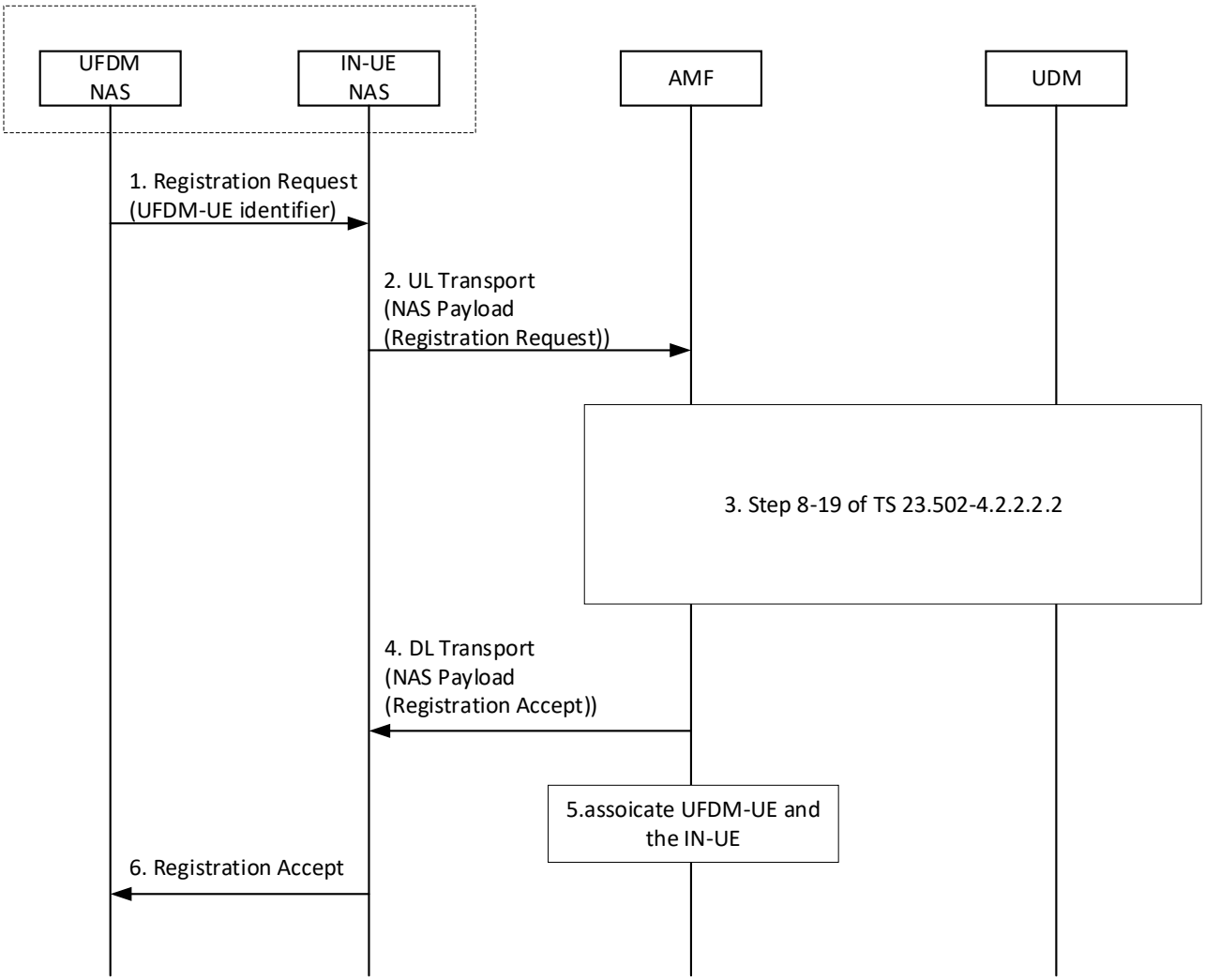


Figure 6.12.2.2-1: UFDM registration

1. The UFDM-UE NAS sends a "Registration Request" to the IN-UE NAS. The UFDM-UE may use the SUPI that's obtained through the provisioning procedure as the initial UE identifier.

2. If the IN-UE is not already in CM-CONNECTED state, it initiates the signalling procedure to enter CM-CONNECTED state, then it sends the UFDM-UE's Registration Request as the payload of its own NAS message, e.g. UL Transport, to the serving AMF.
3. The AMF and other NFs handles the UFDM-UE's Registration request as described in TS 23.502 [5]-clause 4.2.2.2.2.
4. The AMF sends the Registration Accept as the payload of the IN-UE's NAS message, e.g. DL Transport.
5. The AMF associate the UFDM-UE identifier with the IN-UE identifier. By maintaining this association, the AMF is able to target the IN-UE as the recipient for signalling/data of the UFDM-UE.
6. The IN-UE forwards the Registration Accept message to the UFDM-UE.

6.12.3 Impacts on services, entities and interfaces

UE:

- Supports UFDM instantiation and management.
- Supports forwarding UFDM NAS signalling as payload of IN UE NAS messages.
- Supports AIoT data forwarding as UFDM data.

AMF:

- Supports handling UFDM NAS messages as IN NAS payload.
- Supports handling AIoT Device Provisioning procedure.
- Supports maintaining association between UFDM UE and IN UE.

UDM/UDR:

- Supports validation of AIoT device identifier and credential.
- Supports AIoT Device Provisioning.

6.13 Solution #13: Multi-level Device ID management

6.13.1 Description

When it comes to Ambient IoT Device ID Management, it is likely that different Device ID formats are expected depending on the procedures to be followed between Ambient IoT Device, Reader, Controller and Application Function.

On highest level, a Permanent ID is expected to enable uniquely identifying Ambient IoT Device by the 3GPP system. Such ID can entail MNO ID, Enterprise (Owner) ID, Instance ID (see solution #2 in TR 23.700-13).

However, such long Permanent ID can not be conveyed conveniently on all AS level procedures (e.g. between Ambient IoT Device and Reader) or non-AS level procedures (e.g. between Reader and Controller) due to large overhead or security issues.

As a result, it is expected that the Permanent ID of the Ambient IoT Device to be mapped to other temporary IDs across different entities / procedures.

This solution assumes the same principles on temporary ID assignment, mapping and interpretation is applicable to Ambient IoT Device, Reader, Controller irrespective of topology types.

6.13.2 Procedures

At least, two levels of temporary IDs are proposed:

- Long temporary ID: A long temporary Device ID to be used between the Controller and the Reader for command procedures (similar to paging) for certain Ambient IoT Device(s):
  - A set of long temporary IDs can be assigned and pre-provisioned by the Controller to the Reader.
  - The Controller maintains the mapping between long temporary ID and unique Ambient IoT Device identifier (e.g. based on the Permanent ID) per Device based on notification from the Reader.
  - Only one Reader maintains the mapping of long temporary ID to short temporary ID per Ambient IoT Device.
  - If CN paging fails, the Controller will inquire other Readers. The Controller pages the Ambient IoT Device using the device's permanent (unique) ID at the last known Reader. On successful paging response / notification from one of the Readers (incl. the mapping of long temporary ID for paged Ambient IoT Device), the Controller will accordingly update the mapping between long temporary ID and unique Ambient IoT Device identifier.

NOTE 1: Long temporary ID does not need to be exposed to the Ambient IoT Device.

- Short temporary ID: A short temporary Device ID to be used between the Reader and Ambient IoT Device for exchanging inventory request/ responses or command (request)/ responses.
  - The short temporary Device ID can be assigned and provisioned by the Reader to the Ambient IoT Device for message exchanges between the Reader and the Ambient IoT Device.
  - The Reader maintains mapping between long temporary ID and short temporary ID per Ambient IoT Device.
  - The Reader notifies the Controller on mapping of long temporary ID per ambient IoT Device (e.g. based on the Permanent ID).

NOTE 2: The exact format and further details of short temporary ID is in remit of RAN WGs.

NOTE 3: The temporary IDs can be reassigned (when needed) to limit the pool size of temporary IDs. However, the temporary IDs must be locally unique in the reader (for short temporary IDs) and in the Controller (for long temporary IDs).

NOTE 4: Long temporary ID validity should be at minimum equivalent to short temporary ID validity at AS level. However, even longer validity for Long temporary ID can be considered to minimize the signalling towards the Reader.

NOTE 5: Security aspects on temporary IDs will be coordinated with SA WG3.

6.13.3 Impacts on services, entities and interfaces

Controller:

- Assign and pre-provision a set of long temporary IDs per Reader.
- Use long temporary ID as part of command procedures towards the Reader(s) for certain Ambient IoT Devices.
- Maintain the mapping between long temporary ID and unique Ambient IoT Device identifier (e.g. based on the Permanent ID) based on the notification from the Reader.

Reader:

- Assign and provision short temporary ID per Ambient IoT Device.
- Map and Maintain mapping of the long temporary ID to short temporary ID per Ambient IoT Device.
- Notify the Controller on mapping of long temporary ID per Ambient IoT Device (e.g. based on the Permanent ID).

Ambient IoT Device:

- Can interpret and respond to its assigned short temporary ID towards the Reader.

6.14 Solution #14: AIoT Device ID with consideration on credential holder, service provider and mobile network

6.14.1 Description

This solution addresses the KI#2 and the aspect about how to identify Ambient IoT Device and how to format the identifier.

In this solution, the following terms have been used:

- AIoT Device ID: the identifier used to identify the AIoT Device;
- Serving network: PLMN network which includes the Reader and the radio/core network;
- Credential holder: This domain holds the credential associated with the AIoT Device ID;
- Service provider: This domain holds the application function for Inventory and/or Command.

The following stakeholders have been considered:

- Case 1: AIoT Device ID and its associated credential are managed by network (PLMN network operator):
  - Case 1.1: AIoT Device ID and its associated credential are managed by the serving network (PLMN network operator). Service provider can belong to the same serving network (PLMN network operator) or 3rd party.
  - Case 1.2: AIoT Device ID and its associated credential are managed by one network (PLMN network operator); the Credential holder is another PLMN network which is different from the serving network. Service provider may belong to the same serving PLMN network, or Service provider may belong to Credential holder associated to the PLMN network or 3rd party. In this case, the serving network and the credential holder has the "roaming like agreement" to ensure that the AIoT Device ID can be e.g. verified or authenticated.
- Case 2: AIoT Device ID and its associated credential are managed by the 3rd party:
  - Case 2.1: AIoT Device ID and its associated credential are managed by service provider (3rd party); the serving network can be PLMN network. In this case, the Credential holder associated with AIoT Device ID is the service provider.
  - Case 2.2: AIoT Device ID and its associated credential are managed by a 3rd party other than the service provider (3rd party). In this case, the serving network can be a PLMN network.

Editor's note: It is FFS how to consider AIoT Device ID managed in the SNPN network.

In case 1, since the AIoT Device ID and its associated credential are managed by the network (PLMN network operator), then AIoT Device ID has the information to identify the network. The format can be designed as shown in the figure 6.14.1-1. AIoT Device Identification Number is required to be unique in the PLMN network.

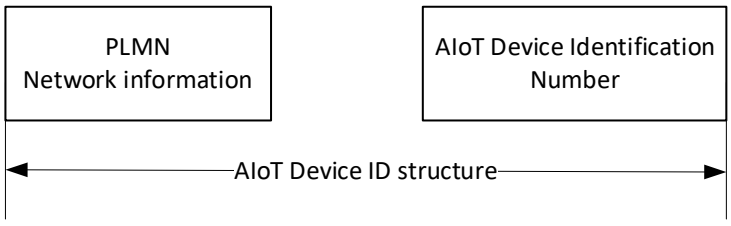


Figure 6.14.1-1 AIoT Device ID structure for case 1

In case 1, the AIoT Device ID can also be in NAI like format <username> @ operator where the <username> part is assigned by the operator and the operator info can be used to identify the operator. In this format, the <username> part is required to be unique @operator.

NOTE: NAI like format does not mean it to be in the format of string.

In case 2, since the AIoT Device ID and its associated credential are managed by 3rd party (service provider or e.g. AIoT Device manufacturer), then the AIoT Device ID has the information to identify the 3rd party (service provider or

other 3rd party e.g. AIoT Device manufacturer). The format can be NAI like format<username> @ 3rd party (SP or and other 3rd party e.g. AIoT Device manufacturer). The username part is assigned by the 3rd party.

6.14.2 Procedures

This clause is to describe the high-level procedure including configuration and provisioning for 2 cases as specified in clause 6.14.1. Other details how to utilize the AIoT Device ID are specified in other solutions.

Case 1:

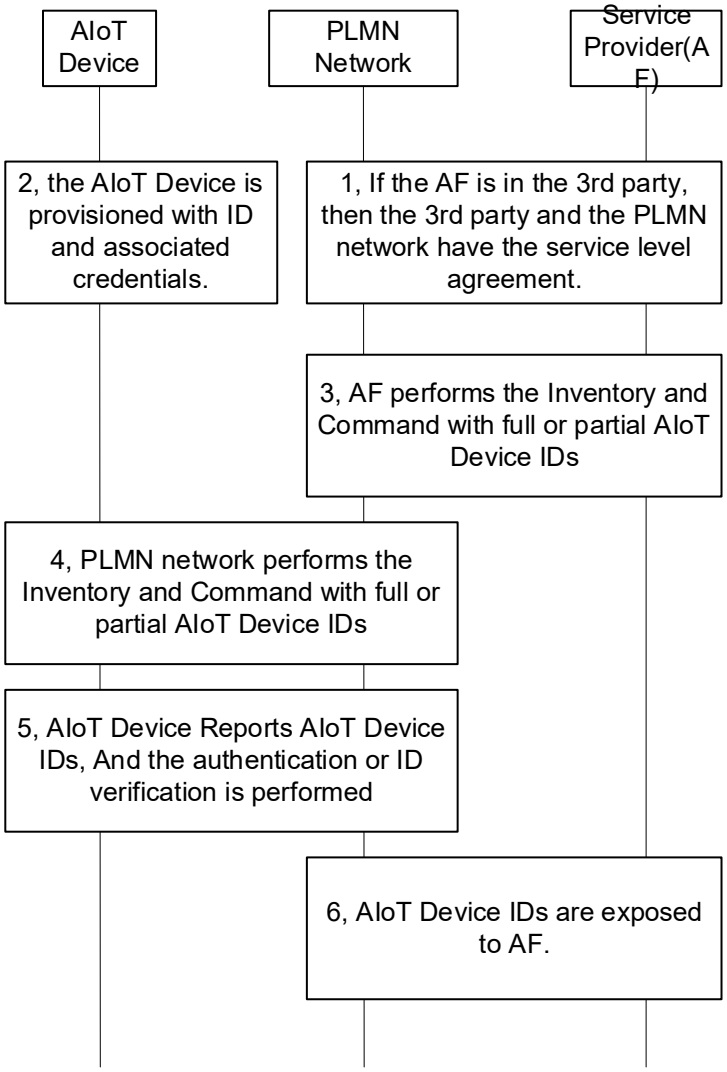


Figure 6.14.2-1: High-level procedure for case 1

1. If the AF is in the 3rd party, then the 3rd party and the PLMN network have the service level agreement to ensure that exposure of AIoT Device ID is also secure.
2. The AIoT Device is provisioned with AIoT Device ID and associated credentials.
3. The AF performs the Inventory or Command with full or partial AIoT Device IDs to the PLMN network. The partial AIoT Device ID is some part of full AIoT Device ID. The full AIoT Device ID is <username> @ operator or full information as shown in figure 6.14.1-1.

Editor's note: It is FFS what is the criteria to use the partial AIoT Device ID.

4. The PLMN network performs the Inventory or Command with full or partial AIoT Device IDs as in step 3 to AIoT Devices.

5. The involved AIoT Devices responds with the AIoT Device IDs together with the Inventory Response or the Command Response to the PLMN network. In this step, the PLMN network can perform the Authentication procedure or the AIoT Device ID verification procedure to check the AIoT Device valid or not.

Editor's note: The verification of AIoT Device ID and authentication requires coordination with SA WG3.

6. The PLMN network exposes the AIoT Device IDs to the AF.

Case 2:

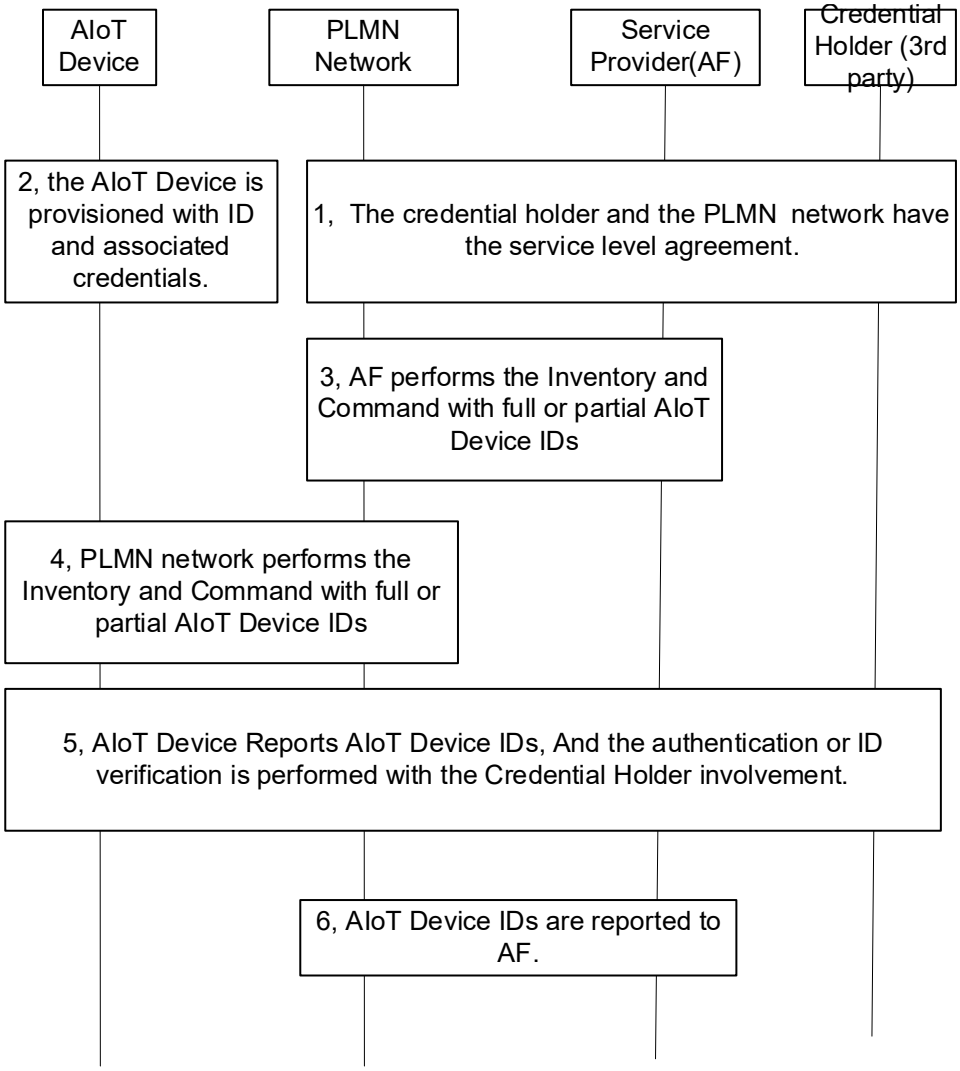


Figure 6.14.2-2: High-level procedure for case 2

1. The Credential holder and the service provider may be the same. The credential holder domain and the PLMN network have the service level agreement to ensure that the AIoT Device ID can be e.g. verified or authenticated.

2. The AIoT Device is provisioned with AIoT Device ID and associated credentials by the credential holder (service provider and other 3rd party e.g. AIoT Device manufacturer).

3. The AF performs the Inventory or Command with full or partial AIoT Device IDs to the PLMN network. The partial AIoT Device ID is some part of full AIoT Device ID. The full AIoT Device ID is <username> @ 3rd party.

4. The PLMN network performs the Inventory or Command with full or partial AIoT Device IDs as in step 3 to AIoT Devices.

5. The involved AIoT Devices responds with the AIoT Device IDs together with the Inventory Response or the Command Response to the PLMN network. In this step, the PLMN network can perform the Authentication procedure or the AIoT Device ID verification procedure with the involvement of credential holder (service

provider and other 3rd party e.g. AIoT Device manufacturer) to check the AIoT Device valid or not. In this step, if the service provider is not the credential holder, then it is not involved in this step.

Editor's note: The verification of AIoT Device ID and authentication requires coordination with SA WG3.

6. The valid AIoT Device IDs are reported to the AF.

6.14.3 Impacts on services, entities and interfaces

Editor's note: This clause captures impacts on existing services, entities and interfaces.

6.15 Solution #15: Ambient IoT Device States

6.15.1 Description

6.15.1.1 Overview

This solution addresses KI#2, KI#3 and proposes new states and sub-states for AIoT devices that may assist the AIoT services the AIoT device supports in any particular moment. In particular, the solution consists of the following aspects:

- A new set of states or mode of operations are defined for the AIoT device, which can be transitions to/from based on network requests, where these requests may come from the AF.
- A new service exposure is proposed that enables the AF to trigger a change in the AIoT device state (or mode of operation). This may also be performed by the network due to local configuration.
- The network and AIoT device can confirm the result of the state change (or change in mode of operation), or provide a report on the current state the AIoT device is currently in.

In this solution, an AIoT device state may also refer to an AIoT device mode of operation and/or to a service operation that the AIoT device can take while being in such state. As such the term state is not to be considered as a restriction that binds an AIoT device to a particular state but rather as an example of how the AIoT device may perform or behave in certain modes or conditions or based on service operations or expectations.

Note that an AIoT device may also be referred to as a UE, or an AIoT device may be considered a special type or category of a UE. A UE may have at least AIoT capabilities, or an AIoT device can be a UE with restricted capabilities.

Clause 6.15.1.2 describes new AIoT device states proposed by this solution and clause 6.15.2 describes a procedure to toggle the state of an AIoT device.

6.15.1.2 New AIoT Device States

AIoT device states can be useful for the application and/or the network to be aware of which AIoT devices are already operating and which function they may be performing, which may in turn assist in their management and monitoring. As an example, the reliability of inventory taking results would be higher when all the AIoT devices that are meant to report identifiers, location, etc. are in a state validated by the network. Furthermore, an appropriate sub-state supporting event-based reporting allows an AIoT device to know how to behave after the event is detected, i.e. whether information regarding certain event should e.g. be reported, stored or ignored.

An AIoT device that has not yet been registered with the network may be in an INVALIDATED state. In this state, the AIoT device needs to be validated e.g. by going through authentication or other security procedures, or by its identity being validated. The AIoT device may also be validated as part of the registration process of the device with the network.

After a successful security and/or validation procedure, the AIoT device may enter a new VALIDATED state, where this is the main state of operation of this device.

Moreover, the AIoT device may operate on (or switch to) any of the following sub-states:

- REPORT-ONLY (or READ-ONLY): in this sub-state the AIoT device is only supposed (or allowed) to report data, or read data that has been collected and/or stored, and report the data.

- In this sub-state the AIoT device may report data either due to the AIoT device needing to report data e.g. periodically, or based on a solicitation (or command, trigger, and/or indication, etc.) from the network. This data reporting may not necessarily be due to a threshold being crossed (e.g. the data reporting may not necessarily be due to occurrence of an event).

- RECEIVE-ONLY (or WRITE-ONLY): in this sub-state, the AIoT device may only receive data and/or commands from the network, and/or from an application function (AF) via the network, but this AIoT device does not report any data itself. Additionally, the AIoT device may be permitted to store this received data. The AIoT device may acknowledge the receipt of the data and/or command but may not send data. As an example, the AIoT device may toggle to a different sub-state, or stay in the same sub-state after receiving the data.

Editor's note: Whether separate dedicated REPORT/READ-ONLY and RECEIVE/WRITE-ONLY states are required is FFS.

Editor's note: Whether REPORT/READ-ONLY state permits data and/or service requests being received in the device is FFS.

- RECEIVE-and/or-EVENT-REPORT: in this sub-state the AIoT device only receives data and/or commands but can only report if particular events and/or conditions have occurred. For example, the event may be that a certain threshold has been crossed in relation to a parameter that has been sensed or monitored. In one example, upon the observed parameter exceeding a pre-configured threshold, the AIoT device enters or change sub-state to EVENT-REPORT in order to report the occurrence of this event to the network and/or the AF:

    - In this sub-state, the AIoT device may not report even if solicited to do so as long as the event in question has not been detected. This differentiates this sub-state from the READ-ONLY or REPORT-ONLY sub-states.

Any of the functions above can be combined into one sub-state as well. For example, the WRITE-ONLY behaviour may be part of the RECEIVE-ONLY sub-state.

Editor's note: The name of the above states/sub-states and whether additional states/sub-states incl. energy-related device states/sub-states can be defined is FFS.

Editor's note: Further details on how the above states may assist the service operations of AIoT devices is FFS.

The network may be configured to inform the AIoT device which state to start with, where this may be based on local policies or based on subscription information. As such, after validation of the AIoT device, the network may send a message, which may be part of the validation process or the registration process, to inform the AIoT device which state the AIoT device should start operating in. Alternatively, the AIoT device may indicate the desired state, and the network may verify whether the UE can operate in that state or not.

Editor's note: Whether and how this solution can be applied to a group of AIoT devices is FFS.

6.15.2 Procedure for State Toggling in AIoT Devices

An AIoT device may operate in any of the states described in clause 6.15.1.2. This clause proposes a method by which the network can toggle the AIoT device state by issuing a command such that the AIoT device will transition from one state to another. Toggling the state of an AIoT device may be useful e.g. after the AIoT device has been validated by the network for the AIoT device to know how it should respond to an inventory or read/write command. For example, the AIoT may reject or ignore such request if it is in INVALIDATED state described in clause 6.15.1.

This proposal is based on an interaction between an AF and the network, but it could also be locally based on policies of the network. The AF for example can have preference for a device to operate in a particular state only based on the usage of the device.

The main proposal in this clause is that the network should inform the AIoT device to change its expected behaviour or set of functions, where each set of functions or expected behaviour may correspond to a well-known action or behaviour. A network entity (e.g. AMF) subscribes to the UDM for getting events related to a requirement to toggle an AIoT device state. NAS related signalling is supported, which could be a specific version of the NAS protocol for AIoT devices.

The dynamicity of the state toggling requests can be adjusted by the AF according to criteria such as minimization of complexity, energy consumption, etc.

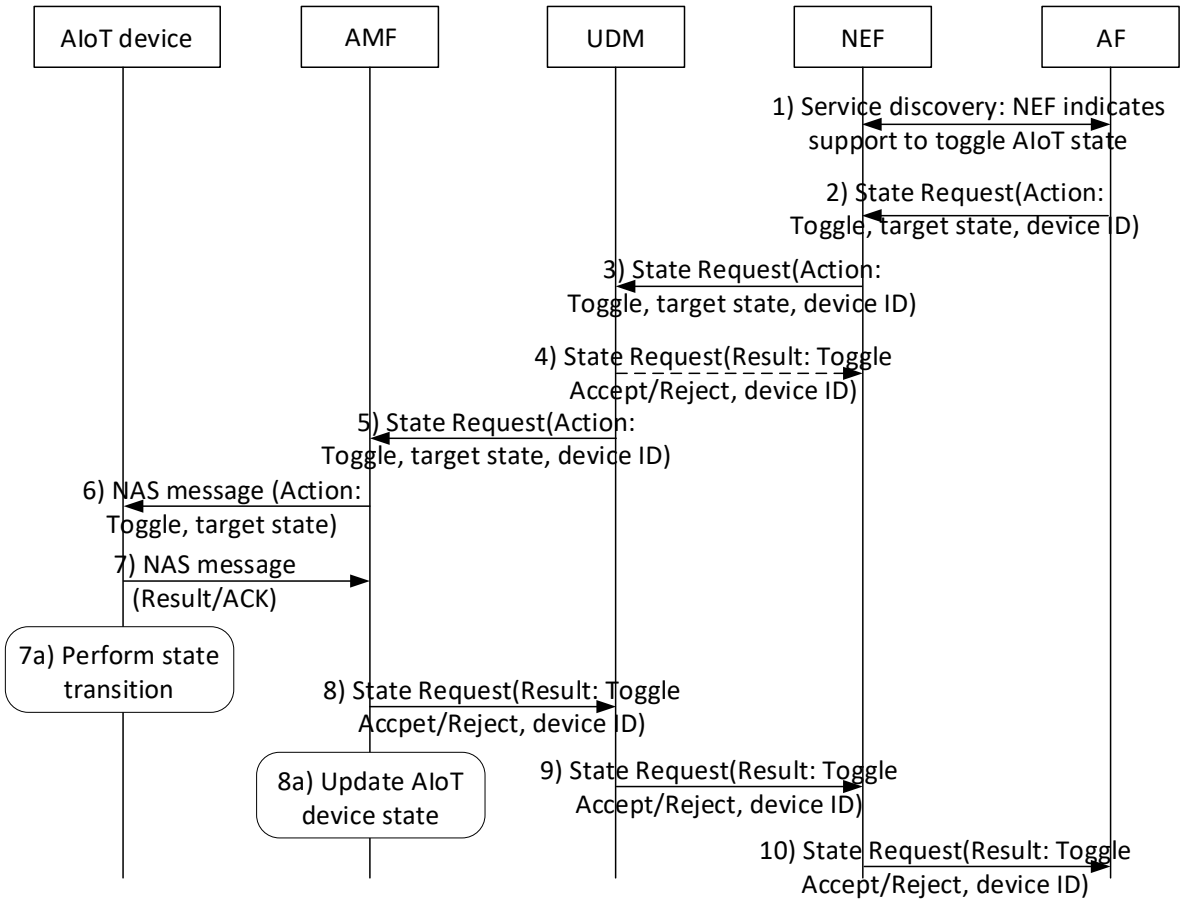


Figure 6.15.2-1: Procedure to support toggling of states in an AIoT device

The procedure in Figure 6.15.2-1 to support toggling of states in an AIoT device is described step by step below.

1. The AF discovers the capability of the NEF exposure service, in particular the service to toggle the AIoT device state.

2. The AF provides a request to the NEF to toggle the AIoT device's state. The AF identifies an AIoT device or group of AIoT devices in the request. The request also provides a clear indication that what is needed is to toggle an AIoT device(s)'s state(s) and as such the target state(s) is/are indicated. The current AIoT device(s)'s state(s) may also be indicated if necessary.

NOTE 1: A single AIoT device will be assumed in this solution hereafter but the solution can still apply to a group of AIoT devices.

3. The NEF communicates the request to toggle the AIoT device state with the UDM in order to verify if the AF is allowed to place such a request. The NEF may indicate the details of the request to toggle the AIoT device state to a target state, e.g. target state and/or current state. The NEF may also provide the identity of the AF.

4. The UDM verifies if the request can be granted based e.g. on the AF identity, the type of the request, etc. The UDM may indicate if the request is not authorized, or if the request can be authorized based on the above criteria.

5. The UDM determines which entity is responsible for this request, e.g. AMF, and communicates with such responsible entity to provide the request to toggle an AIoT device state indicating the target state to use.

Editor's note: Whether the AMF and/or other new or existing entities are responsible for this request is FFS.

Editor's note: It is FFS whether the AMF or any other 5GC NF stores information related to AIoT devices states.

6. The AMF sends a NAS message to the AIoT device indicating the new state the AIoT device should transition to. Alternatively the request may be to report the current state of the AIoT device.

Editor's note: Whether a protocol different from NAS may be used to indicate the new state to the AIoT device is FFS.

7. The AIoT device reports the result of the operation e.g. indicating if any state transition has been performed or reporting the current state of the AIoT device.

   7a. The AIoT device transitions into a new state based on the request received from the network. This step assumes that the AIoT device has successfully toggled its state or that it can successfully do so. Otherwise if this is not possible then the AIoT device does not change its state.

NOTE 2: Step 7a may occur before the AIoT device reports the result of the operation to the network in step 7.

8. The responsible entity in step 5 (e.g. AMF) sends the result of the request to the source NF i.e. the UDM. The result may be that the AIoT device has changed its state or not, and the result may contain the AIoT device's current state (which may be a new state e.g. in the case that the AIoT device has changed its state, or any other state such as the state that the AIoT device is in and that may not have changed based on the request from the network).

8a. The responsible entity in step 5 (e.g. AMF) updates the AIoT device's state or context to reflect the new AIoT device state, based on the operation or result indicated by the AIoT device. If the operation or result is successful, then the responsible entity in step 5 (e.g. AMF) can update the AIoT device's state, otherwise the AIoT device state is not updated and optionally a "failure to update" indication may be stored.

9. The UDM indicates the result of the request to update a AIoT device state (or to report the current state of the AIoT device) to the NEF. This may be based on the result received from the responsible entity in step 5 (e.g. AMF).

10. The NEF indicates the result of the operation e.g. the result to toggle the AIoT device state or to report the current AIoT device state.

6.15.3 Impacts on services, entities and interfaces

Editor's note: Impacts are FFS.

6.16 Solution #16: AIoT Registration/Onboarding

6.16.1 Description

This solution addresses the part of KI#2 related to the AIoT device registration/onboarding. The first bullet and NOTE 1 in solution#1 of clause 6.1.2 mention the onboarding and parameter provisioning. This solution provides a procedure to achieve this.

The relation between this solution and solution 1, is that this solution will provide means to securely authenticate the AIoT device and provision the AIoT with necessary parameters to control the temporary ID as described in solution 1 i.e. an efficient way to generate new temporary ID and to enable AIoT device privacy protection. The proposed onboarding procedure requires minimum three messages to be sent over the radio interface, but the following Inventory and command procedures can be achieved with only two messages (one DL trigger + one UL) and fulfil the SA WG1 requirement in clause 5.2.6 of TS 22.369 [2]:

> *The 5G system shall be able to provide a mechanism to protect the privacy of information (e.g. location and identity) exchanged during communication between an Ambient IoT device and the 5G network or an Ambient IoT capable UE.*

NOTE 1: The actual number of physical messages sent over the radio interface between AIoT device and Reader depends on the RAN design. The numbers used above should be seen as the number of logical messages between the AIoT device and Reader.

As the CN NF would get the information of all AIoT devices that the CN NF performed this registration procedure with, it can also be seen as an initial Inventory collection by the network. This Inventory information can be exposed to an AF subscribing to such exposure service.

The solution is based on the following assumptions:

- The AIoT device is not capable to initiate the registration procedure. The Network will trigger AIoT devices to register.

- The AIoT reader (UE or Base station) periodically or on-demand broadcasts a message, Registration poll message, that triggers nearby unregistered AIoT device(s) to respond. The Registration poll message is assumed to be non-AIoT device specific to allow all unregistered AIoT device(s) to respond in an area.

- An AIoT device does not respond to the Registration poll message if it is already registered in the network or the PLMN is not allowed.

NOTE 2: There may not be a requirement to respond to a Registration poll message when already registered in the network. When concluding KI#2 it can be decided if an AIoT device should respond Registration poll message if it has been a long time since last communication with the network i.e. Inventory or Command.

If the AIoT device is registered to a network but have not received any Registration poll message from this network for some time, then the AIoT device should consider itself unregistered and perform registration/onboarding with a new network after receiving a Registration poll message from this network.

NOTE 3: It cannot be assumed that an AIoT device has a running clock to track time. However, an AIoT device could determine the lapsed time based on broadcasted time by the Reader.

- The AIoT device responds with a Registration Request that includes e.g. device owner ID and concealed device ID. This will allow the CN NF to retrieve further information to proceed with the registration/onboarding of the AIoT device. If the CN NF was not able to retrieve the required information or the AIoT device was not possible to authenticate and authorize, the AIoT device will be rejected to register in the network and shall refrain from trying to register again in this network.

- During the registration procedure the CN NF will provision the AIoT device with e.g. security parameters, initial TempID or parameters to derive the initial Temp ID i.e. parameters for the Temp ID generation algorithm as discussed in clause 6.1.

- An AIoT device has two states: Unregistered or Registered. The AIoT device will remain in Unregistered state until the AIoT device receives the Registration Accept message.

6.16.2 Procedures

CN NF is used to name the network function that performs the necessary task to register/onboard an AIoT device to the network. The CN NF will be defined together with the system architecture design to support the Ambient IoT in 5GC.

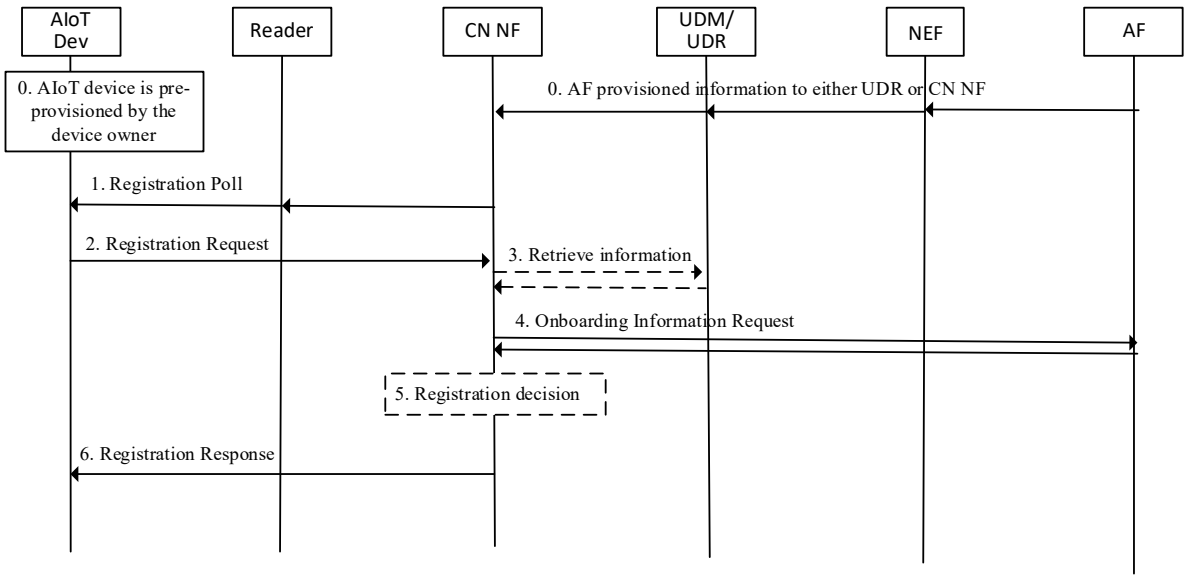


Figure 6.16.2-1: Registration/onboarding procedure for an AIoT device.

0. It is assumed that the AIoT device is pre-provisioned by the device owner with information to be used when register/onboarding to a network. The information includes a device unique identifier (unique at owner level, not necessary globally unique), device owner ID and security IE.

NOTE 1: Whether or not an AIoT device is pre-provisioned with a list of preferred network IDs can be decided when concluding KI#2.

It is assumed that the device owner and the MNO has an SLA. A device owner AF may have provisioned the network with information needed to onboard an AIoT device(s) or a URL or FQDN to establishing IP connection to an AF that holds the device unique onboarding information.

1. The AIoT reader (UE or base station) broadcasts a Registration poll message that includes e.g. the PLMN ID or NID. The broadcast can either be an immediate broadcast triggered by CN NF or a scheduled/periodic broadcast configured by the CN NF.

Editor's note: The format of the Registration poll message needs to be defined in cooperation with RAN WGs.

2. An AIoT device that is not registered in the network responds with a Registration Request (device owner ID, concealed device ID). The Reader forwards the Registration request message to the CN NF.

NOTE 2: Whether or not an AIoT device checks the network ID to a list of preferred network IDs stored in the device can be decided when concluding KI#2.

Editor's note: How the concealed device unique ID is created will be defined by SA WG3.

3. The CN NF may hold AIoT device owner information or can retrieve AIoT device owner information from the UDM/UDR using the received device owner ID. In case the AF has provisioned the network with the device unique information needed to onboard the AIoT device, then step 4 can be skipped.

4. The CN NF uses the retrieved AIoT device owner information to establish a secure IP connection with the AF to retrieve the device unique onboarding information by providing the concealed device ID to the AF. The AF must be able to un-conceal the device ID and verify that the device ID is part of the owner's inventory list. The AF provides the CN NF with authentication/authorization result and device unique information needed to onboard the AIoT device e.g. device security information.

5. In case the AF have provisioned the network with device unique onboarding information in step 0, then the CN NF uses the retrieved information from UDM/UDR to authenticate and authorize the AIoT device.

6. The CN NF sends a Registration response (Accept/Reject, security parameters). The AIoT device uses the security parameters together with the pre-provisioned security IE to e.g. derive an initial Temp ID as proposed in solution 1. Additional AIoT configuration can be provision at this step.

6.16.3 Impacts on services, entities and interfaces

Impacts on existing entities:

CN NF:

- Configure/trigger the Reader to broadcast Registration poll message.
- Retrieve device owner information and device unique onboarding information.
- Accept or reject the AIoT device registration/onboarding request.

AIoT device:

- Receive a Registration poll message.
- Transmit a Registration Request message to the Reader.
- Receive the Registration accept message with configuration parameters.
- Manage state transition between Unregistered and Registered.

AIoT Reader (UE or Base station):

- Broadcast Registration poll (on-demand or periodically configured).
- Transparently forward any messages between the AIoT device and the CN NF.

UDM/UDR:

- Data management of device owner information.
- Be provisioned with device owner information.

6.17 Solution #17: Enable Ambient IoT Management within 5GC to support AIoT services

6.17.1 Description

6.17.1.1 General

This solution addresses the key issues in terms of architecture enhancements, Ambient IoT (AIoT) Device management and AIoT services, i.e. for KI#1, KI#2 and KI#3.

6.17.1.2 Architecture Principles

This solution proposes to introduce enhancements in the 5GS to support AIoT Device management and AIoT services. In particular,

- AIoT specific NAS, to support transfer of AIoT data (AIoT Device Operation commands and responses) between AIoT Device and 5GC directly via the gNB.
- Ambient IoT Management Function (AIoTMF), responsible for the logic for handling of AIoT services including:
  - execute the AIoT service request (e.g. inventory, read) in the network and handle any corresponding AIoT specific NAS messages,
  - support inventory and message routing for AIoT Devices,
  - authorize the AIoT service request,
  - perform validation of the AIoT Device Identifier (ID) and secure AIoT Device operations if needed,
  - collect AIoT data and aggregate the reporting,
  - collect charging information if required by SA WG5.
- UDM enhancements to manage subscription-like information for AIoT Devices and service control information for 3rd party application:
  - An AIoT Device ID is allocated to each AIoT device that has subscription-like information provisioned in the UDM. The AIoT Device ID shall be unique within a mobile network, and it is used to locate the subscription-like information. The format of the AIoT Device ID will be defined by other solutions.
  - The subscription-like information for an AIoT Device is different from the UE subscription data. The subscription-like information contains: AIoT Device ID, security materials, and device status information e.g. last serving Reader node, whether the AIoT Device is disabled, etc.
  - As part of a provisioning process for an AIoT Device, the AIoT Device is provided with its AIoT Device ID, security materials and security settings which control what operations on which memory regions require what levels of security to be applied.
  - The service control information is used to authorize the AF's device operation request. The service control information, based on service agreements between the network operator and the 3rd party AF, may contain: AF ID, allowed service area, allowed service operations, allowed AIoT Device ID information etc.
- NEF enhancements to expose the 5GS AIoT capability so as to allow 3rd party AF to consume the AIoT services.

Figure 6.17.1.2-1 illustrates the enhancements to the 5GC to support AIoT Device management and AIoT services for topology 1 and topology 2.

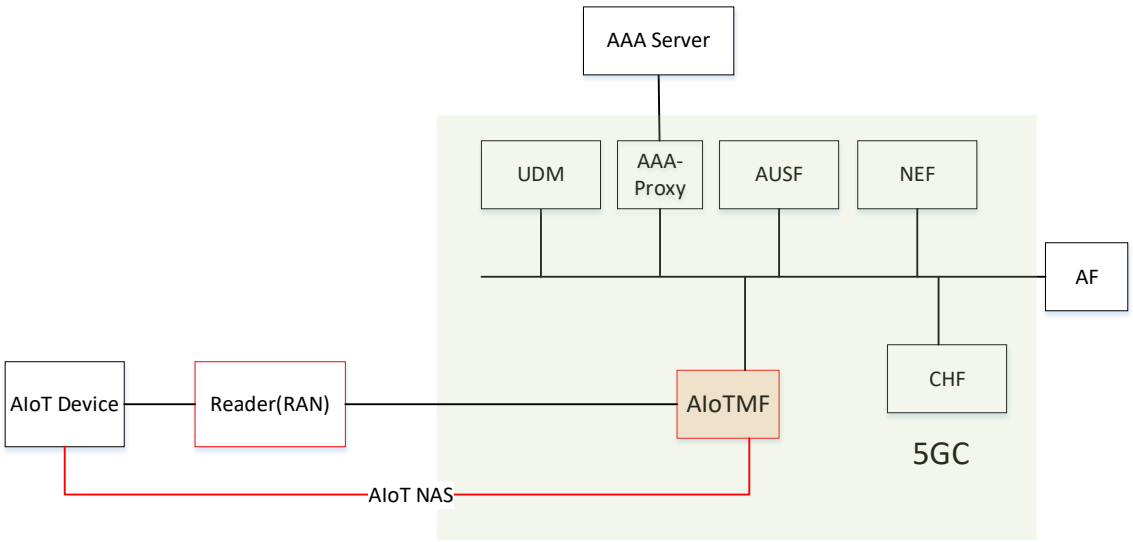


Figure 6.17.1.2-1: 5GC enhancements to support Ambient IoT for topology 1

NOTE 1: From AIoT Device point of view, the interfaces with the network are the same for both Topology 1 Topology 2.

NOTE 2: Instead of UDM, the AAA server, which is external to 5GC, can manage subscription-like information for AIoT Devices and service control information for 3rd party.

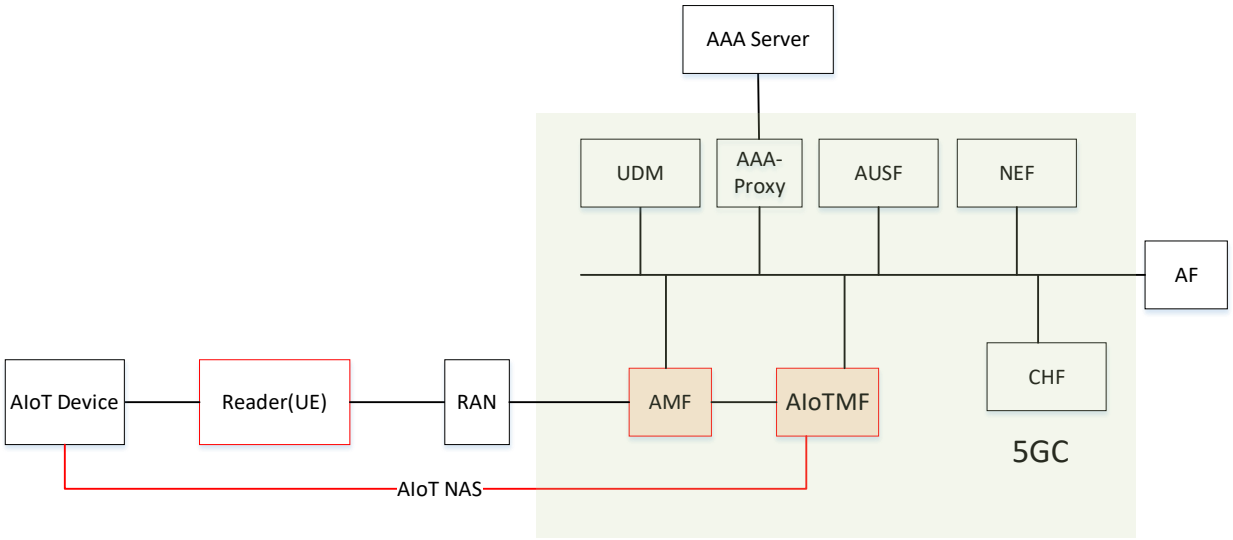


Figure 6.17.1.2-2: 5GC enhancements to support Ambient IoT for topology 2

NOTE 3: This architecture in figure 6.17.1.2-2 only shows the AMF that is needed to support UE reader management of topology 2, details of support for topology 2 can refer to other solutions in this TR, and the rest of the solution focuses on illustration of topology 1.

6.17.1.3 Function Principles

Inventory: collect the AIoT Device ID(s) of all or a subset of AIoT Devices discovered and selected with MASK.

AIoT Service: the network capability that utilizes the NEF exposure capability towards AF and utilises the AIoT Device Operations towards AIoT Device to support AIoT use cases.

AIoT Device Operation: the operations the network supports to communicate with AIoT Device e.g. inventory, read, write, disable.

MASK: a full or partial AIoT Device ID, which will be broadcasted by a reader in order to select a specific group of AIoT Devices to perform random access and report their Device IDs to the network. The AIoT Device whose Device ID matches the MASK will perform random access to report the Device ID. This MASK can also reflect the information fully or partially of a 3rd Party-defined Identifier as described in step 3 and step 5 of Figure 6.17.2.2-1.

Task ID: an identifier generated by 5GC which is used to identify the AIoT Service requested by a 3rd party/AF. To deliver the information between AF and an AIoT Device over 5GS, the 5GC uses the Task ID to maintain the association within the 5GC for a specific AF's AIoT service request from the time it accepts the AF's AIoT service request to the time the AF's AIoT service request is completed.

The 5GS shall support AIoT specific NAS to deliver the UL or DL data between AIoT Device and 5GC. The UL data includes the AIoT Device ID, Results of the Commands, security information, and Indication of cached data etc. The DL data includes the Commands and security information etc. There is a AIoT-NAS connection between AIoT Device and 5GC, and this AIoT-NAS connection is per device level. The AIoT NAS message is transferred over AIoT-NAS connection between AIoT Device and 5GC.

NOTE 1: AIoT Devices have reduced functionality compared to other types of UE (e.g. NR, eMTC or NB-IoT). An AIoT Device is more akin to a memory device (for example a flash memory card) that utilize basic memory comparison operation (match a bit sequence (e.g. the MASK) against memory location Y for length Z) to support identification, and then further operations for inventory, reading and writing data.

The physical memory of the AIoT Device can be logically viewed as containing multiple different types of information, e.g. 3GPP Information and Application Information: 1) The 3GPP Information is information that is known to and used by the 3GPP system. Examples of 3GPP Information include AIoT Device ID, Security Materials etc. 2) The Application Information is information that depends upon the application to use. Examples of use of the Application Information include the device having a fixed application defined memory format, where certain locations in the memory are known to contain information read/written by the application, sensor reading(s) or it can be used for more complex message passing techniques, e.g. being used as a mailbox. The contents and use of the Application Information is not expected to be defined by 3GPP, however 3GPP needs to define the operation primitives that allow reading, writing etc of the memory without detailed knowledge of its contents. The mapping of information / fields and their contents into physical memory is to be defined by Stage 3.

The AIoTMF and AIoT Device shall support to store a temporarily device context. The temporarily device context is kept in the AIoT Device and AIoTMF and is only used while energy is available to the AIoT Device, and it is used to record any information (e.g. security context) required for information transfer over AIoT-NAS connection (in particular for multiple-rounds of information exchange for example step 10 to step 14 in figure 6.17.2.2-1) between the AIoT Device and AIoTMF. The temporarily device context helps avoid re-generation of e.g. security context each time a NAS message is transferred. The temporary device context is generated in the AIoT-NAS interaction procedure between the device and 5GC. The temporary device context on the device side is only available for the time when the device has stored energy, the temporary device context on the 5GC side is removed at the completion of the AIoT-NAS interaction.

NOTE 2: As agreed by RAN, the AIoT Device memory can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing AIoT Device ID, etc. and 2) volatile memory (e.g. registers or RAM) for temporarily keeping any information required for its operation only while energy is available.

NOTE 3: The details of temporary device context depend on SA WG3 decision.

The 5GS shall support AIoT NGAP to control the behaviour of Reader, e.g. start of the inventory, continue of the inventory, end of the inventory. In addition, the AIoT NGAP provides transport of the AIoT NAS messages between Device and AIoTMF. The 5GS shall support AIoT-NGAP connection management for AIoT Service, this is used to transfer AIoT NAS message and AIoT inventory AIoT-NGAP message, e.g. reporting of RAN's AIoT information, requesting inventory for AIoT Device.

The high-level AIoT Protocol Stack is illustrated in the Figure 6.17.1.3-1.

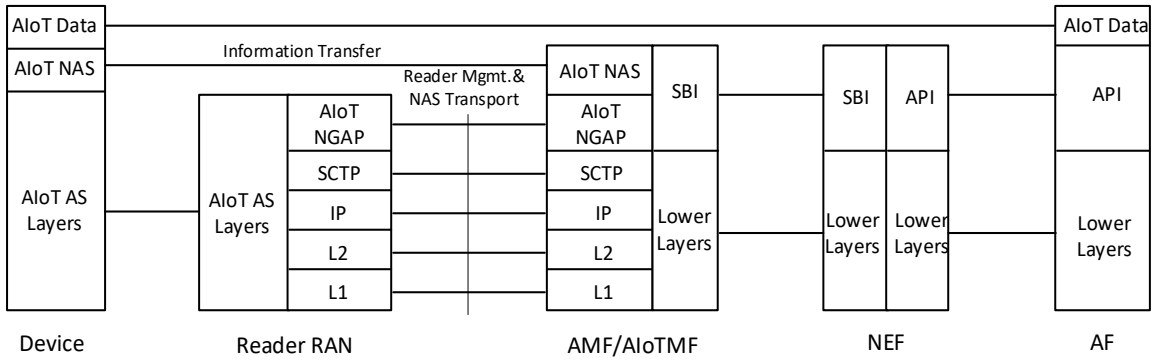


Figure 6.17.1.3-1: AIoT Protocol Stack (Topology 1)

NOTE 3: AIoT AS and AIoT NGAP protocols will be defined by RAN, and it is assumed that the AIoT AS and AIoT NGAP message can transport NAS messages.

NOTE 4: The AIoT NAS needs to be lightweight to adhere to the nature of the AIoT Devices (e.g. ultra-low complexity power, cost and resource-constrained) while addressing e.g. confidentiality, integrity, etc. For example, Security header type, Message authentication code, and Sequence number are only needed if NAS security protection (e.g. Integrity protected, Integrity protected and ciphered) is required.

AIoT Data: This is used to transfer application specific AIoT data between AIoT Device and application. This layer is optional and supports customization and extension of advanced functions on top of the basic standard functions to meet diverse requirements in different scenarios.

AIoT NAS: This is used to indicate the operation commands, as well as transfer of the application specific AIoT data between AIoT Device and 5GC. AIoT NAS message includes e,g. Inventory Response, Read Service Request, Write Service Request, Disable Service Request, Command Service Accept, Command Service Reject. 5GC executes the AIoT service request from AF and generates the corresponding NAS messages, then triggers the reader(s) to perform inventory for AIoT Devices and transfer of the AIoT NAS messages between the AIoT Device and 5GC.

NOTE 5: It is assumed that the 5GC needs to indicate to the Reader the different device operations so the Reader can, if required, apply different handling for different AIoT Device operation commands e.g. Read or Write or Disable. This will be further studied by RAN groups.

NEF exposure is used by the AF to invoke AIoT service (e.g. Inventory, Commands: Read, Write, Disable). Along with this AIoT service request, the AF is implicitly subscribed to receive notifications for its AIoT service request. At the same time, the NEF exposure can be used to deliver the UL or DL data for AIoT Devices between AF and 5GC. The DL data includes the Inventory Parameters and Command Parameters, and the UL data includes the Results of the Inventory and Command, Indication of cached data.

6.17.2 Procedures

6.17.2.1 General

This clause provides end to end high-level information flow for support of AIoT service (Inventory, Command [Read/Write], Disable) over 5GS as specified in clause 6.17.2.2, as well as the NEF service operations as specified in clause 6.17.2.3.

6.17.2.2 AIoT Service High-Level Information Flow

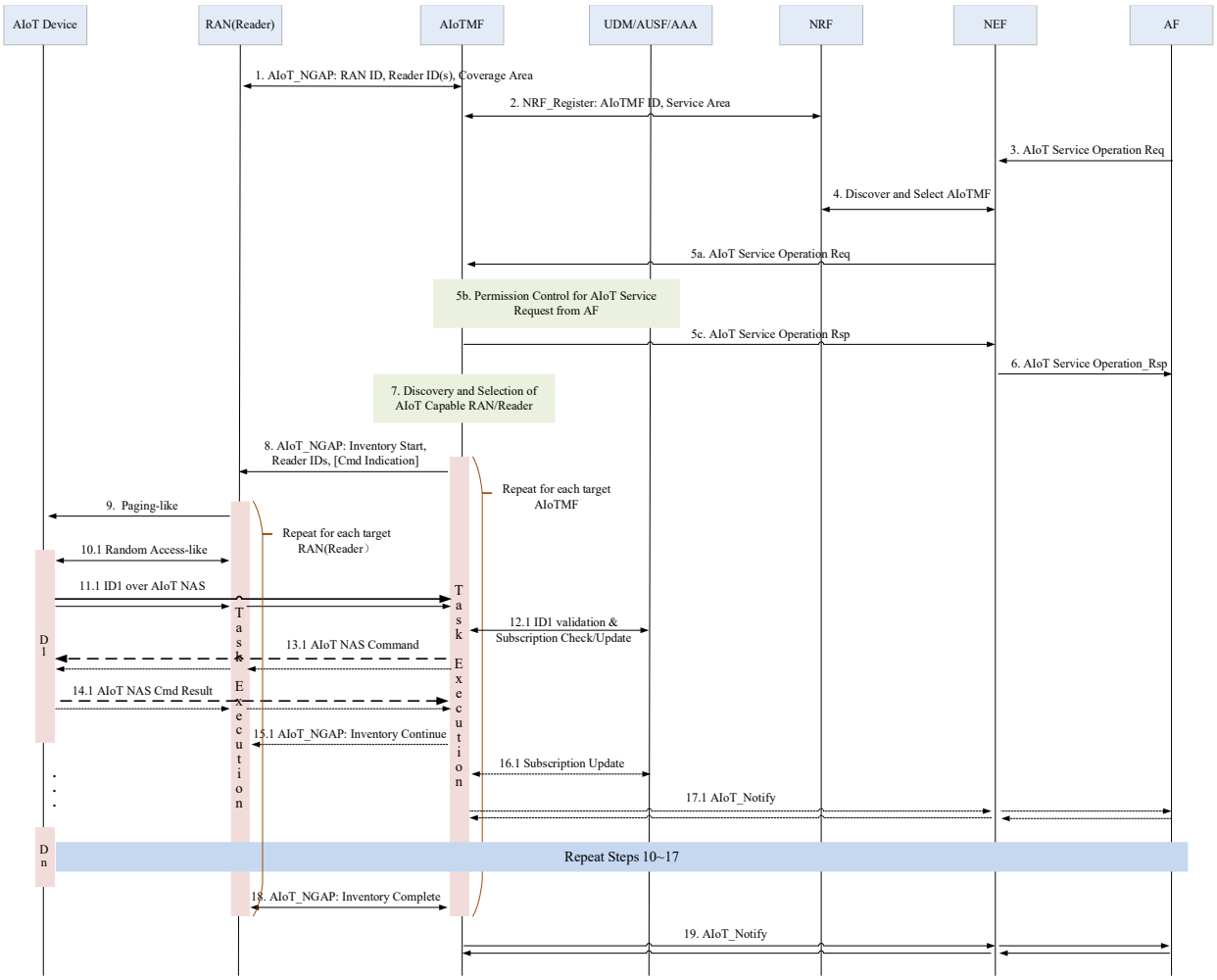


Figure 6.17.2.2-1: AIoT Service High-Level Information Flow

1. During the node-level AIoT-NGAP association establishment procedure, the RAN reports the RAN's AIoT information (e.g. RAN ID, Reader ID(s), coverage area of the Reader(s)) to the AIoTMF.

2. Upon reception of the RAN's AIoT information, the AMF/AIoTMF aggregates the AIoT information of all connected RANs and registers its AIoT information into the NRF (AIoTMF ID, coverage area for AIoT service).

3. AF sends AIoT service operation request to NEF, carrying the AF ID, device information, and reporting control information.

   The device information includes one or more of the following:

   - device area information, indicates the requested area where the inventory service will be performed.

   - device filter information: indicates complete or partial of 3rd Party-defined Identifier, and is used to compose the MASK used in step 5 to filter the Ambient IoT Devices to be inventoried.

   The reporting control information indicates when and how to send the reports, e.g. immediate reporting of new result or progress, periodic reporting, one-time reporting at the end of the AIoT service procedure, and it can also indicate to report cached data along with the AIoT Device ID etc.

   The AIoT service operation can be AIoT_Inventory service operation or AIoT_ Read service operation or AIoT_Write service operation or AIoT_Disable service operation as specified in clause 6.17.2.3. If the AIoT service operation is AIoT_ Read or AIoT_Write service operation, the AIoT service operation additionally includes the AIoT Device operation related parameters (e.g. memory location to start, length to read/write or data to write). If the AIoT service operation is AIoT_Disable service operation, the AIoT service operation additionally includes the AIoT Device Disable operation command related parameters (e.g. signature and password).

4. NEF invokes NRF to discover and select the AIoTMF(s) using the AF requested device area information. The NRF determines the target AIoTMF(s) based on the AF requested device area information and the AIoT information of the registered AIoTMF in step 2.

5. NEF sends the AIoT service operation request to each of the selected AIoTMF. The AIoTMF may obtain the service control information from UDM.

   The AIoTMF performs the permission control of the AIoT service operation request based on the operator policy, service agreement between the AF and the operator, and information obtained from the UDM, e.g. based on service area which allows the AF to request the service operation, or subscribed service operation(s) that allows the AF to invoke. If not authorized, the AIoTMF rejects the AIoT service operation request with an appropriate cause code, and step 7 onwards are skipped. Once the AIoT service operation request is authorised, the AIoTMF generates a Task ID corresponding to this AF service operation request, as well as the associated device MASK information based on device information (received in step 3), information obtained from the UDM, and/or local configuration.

   If the AIoT service operation is AIoT_Disable service operation, the AIoTMF additionally determines the AIoT Devices that have subscription-like information in UDM/AAA and match with the device information (received in step 3). The AIoTMF performs validation of the signature provided by the AF, and if validated, it stores the password and sets the device status in UDM/AAA to "TO BE DISABLED" for the determined AIoT Devices. With this "TO BE DISABLED" indication, the 5GC won't provide AIoT service to AF for those AIoT Device.

6. NEF sends the AIoT service operation response to the AF, containing the accept or reject result for the AIoT service operation request.

7. The AIoTMF selects the Reader(s) and corresponding Ambient IoT capable RANs, taking the device information (e.g. AF requested device area information) and information obtained from the UDM/AAA (e.g. allowed service area) into account.

   The AIoTMF may also use the last serving Reader in the subscription-like information to assist with determining which Readers to use for an AFs request targeting for a specific AIoT Device. If the last serving reader is within the AF requested device area information, then the AIoTMF may attempt the AF requested operation using only the last serving reader. If the operation was unsuccessful with just the last serving reader, then the AIoTMF may then use any other readers indicated by the AF requested device area information (if available).

8. For each of the selected RAN, the AIoTMF sends the Inventory Start NGAP message (node-level AIoT NGAP message) with the device MASK information, selected Reader ID(s) and the Task ID generated in step 5.

   If the AIoTMF receives the AIoT_Read or AIoT_Write or AIoT_Disable service operation request, the Inventory Start NGAP message additionally includes the indication that transfer of AIoT NAS Command message subsequently.

   If no "command indication" received, Reader RAN will continue to the next matched AIoT device and step 13 - step 16 are skipped.

   If the AIoTMF receives indications to report cached data along with the Device ID, the Inventory Start NGAP message also includes the indication of data reporting.

9. Upon reception of the Inventory Start NGAP message, RAN/Reader executes inventory Task by triggering the Paging-like procedures with the device MASK information and indication of data reporting (if received in step 8) towards the AIoT devices.

10. If an AIoT device matches the device MASK information, the AIoT device performs random access like procedures to establish the AS connection with RAN.

11. The AIoT Device sends the truncated AIoT Device ID in AIoT Inventory Response NAS message. The truncated AIoT Device ID is a partial AIoT Device ID without MASK. The AIoT Inventory Response NAS message is encapsulated within the A-Uu message sent to Reader/RAN, and the RAN/Reader relays the AIoT Inventory Response NAS message over AIoT-NGAP message (Task ID, RAN NGAP ID, AIoT NAS) to AIoTMF.

    If the AIoT Device has cached data, the AIoT Device also includes the Indication of the cached data in the AIoT Inventory Response NAS message, or if the AIoT Device receives the indication of data reporting in step 9, the AIoT Device includes the cached data in AIoT Inventory Response NAS message.

    Based on the received indication that transfer of NAS Command message in step 8, the Reader RAN allocates the RAN NGAP ID and keeps the NGAP context for this selected AIoT Device.

12. The AIoTMF constructs the entire AIoT Device ID by adding the MASK information on the partial AIoT Device ID. The AIoTMF performs validation based on the device subscription-like information received from the UDM/AAA, including Device ID validation. The AIoTMF updates the device status for this AIoT Device in the UDM/AAA, e.g. last serving node.

    If the device status indicates "TO BE DISABLED" for AIoT_Inventory or AIoT_Read or AIoT_Write service operation:

    - In the case of AIoT_Inventory service operation, the AIoTMF records the AIoT Device ID and the serving Reader node for this AIoT device. In a later time, the AIoTMF performs the step 7~16 to disable this AIoT Device.

    - In the case of AIoT_Read or AIoT_Write service operation, the AIoTMF in step 13 proceeds as AIoT_Disable service operation instead of AIoT_ Read or AIoT_Write service operation to disable this AIoT Device.

    If the device status indicates "DISABLED", the AIoT Device ID of this AIoT Device is discarded and subsequent steps are not pursued for this AIoT Device.

13. [Conditional] If the AIoT service operation is AIoT_Read or AIoT_Write or AIoT_Disable service operation, the AIoTMF constructs the corresponding AIoT Command NAS messages (e.g. AIoT Read Service Request NAS message, AIoT Write Service Request NAS message, AIoT Disable Service Request NAS message), and the AIoT Device operation command related parameters as described in step 3 are payload of the AIoT Command NAS message.

    The AIoTMF sends the AIoT Command NAS message to the AIoT Device. This AIoT Command NAS message is sent to RAN/Reader over AIoT- NGAP message (Task ID, RAN NGAP ID, CN NGAP ID, AIoT NAS), and then the AIoT Command NAS message is relayed by Reader RAN to the AIoT Device over A-Uu message.

14. [Conditional] If the AIoT Device receives an AIoT Command NAS message (e.g. Read Service Request, Write Service Request, Disable Service Request), the AIoT Device generates the results of the AIoT Device operation command and sends the AIoT Command Service Accept or Reject NAS message to AIoTMF depending on whether the command was successful or not. The results (including appropriate cause value if failed) of the AIoT Device operation command are carried as payload of the AIoT Command Service Accept or Reject NAS message. The AIoT NAS message is sent to Reader RAN over A-Uu message and then the AIoT NAS message is relayed by Reader RAN to AIoTMF over AIoT-NGAP message (Task ID, RAN NGAP ID, CN NGAP ID, AIoT NAS).

    If the AIoT Device receives the AIoT Disable Service Request NAS message, the AIoT Device needs to perform validation with the received password.

NOTE: Other security mechanism than the password based approach to support AIoT service needs coordination with SA WG3.

15. [Conditional] If the AIoT service operation is AIoT_ Read or AIoT_Write or AIoT_Disable service operation, and when the AIoTMF completes the device operation on the selected AIoT device, the AIoTMF sends the Inventory Continue NGAP message (Task ID, RAN NGAP ID, CN NGAP ID) to the RAN/Reader. Upon reception of this AIoT-NGAP message, the RAN/Reader releases the NGAP context for the current AIoT device and continues looking for next matched AIoT device.

16. [Conditional] If the AIoT service operation is AIoT_Disable service operation, and when the Disable Operation is completed successfully for one AIoT device, the AIoTMF sets the device status of this AIoT device in UDM/AAA to " DISABLED". With this " DISABLED" indication, the 5GC won't provide AIoT service to AF for this AIoT Device and the 5GC can delete the subscription-like information immediately or after a configured time period.

17. [Conditional] If the reporting control information matches (e.g. immediate reporting of new result or progress, periodic reporting), the AIoTMF reports the AIoT Device ID (s), results of the AIoT Device operation command and if available, the reported data or indication of cached data to NEF by sending the AIoT_ Notify message (list of Device IDs, results of the AIoT Device operation command, reported AIoT data or indication of cached data). The NEF forwards the received information to AF by sending the AIoT_ Notify message.

    Reader RAN repeats steps 10~17 if there is a next matched AIoT Device.

18. The RAN/Reader sends the Inventory Complete NGAP message (Task ID) to AIoTMF in case of e.g. there is no further Ambient IoT Device performing random access, or AIoTMF proactively sends the Inventory Complete

NGAP message (Task ID) to RAN/Reader in case of e.g. all target Ambient IoT Device(s) have been inventoried. The RAN Reader confirms the Task indicated by Task ID is completed.

19. The AIoTMF completes the Task for AIoT service operation request for each selected RAN/Reader. The AIoTMF reports the progress of the AIoT service operation request to the NEF by sending the AIoT_ Notify message, the NEF informs the AF of the progress of the AIoT_Inventory request by sending the AIoT_Notify message.

    If the reporting control information matches e.g. one-time reporting at the end of the AIoT service procedure, the AIoTMF can further aggregate the Device ID(s) and/or results of the AIoT Device operation command and reported AIoT data or indication of cached data. The AIoTMF further performs filtering and then includes into AIoT_ Notify message the list of Device IDs, reported AIoT data or indication of cached data, results of the AIoT Device operation command etc.

6.17.2.3 NEF AIoT Service

The NEF supports a new Nnef_AIoT service with the operations described in table 6.17.2.3-1.

Table 6.17.2.3-1: AIoT Service provided by NEF

Service Name	Service Operations	Operation Semantics	Example Consumer(s)
Nnef_AIoT_Service	Inventory	Request/Response	AF
	Read	Request/Response	AF
	Write	Request/Response	AF
	Disable	Request/Response	AF
	Notify	Subscribe/Notify	AF

Inventory Service Operation: The inventory operations for the Nnef_AIoT service allows the consumer to request to perform an inventory device operation for an AIoT device or a group of AIoT devices.

Read Service Operation: The read operations for the Nnef_AIoT service allows the consumer to request to perform a read or write device operation for an AIoT device or a group of AIoT devices.

Write Service Operation: The write operations for the Nnef_AIoT service allows the consumer to request to perform a write device operation for an AIoT device or a group of AIoT devices.

Disable Service Operation: The disable operations for the Nnef_AIoT service allows the consumer to request disabling RF transmission from an AIoT device or a group of AIoT devices. If the AIoT Device is disabled then the network may also discard any stored state for the AIoT Device and no further operations for that AIoT Device will succeed.

Notify Service Operation: The notify operations for the Nnef_AIoT service allows the consumer to receive the status and results of the requested service operation.

6.17.3 Impacts on services, entities and interfaces

UDM:

- Support subscription-like information for AIoT Devices and service control information as in clause 6.17.1.2.

NEF:

- Support AIoT services as in clause 6.17.2.3.

(New) AIoTMF:

- Support logic handling of AIoT services and AIoT NAS as in clause 6.17.1.2.

AIoT NAS:

- Support transfer of AIoT data (AIoT Device inventory responses, operation commands and responses) between AIoT Device and 5GC directly.

Reader RAN:

- Support AIoT NGAP as in clause 6.17.1.3 and Paging-like procedures with the device MASK information and random-access procedure.

AIoT Device:

- Support AIoT NAS as in clause 6.17.1.3.

6.18 Solution #18: Solution for support of Ambient IoT services

6.18.1 Description

This solution resolves Key Issue #3 about the Support of Ambient IoT Services. The following two connectivity topologies as defined in TR 38.848 [7] are to be studied:

- Topology 1: BS <--> Ambient IoT Device;

- Topology 2: BS <--> intermediate node <--> Ambient IoT Device: Only a UE can act as an intermediate node that is under the network control.

The third Ambient IoT AF triggers the Ambient IoT service toward ambient IoT devices via 5GS. In some Ambient IoT services, such as inventory, read, and others, one Ambient IoT service request from AF will cause numerous responses from ambient IoT devices, which may cause a severe signal storm in the 5GS. In order to mitigate the signal storm impact, we suggest that Base Station Reader supports response signal packet aggregation at the IP transportation level if Base Station Reader can't look into the response packet content. If Base Station Reader can decode the response packets, Base Station Reader supports aggregation at the content level. At the same time, the AMF or new Ambient IoT NF supports content-level aggregation too to significantly reduce signal storm impact.

Besides, this solution aims to solve the issue of how the serving Base Station Reader or UE allows access from the different operators' ambient IoT devices. The third ambient IoT AF has the information on the serving operator and the operators list of who provides ambient IoT devices in each location. The third ambient IoT AF includes EPC, an operator ID list, and location info in the service request toward the serving operator NEF. Based on SLA agreement between the third ambient IoT AF and serving operator, the NEF will check the operator ID list to allow it or not. If allowed, then serving operator NEF will forward the both EPC and operator ID list to Base Station Reader or UE. Base Station Reader or UE will utilize both EPC info and the operator ID list to identify the targeted ambient IoT devices in the service operations. Once the ambient IoT devices are matched by both EPC info and the Operator ID list (the device ID contains Operator ID info), they will respond to the request. Please note that if only EPC is matched, but operator ID is not matched, the ambient IoT device should not respond to the request. In this way, serving operator can control and allow the access from the different operators' ambient IoT devices.

The principles related to Ambient IoT services are depicted below:

- The Ambient IoT device has a unique ambient device ID which consists of MNO operator ID, Company info, and instance ID. Device ID will not be exposed to the third Ambient IoT AF.

- Ambient device ID is stored in ambient IoT device.

- The third Ambient IoT AF will use EPC , location info and operator ID list to perform flexible Ambient IoT service. EPC is used to define a specific ambient IoT device or a group of ambient IoT devices. Operator ID list is used to allow access from the different operators' ambient IoT devices.

- To mitigate signal storms in the 5GS, the UE, Base Station Reader, and AMF/New Ambient IoT NF can perform aggregation based on transaction ID.

- Both Leveraging existing AMF and Introducing new Ambient IoT NF apply to this solution.

6.18.2 Procedures

6.18.2.1 Ambient IoT service - Network initiated Ambient IoT service procedure

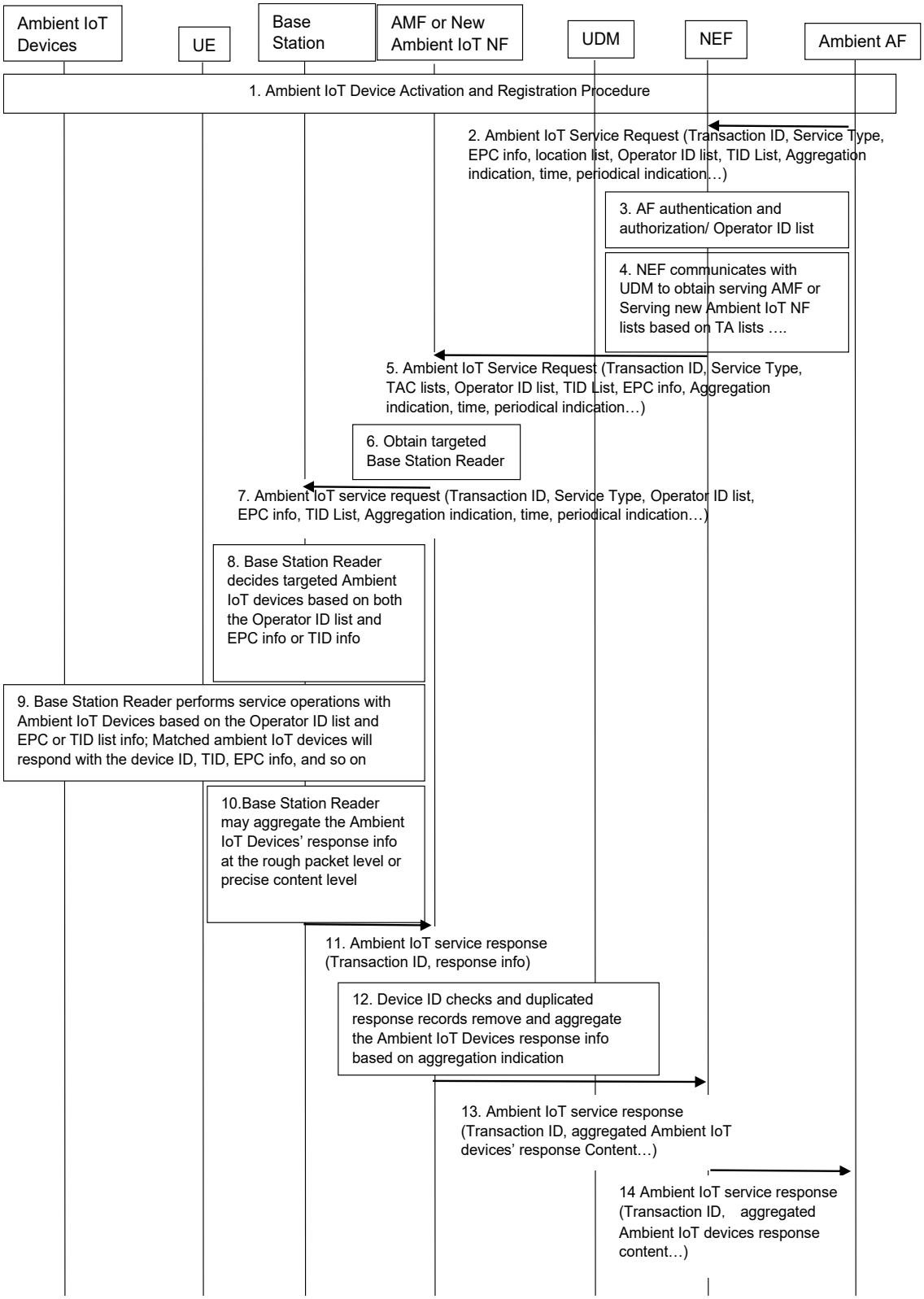


Figure 6.18.2.1-1: Network Initiated Ambient IoT service procedure

1. Ambient IoT devices perform activation and registration procedures. Ambient IoT devices will have the EPC info and device ID, and UDM will have the Ambient IoT device profile info indexed by device ID after successful Ambient IoT devices activation and registration procedures. Besides, the device ID contains operator

ID info, Company info, and instance ID. Ambient IoT device ID will not be exposed to the third AF, and it is used in Ambient IoT devices and 5GS as internal ambient IoT device identification for access, routing, and so on.

2. The third Ambient AF has the store location information and relevant AIoT serving operator info. The third Ambient AF launches an Ambient IoT service request towards NEF belonging to the AIoT serving operator. In the Ambient IoT service request, the third Ambient AF will provide Transaction ID, Service Type, EPC info, Location info, Operator ID list, Aggregation indication, time, periodical indication, and other parameters to NEF:

   - Service Type: This information is used to define Ambient IoT service types, such as Inventory, Read, Write, and so on.

   - EPC info: This information is used to define the targeted ambient IoT devices in the service from the products' perspective. it could be one specific ambient IoT device or a group of ambient IoT devices

   - Operator ID list: This information is used to define the targeted ambient IoT devices in the service from the operator's perspective, supporting multiple operator ambient IoT device access. It could be one specific operator or a list of operators.

   - TID list: The information can be used to define the targeted ambient IoT devices, such as in the write or read service command.

   - Location info: This information is used to define locations where service happens.

   - Aggregation indication: It is used to tell the 5GS how to handle the response messages -- aggregation response or not

   - Time: when the service will be carried out exactly

   - Periodical indication: the service operation will be executed periodically.

3. NEF will perform the below actions towards the third Ambient AF:

   - Perform authentication to the third Ambient AF to decide whether it is allowed to access 5GS or not.

   - Check the authorization to decide whether the third Ambient AF is allowed to perform certain Ambient IoT service operations or not.

   - Check the authorization to see whether multiple Operators' ambient IoT device access is allowed or not.

   - Converting location information to TA List information.

4. NEF will obtain serving AMF or serving New Ambient IoT NF based on TA lists

5. NEF forwards Ambient IoT service requests to serving AMFs or Serving New Ambient IoT NFs. Both leveraging existing AMF and introducing new ambient IoT NF can be applied to this solution.

6. AMF or New Ambient IoT NF determines the serving Base Station Readers based on TA lists. AMF or New Ambient IoT NF decides to perform aggregation operations in terms of aggregation indication.

7. AMF or New Ambient IoT NF forwards the Ambient IOT service request to the serving Base Station Readers.

8. Base Station Reader decides what kind of service operation to perform based on Service Type, and Base Station Reader decides to perform aggregation operations in terms of aggregation indication. Base Station Reader decides when to execute service operations based on time parameters, whether to perform periodical operations based on periodical indication and so on.

9. Base Station Reader performs service operations toward the serving ambient IoT devices. Both EPC info and the Operator ID list will be used in the service operation to identify the targeted ambient IoT devices. Once the ambient IoT devices are matched by both EPC info and the Operator ID list (the device ID contains Operator ID info), they will respond with the full device ID, TID, EPC info, and so on based on service operation type.

10. Base Station Reader will perform the response packet aggregation operation at the rough packetlevel based on transaction ID in terms of aggregation indication to mitigate signal amounts between Base Station Reader and 5GC if Base Station Reader can't decode the response packets to look into the content. If Base Station Reader can decode the response packets to look into the content, Base Station Reader should perform the aggregation operation at the content level based on the transaction ID in terms of aggregation indication to further deduce

signal amounts significantly between Base Station Reader and 5GC. Base Station Reader can decide the aggregation waiting time based on local configuration or AF indication.

11. Base Station Reader responds with an Ambient IoT service response message with parameters-transaction ID, Aggregated Inventory content, and so on-to AMF or the new Ambient IoT NF.
12. The AMF or new Ambient IoT NF will implement the Ambient IoT device check, remove duplicate Ambient IoT device records, and implement an aggregation operation at the content level based on the aggregation indication.
13. The AMF or new Ambient IoT NF continues to forward the response to NEF.
14. NEF will forward the response to the third Ambient AF.

Editor's note: This solution focuses on Topology 1. How to expand the solution to support Topology 2 is in FFS.

Editor's note: The security concern about how to access Ambient device is in FFS.

6.18.3 Impacts on services, entities, and interfaces

This solution impacts the following entities:

NEF:

- Capability to support the selection of the serving AMF or serving new Ambient IoT NF based on location information.
- Capability to aggregate the retrieved Ambient IoT devices' EPC contents based on transaction ID.
- Capability to support Ambient IoT service-related procedures.
- Capability to authentication and authorization to the third ambient IoT AF.
- Capability to support verification on operator ID list.

AMF or New Ambient IoT NF:

- Capability to support Ambient IoT service-related procedures.

UDM:

- Capability to manage the subscription information of Ambient IoT devices indexed by ambient device ID.

UE:

- Capability to support ambient IoT service procedures in terms of EPC and operator ID list
- Capability to aggregate the retrieved Ambient IoT devices' response at content level.
- Capability to support Ambient IoT service-related procedures with 5GC.

Base Station Reader:

- Capability to support ambient IoT service procedures with Ambient IoT devices in terms of EPC and operator ID list.
- Capability to aggregate the Ambient IoT devices' response at IP transportation level or content level.
- Capability to support Ambient IoT service-related procedures with 5GC.

6.19 Solution #19: AIoT communication for connectivity topology 1

6.19.1 Key Issue mapping

This solution addresses the following requirements:

- from Key Issue #1 the requirement:
  - *System architecture identified along with the solutions for KI#2 and KI#3.*
- from Key Issue #3 the requirements:
  - *Study how to support information transfer for Ambient IoT services and related system functionality, including the information transfer for an Ambient IoT device and for a group of Ambient IoT Devices.*

NOTE: Including whether there is a need to support session based transfer between Ambient IoT Device and the network considering the device types and capabilities.

  - *Study which of the enabled Ambient IoT services are exposed to AF and how, e.g. for the case AF requests Ambient IoT service for an Ambient IoT Device and for a group of Ambient IoT Devices.*

6.19.2 Functional Description

This solution focuses on the Topology 1. It is proposed that the AIoT is a new radio access technology (RAT) type supported in the Base Station (BS). The BS nodes supporting the AIoT RAT establish N2' association with the 5GC (specifically with the AIoT function, AIoTF). The N2' association is independent of the N2 association for the communication via NR RAT or LTE RAT with the AMF.

This solution proposes that a new NF called Ambient AIoT function (AIoTF) is introduced. The AIoT application function or server (AIoT AF/AS) establishes an association with the AIoTF and transmits the AIoT data/information to the BS AIoT Reader via the AIoTF. The assumed architecture is shown in Figure 6.19.2-1.

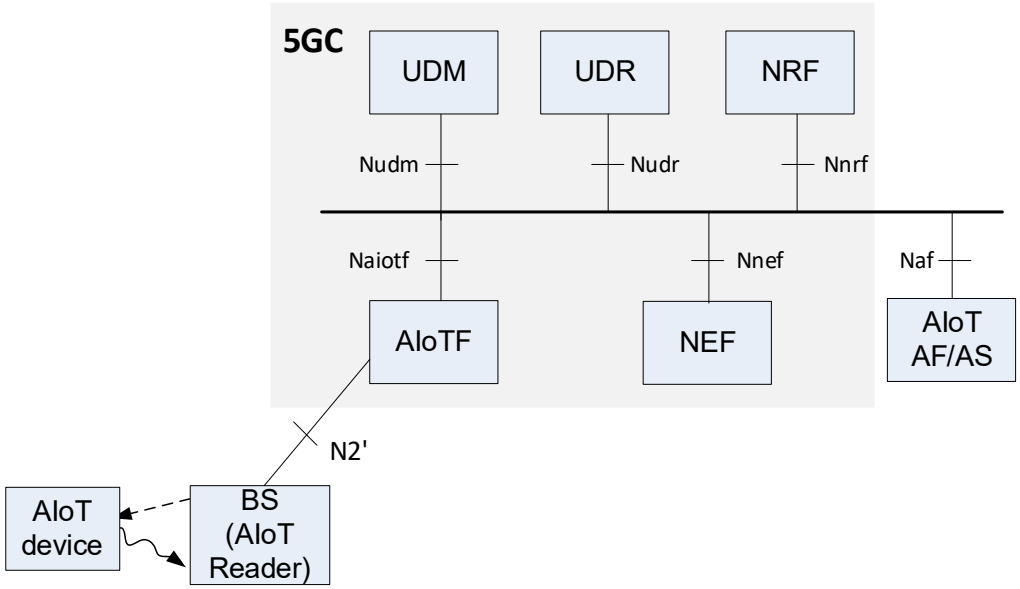


Figure 6.19.2-1: Assumed architecture for topology 1

The AIoTF implements the following functionalities:

- Receive a request to establish an N2-like association (e.g. called N2' association) from an AIoT capable BS acting as "reader" for AIoT devices. The BS can indicate the AIoT radio coverage area, its location etc.
- Store the AIoT service parameters and configuring the BSs.

- Establishing an AIoT session with the AIoT AF/AS for transmission of AIoT data/signalling in a target transmission area.
- Selecting appropriate AIoT capable BS(s) for the transmission of AIoT data/signalling to the target transmission area.
- Receiving and transmitting AIoT data/signalling from/to the AIoT application server.
- Receiving and transmitting AIoT data/signalling from/to the BS acting as AIoT reader.
- Perform store-and-forward functionality for the AIoT data/signalling.
- To verify the identity of the AIoT device(s) and the security protection of the AIoT data/signalling.
- Creating charging data for the AIoT data/signalling and transmit the data to the CHF.

6.19.3 Procedures

6.19.3.0 General

Two procedures are shown to explain the concept of this solution:

- the procedure for N2' association establishment for AIoT data/signalling transmission for connectivity topology 1; and
- the procedure for and AIoT service-based session establishment between the AIoTF and the AIoT AF/AS and AIoTF and the BS AIoT Reader via the N2' association.

6.19.3.1 N2' association establishment

The Figure 6.19.3.1-1 describes the signalling flow of the N2-like (N2') association establishment between of AIoT capable BSs for reader functionality (called "BS AIoT Reader)" and the AIoTF.

The Figure 6.19.3.1-1 shows AIoTF1 and AIoTF2 to express the optional flexibility that a BS AIoT Reader can establish N2' associations to multiple AIoTFs.

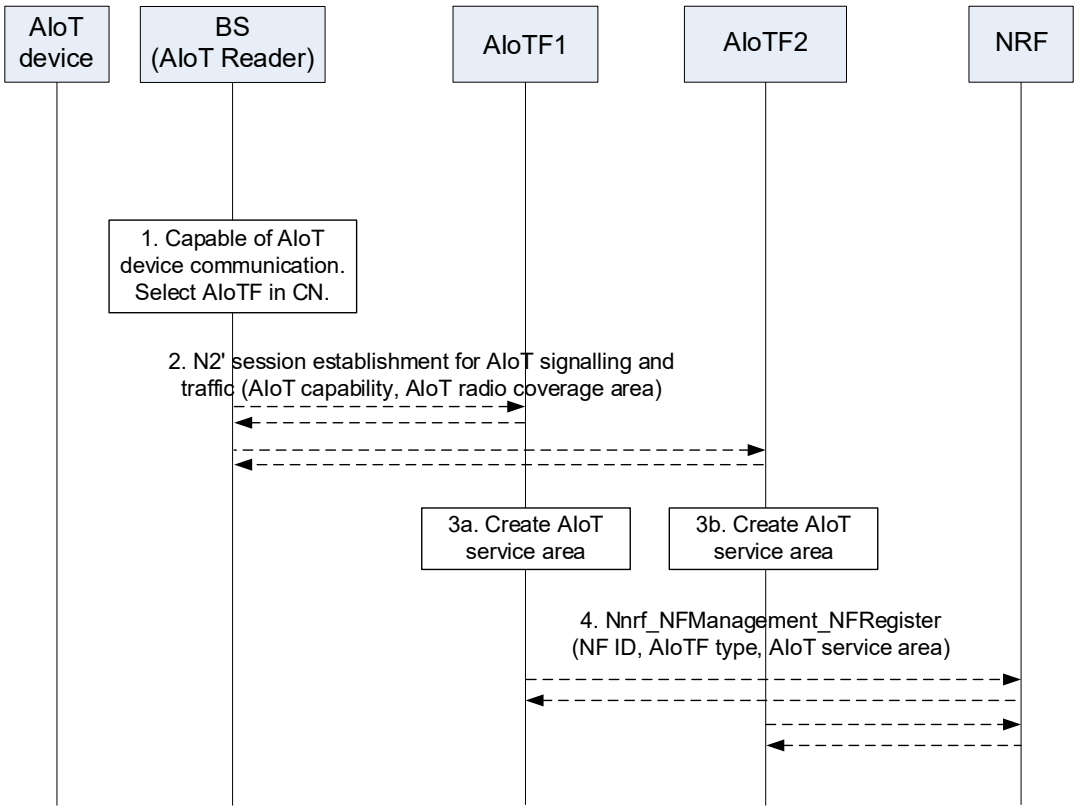


Figure 6.19.3.1-1: Signalling flow of the N2' association establishment between of AIoT capable BS node and AIoTF

The detailed description of the steps is provided as follows:

1. The BS implements capability for AIoT transmission to AIoT devices (e.g BS AIoT Reader). The BS AIoT Reader stores various parameters for AIoT communication, e.g. power mode operation for AIoT, whether the BS AIoT is mobile or stationary, supported frequency bands for AIoT transmission, range of communication, etc. Each BS AIoT Reader is configured with an identifier, location information (e.g. GPS coordinates) and/or radio coverage area.

NOTE 1: The BS AIoT Reader is configured by the RAN OAM system.

The BS AIoT Reader can implement an AIoT client functionality which may receive and transmit 1) AIoT data/signalling from/to the AIoTF and 2) transmit and receive AioT data/signalling to/from the AIoT devices.

NOTE 2: The RAN architecture for AIoT is in the remit of the RAN WGs, e.g. whether the AIoT RAN includes one or more BS AIoT readers organized in hierarchical manner.

2. The BS AIoT Reader selects an AIoTF to register with. The BS AIoT Reader may be preconfigured (e.g. from the OAM system) with the AIoTF selection information. The BS AIoT Reader initiates the establishment of TNLA and N2-like association (e.g. called N2') with the AIoTFs.

   The figure shows that the BS AIoT Reader establishes N2' associations and registers with the AIoTF1 and AIoTF2.

3a-3b. The AIoTF1 and AIoTF2 create its AIoT service area which is a combination of the AIoT radio coverage area received from the BS AIoT Readers.

4. The AIoTFs register with the NRF. For example, the AIoTFs use Nnrf_NFManagement_NFRegister service operation and indicate to the NRF the AIoTF type, the supported AIoT service IDs it is configured to serve (please refer to clause 6.19.3.2), the AIoT service area, AIoT capability (e.g. CP transmission or UP transmission), etc.

6.19.3.2 AIoT session establishment between the AIoTF and the AIoT AF/AS and AIoTF and the BS AIoT Reader

The Figure 6.19.3.2-1 describes the signalling flow of the AIoT session establishment between the AIoTF and the AIoT AF/AS and between the AIoTF and the BS AIoT Reader via the N2' association. Such AIoT session is established for a particular AIoT service, e.g. inventory service or command/response service to AIoT devices associated with the AIoT service, and for a specific service area.

NOTE 1: In one example, the AIoT session can be used to allow the AIoT AF/AS to send a single request for AIoT service (e.g. read the AIoT sensors in a certain area), enable the AIoTF to create and send (e.g. periodically or a single request)the AIoT signalling to the AIoT devices in the target area, and the AIoTF reports the retrieved AIoT data to the AIoT AF/AS. In another example, if the AIoT AF/AS sends the AIoT data for each AIoT device, then the AIoT session is established to avoid the AIoTF selection in a target area for each AIoT data to a particular area.

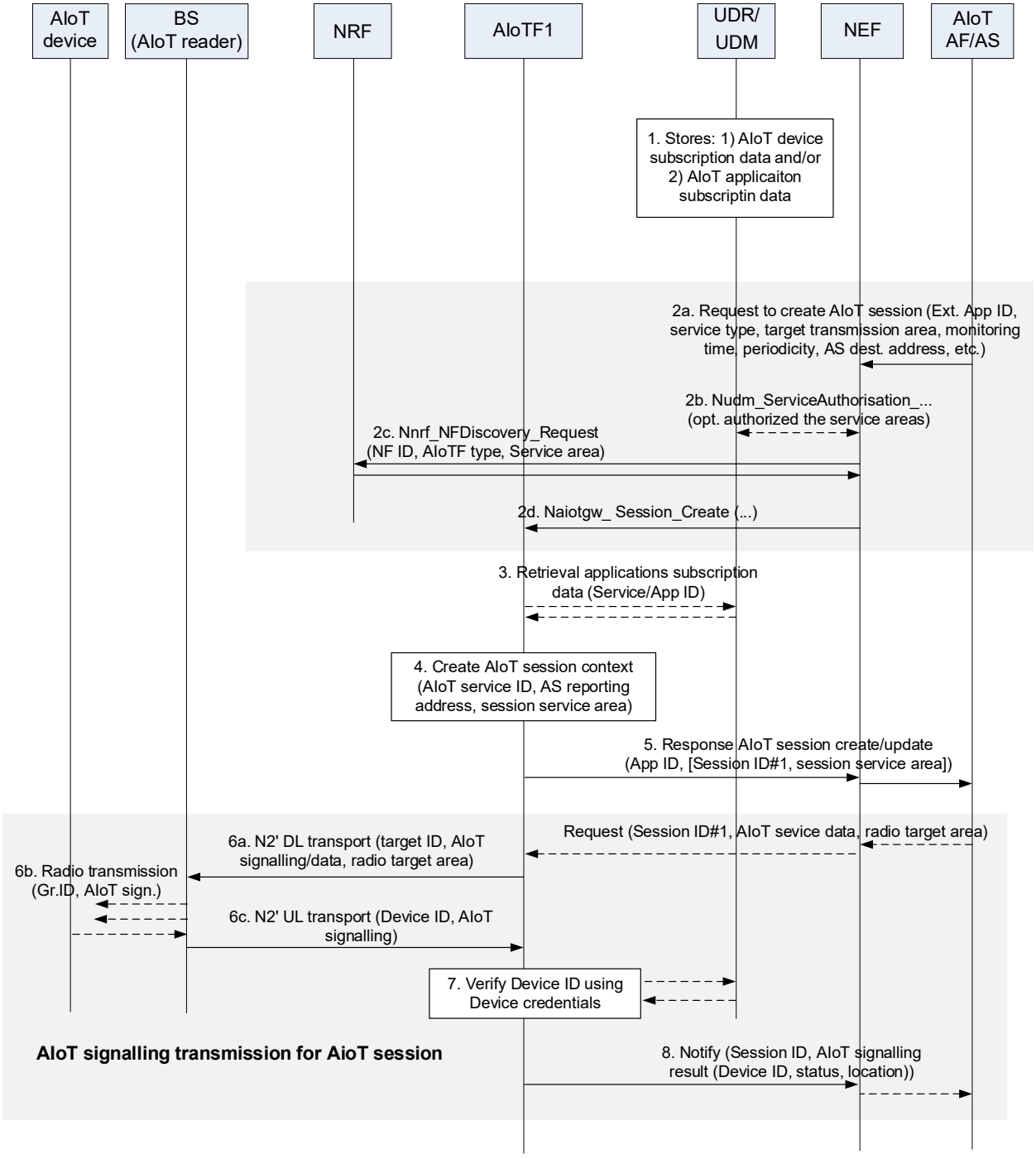


Figure 6.19.3.2-1: Signalling flow for AIoT session establishment between the AIoTF and the AIoT AF/AS and AIoTF and the BS AIoT Reader via the N2' association for a particular target area

The detailed description of the steps is provided as follows:

1. The UDM/UDR stores AIoT device subscription data and (AIoT) application subscription data. The application subscription data contains the configuration information for the AIoT application (or also called AIoT service). The AIoT device subscription data can include individual device ID and corresponding credentials.

   The whole step 2, including steps 2a, 2b, 2c, and 2d, describes a request to create (or update) an AIoT session between the AIoT AF/AS and the AIoTF for a particular target area.

2a. The NEF expose a new service to AFs offering the AIoT session establishment. Such service can be called Nnef_IoTsession and can have the service operations Create, Update and Delete. The AIoT AF/AS uses this service and send a request including parameters like: external AIoT service/application ID, service type, target transmission area, application/service description parameters (e.g. service monitoring time, periodicity for monitoring, reporting, transmission time to device, UP/CP reporting, AIoT AS destination address, etc.).

   The target transmission area is the area to which the to which AIoT data/signalling is to be transmitted. The target transmission area is mapped to one or more service areas of the AIoTFs.

2b. The NEF authenticates and authorizes the service request from the AIoT AF/AS.

2c. The NEF selects serving AIoTFs depending on 1) how the target transmission area (received in step 5a) maps to the AIoTF's AIoT service areas and 2) what is the requested service type (e.g. inventory service or read/write service (also referred as command service)).

2d. The NEF forwards the request message for AIoT session establishment to the selected AIoTF. The AIoTF can expose a service like Naiotgw_Session_Create. The NEF uses the AIoT service/application ID corresponding to the external identifier from step 2a.

3. The AIoTF can retrieve from the UDM/UDR application subscription data by using the internal AIoT service/application ID as input parameter. The application subscription data can include credentials for network layer security to security protect (e.g. encryption and/or integrity protection) of the AIoT data/signalling transmitted to the AIoT device(s) belonging to the AIoT service ID.

4. The AIoTF creates a context for the AIoT service/application ID. The context may include the session service area, application/service description parameters, AIoT AS reporting address.

   The session service area is the interclause of the target transmission area and the AIoTF-specific AIoT service area. If the target transmission area is smaller than the AIoT service area, the AIoTF uses the target transmission area as session service area. The AIoTF find the appropriate BS AIoT Reader(s) to which AIoT data/signalling is transmitted and which are part of the session service area. If the target transmission area is larger than the AIoT service area, then the AIoTF selects its whole AIoT service area as session service area.

5. The AIoTF sends a response to the AIoT AF/AS to acknowledge the AIoT session create/update request from step 2a. The response message can include the AIoT service/application ID, Session ID#1, session service area, etc.

Editor's note: It is FFS upon which events the AIoT session between the AIoT AF/AS and the AIoTF can be modified or deleted.

The following steps 6a, 6b, 6c, 7 and 8 build a procedure for AIoT signalling transmission for the AIoT session. The AioT signalling to the AioT devices is created by the AioTF and the replies from the AioT devices are processed in the AIoTF. The AIoTF sends one or more reports to the AIoT AF/AS for the AIoT session.

6a. In one option, the AIoTF can create an AIoT signalling/data to be transmitted to the AIoT devices. In another option, the AIoTF can receive DL AIoT data to be transmitted from the AIoT-AS. The AIoTF uses a N2' procedure to carry the AIoT signalling/data to the BS AIoT Reader. The Downlink AIoT signalling transport message may include the target AIoT device ID, AIoT signalling, radio target area. The AIoT signalling may include an indication for inventory service to check the AIoT device ID and possibly type of good to which the AIoT device is attached.

NOTE 2: In one example, the identification of the type of good or product can be bound to the AIoT device identity as described in Annex A "Overview of the Electronic Product Code". In another example, a separate good or product identity ID, to which the AIoT device is attached, may be used, e.g. as described in GS1 organisation [11].

NOTE 3: The identification of the AIoT device and the applied security is not in the scope of this solution.

6b. The BS AIoT Reader transmits the AIoT signalling to the AIoT device(s). The AIoT device can backscatter the device ID and type of good to which the AIoT device is attached.

The AIoT device may verify the ID and security protection of the received DL AIoT signalling.

6c. The BS AIoT Reader receives the reply from the AIoT device (e.g. from the backscattered signals. The BS AIoT Reader creates a report transmits the report to the AIoTF.

7. The AIoTF processes the message received from the BS AIoT Reader. The AIoTF may verify the device ID and/or other parts of the AIoT signalling using AIoT device credentials retrieved from the UDM/UDR (e.g. as part of the AIoT device subscription data).

8. The AIoTF creates and transmits a notification message (which can be also a bulk message) to the AIoT AF/AS including the Session ID, AIoT service/application ID, AIoT signalling result like status of the device (e.g. type of good to which the device is attached, ID verification failed/succeeded), AIoT device location.

6.19.4 Impacts on existing services, entities and interfaces

The following NFs and interfaces are impacted:

- AIoTF: a new NF with the functionality described in clause 6.19.2.
- BS AIoT Reader: 1) supporting AIoT capabilities to communication with the AIoT devices and 2) capabilities to select an AIoTF and to establish N2' association with the AIoTF.
- NEF: expose a new API for AIoT session creation.
- UDM/UDR: store AIoT device subscription data and AIoT application subscription data.

6.20 Solution #20: E2E handling for interaction of AIoT Device related information

6.20.1 Description

The solution is to address KI#1#2#3.

The interaction for AIoT Device related information may happen between AIoT device and AIoT function or AF. Those AIoT Device related information includes:

- Device UL info which is from AIoT device and will be sent to AIoT function or AF.
- Device DL info which is from AIoT function or AF and will be sent to the AIoT device.

Generally, both Device UL info and Device DL info are transparent to the reader.

The legacy procedure for UE information interaction is as the following.

- Step1: After RRC establishment, the NG RAN node trigger initial UE message to the send the UE initial NAS information to the AMF
- Step2: Based on the initial UE message, AMF setup UE association between the NG RAN node and the AMF. The subsequent UL and DL UE NAS PDU delivery use the UE association related signalling.

The UE association between RAN and AMF as described as the following:

9.3.3.1 AMF UE NGAP ID

This IE uniquely identifies the UE association over the NG interface, as described in TS 38.401 [14].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF UE NGAP ID	M		INTEGER (0..2^{40}-1)	

9.3.3.2 RAN UE NGAP ID

This IE uniquely identifies the UE association over the NG interface within the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN UE NGAP ID	M		INTEGER (0..2^{32}-1)	

W.R.T the Ambient IoT, there is also per AIoT device interaction although there is no RRC states for AIoT device, but the interaction for the AIoT device related information may happen in several rounds between AIoT device and AIoT Function or AF. E.g. authentication firstly and further read and/or write. Operation. But those detail can be transparent to the reader. So there also is requirement for per AIoT device association between the reader and the AIoT function to associate and ensure the interaction for AIoT Device related information.

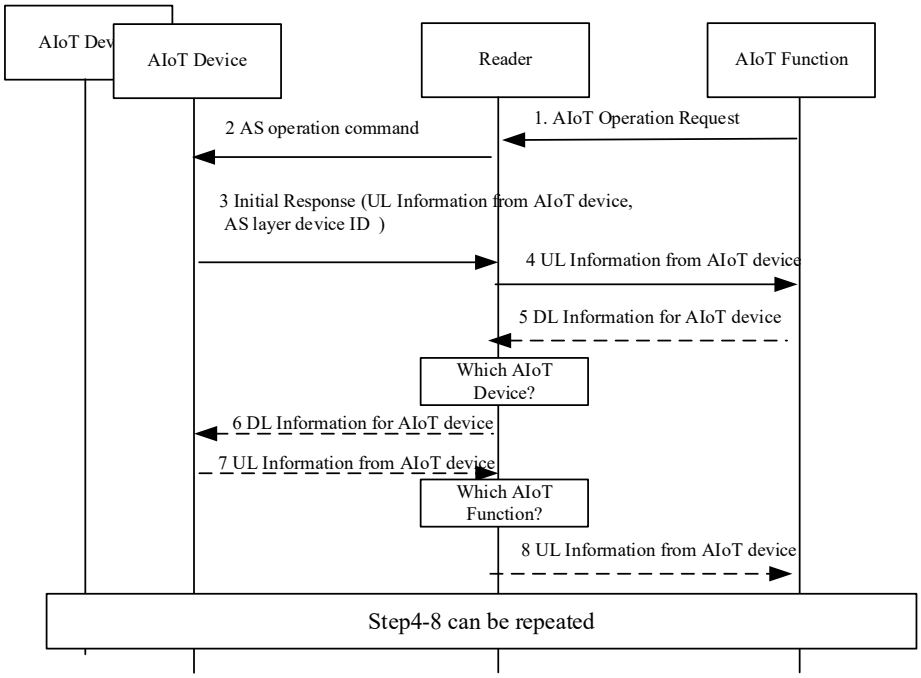


Figure 6.20.1-1: interaction for AIoT Device related information

NOTE: The AS layer ID for AIoT Device is up to RAN WG decision. It is assumed the High layer ID (the layer above AS, e.g. NAS layer ID, or Application layer ID) for AS Device is transparent to the reader.

However, there might be thousands of AIoT devices that match the AIoT operation and it is not expected to setup thousands of per AIoT device association between the reader and AIoT function. And it delays much to setup the per AIoT device association after the reader receives the initial response from the AIoT device, since the association setup delay is probably longer than the delay to finish the interaction with an AIoT device.

Hence, it is proposed the following for the per AIoT device association between the reader and AIoT function:

- The AIoT function trigger to setup a list of per AIoT device association between the reader and AIoT function in advance.
- For each new discovered AIoT device (e.g. receiving the initial response from the AIoT device), the reader binds the AIoT device to a pre-setup per AIoT device association. The binding exists during the interaction with AIoT devices but doesn't change the states of AIoT device, i.e. it is still no RRC state.
- After the interaction with the AIoT device finished, the binding is released and the per AIoT device association become available again and can be reused to bind another new discovered AIoT device.
- The pre-setup per AIoT device association doesn't necessary to be the exact number for target AIoT devices that match the AIoT operation. E.g. pre-setup per AIoT device association is 50, but the potential AIoT devices of the AIoT operation may be 1000. Since the AIoT devices discovery procedure by the reader is one by one and number of AIoT devices under the reader's operation in the meantime is limited, so the limited number of pre-setup per AIoT device association are sufficient and efficient to support the AIoT operation.

As shown figure 6.20.1-2, for one AIoT operation, the AIoT function can setup separate association with multiple readers for one AIoT operation session. And under each reader, a list of per AIoT Device association is pre-setup and will be used to associate and ensure the information related the discovered AIoT Devices.

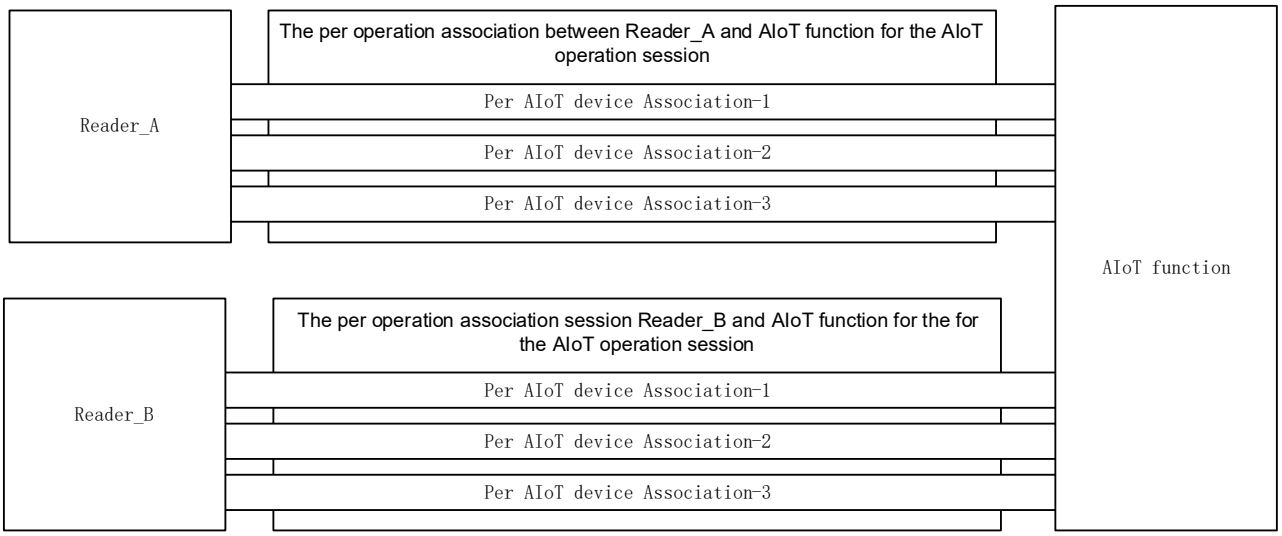


Figure 6.20.1-2: Multiple per AIoT device Association within an AIoT operation

6.20.2 Procedures

6.20.2.1 The interaction for AIoT Device among AIoT device, reader and AIoT Function

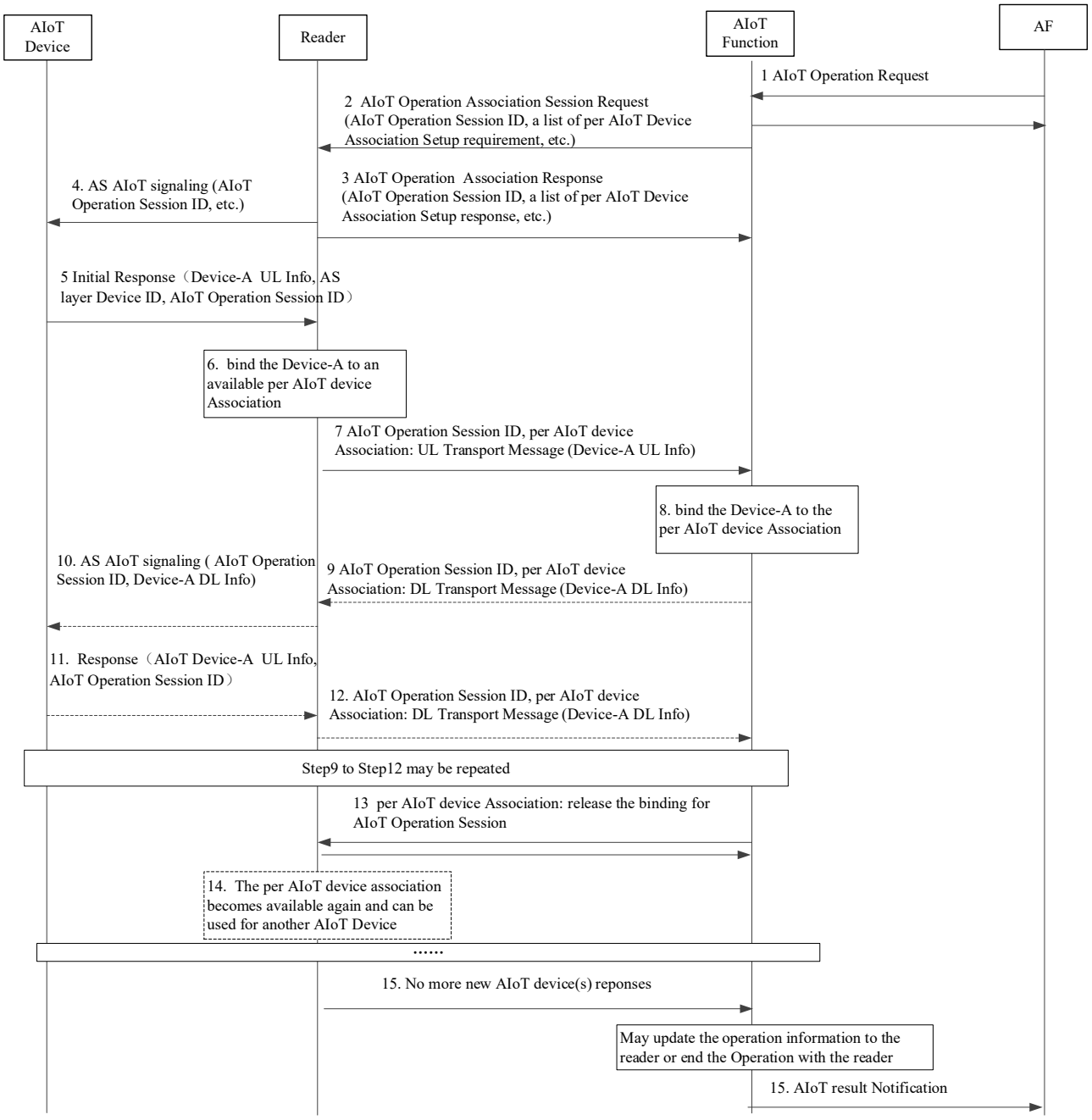


Figure 6.20.2.1-1: Service Flow

1. An AF requests AIoT operation to the AIoT Function directly or via NEF.
2. AIoT Function allocates an AIoT operation session ID used for the once invoked AIoT operation request and it is used to associate the information related to AIoT operation during the AIoT operation session. The AIoT function requires the AIoT operation association setup to the selected readers, including: the AIoT operation session ID, AIoT operation information, a list of per AIoT device association setup requirement, etc.
3. The reader accepts to setup the AIoT operation association and a list of per AIoT device associations.
4. The reader uses the AIoT operation information to trigger AS AIoT signalling, including the AIoT operation session ID.

5. The AIoT Device-A feedback the initial response (Device-A UL Info, AS layer Device ID, AIoT operation session ID).

NOTE 1: Device-A presents one AIoT device. The AIoT operation session ID in step5 is the AIoT operation session ID received in step4. It is used by the reader to identify the related AIoT operation session and AIoT function that should receive the Device-A UL Info.

6. The reader binds new discovered Device-A to an available per AIoT device association.

NOTE 2: There might be multiple bind per AIoT device associations ongoing in the meantime.

7. The reader uses the bind per AIoT device association to forward the Device-A UL Info to the AIoT function which is determined via the AIoT operation session ID.

8. The AIoT function receives the Device-A UL Info from per AIoT device association and binds Device-A and the AIoT device association together.

9. If there is further info for Device-A, the AIoT function uses the bind per AIoT device association to forward the Device-A DL Info to the reader.

10. The reader forwards the received Device-A DL Info to Device-A based on the binding between Device-A and per AIoT device association.

11-12: The reader forwards the received Device-A UL Info to the AIoT funcion based on the binding between Device-A and per AIoT device association.

NOTE 3: Step9 to step 12 may be repeated.

13-14. After finished the interaction with the AIoT device, the AIoT function can trigger to release the binding and the per AIoT device association becomes available again and can be used for another additional discovered AIoT device.

15. If the reader doesn't receive new AIoT device response any more, the reader notifies the situation to the AIoT function. Based on the received notification, the AIoT function may update the AIoT operation information to the reader or end the AIoT operation with the reader. The AIoF Function may also notify the situation to the AF and let AF to determine update the AIoT operation information or end the AIoT operation.

16. When the AIoT operation ends, the AIoT function reports to AIoT operation result to the AF.

6.20.2.2 The interaction for AIoT Device among AIoT device, reader, AIoT Function and AF

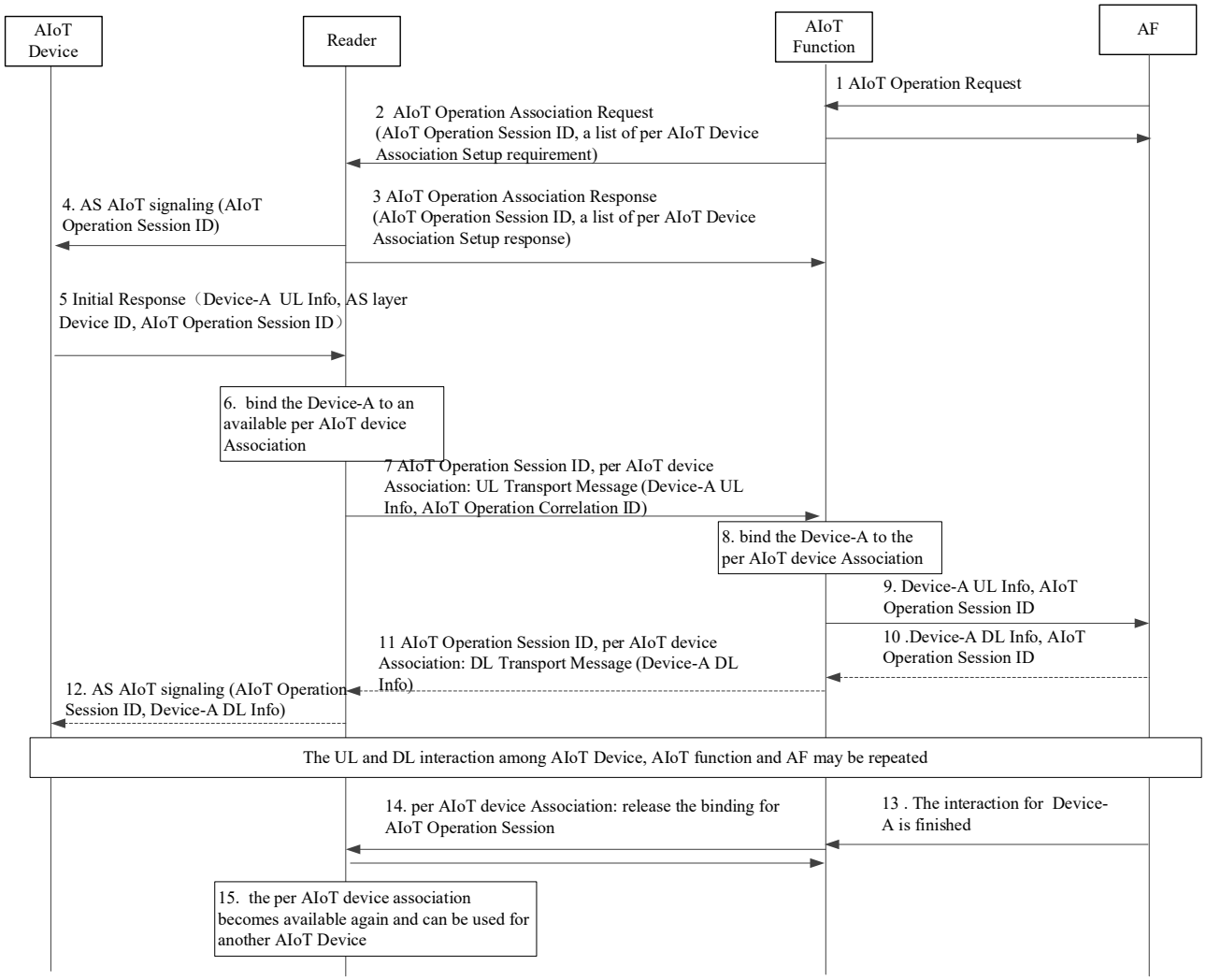


Figure 6.20.2.2-1: Service Flow

Steps 1-8 are the same as steps1-8 as described in clause 6.20.2.2-1.

9. If Device-A UL Info is for AF, the AIoT function forward it the AF with the AIoT operation session ID.

NOTE 1: Device-A present one AIoT device.10. Optionally, the AF may send to the AIoT function the Device-A DL Info with the AIoT operation session ID, or the AF notifies the AIoT function that the interaction with AIoT Device-A is finished.

11. The reader forwards the received Device-A DL Info to the reader based on the binding between Device-A and per AIoT device association.

12. The reader forwards the received Device-A DL Info to Device-A based on the binding between Device-A and per AIoT device association.

NOTE 2: The UL and DL interaction among AIoT Device, AIoT function and AF may be repeated.

13. After finished the interaction with the AIoT device, the AF notifies the AIoT function that the interaction with AIoT Device-A is finished.

14-15. The AIoT function can trigger to release the binding and the per AIoT device association becomes available again and can be used for another additional discovered AIoT device.

If the reader doesn't receive new AIoT device response any more, the reader notifies the situation to the AIoT function. Based on the received notification, the AIoT function may update the AIoT operation information to the reader or end

the AIoT operation with the reader. The AIoT Function may also notify the situation to the AF and let AF to determine update the AIoT operation information or end the AIoT operation.

6.20.3 Impacts on services, entities and interfaces

Editor's note: This clause captures impacts on existing services, entities and interfaces.

6.21 Solution #21: Architecture enhancement to support AIoT services for topology 2

6.21.1 Description

The solution applies to the Key Issue#1 "Architecture support of Ambient IoT Devices", the Key Issue #2 "Identification, Subscription, Registration and Connection management" and the Key Issue #3 "Support of Ambient IoT Services". This solution applies to Topology 2 as defined in TR 38.848 [7].

The Figure 6.21.1-1 illustrates the system architecture to support AIoT services for Topology 2.

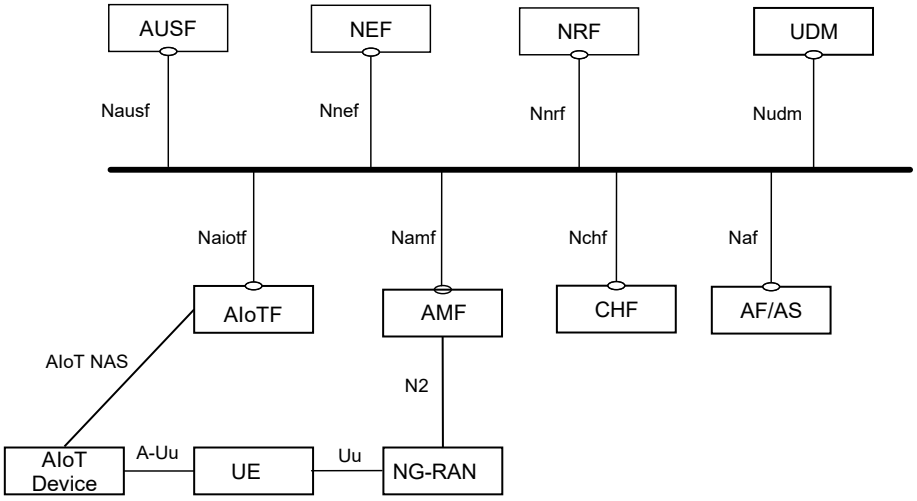


Figure 6.21.1-1: System Architecture to support AIoT Services for Topology 2

The system architecture and functions as defined in TS 23.501 [4] are used as basis to support the Ambient IoT services (e.g. Inventory, Command) with the following clarifications:

- AIoTF is introduced to support transfer of service request from AF (via NEF) to UE, and transfer of service response from UE to AF (via NEF).
- NEF is enhanced to support transfer of service request from AF to AIoTF, and transfer of service response from AIoTF to AF.
- UDM is enhanced to support authorize AIoT service request from AF, and AIoT service response from UE.
- AMF is enhanced to support authorization of UE communication with AIoT Device, and to support transfer of service request from AIoTF to UE and transfer of service response from UE to AIoTF.
- UE is enhanced to support communication with AIoT Device and interaction with AIoTF.

The Figure 6.21.1-2 illustrates the protocol stack to support AIoT services for Topology 2.

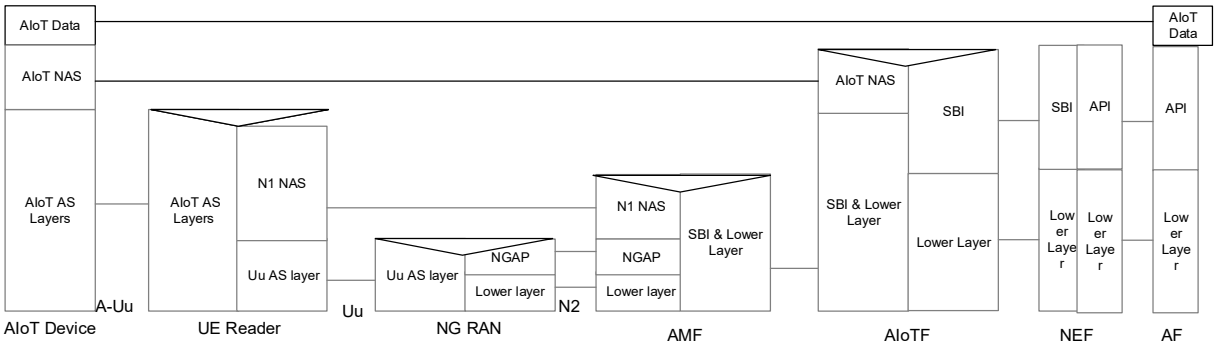


Figure 6.21.1-2: Protocol stack to support AIoT Services for Topology 2

As shown in the Figure 6.21.1-2, it is assumed AIoT Device communicates with UE via a new interface AIoT and both Access Stratum layer and Non-Access Stratum layer are supported. The UE communicates with the network using the existing Uu interface.

NOTE: The Access Stratum layer functionality is to be defined by RAN WGs.

For topology 2, the UE performs the Reader functions to send AIoT service request (e.g. Inventory, Command) to the AIoT Devices and receive AIoT service response (e.g. Device ID, AIoT data) from the AIoT Devices. The UE also acts as an intermediate node which is under the network control. This solution proposes a control plane based solution for core network to select the UE and to transfer the AIoT service request/response between UE and AF.

The AF requests service operation for an AIoT Device or a group of AIoT Devices or all of AIoT Devices via the NEF, and the NEF forwards the requested service operation to the AIoTF. The AIoTF selects the UE and delivers the requested service operation to the UE. The UE performs service operation (e.g. Inventory, Command) with the AIoT Devices and transfers the AIoT information (e.g. Device ID, AIoT data) received from the AIoT Devices to the AIoTF. The AIoTF then sends the AIoT information the AF via the NEF.

6.21.2 Procedures

Depicted in Figure 6.21.2-1 is the procedure to support AIoT Services for Topology 2.

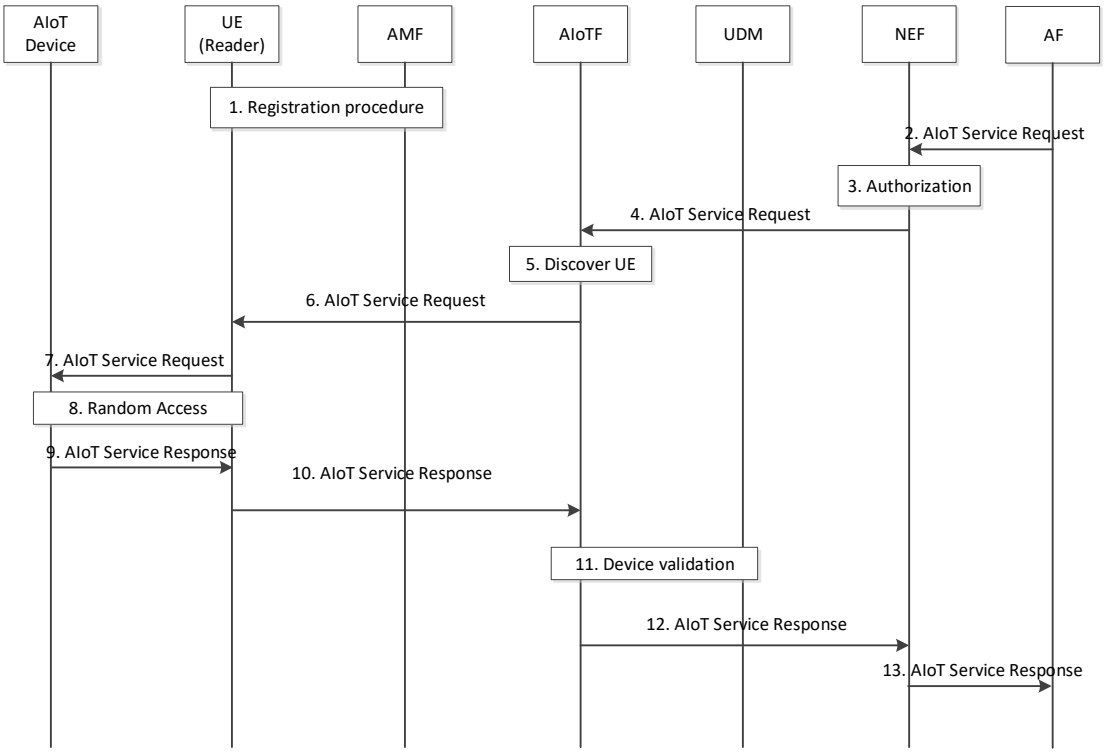


Figure 6.21.2-1: Procedure to support AIoT Services for Topology 2

1. UE performs Registration procedure as defined in TS 23.502 [5]. The UE includes AIoT Communication Indication in the Registration Request message and the AIoT Communication Indication indicates whether UE supports communication with AIoT Devices. If the UE is authorized to communicate with the AIoT Devices, the AMF includes authorized information in the Registration Accept message to the UE.

2. AF sends an AIoT Service Request (AF Identifier, UE ID, Device ID, requested service operation, area information) message to the NEF. The requested service operation indicates the service (e.g. Inventory, Command) the AF requested for the Ambient IoT Device(s). Device ID is used to identify one/group/all of Ambient IoT Devices. The area information is used to indicate the area the service operation will be applied, and the area information could be geographic area. The UE ID is the GPSI of UE. If the AF knows the UE is in the target area (e.g. UE may report its location information via application layer), the UE ID can be included in the Request message.

   The service operation could be an end-to-end parameter between AF and Ambient IoT Device which means it is transparent to CN, NG-RAN and UE, it also could be a parameter that can be understood by CN and UE. The following steps further clarify how these two options work.

3. The NEF checks if the AF is authorized to request the AIoT service and requests for the serving AIoTF of the UE if UE ID is provided in step 2.

4. The NEF selects the AIoTF either using the serving AIoTF retrieved from UDM in step 3 or mapping the area information to the corresponding AIoTF, and forwards the requested service operation to the AIoTF using AIoT Service Request (Device ID, UE ID, requested service operation) message. The UE ID is SUPI of UE.

5. The AIoTF selects the UE (if not provided in step 4) for AIoT service operation. The UE Reader can also be selected using other solutions (e.g. solution #11).

6. The AIoTF may derive the service operation based on the requested service operation and include the service operation in a AIoT Service Request message (Device ID, service operation) to the UE via the AMF, or the AIoTF may include the requested service operation as a container in a AIoT Service Request message (Device ID, service operation) to the UE via the AMF. If the UE is in CM_IDLE state, the network initiated service request procedure is performed before sending the AIoT Service Request message.

7. The UE (Reader) performs AS procedure (e.g. paging-like procedure) with Ambient IoT Devices. The UE may include an AIoT-NAS message (Device ID, service operation) into the AS message to the Ambient IoT Devices.

8. For each Ambient IoT Device, if the Ambient IoT Device receives AS message and is matched with the Device ID, the Ambient IoT Device initiates the Random Access like procedure.

9. The Ambient IoT Device sends and AIoT-NAS message (Device ID, AIoT data) over AS message. If the service operation is Inventory, the Device ID is included in the AIoT-NAS message. If the service operation is Command (e.g. Read), the AIoT data along with the Device ID are included in the AIoT-NAS message if requested by the service operation.

NOTE 1: The AS procedures in steps 7, 8 and 9 are to be defined by RAN WGs.

NOTE 2: The security protection of AIoT-NAS message including group security is to be designed by SA WG3.

10. The UE sends an AIoT Service Response message (Device ID, AIoT data) to the AIoTF via the AMF.

11. The AIoTF performs validation based on the device subscription-like information received from the UDM, including Device ID validation.

12. The AIoTF responds to the NEF using AIoT Service Response (Device ID, AIoT data) message.

13. The NEF responds to the AF using AIoT Service Response (Device ID, AIoT data) message.

6.21.3 Impacts on services, entities and interfaces

AIoTF:

- AIoTF is introduced to support transfer of service request from AF (via NEF) to UE via AMF, and transfer of service response from UE via AMF to AF (via NEF).

NEF:

- NEF is enhanced to support transfer of service request from AF to AIoTF, and transfer of service response from AIoTF to AF.

UDM:

- UDM is enhanced to support authorize AIoT service request from AF, and AIoT service response from UE.

AMF:

- AMF is enhanced to support authorization of UE communication with AIoT Device, and to support transfer of service request from AIoTF to UE and transfer of service response from UE to AIoTF.

UE:

- UE is enhanced to support communication with AIoT Device.

6.22 Solution #22: AF Based Solution for Topology 2

6.22.1 Description

6.22.1.0 General

This solution proposes an AF based solution, which addresses KI#1, KI#2 and KI#3.

In this solution, it is AF who is in charge of the intermediate UE for the Ambient IoT operations, including communicating with and determining intermediate UEs, sending operation commands to and receiving results from intermediate UEs. As the licensed spectrum is owned by MNO, it is proposed to let network provide the radio resource information towards the intermediate UEs about the spectrum information for the over-the-air interface between Intermediate UEs and AIoT devices.

Editor's note: It is FFS about the improvements by locating more functions in CN, e.g. AF get locations of Intermediate UEs from CN but not from UEs; CN may determine Intermediate UEs on behalf of AF.

6.22.1.1 Reference Architecture

The Figure 6.22.1.1-1 illustrates the architecture for AF based solution:

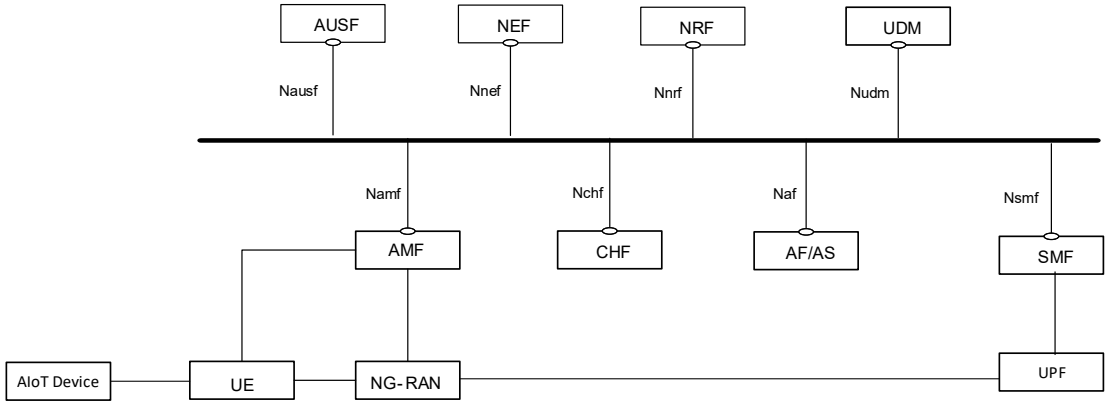


Figure 6.22.1.1-1: System Architecture of AF Based Solution

This solution focuses on Topology 2.

Editor's note: It is FFS whether and how the solution can be evolved for Topology 1.

The functional entities defined in TS 23.501 [4] are reused with the exception for the following additions:

- UDM/UDR: The authorization information of Intermediate UE for AIoT is stored in UE subscription data

- AMF: Receive AIoT capability information from UE and authorize based on the subscription data in UDM/UDR.
- NG-RAN: Provide spectrum information towards authorized intermediate UE.
- Intermediate UE:
  - Provide Ambient IoT capability information to AMF and receive the authorization information
  - Receive the instruction from AF and perform Ambient IoT operations (e.g. inventory, command, etc.) on the proper spectrum. The radio resource information is received from NG-RAN
  - Receive command results from AIoT devices and send to AF.
- AF:
  - UE AIOT layer interactions with Intermediate UEs
  - Determine Intermediate UEs, send the instruction to the Intermediate UEs, and receive responses from Intermediate UEs.

Editor's note: Further involvement of the CN is FFS, including whether and how to perform access control for devices, and how to perform charging towards CHF.

6.22.1.2 Protocol Stack

The Figure 6.22.1.2-1 illustrates the protocol stack for AF based solution over user plane:

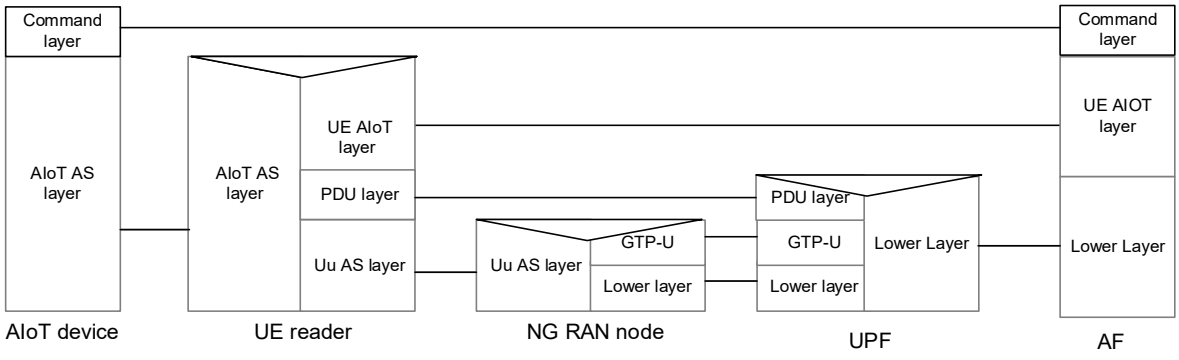


Figure 6.22.1.2-1: Protocol Stack for AF Based Solution (over user plane)

Within the protocol stack:

- UE AIoT layer: between AF and UE reader. AF provides Ambient IoT operation commands to Intermeidate UEs via UE AIoT layer.
- Command layer: The command layer protocol between AIoT devices and AF, including, e.g. read, write, execute, etc.
- Uu AS layer: On top of the existing Uu AS layer, radio resource information request from UE reader to NG-RAN.

It is assumed that the end-to-end protection is implemented between AF and AIoT devices.

NOTE 1: The UE AIOT layer and Command layer are assumed to be defined by SA WG2 (Stage 2 aspects) and CT WG1 (Stage 3 aspects) but can be determined during conclusion phase.

NOTE 2: The Details of the protection between AIoT device and AF and the protection between AIoT device and UE reader are assumed to be addressed by SA WG3.

The UE AIOT layer messages between AF and UE reader can be transferred via PDU session user plane. AF delivers downlink messages towards Intermediate UE and receives uplink messages from Intermediate UE via UPF.

The protocol stack for AF Based Solution over control plane is similar, with the involvement of CN control plane NFs (AMF, SMF, NEF) instead of user plane NF (UPF). Over control plane, UE AIOT layer is over the NAS layer in UE reader. AF communicates with UE reader via control plane, by utilizing the Control Plane CIoT 5GS Optimisation. The downlink messages are sent by AF, and delivered through NEF, SMF, AMF, NG-RAN towards Intermediate UE. The uplink messages are sent by the Intermediate UE, and delivered through NG-RAN, AMF, SMF, NEF towards AF.

6.22.2 Procedures

NOTE: The message names in the procedures below are descriptive. It is assumed that the names are updated with corresponding SBI based names where applicable during the normative phase.

6.22.2.1 AIoT Service Authorization for Intermediate UE

The Registration procedure for UE is performed as defined in clause 4.2.2.2 of TS 23.502 [5] with the following additions:

- UE includes the AIoT Intermediate node capability as part of "5GMM capability" in Registration Request message.
- The AMF obtains the AIoT Subscription data as part of the user subscription data from UDM using Nudm_SDM service
- The AMF determines whether the UE is authorized to work as Intermediate UE for AIoT based on UE's AIoT Intermediate node capability and the AIoT Subscription data. The AMF includes the authorization information as part of UE context in NGAP message sent to NG-RAN.

In Service Request procedure, N2 Handover procedure, Xn Handover procedure, and when receiving Subscriber Data Update to AMF, the AMF includes the authorization information in NGAP message sent to NG-RAN.

6.22.2.2 Inventory Procedure

The Inventory procedure can be initiated by the AF to discover one or more AIoT devices in a specific area via Intermediate UEs.

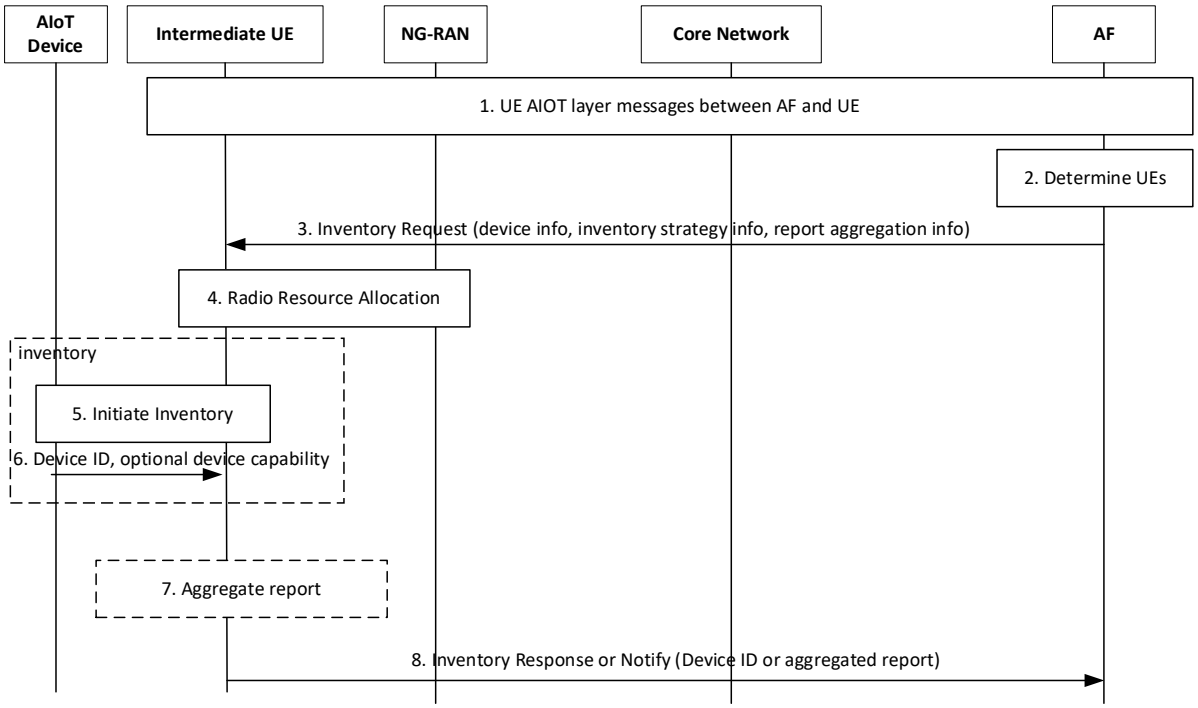


Figure 6.22.2.2-1: Inventory Procedure

1. Via the Intermediate UE's PDU session user plane or control plane, the Intermediate UE communicates with the AF to transport UE AIOT layer messages, including location information reporting and reader capability information.

2. The AF determines Intermediate UEs based on the location information and reader capability information reported by the Intermediate UEs in step 1, as well as the area AF intended to perform inventory.

Editor's note: It is FFS on whether and how the posture information of Intermediate UEs can be collected and the impacts on the determination of Intermediate UEs.

3. The AF sends Inventory Request towards the selected Intermediate UEs via Core Network and NG-RAN over PDU session user plane or control plane. The Inventory Request includes the device information, inventory strategy information.

- The device information could be device ID, device group ID, and/or device type. The device type refers to type 1, 2A or 2B in TR 38.769 [8].

NOTE 1: It's up to RAN to determine whether the device type is useful or not for Intermediate UE, based on the assumption of harmonized air interface.

- The inventory strategy information contains, e.g. the inventory frequency and inventory period to guide the reader to perform the inventory periodically. It also indicates whether all the targeted devices need to respond (full inventory), or only those who haven't performed the inventory procedure (delta inventory) should respond. For flexibility, the delta inventory may require the device to respond if it is reading by a different reader, even if the device has performed the inventory procedure, which requires the AIoT devices to keep the reader ID.

- The report aggregation info indicates whether the reports need to be aggregated or not for a specific aggregation period, and whether the reports are needed after the aggregation period.

4. The Intermediate UE interacts with NG-RAN for radio resource allocation

NOTE 2: The detail of this function is assumed to be defined by RAN.

5. The Intermediate UE initiates inventory based on device information as well as the inventory strategy information provided by the AF. To be able to differentiate readers in delta inventory, the Intermediate UE may provide reader identity information to enable the AIoT devices to understand they are read by which Intermediate UE. Considering the mobility of the Intermediate UE (e.g. when the Intermediate UE and some devices move together to another room, in the following delta inventory, the devices can regard the Intermediate UE in a new room as a different reader than the one they have responded), the reader identity information can be a combination of an application layer reader ID and the location information.

6. The AIoT Device reports the device ID. If the Inventory procedure indicates only who haven't performed the inventory procedure should respond, and if the AIoT Device has performed the inventory procedure towards this reader, it should skip the reporting.

7. The Intermediate UE may perform aggregation for the device ID, based on the report aggregation information provided by the AF. Within the aggregation period, the Intermediate UE will buffer the device IDs reported from the AIoT devices. The Intermediate UE may stop buffering and send report immediately, if it determines no further report from devices. When the aggregation period expires, the Intermediate UE sends the report. For those device ID report after the aggregation period, if it is needed by the AF, the Intermediate UE sends the report. Otherwise, it will be dropped.

8. The Intermediate UE sends Inventory Response or Notification Request towards the AF for the device ID or the aggregated device ID information.

   The Intermediate UE may perform periodic inventory following the instructions from AF, as described in Solution#8.

6.22.2.3 Command Procedure

The Command procedure is initiated by the AF to request one or more AIoT devices in a specific area to execute a command via Intermediate UEs. The device may or may not send back the command results depends on the command.

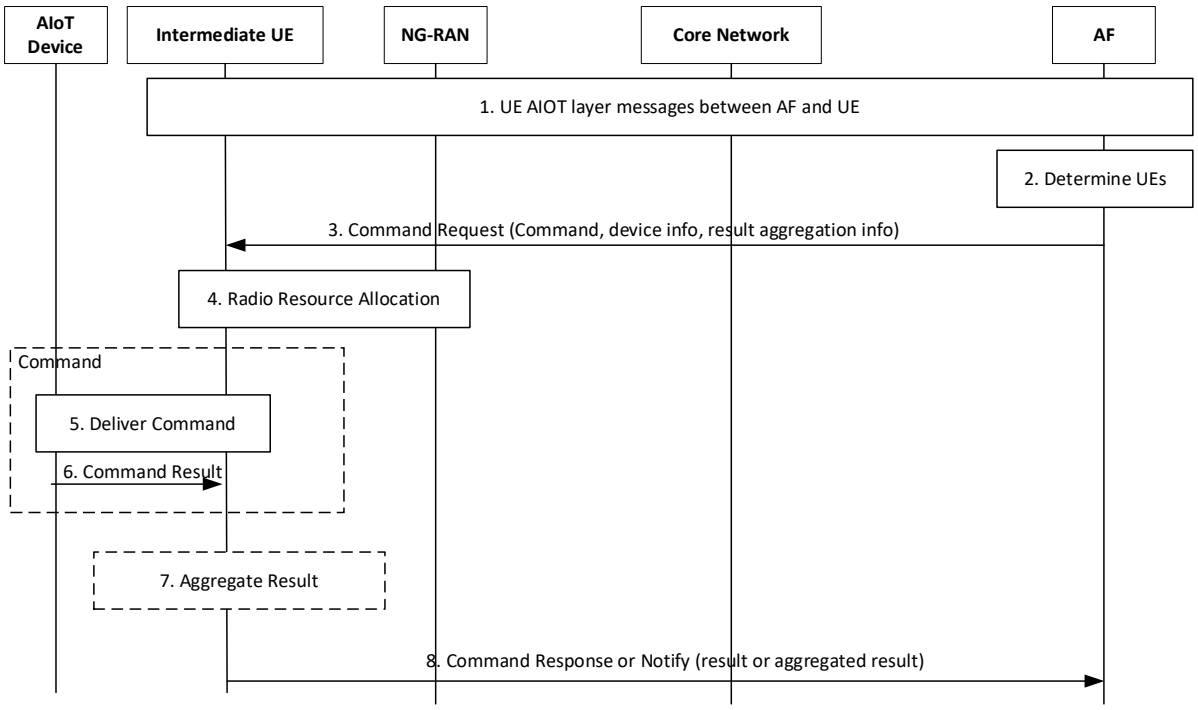


Figure 6.22.2.3-1: Command Procedure

1. Via the Intermediate UE's PDU session user plane or control plane, the Intermediate UE communicates with the AF to transport UE AIOT layer messages, including location information reporting and reader capability information.

2. The AF determines Intermediate UEs based on the location information and reader capability information reported by the Intermediate UEs in step 1, as well as the area AF intended to perform command.

3. The AF sends Command Request towards the selected Intermediate UEs via Core Network and NG-RAN over user plane or control plane. The Command Request includes the command, device information, result aggregation information.

   - The command is the command to be executed in the device, including read, write, enable, disable, or other application specific command.

   - The device information could be device ID, device group ID, and/or device type. The device type refers to type 1, 2A or 2B in TR 38.769 [8].

NOTE 1: It's up to RAN to determine whether the device type is useful or not for Intermediate UE, based on the assumption of harmonized air interface.

   - The result aggregation info indicates whether the results need to be aggregated or not for a specific aggregation period, and whether the reports are needed after the aggregation period.

4. The Intermediate UE interacts with NG-RAN for radio resource allocation

NOTE 2: The detail of this function is assumed to be defined by RAN.

5. The Intermediate UE delivers the command to the AIoT devices.

6. The AIoT Device executes the command and send back the result to the Intermediate UE if needed.

7. The Intermediate UE may perform aggregation for the result, based on the report aggregation information provided by the AF. Within the aggregation period, the Intermediate UE will buffer the results from the AIoT devices. The Intermediate UE may stop buffering and send report immediately, if it determines no further results from devices. When the aggregation period expires, the Intermediate UE sends the report. For those results after the aggregation period, if it is needed by the AF, the Intermediate UE sends the report. Otherwise, they will be dropped.

8. The Intermediate UE sends Command Response or Notification Request towards the AF for the result or the aggregated results.

6.22.3 Impacts on services, entities and interfaces

The following NFs are impacted:

- **UDM/UDR:**
  - Store the authorization information of Intermediate UE for AIoT.

AMF:

- Receive AIoT capability information from UE and authorize based on the subscription data in UDM/UDR.

NG-RAN:

- Provide spectrum information towards authorized intermediate UE.

- **Intermediate UE:**
  - Provide Ambient IoT capability information to AMF and receive the authorization information.
  - Receive the instruction from AF and perform Ambient IoT operations (e.g. inventory, command, etc.) on the proper spectrum. The radio resource information is received from NG-RAN.
  - Receive responses from AIoT devices and send to AF.
- **AF:**
  - UE AIOT layer interactions with Intermediate UEs.
  - Determine Intermediate UEs, send the instruction to the Intermediate UEs, and receive responses from Intermediate UEs.

6.23 Solution #23: UE reader selection for inventory procedure

6.23.1 Description

6.23.1.1 Definition

AIoT device capability: e.g. inventory frequency, etc. Different AIoT devices may support different inventory frequency.

reader: part of the intermediate node UE, which can communicate with the AIoT device via radio interface.

UE reader capability: e.g. inventory frequency, etc. Different readers may support different inventory frequencies, and the inventory frequency supported by the reader may be different from the inventory frequency supported by the AIoT device as well. To execute the inventory related procedure between the device and the reader, the inventory frequency used by the device and the reader should match with each other.

6.23.1.2 General

This solution applies to Key issue #3 in terms of selecting the reader used for Ambient IoT inventory service.

This solution applies to the following scenario:

- The AF (i.e. third party) uses a bulk of AIoT devices that have the same AIoT device capabilities, e.g. the same inventory frequency, etc.;
- the AF holds the AIoT device information (e.g. device capabilities, etc.).

This solution applies to topology 2.

This solution proposes to select the appropriate UE readers before performing the inventory procedure.

6.23.1.3 Architecture and Principles

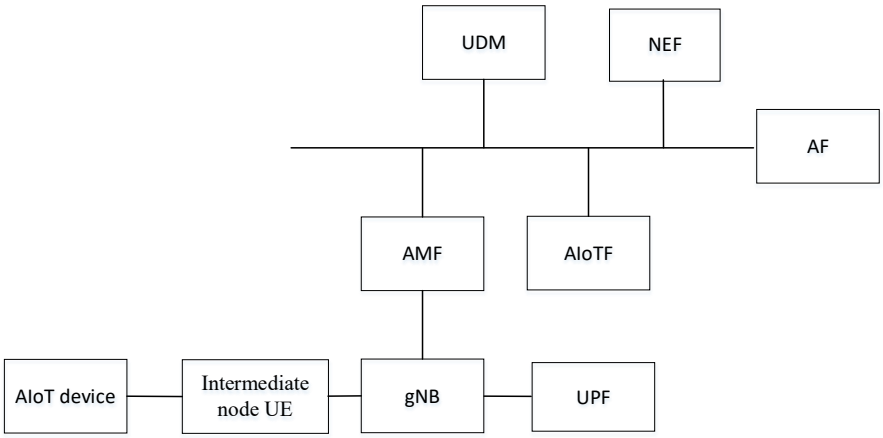


Figure 6.23.1.3-1: Reference architecture

The principles of the solution:

- AF provides the AIoT device capability to the Ambient IoT Function (AIoTF);
- the selection of the intermediate node UE will be based on: the AIoT device capability, the inventory area wherein the AIoT devices are supposed to be, and the the UE reader capability.

- AIoTF is responsible for:
  - mapping the third party information (e.g. the inventory area information) to 3GPP information (e.g. cell ID, etc.);
  - selecting the appropriate intermediate node UE.

6.23.2 Procedures

This clause provides how to select the UE readers that will be used for the inventory procedure.

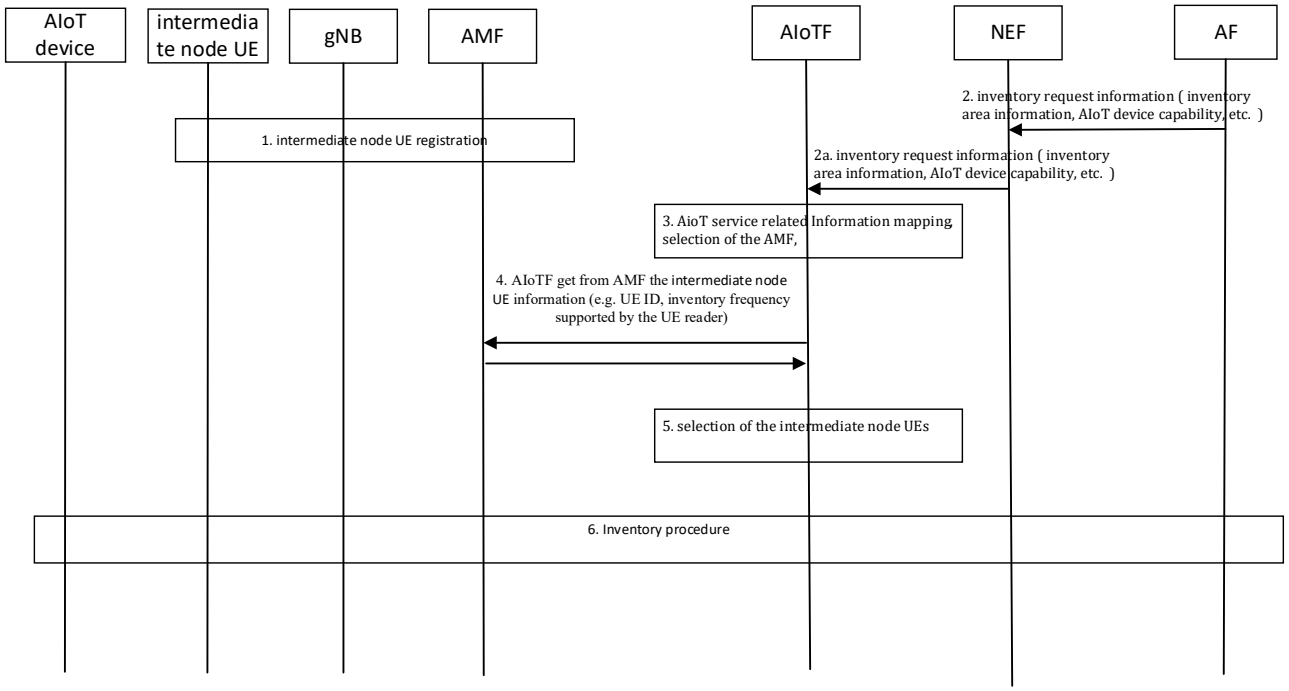


Figure 6.23.2-1: UE reader selection procedure

1. The intermediate node UE performs the registration procedure. The AIoT UE reader capabilities (e.g. supported inventory frequency) will be transmitted to the AMF via the NAS message (e.g. the Registration Request message).

2. The AF sends to the NEF the inventory request information that includes: inventory area information, AIoT device capabilities (e.g. supported frequency, etc.), etc. The inventory area information defined by the third party indicates the area where the AIoT devices are supposed to be.

2a. The NEF maps the inventory area information to 3GPP based location information (e.g. TA(s), cell ID(s), etc.), and then selects the AIoTF(s) based on the mapped location information. The mapping between AF requested inventory area information and 3GPP defined location information is stored in NEF locally. The NEF sends to each of the selected AIoTF the inventory request information received from the AF with the mapped location information.

3. The AIoTF selects the AMF(s) based on the mapped location information provided by NEF in step 2a. If multiple AMFs serve the same area, then all AMFs serving the area or a subset of the AMFs could be selected based on the AMF load information.

4. The AIoTF get from the AMF the information of the intermediate node UEs that are available in the cell with the cell ID mapped in step 3. The information of the intermediate node UE includes: e.g. UE ID, inventory frequency;

5. The AIoTF selects the appropriate intermediate node UEs that will be engaged in the inventory procedure. The inventory frequency of the intermediate UE should overlap with the inventory frequency supported by the AIoT device.

6. The inventory procedure is carried out. The radio configuration (inventory frequency, etc.) of the intermediate node, e.g. how RAN to schedule the radio resource to reader UE, is up to RAN WGs.

NOTE 1: It depends on operator's policy or configuration that there can be different selection criteria of the intermediate node, e.g. based on the frequency capability of the AIoT device and the intermediate node.

NOTE 2: Other aspects of the inventory procedure can be carried out using other solutions (e.g. solution #11).

6.23.3 Impacts on services, entities and interfaces

NEF:

- NEF is enhanced to support transfer of AIoT service request from AF to AIoTF.

AIoTF:

- AIoTF is introduced to support mapping the AIoT related information from the third party defined format to the 3GPP defined format;
- Supports selecting intermediate node.

AMF:

- Supports providing the information of the intermediate node UE to the AIoTF.

UE:

- UE is enhanced to support indicating its capability to support AIoT services to AMF (e.g. in registration message).

6.24 Solution #24: Availability Reporting without Ambient IoT Device Timers

6.24.1 Functional Description

This solution focuses on the registration management and/or connection management aspects of Key Issue #2 "Identification, Subscription, Registration and Connection management".

The main principle of the proposal is that the architecture should support a way for the Ambient IoT Device to determine when it needs to perform a control plane procedure without requiring the Ambient IoT device to always run timers (e.g. a periodic registration timer).

In this solution, the network can send a "check-in" time value to the Ambient IoT Device in any procedure. The Ambient IoT Device stores the time value while it sleeps and does not have to run any timers. When the Ambient IoT Device wakes up from a sleep state, the Ambient IoT Device will read time information that is broadcasted from the network or an intermediate node. If the broadcasted information indicates that the time is past the "check-in" time, then the Ambient IoT Device will initiate a control plane procedure.

This approach will help ensure that the Ambient IoT Device will not go a long period of time without announcing its availability (e.g. registering) and will also help ensure that the Ambient IoT Device does not initiate control plane procedures too often.

6.24.2 Procedures

Figure 6.24.2-1 shows the procedure.

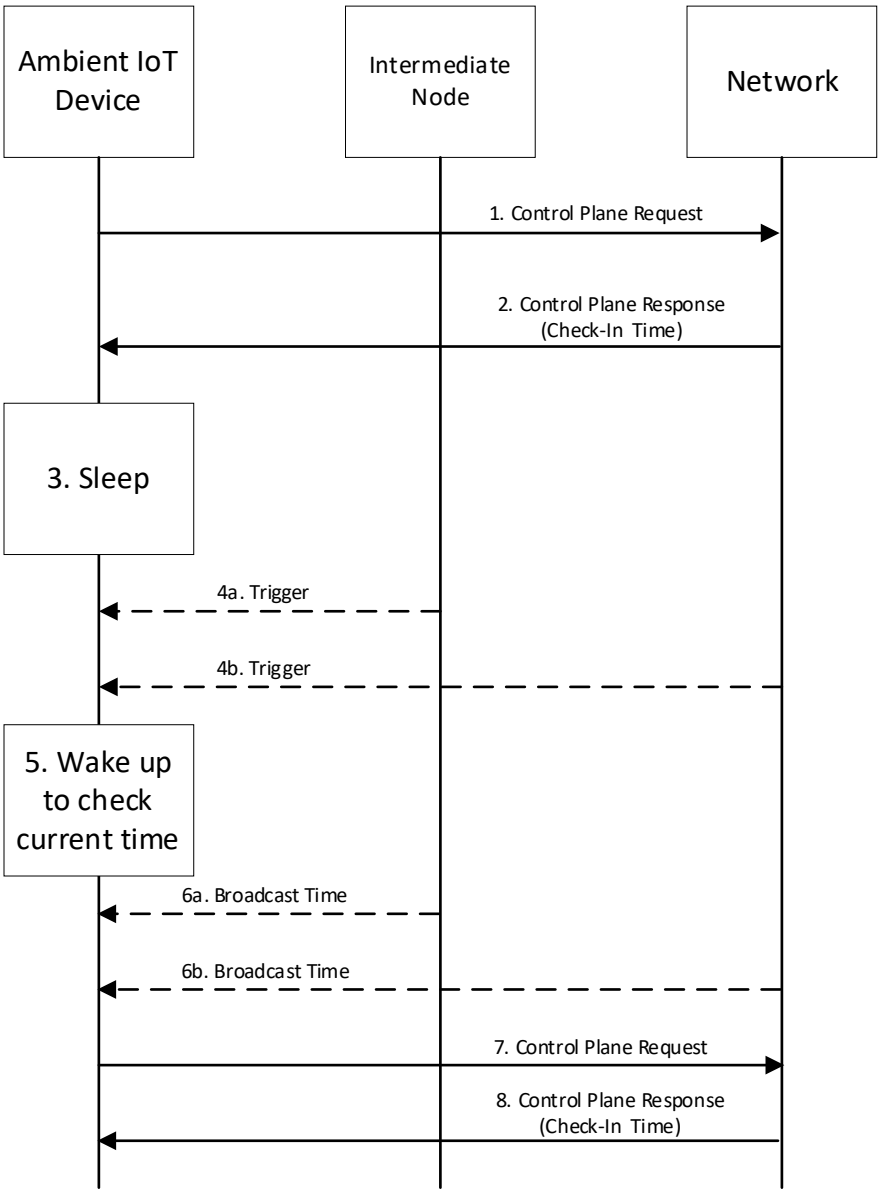


Figure 6.24.2-1: Procedure for Ambient IoT Device Availability Reporting

1. Upon network or IN trigger (e.g. activation signal, inventory request, not shown in the figure) the Ambient IoT Device sends a control plane request to the network. For example, it may be a device Registration request or a Inventory report. This control plane request informs the network that the Ambient IoT Device is still present in the network.

2. The Ambient IoT Device receives a control plane response from the network. The response includes an absolute time value. The time value indicates to the Ambient IoT Device that the device should send another control plane request sometime after the time value. The network node that assigns the time value depends on the overall architecture of the network that the Ambient IoT Devices connect to, but would most likely be the network function where control plane messages terminate and/or the network function that keeps track of Ambient IoT Device's presence in the network.

   The Ambient IoT Device is allowed to initiate a control plane request before the time value, for example, to report a detected event. However, the time value will help ensure that the Ambient IoT Device will not go a long period of time without announcing its availability (i.e. registering) and will also help ensure that the Ambient IoT Device does not initiate control plane procedures too often.

3. The Ambient IoT Device enters the sleep state. Timers do not need to run in the Ambient IoT Device and no real-time clock needs to be maintained by the Ambient IoT Device.

4. The Ambient IoT Device is energized or triggered. The trigger could be an activation signal that is received from the Network or Intermediate Node.

5. The Ambient IoT Device wakes up because of the trigger.
6. The Ambient IoT Device reads time information that is broadcasted by the Network or Intermediate Node. If the time information indicates that the current time is earlier than the time value that was received in step 2, then the Ambient IoT Device can return to the sleep state of step 3. If the time information indicates that the current time is equal to or later than the time value that was received in step 2, then the Ambient IoT Device indicates its availability to the network by proceeding to step 7.
7. In response to 4a/4b trigger, the Ambient IoT Device sends a control plane request to the network.
8. The network updates the device's availability info (e.g. its current area, associated Reader, etc.). The network may take the opportunity to send data in a response or new request to the device. The response or new request may include a new time value.

NOTE: If the network doesn't receive the device's check-in message long after the check-in time, the network may consider the device not present in the network.

6.24.3 Impacts on existing services, entities and interfaces

Network:

- Sends a time value to the Ambient IoT Device.

RAN/Intermediate Node UE:

- Broadcast time information.

Ambient IoT Device:

- Receives and stores a time value from the network.
- Reads broadcasted time information.
- Compares the broadcasted time information with the value that was received from the network and uses the comparison to detect if it needs to send a control plane request.

6.25 Solution #25: Subscription Management for Ambient IoTs

6.25.1 Description

This solution is targeting KI#2 Identification, Subscription, Registration and Connection management. It is related with KI#1 as well.

Regarding the subscription management for Ambient IoTs, the following should be considered.

- No USIM (UICC): No Root Key (e.g. LTE K):
  - Not available existing EPS-AKA, EAP-AKA', 5G AKA
- Less Computation Power:
  - (May) Not available EPS AKA, EAP-AKA', 5G AKA algorithm
  - (May) Not available Key derivation with Key Derivation Function (KDF)
- Huge Number of UEs:
  - (May) Not available individual authentication of each UE

Considering the above requirements, subscription of several millions of ultra-low complexity, power, cost and resource-constrained devices to MNO is a difficult problem we need to solve. For this issue, we propose a 3rd Party Service Provider(AF)-based on-line group subscription via NEF.

Editor's note: Details of the usage of this solution will be added later.

An example of the on-line group subscription via NEF is depicted in Figure 6.25.1-1.

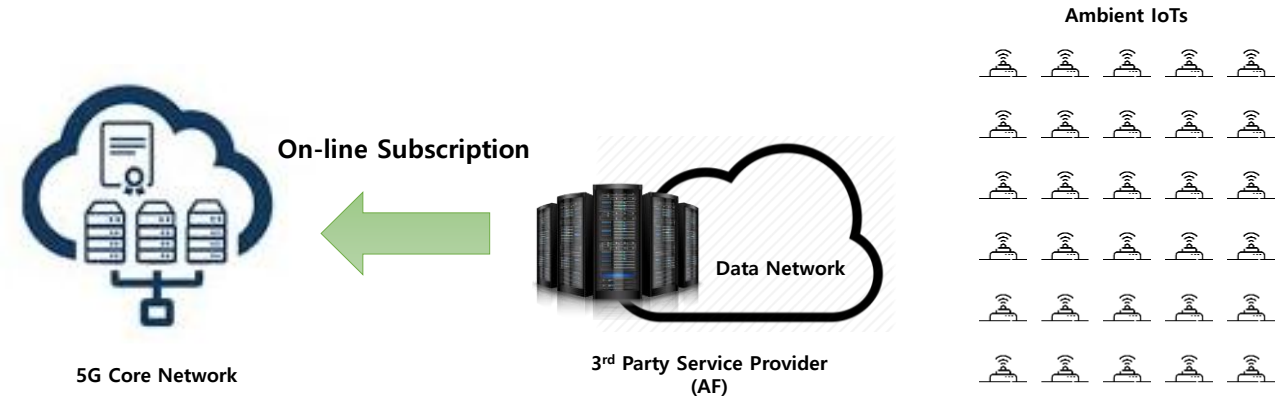


Figure 6.25.1-1: Online-subscription Management from 3rd Party Service Provider (AF)

The AF is a 3rd Party Service Provider, which may have business operator using Ambient IoT.

For Ambient IoT devices, the selection of NFs and the related PDU Session setup procedures are be burden, as the number of devices is very large. When Registration happens some of those parameters are configured based on the subscription date described below. After registration, the data path for the devices are configured according to the AM/SM subscription data stored in UDM during the Subscription Procedure.

The subscription data may include the following:

- Access and Mobility Subscription data which may include AF ID, Usage ID, List of Ambient IoT ID(s), SMF ID
- Session Management Subscription data which may include AF ID, Usage ID, List of Ambient IoT ID(s), NEF ID for NIDD, NIDD Information

Editor's note: Details of subscription data will be added later.

In addition to AM and SM subscription data, the following information may be needed for online-subscription management for Ambient IoT(s).

Table 6.25.1-1

Subscription data type	Field	Description
Online-subscription Management	AF ID	ID of 3rd Party Service Provider (AF)
	List of Usage ID(s)	ID of TASK of Ambient IoTs. It may have NAI format (host part + realm part)
	List of Ambient IoT ID(s)	List of each Ambient IoT ID(s) which take part in common TASK identified by Usage ID
	Authentication/Authorization information for AF	Information for EAP-AKA between 5GC and AF
	List of Credentials of Ambient IoTs	Credential information of Ambient IoTs
	TASK Information	Description of TASK identified by Usage ID
	Online-subscription Management (OM) States of Ambient IoTs	OM states of each Ambient IoT (either OM_Subscribed or OM_Unsubscribed)
	Registration Management (RM) States of Ambient IoTs	RM states of each Ambient IoT (either RM_Registered or RM_Deregistered)
	Other information	Other information regarding subscription

Regarding this online-subscription management, there are OM-UNSUBSCRIBED state and OM-SUBSCRBIED state in UDM and AF. The changes between these two states are depicted in Figure 6.25.1-2. After subscription accept (that is Subscription/Registration Update Accept), transition into OM-SUBSCRIBED state. After unsubscription, transition into OM-UNSUBSCRIBED state.

In OM-UNSUBSCRIBED state, the followings are information is stored in the UDM.

- No subscription information in Access and Mobility Subscription data and Session Management Subscription data.
- Only Online-subscription Management Subscription data with OM-UNSUBSCRIBED state:
  - These information can be stored during Service Level Agreement(SLA) between MNO and 3rd party service provider.
- May not be assigned with Credential information.

In OM-SUBSCRIBED state, the followings information is stored in the UDM.

- Subscription information in Access and Mobility Subscription data, Session Management Subscription data and Subscription Management Subscription data with OM-SUBSCRIBED state.
- Assigned with Credential information.

NOTE: AF performs bulk device provisioning (Group Registration) in OM-SUBSCRIBED state only, as the device information including credentials are in the 5GC in this state.

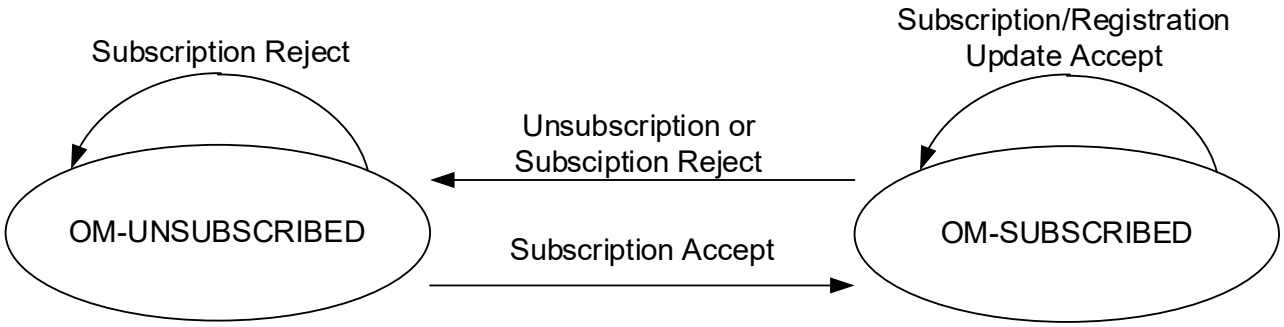


Figure 6.25.1-2: Online-subscription Management state model in AF and UDM

6.25.2 Procedures

Overall Procedures of Subscription for Ambient IoT is as depicted in Figure 6.25.2-1.

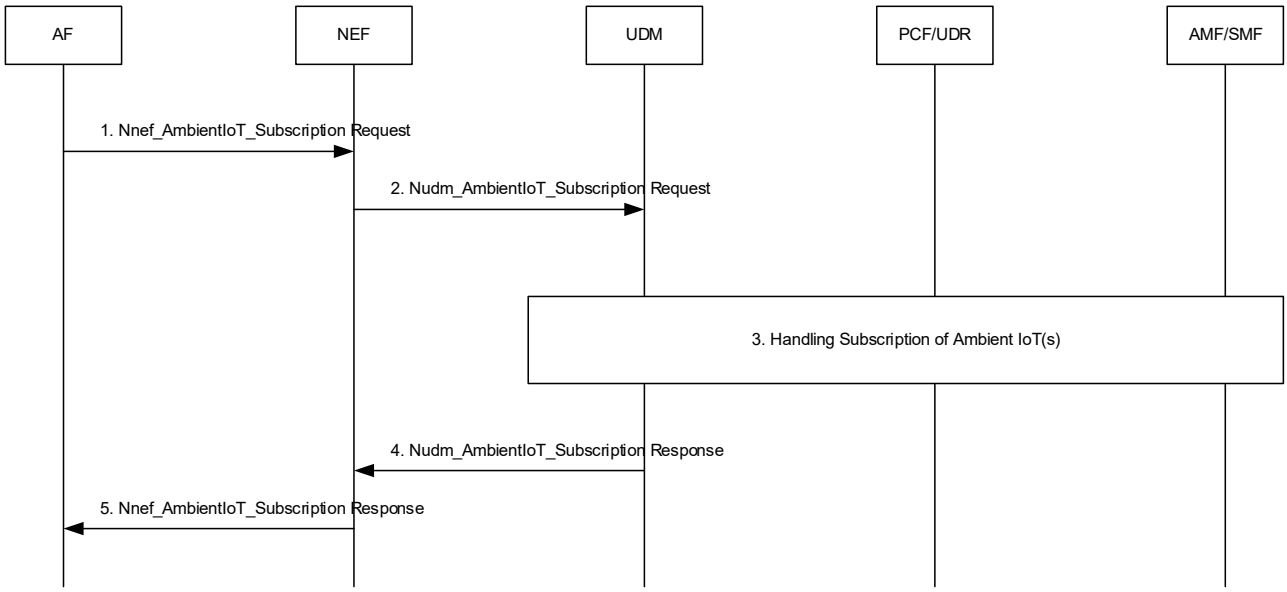


Figure 6.25.2-1: Subscription procedures for Ambient IoT Devices

1. AF sends AmbientIoT Subscription Request to NEF. It contains AF ID, Usage ID, List of Ambient IoT ID(s) which take part in this TASK, TASK information.
2. NEF sends AmbientIoT Subscription Request to UDM.

3. Handling subscription of Ambient IoT(s) may include the following:
   - Allocates Authentication/Authorization information AF (Usage).
   - Allocates Credentials for Ambient IoTs.
   - Changes OM states of Ambient IoTs as OM-SUBSCRIBED.
   - Determines suitable AMF and SMF via NRF with the help of OAM.
   - Update Access and Mobility Subscription data and Session Management Subscription data.
4. UDM sends NEF AmbientIoT Subscription Response. It may include the following:
   - Authentication/Authorization information for AF with TASK.
   - List of credentials of Ambient IoTs.
   - Shared PDU Session information.

Editor's note: Details of Shared PDU Session Information will be described later.

   - Selected AMF ID.
   - Other information.
5. NEF sends AF AmbientIoT Subscription Response.

Overall Procedures of Unsubscription for Ambient IoT is as depicted in Figure 6.25.2-2.

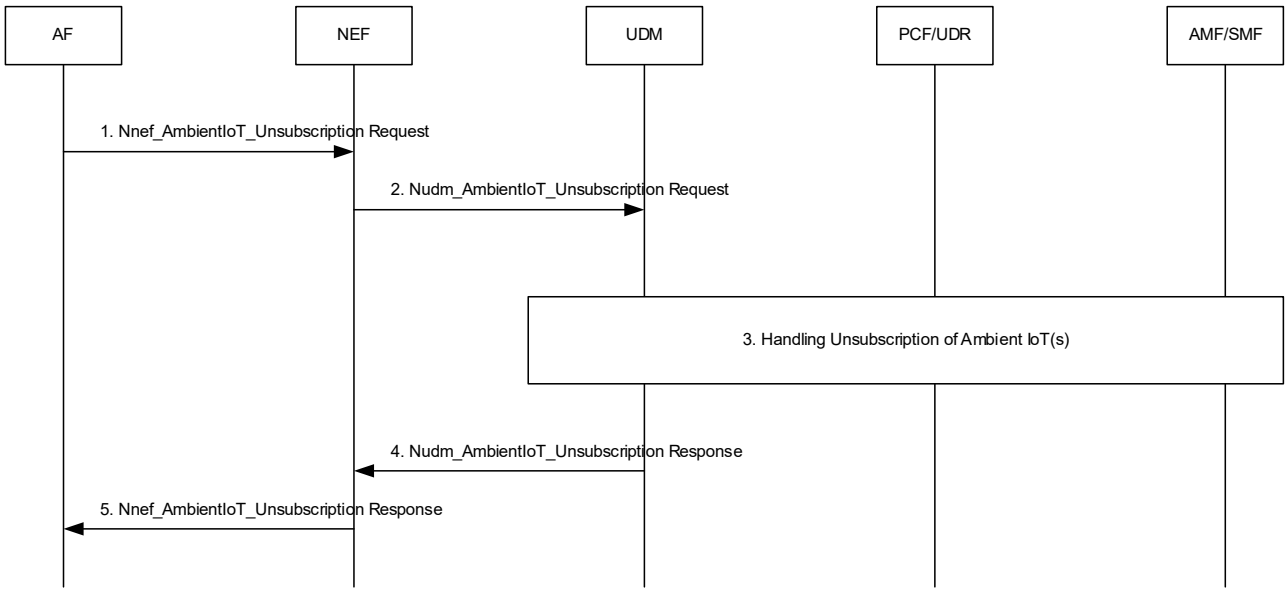


Figure 6.25.2-2: Unsubscription procedures for Ambient IoT Devices

1. AF sends AmbientIoT Unsubscription Request to NEF. It contains AF ID, Usage ID.
2. NEF sends AmbientIoT Unsubscriptin Request to UDM.
3. Handling Unsubscription of Ambient IoT(s) may include the following:
   - Deallocates Authentication/Authorization information AF(Usage).
   - Deallocates Credentials for Ambient IoTs.
   - Removing all the information regarding PDU Session identified by combination of AF ID and Usage ID.
   - Changes OM states of Ambient IoTs, OM-UNSUBSCRIBED.
   - Removes Access and Mobility Subscription data and Session Management Subscription data.

4. UDM sends NEF AmbientIoT Unsubscription Response. It includes Result of Unsubscription Request.
5. NEF sends AF AmbientIoT Unsubscription Response.

6.25.3 Impacts on services, entities and interfaces

AF:

- Performs Online-subscription management for Ambient IoTs with the PLMN.

UDM:

- Supports Online-subscription management of Ambient IoTs.

AMF/SMF:

- Supports pre-Configuration based on the request from the AF.

6.26 Solution #26: Unified API for Topology 1 and Topology 2 for an AF

6.26.1 Description

6.26.1.1 General

There are two connectivity topologies are studied:

- Topology 1: BS <--> Ambient IoT Device;
- Topology 2: BS <--> intermediate node <--> Ambient IoT Device: Only a UE can act as an intermediate node which is under the network control.

Different topologies will use different types of reader (BS Reader or UE Reader) to interact with an Ambient IoT Device. When an AF request is made, according to the topology, 5GC triggers the corresponding procedure to complete the AF request. In order to enable flexible deployment, a unified interface between AF and 5GC is proposed so that based on the unified API message sent from the AF, 5GC can determine the corresponding reader type(s) and the following procedure(s).

To align the topologies a Target Set is used, which can be used to identify which readers an AF request targets. In each Target Set, there is 1 or more readers of any type (i.e. the Target Set can contain only BS Readers or only contain UE Readers or a mixture of both reader types). The 3rd Party and operator co-ordinately determine which readers are in which Target Set, based on the deployment location of readers, and allocate the Target Set ID as part of the service agreements between the operator and the 3rd party. For example, A Target Set can be configured based on location (e.g. readers located by warehouse door, or a floor of a building, etc), while a different Target Set can be configured to be a specific reader (e.g. specific mobile UE Reader, etc). One reader can be included in multiple Target Sets. The Target Set information is pre-configured in the 5GC and the 5GC can use the Reader information in the Target Set to locate the Reader that is going to handle the AF AIoT service request.

The coordination between the operator and 3rd Party/AF to determine which readers are in a Target Set is not in scope and likely to be a deployment specific service offered by the operator.

Target Set: 1 or more readers which are grouped to perform the Ambient IoT service operation.

Target Set ID: An identifier of the Target Set.

The AF makes a request and includes a Target Set ID, which then allows 5GC to determine which Readers are used for the AF request. The AF does not request individual readers and the readers do not need to know which Reader Set(s) they are within.

The Target Sets can be constructed in coordination between the operator and the AF, and allows the AF to target specific cases. How the Target Sets are constructed, what they represent and how the Target Set IDs are shared between an operator and an AF are out of scope. Examples of uses of the Target Sets is provided in clause 6.26.1.2.

When an AF request is made, 5GC determines the reader(s) to use for the request based on the Target Set ID included in the request from the 3rd Party AF and therefore knows the reader type (i.e. BS Reader or UE Reader). If the reader is a BS Reader, procedure such as those in other solutions are performed. If the reader is a UE Reader, procedures such as those in other solution are performed. If the Target Set includes both BS Readers and UE Readers, the procedure corresponding to each reader type will be performed respectively.

NOTE: The procedure to support AIoT service between 5GC and AIoT Device via Reader is not in this solution scope.

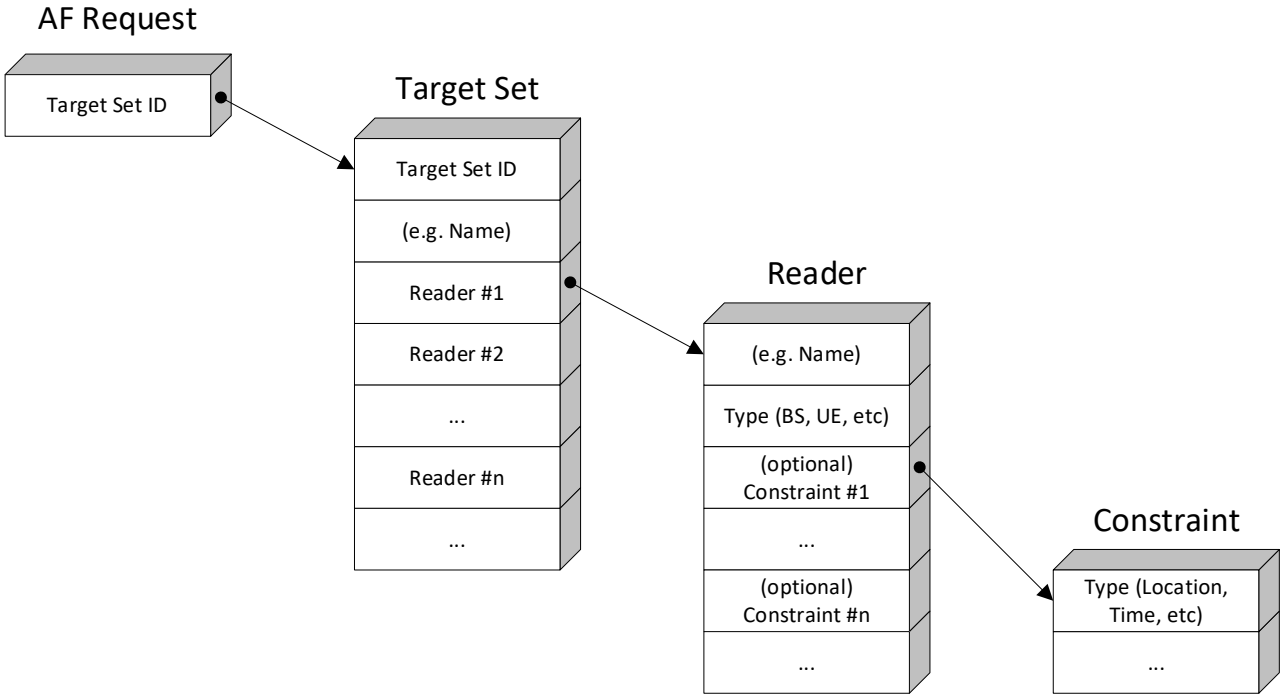


Figure 6.26.1.1-1 Target Set Overview

As can be seen from Figure 6.26.1.1-1, a Target Set contains a list of Readers, and for each Reader its type is identified and any additional constraints which determine whether the Reader can be used. The constraints have to be satisfied in order for Reader to be used. If a constrain it not met, the Reader is not considered for the operation.

The constraints could be aspects like location, time, etc. For example, if there is a mobile reader, but it should only be used if it is in a specific location (e.g. while in a specific location), or used at a specific time of day (e.g. during a shift), then these constraints can be checked by 5GC.

The exact details of constrains are expected to be operator policy or deployment specific.

6.26.1.2 Example Target Sets

The Target Sets can be constructed between the operator and the AF and allows the AF to target specific cases. For example, consider deployment such as shown in Figure 6.26.1.2-1, there can be many different areas an AF may wish to target with a, for example, inventory.

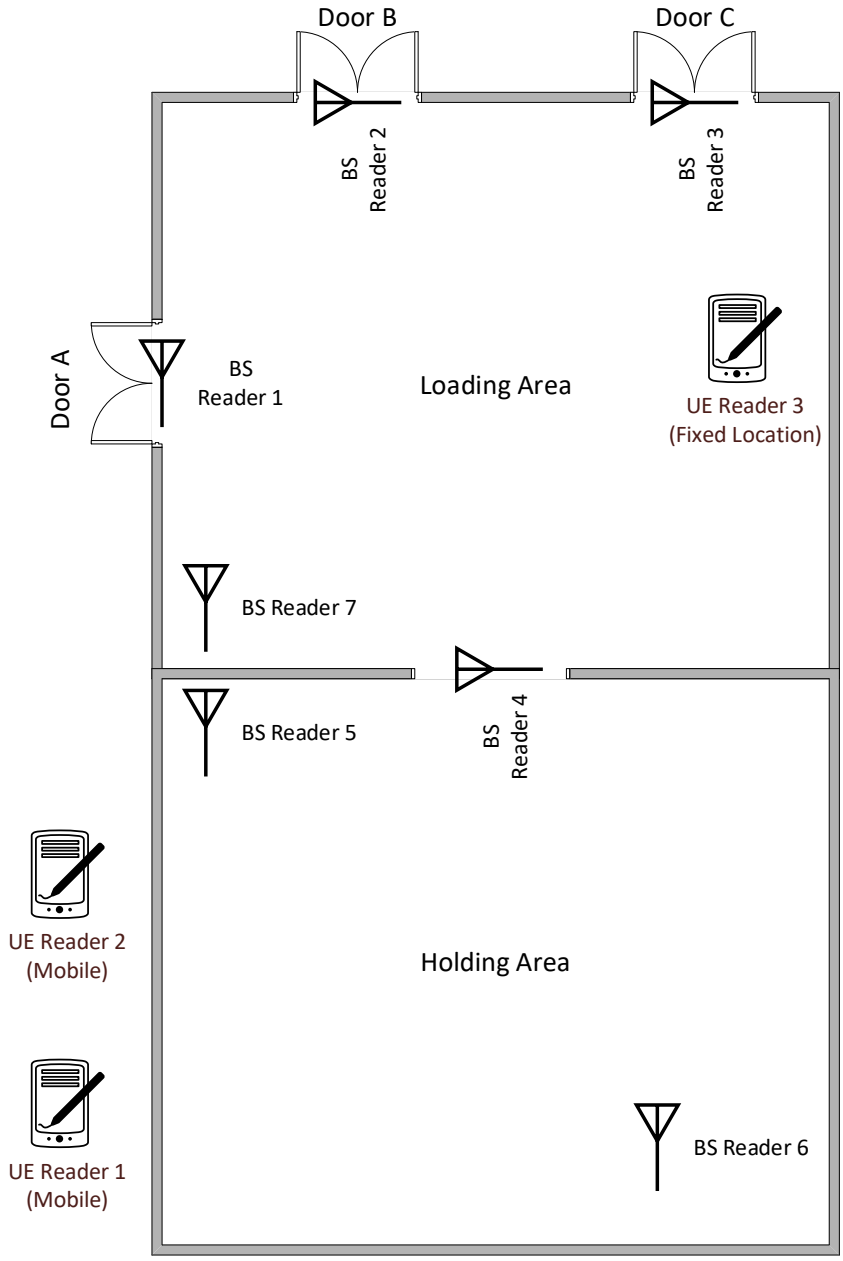


Figure 6.26.1.2-1: Example Target Sets Plan

For example, the AF/Operator could create the following Target Sets, based on known reader locations and constraints.

Table 6.26.1.2-1: Example Target Sets

Target Set ID: (Name)	Reader(s)	Constraints
Target Set ID 1 ("Holding Area")	BS Reader 5	-
	BS Reader 6	-
	UE Reader 1	Must be in Holding Area
	UE Reader 2	Must be in Holding Area
Target Set ID 2 ("Door A")	BS Reader 1	-
Target Set ID 3 ("Loading Area Exits")	BS Reader 1	-
	BS Reader 2	-
	BS Reader 3	-
	BS Reader 4	-
Target Set ID 4 ("Mobile Reader 1")	UE Reader 1	Between 9am and 9pm
Target Set ID 5 ("Mobile Reader 2")	UE Reader 2	Between 9am and 9pm
Target Set ID 6 ("Loading Area")	BS Reader 7	-
	UE Reader 3	-

In the above example Target Sets, if an Inventory indicated:

- Target Set ID 1 then BS Reader 5 & BS Reader 6 would always be used, and if UE Reader 1 and/or UE Reader 2 were in the Holding Area would additionally be used.
- Target Set ID 2, then just uses BS Reader 1. This could be used to determine when AIoT Device moves into/out of the Loading Area (and similar principle can be applied to other doors with other Sets).
- Target Set ID 3, then BS Reader 1, BS Reader 2, BS Reader 3 and BS Reader 4 would be used. This could be used, for example, to determine AIoT Devices entering/leaving an area.
- Target Set ID 4, then only UE Reader 1 would be used if it was between 9am and 5pm, otherwise the Target Set is empty and the operation will fail. This can be used when, for example, the operation is initiated by the user of the mobile UE Reader 1.
- etc.

NOTE: When readers are deployed the location can be know if they are fixed and therefore, for example, location constrains would not be required as shown for UE Reader 3 in Target Set ID 6 in Table 6.26.1.2-1.

If, for example, GPS location descriptions were used to select BS Readers, it can be seen that they would have to have extremely high accuracy, BS Reader 5 and BS Reader 7 could be situated a couple of centimetres apart in absolute terms, but because of building construction never cover the same area, which could lead to ambiguous or misleading results.

6.26.1.2 NEF IoT Services

6.26.1.2.0 Overview

This solution takes the NEF Inventory service operation captured in Solution #17 as baseline/example to illustrate how the Target Set parameter can be provided to 5GC and how 5GC could additionally provide additional information in the responses. The principle can be applied to any solution. In the example service operations, the Target Set ID is provided to enable 5GC to determine which readers to use to fulfil the requested operation.

The NEF supports a new Nnef_AIoT service with the operations described in table 6.17.2.3-1.

6.26.1.2.1 Inventory service operation

6.26.1.2.1.1 Nnef_AIoT_Inventory service operation

Service operation name: Nnef_AIoT_Inventory

Description: The inventory operations for the Nnef_AIoT service allows the consumer to request to perform an inventory device operation for an AIoT device or a group of AIoT devices via a set of reader(s).

Inputs, Required: None.

Inputs, Optional: Target Set ID

Outputs, Required: Status, Correlation Information.

6.26.1.2.1.2 Nnef_AIoT_Inventory_Notification service operation

Service operation name: Nnef_AIoT_InventoryNotification

Description: Inventory results from a requested inventory operation.

Inputs, Required: Correlation Information.

Outputs, Required: Correlation Information, Status, List AIoT Devices ID(s).

6.26.2 Procedures

The example procedure below shows how 5GC determines the readers requested in the AFs request and depending on type interacts with a BS Reader or UE Reader or both types to fulfil the request.

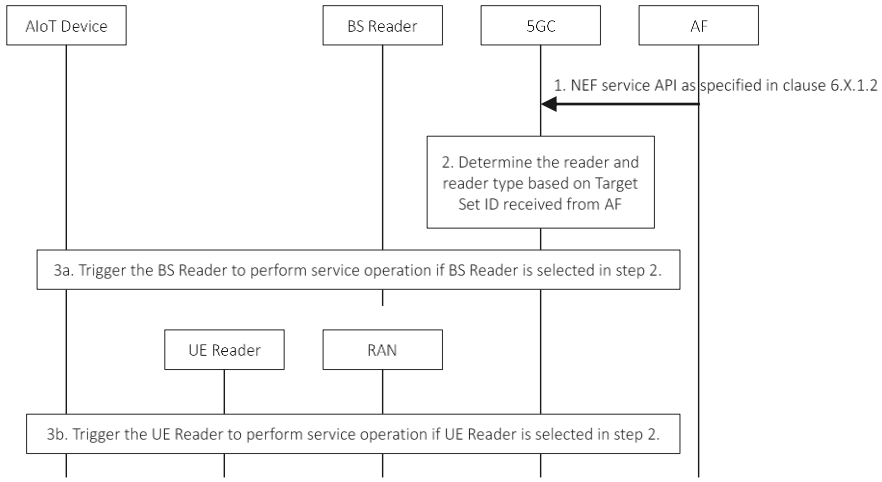


Figure 6.26.2-1: Procedure for unified API between 5GC and AF to support AIoT service

1. AF sends a service request to NEF to trigger an Ambient IoT service operation. A Target Set ID is included in the request as specified.
2. NEF sends a service request to the AIoT function (e.g. AIoTMF as defined in clause 6.17) including the Target Set ID. Based on the Target Set ID, the 5GC determines the reader(s) and the reader type(s) to perform the service operation, including checking whether the reader has any constraints that must be satisfied.
3. Based on the reader type for each reader determined in step 2, following procedure(s) are triggered:
   a) If the reader is BS Reader, a procedure to interact with BS Readers as described in other solutions is used to fulfil the request.
   b) If the reader is UE Reader, a procedure to interact with UE Readers as described in other solutions is used to fulfil the request.

6.26.3 Impacts on services, entities and interfaces

5GC:

- The service operations exposed need to include a Target Set ID

- 5GC needs to determine which readers are used based on the Target Set

RAN/BS Readers:

- None.

UE Reader:

- None.

AIoT Device:

- None.

6.27 Solution #27: AIoT service operation with privacy and authentication protection.

6.27.1 Description

This solution address KI#3 and proposes procedures that provides privacy and authentication protected Ambient IoT service (Command and Inventory) for both the operator and the AIoT device. The AIoT device must know that the Reader (UE or Base station) is an authorized Reader to receive e.g. sensor information from the AIoT device and/or send commands to the AIoT device. Equally important is that the information received by the Reader originates from an authentic AIoT device i.e. the network can detect when a fake AIoT device responds to an Inventory request. Solution 16 (AIoT Registration/Onboarding) and solution 1 (AIoT Temporary Identifier Control) provide important components of such system, but they do not provide the full system solution for providing privacy and authentication protected Ambient IoT services.

Editor's note: Whether the privacy protection described in this solution fulfils SA WG3 requirement on privacy check is up to SA WG3 to determine.

The solution is based on the following assumptions:

- The AIoT device must be registered/onboarded in the network as described in solution 16, and both the AIoT device and Network holds necessary parameters/information used when providing a secure Ambient IoT service.

- The AIoT Reader (UE or Base station) and AIoT device only send the AIoT ID (Temporary ID) once over the radio interface as described in solution 1. This provides e.g. the AIoT device privacy protection and prevent unauthorized AIoT device tracking.

- In case of Inventory service, the CN NF derives the permanent AIoT device ID of the registered AIoT device from the Temporary ID received by the AIoT device i.e. the CN NF can authenticate the Temporary ID belongs to the registered AIoT device. The CN NF knows based on the Inventory request from the AF the expected Temporary IDs to receive as only registered AIoT devices will respond to the Inventory request and the inventory request includes Owner ID or Group ID. In case CN NF changes, it is expected that the AIoT device context can be transferred from the old CN NF to the new CN NF.

Editor's note: There is a risk that the AIoT device and CN NF Temporary ID generation get out-of-synch as documented in solution 1. As the Temporary ID generation algorithm is expected to be deterministic, it would be possible for the CN NF to check further Temporary ID(s), to re-synchronize with the AIoT device. However, as mention in an editor's note in solution 1, details of such re-synchronization should be specified by SA WG3.

- In case of Command service, the AIoT device authenticates the Reader when matching the Temporary ID received by the Reader i.e. the AIoT device only responds when it receives an expected Temporary ID from the reader (as described in solution 1). The CN NF performs the validation of the Command response by checking that the response includes the expected Temporary ID of the AIoT device when using the Temporary ID generation described in solution 1.

- The AF may provide assistance information to the network when invoking an AIoT service. The assistance information can be e.g. the location/area to perform Inventory, expected number of AIoT devices in the area, what type of AIoT device that may respond (Backscatter type or transmission capable type), response type (Device ID, Command Acknowledgement, UL data).

- The CN NF may provide assistance information to the Reader together with the Inventory/Command request.

6.27.2 Procedures

6.27.2.1 Inventory procedure

The name CN NF is used to name the network function that performs the necessary operation for an AIoT device when the network provides the AIoT service operation. The CN NF will be defined together with the system architecture design to support the Ambient IoT in 5GC. The Reader can be a base station Reader or UE Reader.

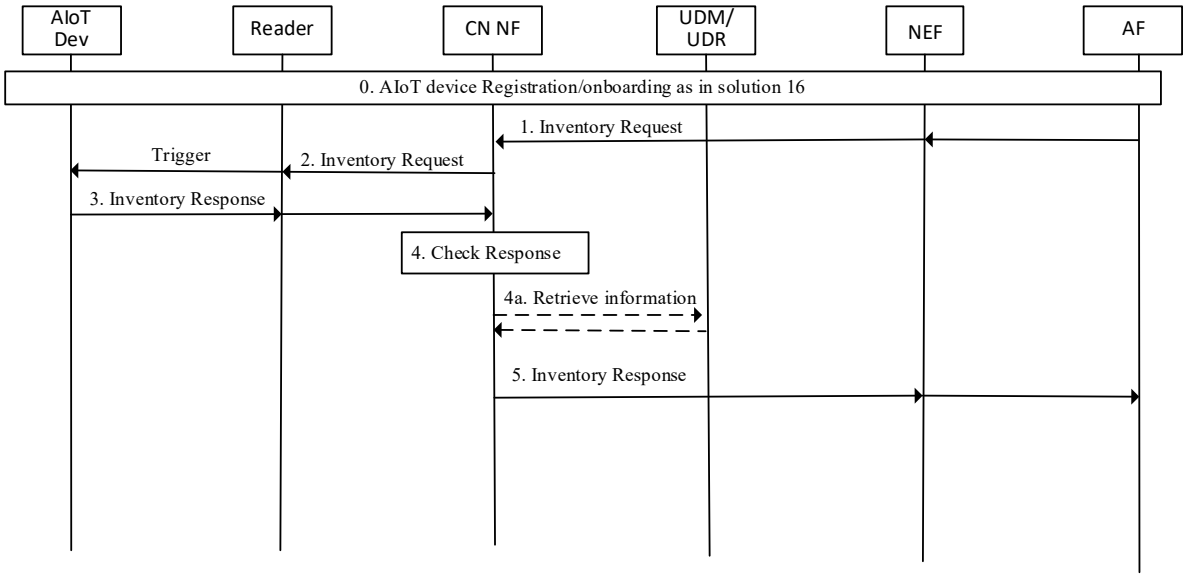


Figure 6.27.2.1-1: Inventory Procedure

0. It is assumed that the AIoT device has performed Registration/Onboarding procedure as described in e.g. solution 16. The AIoT device has the information needed to respond to an Inventory trigger broadcasted by a Reader and the response (Temporary ID) can be authenticated by the CN NF and not tracked by unauthorized Readers/sniffers. The CN NF holds the last known location of the device i.e. the location where the device performed the registration/onboarding or the location where the last interaction with the AIoT device took place.

1. The AF uses the Northbound API to invoke the Inventory service. Legacy procedures is used to check that the AF is authorized to use the service API. The Inventory Request may include e.g. Owner ID, Group ID, Device Type, location/area, expected number of devices, periodicity. The NEF forwards the Inventory request to the CN NF.

2. Based on e.g. location/area information the CN NF selects the Reader(s) and requests the Reader(s) to perform Inventory. This request may include assistance information e.g. expected number of responses and type of devices. The Reader(s) broadcasts an Inventory trigger that includes e.g. Owner ID or Group ID. The format of the Owner ID and Group ID are discussed in other solutions e.g. solution 3.

3. If an AIoT device matches to the Inventory trigger the AIoT device responds with its temporary ID. After responding to the Inventory trigger, the AIoT device generates a new temporary ID locally (as described in solution 1). The Reader may check the number of AIoT devices who has responded. If the number is too low (lower than the expected number of responses or lower than a threshold based on the expected number of responses), the Reader may repeat the Inventory trigger broadcast again. The Reader forwards the response(s) to the CN NF.

4. For each response, the CN NF checks that the temporary ID is valid and belongs to the Owner or Group and retrieves the permanent ID of the AIoT device. The CN NF generates the next expected temporary ID (as described in solution 1) for all AIoT devices that responded and stores these IDs. If the number of responses is too low (lower than the expected number of responses or lower than a threshold based on the expected number of

responses), the CN NF may send an Inventory Request to the same Reader again or to a different Reader (repeat step 2-4).

NOTE: As the AIoT device cannot distinguish whether the repeated Inventory trigger is a new or repeated one, the CN NF may receive duplications. The CN NF removes those before step 5.

If needed to perform this step the CN NF may retrieve information from the UDM/UDR.

5. The CN NF sends the Inventory response to the AF with a list of all devices that responded to the Inventory Request.

6.27.2.2 Command procedure

The name CN NF is used to name the network function that performs the necessary operation for an AIoT device when the network provides the AIoT service operation. The CN NF will be defined together with the system architecture design to support the Ambient IoT in 5GC. The Reader can be a base station Reader or UE Reader.

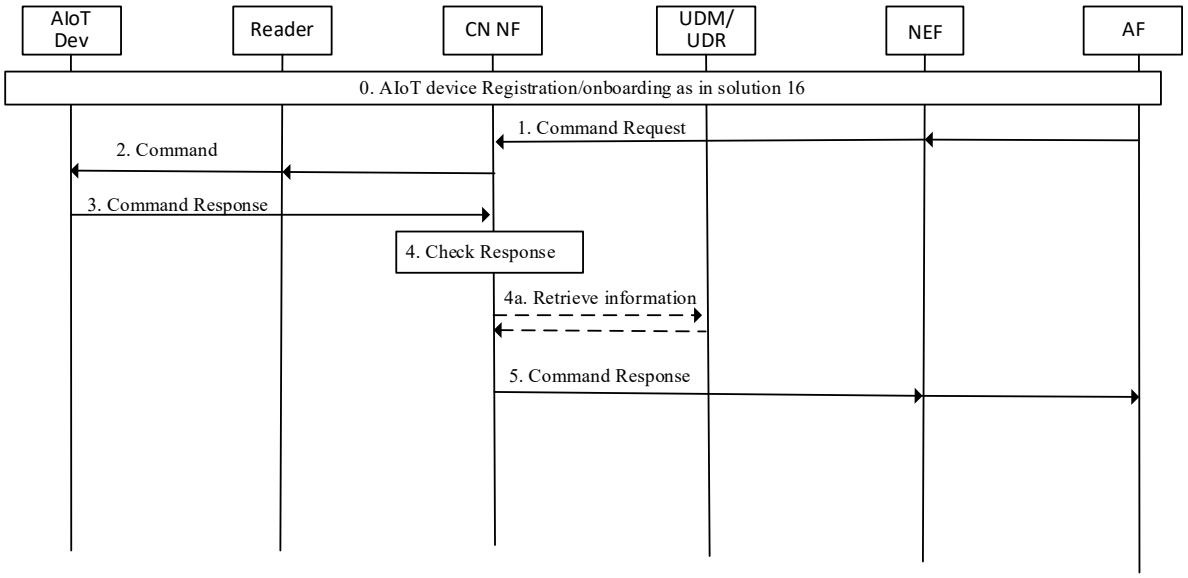


Figure 6.27.2.2-1: Command Procedure

0. It is assumed that the AIoT device has performed Onboarding/Registration procedure as described in e.g. solution 16. The AIoT device has the information to only respond to a Command request from an authorized Reader and the response can be authenticated in network and not tracked by unauthorized Readers/sniffers. The CN NF holds the last known location of the device, the location where the device performed the onboarding/registration or the location where the last interaction with the AIoT device took place.

1. The AF uses the Northbound API to invoke the Command service. Legacy procedures is used to check that the AF is authorized to use the service API. The Command Request may include e.g. Device ID, Device Type, Type of command (read/write), expected size of the response, location/area. The NEF forwards the Command request to the CN NF.

2. The CN NF retrieves (or generate as described in solution 1) the temporary ID related to the device ID received from the AF. Based on e.g. location/area information the CN NF selects the Reader(s) and requests the Reader(s) to send the Command. The request to the Reader includes the temporary ID, Device Type, command type and if possible expected response size in e.g. Bytes. The Reader(s) sends the Command.

3. If an AIoT device matches the Temporary ID in the Command, the AIoT device either responds with an Acknowledgement if the command was a write command or include the requested data if the command was a Read command. In addition, the response includes a new temporary ID as the temporary ID can only be sent once over the radio interface, before sending the response the device generates this new temp ID locally (as described in solution 1), this prevents that an unauthorized reader/system to correlate the Command request with the AIoT device response. The Reader forwards the response to the CN NF.

NOTE: The examples of read/write command above is only provided to show that the AIoT device response can be different depending on the command.

4. The CN NF authenticate the response from the AIoT by checking that the temporary ID belongs to the AIoT device. The CN NF performs the validation of the response by checking that the response includes the expected Temporary ID of the AIoT device when using the Temporary ID generation described in solution 1. The CN NF locally generates the next temporary ID as described in solution 1.

   If needed to perform this step the CN NF may retrieve information from the UDM/UDR.

   If there is no Command response from the AIoT device, the CN NF may select further Reader(s) and send the Command request to the further Reader(s) and repeat step 2-4.

5. The CN NF sends the Command response to the AF.

6.27.3 Impacts on services, entities and interfaces

Impacts on existing entities:

CN NF:

- Receive Inventory/Command request from AF.
- Select and trigger the Reader(s) to broadcast Inventory trigger or send Command message based on the information received by the AF.
- Receive response(s) from the AIoT device(s) and authenticate that the response(s) originate from registered AIoT device(s).
- Optionally repeat the Inventory request to collect more responses.

AIoT Reader (UE or Base station):

- Broadcast the Inventory trigger or send Command message, based on the information in the Inventory Request or Command Request sent by the CN NF.
- Receive the response(s) back from AIoT device(s) and forward the response(s) to the CN NF.
- Optionally repeat the Inventory Trigger to collect more responses.

AIoT device:

- Receive Inventory trigger or Command message.
- Transmit a Response message to the Reader.

6.28 Solution #28: Device-Reader Binding Information

6.28.1 Description

The network stores the binding information between devices and their serving readers. For Topology 1 in which RAN acts as a reader, the Device-Reader binding information includes Device ID and the identity of RAN serving the device; for Topology 2 in which an intermediate node UE acts as a reader, the Device-Reader binding information includes Device ID and the UE ID which serves the device. The binding information can be obtained during the initial inventory process and updated when the reader serving the device changes.

For the following Ambient IoT services such as Command, Read, Write and Disable after the initial inventory process, the AIoTMF can use the Device-Reader binding information to identify the readers serving the devices and accelerate the locating of the devices.

In this solution, a new network function AIoTMF or enhanced AMF (i.e. AIoTMF/AMF) is used to operate Ambient IoT services and to handle Device-Reader binding information.

NOTE: The solution only applies to scenario in which devices do not move frequently and their serving readers have fixed position, such as warehouse and storage.

Editor's note: How the solution applies to a group of devices handling is FFS.

For the handling of large numbers of AIoT devices, there may be multiple AIoTMF/MFs deployed in the network. These AIoTMF/AMFs can share the Device-Reader binding information when one of them fetches the information. When receiving service requests for AIoT devices, AIoTMF/AMF uses the binding information to quickly locate the Readers serving target devices.

6.28.2 Procedures

6.28.2.1 Device-Reader Binding Information Storage

Figure 6.28.2.1-1 depicts a flow diagram for Device-Reader binding information acquirement and storage.

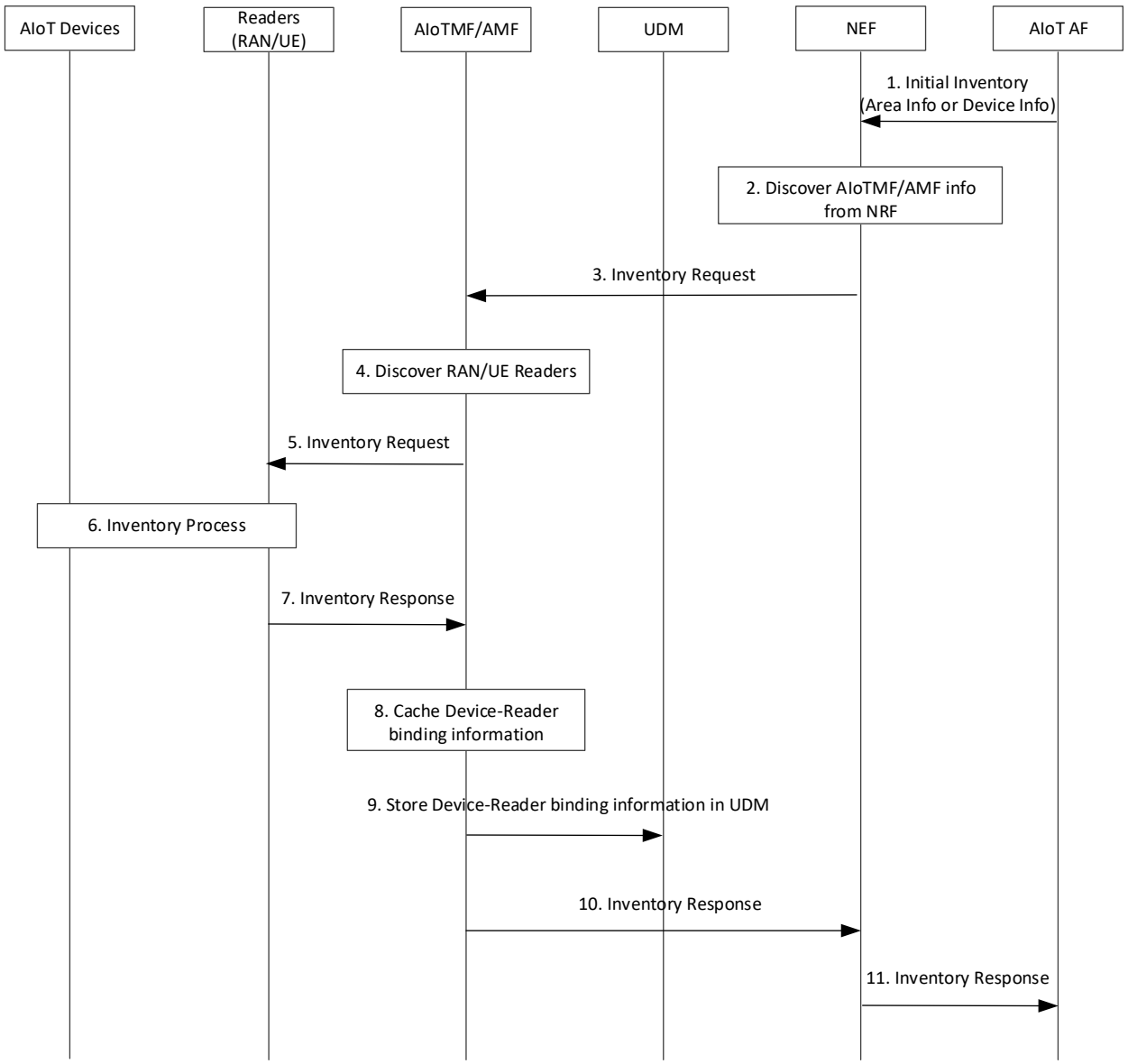


Figure 6.28.2.1-1: Device-Reader binding information storage

The steps in the Device-Reader binding information storage procedure are as follows:

1. AIoT AF initiates inventory service to NEF. The inventory request includes the area information or device information.
2. NEF discovers the AIoTMF/AMF from NRF based on the area information received in the request.

NOTE 1: It is assumed AIoTMF/AMF has already registers its information and coverage area into NRF.

3. NEF sends inventory request to AIoTMF/AMF. If the request from AIoT AF includes device information, NEF may translate it to Device ID.

NOTE 2: The request could be sent to multiple AIoTMF/AMF which are discovered in step.2 based on the area information.

4. AIoTMF/AMF discovers destination readers based on area information or device information received in inventory request. The readers could be RANs for Topology 1 and UEs for Topology.

5. AIoTMF/AMF sends inventory request to readers. The readers could be either RANs or UEs.

6. Readers proceed inventory with the devices within its coverage area.

7. Readers send back the inventory result when finishing the inventory process with multiple devices.

8. AIoTMF/AMF gets the binding information based on which Reader sends back the inventory result of devices and caches the Device-Reader binding information with a configurable timer indicating the valid period according to the inventory result from readers. This cache information could be used in the future Ambient IoT services then there is no need to interact with UDM.

   When there are more than one AIoTMF/AMF deployed in the network, the AIoTMF/AMF will share the Device-Reader binding information with other AIoTMF/AMFs through the interfaces between them. AIoTMF/AMF will update the stored binding information and remove the duplicated information if it exists. It will accelerate locating of the reader serving devices when other AIoTMF/AMFs receiving service requests for AIoT devices.

9. AIoTMF/AMF stores the Device-Reader binding information with the timer into UDM.

10~11. AIoTMF/AMF sends back the inventory result to AIoT AF through NEF.

6.28.2.2 Device-Reader Binding Information Usage

Figure 6.28.2.2-1 depicts a flow diagram regarding the usage of Device-Reader binding information.

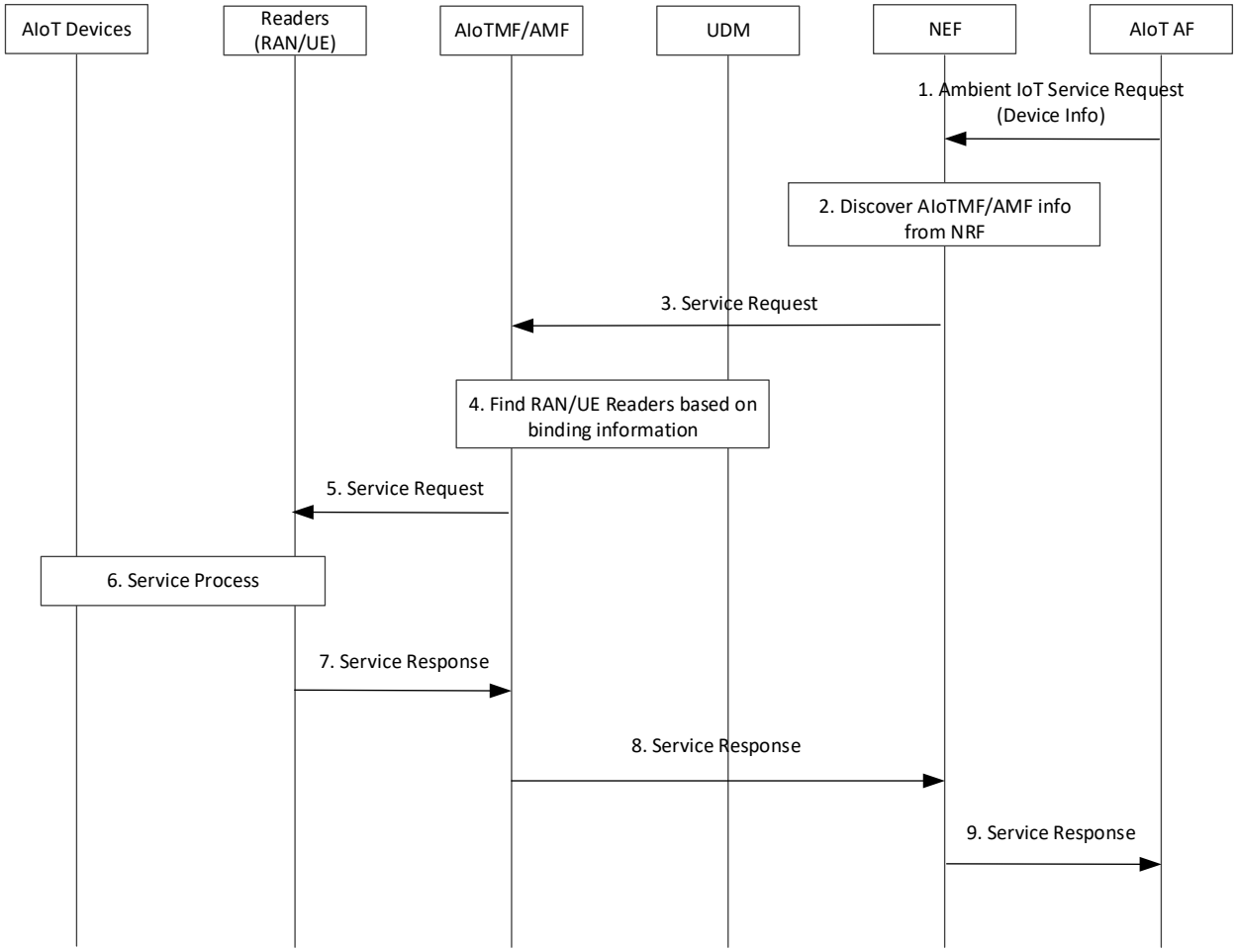


Figure 6.28.2.2-1: Device-Reader binding information usage

The steps in the Device-Reader binding information usage procedure are as follows:

1. AIoT AF sends Ambient IoT service request to NEF; the service could be Command, Read, Write and Disable operations. The service request includes the device information and other service related data.
2. NEF discovers the AIoTMF/AMF from NRF.
3. NEF sends the service request to AIoTMF/AMF.
4. AIoTMF/AMF identifies destination reader based on device information and cached Device-Reader binding information if it exists in AIoTMF/AMF; or interacts with UDM to fetch the binding information.
5. AIoTMF/AMF sends the service request to the identified reader. The reader could be either RAN or UE serving the specific device oriented in the service request.

NOTE 1: If in step. 4 the binding information is not obtained or in step. 5 the service request is not successfully sent to the reader, a new inventory is operated to identify the reader serving the device and this new binding information needs to be cached in AIoTMF/AMF or stored into UDM.

6. The reader proceeds the requested service with the device.
7. The reader sends back the service operation result.

8~9. AIoTMF/AMF sends back the service operation result to AIoT AF through NEF.

NOTE 2: The procedures in this solution only illustrates how to store and use the binding information between devices and serving readers; the mechanism (storing binding information) can be used by Inventory and Command procedures in other solutions for the serving readers locating.

6.28.3 Impacts on Existing Nodes and Functionality

AIoTMF/AMF:

- Fetch and cache Device-Reader binding information.
- Store Device-Reader binding information into UDM.
- Obtain Device-Reader binding information from UDM.
- Share Device-Reader binding information with other AIoTMF/AMFs.

UDM:

- Store and provide Device-Reader binding information.

NEF:

- Receive AIoT service requests from AF and send back service operation result to AF.
- Discovery AIoTMF/AMFs and send AIoT service requests to them.

Reader (RAN/UE):

- Receive AIoT service requests from AIoTMF/AMF and send back service operation result to AIoTMF/AMF.
- Interact with AIoT devices to execute service operation.
- Report Device-Reader binding information to AIoTMF/AMF implicitly.

NRF:

- Store AIoTMF/AMF information.

6.29 Solution #29: Periodic triggered Ambient IoT Devices for registration update in Topology-1

6.29.1 Description

This solution is targeting KI#2 Identification, Subscription, Registration and Connection management. It is related with KI#1 as well.

It is assumed each Ambient IoT device has and is identified with a unique internal ID (e.g. EPC ID), and is capable of DO-DTT transmission only. It is assumed each the BS serves a registration area. The BS has a timer and controls the timing of starting the registration procedure for each AIoT device. AMF keeps the registration area information of the AIoT devices, determines the movement of the AIoT device as well, and controls the device registration information stored in BS. To deal with many AIoT devices, BS has also a mechanism to spread transmissions from AIoT devices.

Editor's note: It is FFS what is registration area in terms AIoT devices.

6.29.2 Procedures

6.29.2.1 Periodic triggered AIoT devices to perform Registration Update

The procedure of Periodic Registration Update that serves also as Mobility Registration Update is as depicted in Figure 6.29.2.1-1.

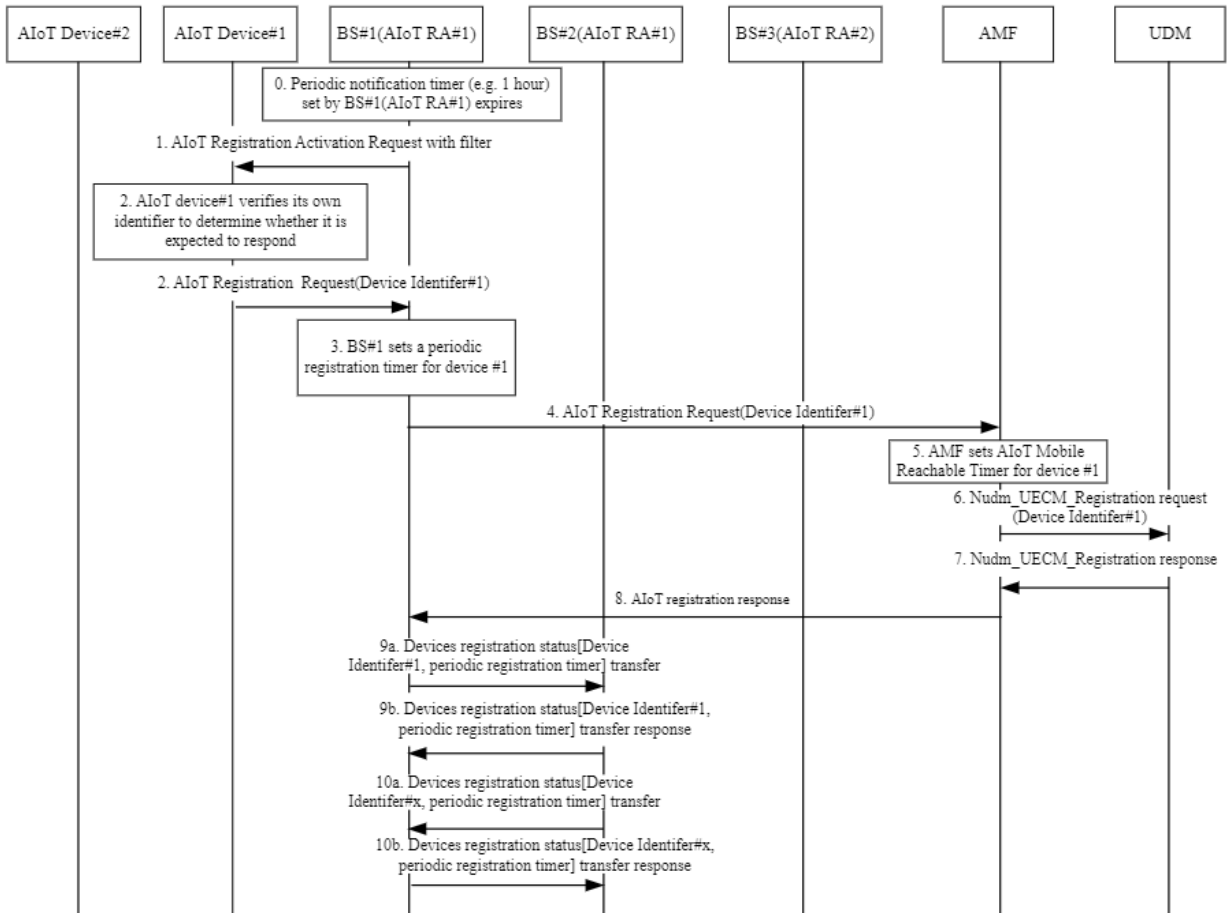


Figure 6.29.2.1-1: Periodic triggered AIoT devices to perform Registration Update

0. Periodic notification timer (e.g. 1 hour) set by BS#1 (AIoT RA#1) expires.

1. BS#1 (AIoT RA#1) broadcasts the AIoT Registration Activation Request. It may include the following to indicate AIoT devices not expected to respond to this request:

   - A list of EPC IDs of AIoT devices which have already registered in BS#1 (RA#1).

   - Optionally, bitmask.

Editor's note: It is FFS whether it is feasible to include the complete list of AIoT Device IDs in the Activation Request Message.

2. AIoT device#1 verifies its own identifier against the list of EPC IDs and the bitmask to determine whether it is expected to respond. If no, then the AIoT device#1 does not respond. If yes, then the AIoT device#1 responds to the request in step1 including its device identifier. (e.g. device identifier #1).

3. BS#1 (AIoT RA#1) sets a periodic registration timer (e.g. 6 hours) for the AIoT device #1.

4. BS#1 (AIoT RA#1) sends the AIoT Registration Request containing device identifier #1 to AMF.

5. AMF notices the AIoT device#1 is situated in RA#1 based on that AMF has received the AIoT Registration Request from the BS#1. AMF stores this RA#1 in the context of the AIoT device #1 and configures AIoT Mobile Reachable Timer for the AIoT device #1 (e.g. 6 hours and 30 minutes) which is longer than the periodic registration timer configured in steps 3.6. AMF sends Nudm_UECM_Registration request containing device identifier #1 to UDM.

7. UDM sends Nudm_UECM_Registration response back to AMF.

8. AMF sends the AIoT Registration Response to BS#1 (AIoT RA#1).

9. The device registration status is transferred by BS#1 (AIoT RA#1) to other BS(s) which make up the AIoT RA#1. BS#2 responds to the massage. It may include the following:

- Device Identifier;
- Periodic registration timer.

10. BS#1 will also receive the device registration status from other BS(s) in the same Registration Area (AIoT RA#1). For example, when another AIoT device#x becomes registered in BS#2 (AIoT RA#1), BS#2 (AIoT RA#1) also transfers the registration status of the AIoT device#x to BS#1 (AIoT RA#1) and BS#1 responds to the transferred message.

Editor's note: It is FFS whether the periodic registration update triggered by BS is needed for Inventory.

Editor's note: It is FFS what is the device registration information stored in BS.

6.29.2.2 Periodic triggered AIoT devices to perform Registration Update (Continued)

The procedure of Periodic Registration update that serves also as Mobility Registration Update continues as depicted in Figure 6.29.2.2-1.

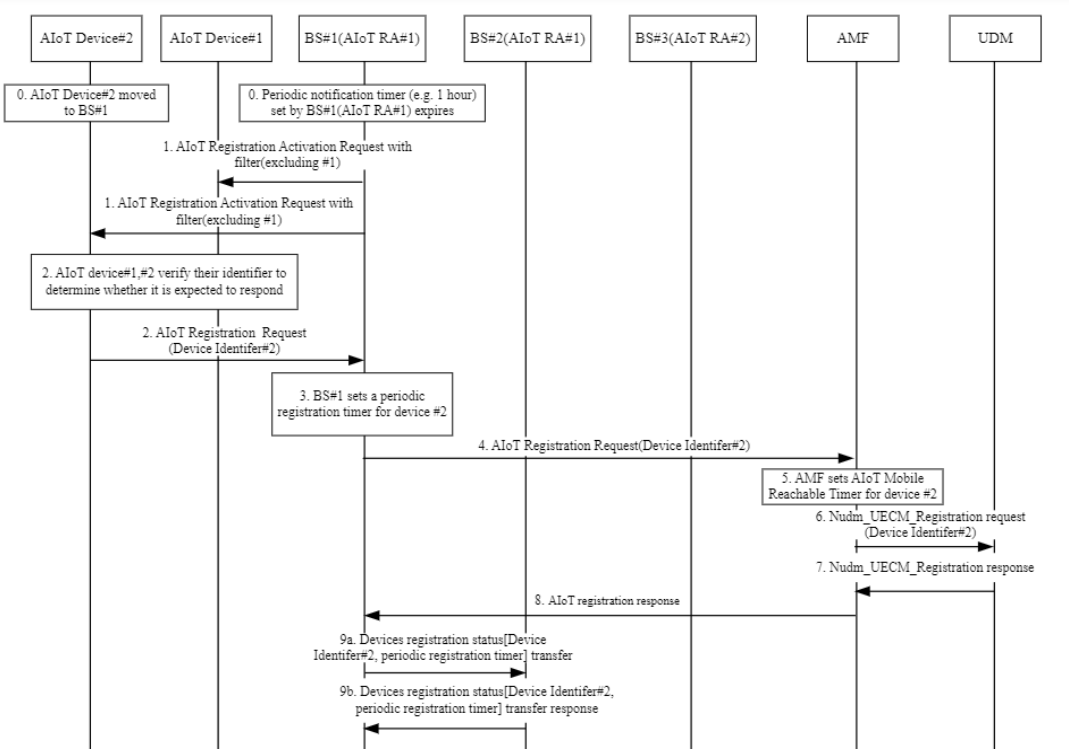


Figure 6.29.2.2-1: Periodic triggered AIoT devices to perform Registration Update (Continued)

0. It is assumed that AIoT device#2 has moved to the coverage of BS#1 (AIoT RA#1). Another periodic notification timer (e.g. 1hour) expiries.

1. BS#1 (AIoT RA#1) sends the AIoT Registration Activation Request again, but this time the message contains EPC ID of the AIoT device#1 for which the periodic registration timer is not expired yet.

2. AIoT device#1 verifies its own identifier against the list of EPC IDs and the bitmask and determines not to respond.

   AIoT device#2 verifies its own identifier against the list of EPC IDs and the bitmask and determines to respond. AIoT device#2 responds to the request including its device identifier. (e.g. device identifier #2).

Afterwards, step 3 and onwards in clause 6.29.2.1 apply.

6.29.2.3 BS relocation according the update of the latest registration area of AIoT device

The procedure of Periodic Registration update that serves also as Mobility Registration Update when Ambient IoT device moves is as depicted in Figure 6.29.2.3-1.

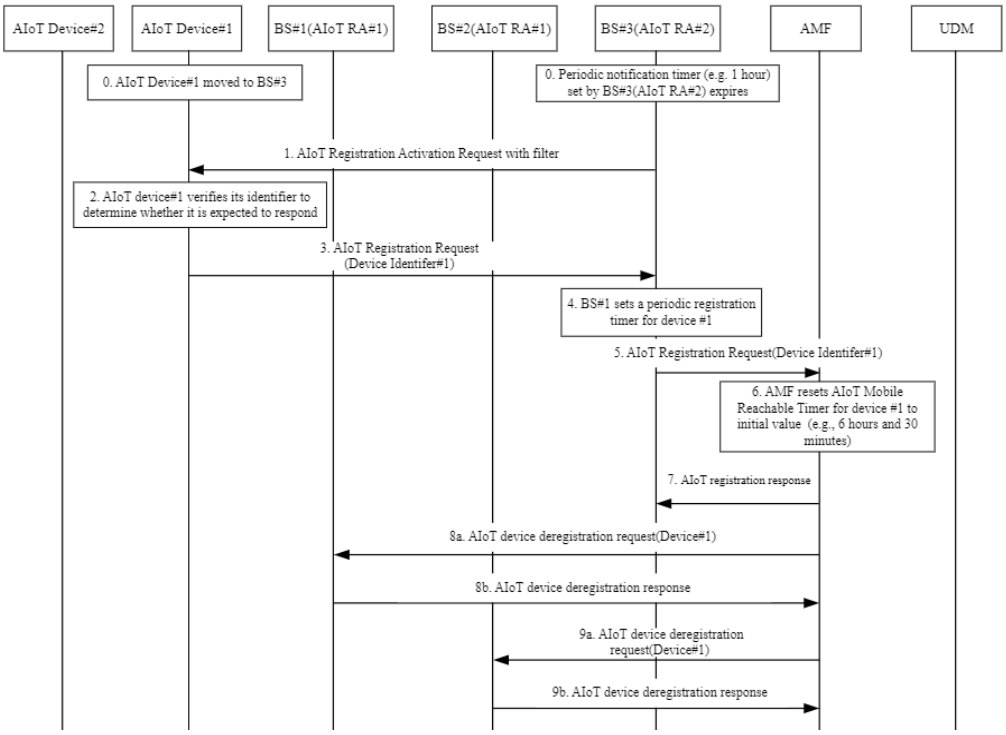


Figure 6.29.2.3-1: BS relocation according the update of the latest registration area of AIoT device

0. It is assumed that after 3 hours, AIoT device#1 moves to BS#3 (AIoT RA#2). Periodic notification timer (e.g. 1 hour) set by BS#3 (AIoT RA#2) expires.
1. BS#3 (AIoT RA#2) sends the AIoT Registration Activation Request.
2. AIoT device#1 verifies its own identifier against the list of EPC IDs and the bitmask and determines to respond.
3. AIoT device#1 responds to the request including its device identifier. (e.g. device identifier #1).
4. BS#3 (AIoT RA#2) sets a periodic registration timer (e.g. 6 hours) for AIoT device #1.
5. BS#3 (AIoT RA#2) sends the AIoT Registration Request containing device identifier #1 to AMF.
6. AMF notices the AIoT device#1 is situated in RA#2 based on that AMF has received the AIoT Registration Request from the BS#3. AMF recognizes that the AIoT device #1, which was registered to AIoT RA#1, is now registered in AIoT RA#2. AMF stores this RA#2 in the context of the AIoT device #1 and resets the AIoT Mobile Reachable Timer for the AIoT device #1 to the initial value (e.g. 6 hours and 30 minutes).

   Since it is the same AMF, notification to UDM can be omitted.
7. AMF sends the AIoT Registration Response to BS#3 (AIoT RA#2).
8. AMF sends the AIoT device Deregistration Request to BS#1 (AIoT RA#1) for AIoT Device#1. BS#1 (AIoT RA#1) sends the AIoT device Deregistration Response to AMF.
9. AMF sends the AIoT device Deregistration Request to BS#2 (AIoT RA#1) for AIoT Device#1. BS#2 (AIoT RA#1) sends the AIoT device Deregistration Response to AMF.

6.29.2.4 Deregistration

The procedure of Deregistration is as depicted in Figure 6.29.2.4-1.

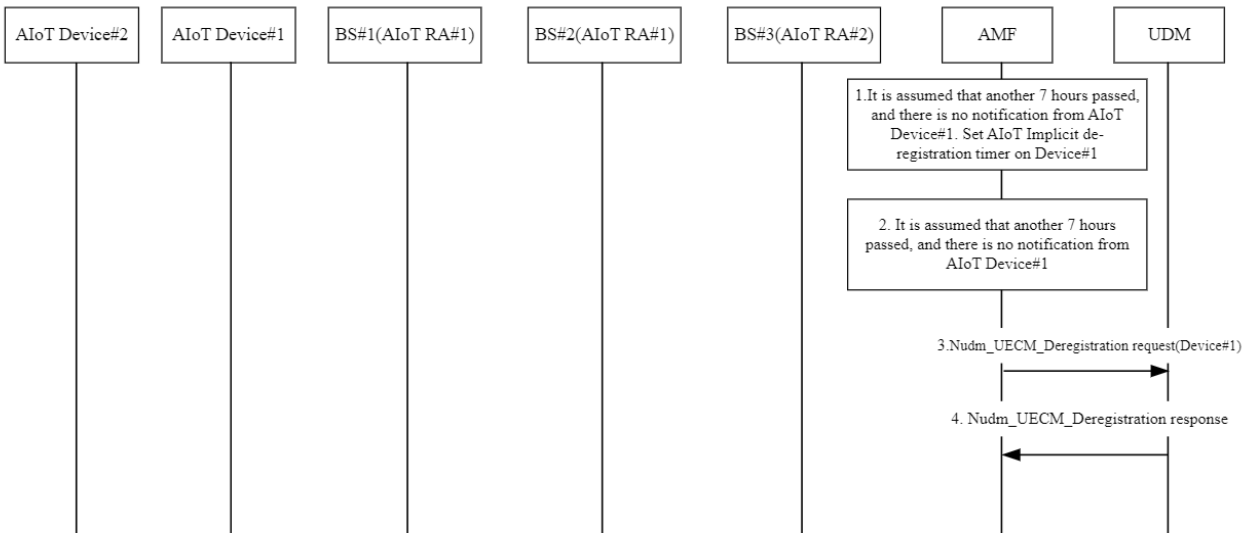


Figure 6.29.2.4-1: Deregistration

1. After that, it is assumed that another 7 hours have passed, and there is no notification from AIoT device#1. Due to the expiration of AIoT Mobile Reachable Timer for AIoT device #1, AMF sets AIoT Implicit De-registration timer (e.g. 6 hours and 30 minutes) for AIoT device #1.
2. After that, it is assumed that another 7 hours have passed, and there is no notification from AIoT device#1. Due to the expiration of AIoT Implicit De-registration timer for AIoT device #1, AMF starts the Deregistration procedure for AIoT device#1.
3. AMF sends Nudm_UECM_Deregistration request to UDM including device identifer#1.
4. UDM sends Nudm_UECM_Deregistration response to AMF.

6.29.3 Impacts on services, entities and interfaces

UDM:

- Supports the registration management for AIoT devices based on the AIoT device identifier.

AMF:

- Storing RA of AIoT device and supports controlling contexts for the registration management stored in BS (i.e. removing them when needed).
- Supports setting AIoT Mobile Reachable Timer and AIoT Implicit de-registration timer per AIoT device.

BS:

- Supports broadcasting the AIoT Registration Activation Request periodically.
- Supports managing periodic registration timer per AIoT device and constructing the content of the AIoT Registration Activation Request accordingly.
- Supports sharing the device registration status among other BSs that share the same registration area.

6.30 Solution #30: Support AIoT Devices using AIOTF for Topology 2

6.30.1 Description

6.30.1.0 Introduction

This solution proposes a complementary solution to Solution #8. It enables the support of Ambient IoT devices for topology 2, which addresses KI#1, KI#2 and KI#3. Together with Solution #8 which focus on Topology 1, the solutions can support AIoT devices for both Topologies.

In this solution, it is AIOTF who is in charge of the intermediate UE for the Ambient IoT operations, including determining intermediate UEs, sending operation commands to and receiving results from intermediate UEs. As the licensed spectrum is owned by MNO, it is proposed to let network provide the radio resource information towards the intermediate UEs about the spectrum information for the over-the-air interface between Intermediate UEs and AIoT devices.

6.30.1.1 Reference Architecture

The Figure 6.30.1.1-1 illustrates the architecture for supporting AIoT Devices using AIOTF for Topology 2:

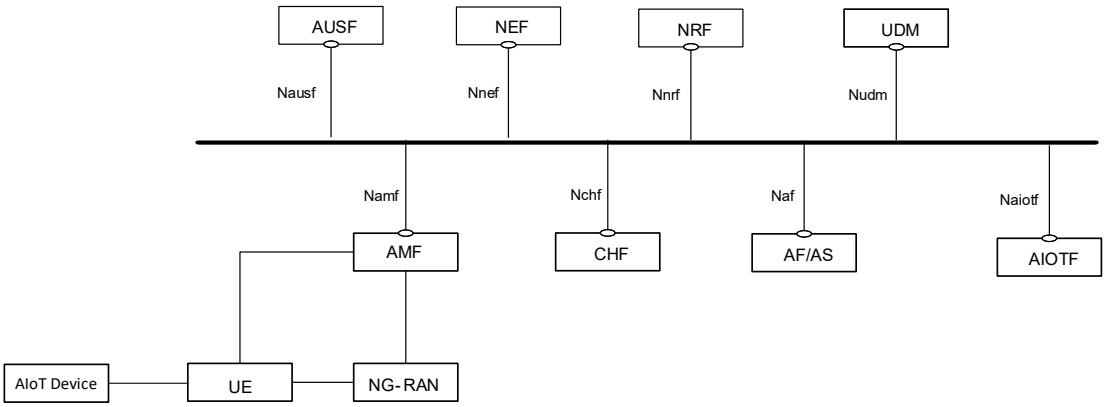


Figure 6.30.1.1-1: System Architecture of Supporting AIoT Devices using AIOTF for Topology 2

This solution focuses on Topology 2, which works together with Solution #8 addressing Topology 1.

The functional entities defined in Solution #8 and TS 23.501 [4] are reused with the exception for the following additions:

- UDM/UDR: The authorization information of Intermediate UE for AIoT is stored in UE subscription data.
- AMF: Receive AIoT capability information from UE and authorize based on the subscription data in UDM/UDR.
- NG-RAN: Provide spectrum information towards authorized intermediate UE.
- Intermediate UE:
  - Provide Ambient IoT capability information to AIOTF and receive the authorization information.
  - Receive the instruction from AIOTF and perform Ambient IoT operations (e.g. inventory, command, etc.) on the proper spectrum. The radio resource information is received from NG-RAN.
- Ambient IoT Function (AIOTF): AIOTF is introduced to support AIoT services, with some AMF's functionalities integrated, which includes:
  - Inventory handling and device context management.

- Command delivery.
- Authentication and authorization for the access, which triggers interaction with AUSF/UDM.
- Collect charging data and interact with CHF for charging.
- Routing the request from AF (via NEF) to Intermediate UEs, for DO-DTT/DT traffic types.
- Routing the response from Intermediate UEs to AF (via NEF) for DO-DTT traffic type.

6.30.1.2 Protocol Stack

The Figure 6.30.1.2-1 illustrates the protocol stack:

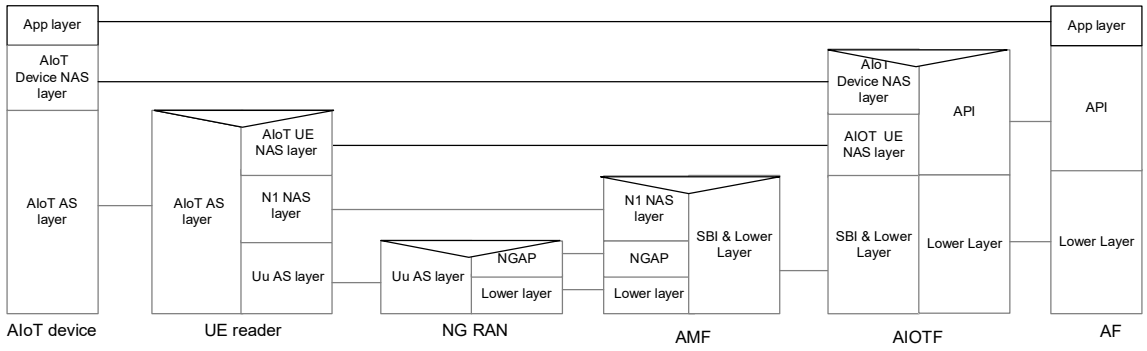


Figure 6.30.1.2-1: Protocol Stack for Supporting AIoT Devices using AIOTF for Topology 2

Within the protocol stack, the AIOT Device NAS layer, AIOT AS layer and App layer are defined as Solution #8.

As UE reader is used in Topology 2, which is different from the RAN reader in Topology 1, the following layer is proposed specific in this solution:

- AIOT UE NAS layer: The NAS protocol between AIOTF and UE readers. Besides the 5GMM NAS and 5GSM NAS, it can be defined as a new NAS (e.g. 5GAIOT NAS) which is piggybacked in 5GMM NAS transport messages. The AIOT UE NAS layer is used to communicate the AIOT specific information to the UE reader e.g. inventory and command.

6.30.2 Procedures

NOTE: The message names in the procedures below are descriptive. It is assumed that the names are updated with corresponding SBI based names where applicable during the normative phase.

6.30.2.1 AIoT Service Authorization for Intermediate UE

The Registration procedure for UE is performed as defined in clause 4.2.2.2 of TS 23.502 [5] with the following additions:

- UE includes the AIoT Intermediate node capability as part of "5GMM capability" in Registration Request message.
- The AMF obtains the AIoT Subscription data as part of the user subscription data from UDM using Nudm_SDM service
- The AMF determines whether the UE is authorized to work as Intermediate UE for AIoT based on UE's AIoT Intermediate node capability and the AIoT Subscription data. The AMF includes the authorization information as part of UE context in NGAP message sent to NG-RAN.

In Service Request procedure, N2 Handover procedure, Xn Handover procedure, and when receiving Subscriber Data Update to AMF, the AMF includes the authorization information in NGAP message sent to NG-RAN.

After NG-RAN receives the authorization information in the NGAP message from the AMF that the UE is authorized as an Intermediate UE, it does not release the UE to RRC_IDLE state afterwards. The UE can be released to RRC_INACTIVE when needed, in which case the NG-RAN assigns a suitable RNA and the NG-RAN need to perform RAN paging before the UE can act as a Intermediate UE.

6.30.2.2 AIOTF Keep Track of Intermediate UE

For authorized Intermediate UEs (Those Ambient IoT Capable UEs which have been authorized by the AMF as described in clause 6.30.2.1), in Initial Registration procedure as defined in clause 4.2.2.2 of TS 23.502 [5] with the following additions:

- The AMF sends Create UE Context Request to the AIOTF for the Intermediate UE to establish the usage of the AIOT UE NAS layer, so that the AIOTF may select the Intermediate UE for AIoT operations.

In Deregistration procedure as defined in clause 4.2.2.3 of TS 23.502 [5], the following additions are performed:

- The AMF sends Release UE Context Request to the AIOTF for the Intermediate UE, so that the Intermediate UE will no longer be used.

The AIOTF can subscribe the UE mobility event of the Intermediate UE towards the AMF to be updated for the location information.

Editor's note: It is FFS whether to add separate procedure from the UE for establishing the NAS connection for the AIOT UE NAS layer, instead of establishing the connection at registration procedure when the Ambient IoT Capable UEs have been authorized Intermediate UE.

6.30.2.3 Application Inventory Procedure

The Inventory procedure can be initiated by the AF to discover one or more AIoT devices in a specific area via Intermediate UEs.

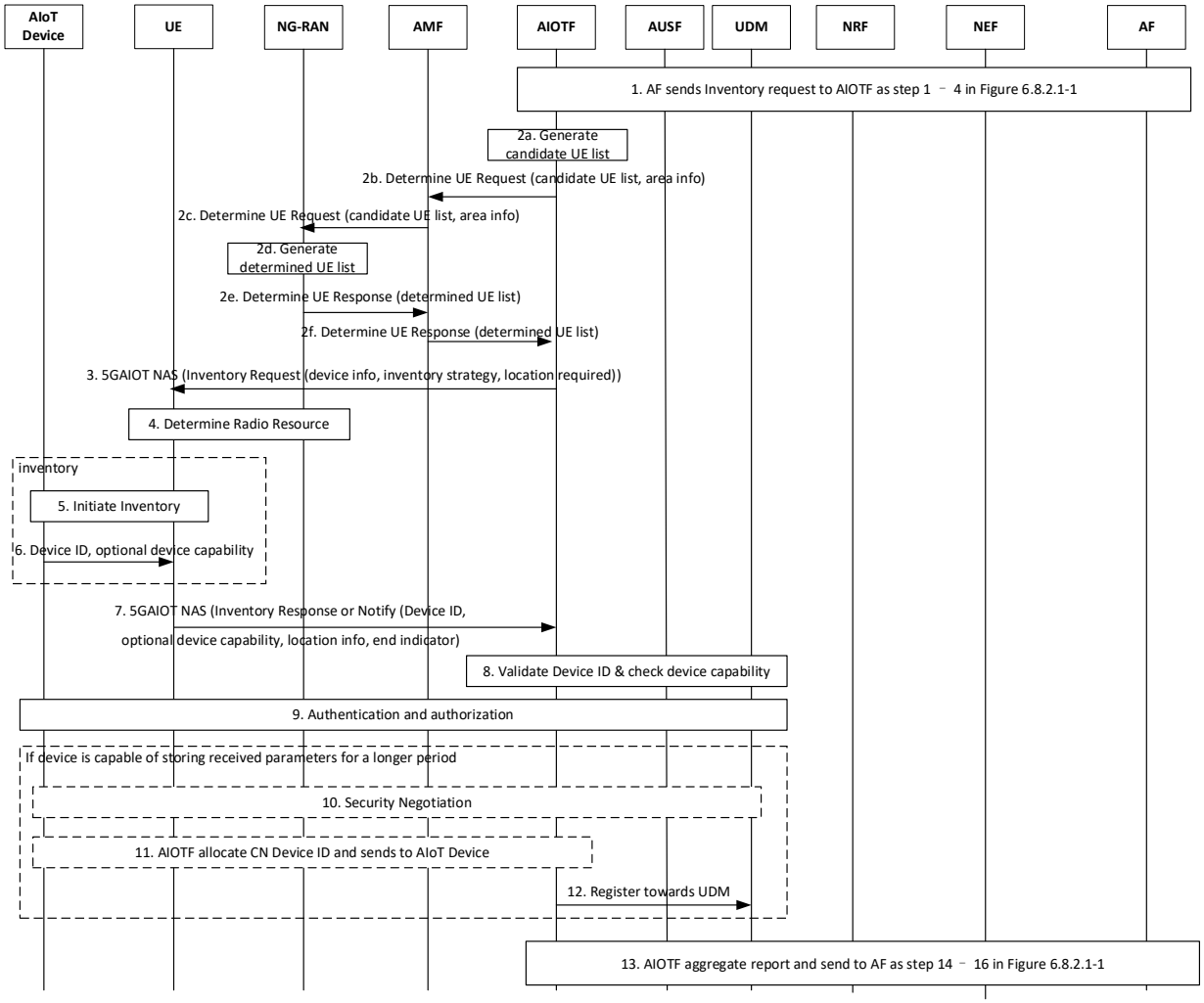


Figure 6.30.2.3-1: Inventory Procedure

1. AF sends Inventory Request to the AIOTF, which is the same as step 1 - 4 in Figure 6.8.2.1-1 in Solution #8.

2a. The AIOTF generates the candidate UE list for the Intermediate UEs to be used, based on the location information for the Intermediate UEs, as well as area information provided by the AF.

2b. The AIOTF sends Determine UE Request to the AMF with the candidate UE list and the area information.

2c. The AMF sends N2 message: Determine UE Request to the NG-RAN with the candidate UE list and the area information. The AMF may send to multiple NG-RANs within the area.

2d. The NG-RAN selects the most appropriate UEs based on the candidate UE list and the area info. The NG-RAN needs to consider the location of the UE, optionally the spectrum support of the UE, to ensure the coverage of the area with less interference among the UEs.

Editor's note: It is FFS whether RAN has enough information to determine suitable UEs.

2e. The NG-RAN sends N2 Message: Determine UE response to the AMF with the determined UE list.

2f. The AMF aggregates the responses and send Determine UE response to the AIOTF with the determined UE list.

3. The AIOTF creates and sends the 5GAIOT NAS message to request each of the Intermediate UE(s) in the determined UE list to perform the Inventory. The 5GAIOT NAS message is delivered via the AMF and the NG-RAN.

4. The Intermediate UE determines radio resource or requests NG-RAN to determine radio resource, which is used between AIoT Devices and Intermediate UE.

5. The Intermediate UE initiates inventory based on device information as well as the inventory strategy information provided by the AF. The Intermediate UE may provide reader identity information to enable the AIoT devices to understand they are read by which Intermediate UE. Considering the mobility of the Intermediate UE, the reader identity information can be a combination of a UE ID and the location information.

6. The AIoT Device reports the device ID to the Intermediate UE. If the Inventory procedure indicates delta inventory only, and the AIoT Device has performed the inventory procedure towards this reader, it should skip the reporting. The device capability index can be provided by AIoT device optionally.

7. The Intermediate UE creates the 5GAIOT NAS message and sends it towards the AIOTF, including the AIoT Device ID and optional device capability index. The Intermediate UE may further provide location information (ULI) of the device, if requested from the AIOTF and allowed by local policy. The Intermediate UE may further provide an end indicator to inform the AIOTF whether it is the last inventory response for the inventory round, e.g. based on timeout in Intermediate UE.

NOTE 1: It's up to RAN WG3 to determine the details and enhancements of the location information, and the ability to provide an end indicator from a reader is assumed to be aligned between UE as reader and RAN as reader.

8. The AIOTF validates the concealed device ID via interacting with AUSF and UDM. The AIOTF may further get the device capability information from the device capability profile stored in the UDM based on the device capability index provided by the device.

9. The AIOTF together with AUSF and UDM, triggers authentication and authorization procedures (i.e. like the authentication request/response between UE and network) towards the AIoT device. In Topology 2, the message between the AIOTF and the AIOT device is transmitted via the Intermediate UE over 5GAIOT NAS.

Based on the device capability information from UDM, if the AIoT device is capable of storing received parameters for a longer period, step 10 - step 11 are executed i.e. the AIOT device is then considered registered:

10. The security mode negotiation and security parameter exchanges are performed (i.e. like the security mode command/complete between UE and network). In Topology 2, the message between the AIOTF and the AIOT device is transmitted via the Intermediate UE over 5GAIOT NAS.

NOTE 2: Further details of the security procedures is to be defined by SA WG3.

11. The AIOTF may further allocates CN device ID and sends to the AIoT device.

12. The AIOTF registers with the UDM (UECM) for the device access.

13. The AIOTF may further aggregate the reported device IDs and send to AF via the NEF, as step 14 - 16 in Figure 6.8.2.1-1.

6.30.2.4 Periodic Inventory

The periodic inventory may be requested by the AF, so that network can trigger the inventory periodically without the request from the AF directly.

Considering the movement of the Intermediate UE, and the inventory is based on area information, the periodic inventory should be initiated by the CN (i.e. the AIOTF), which follows the instructions from the AF. When the period expires in the AIOTF, the AIOTF determines the Intermediate UEs based on the area information from the AF and the location information for the Intermediate UEs. Step 2-10 in Figure 6.30.2.3-1 are executed.

6.30.2.5 Command Procedure

The command procedure is initiated by the AF to request one or more specific AIoT devices or a group of AIoT devices in an area to execute the command. The command can be read, write, enable, disable, or execution request from the AF. The command result from the devices may or may not be required.

Editor's note: The command for enable/disable is to be studied by SA WG3, and to be aligned with SA WG3.

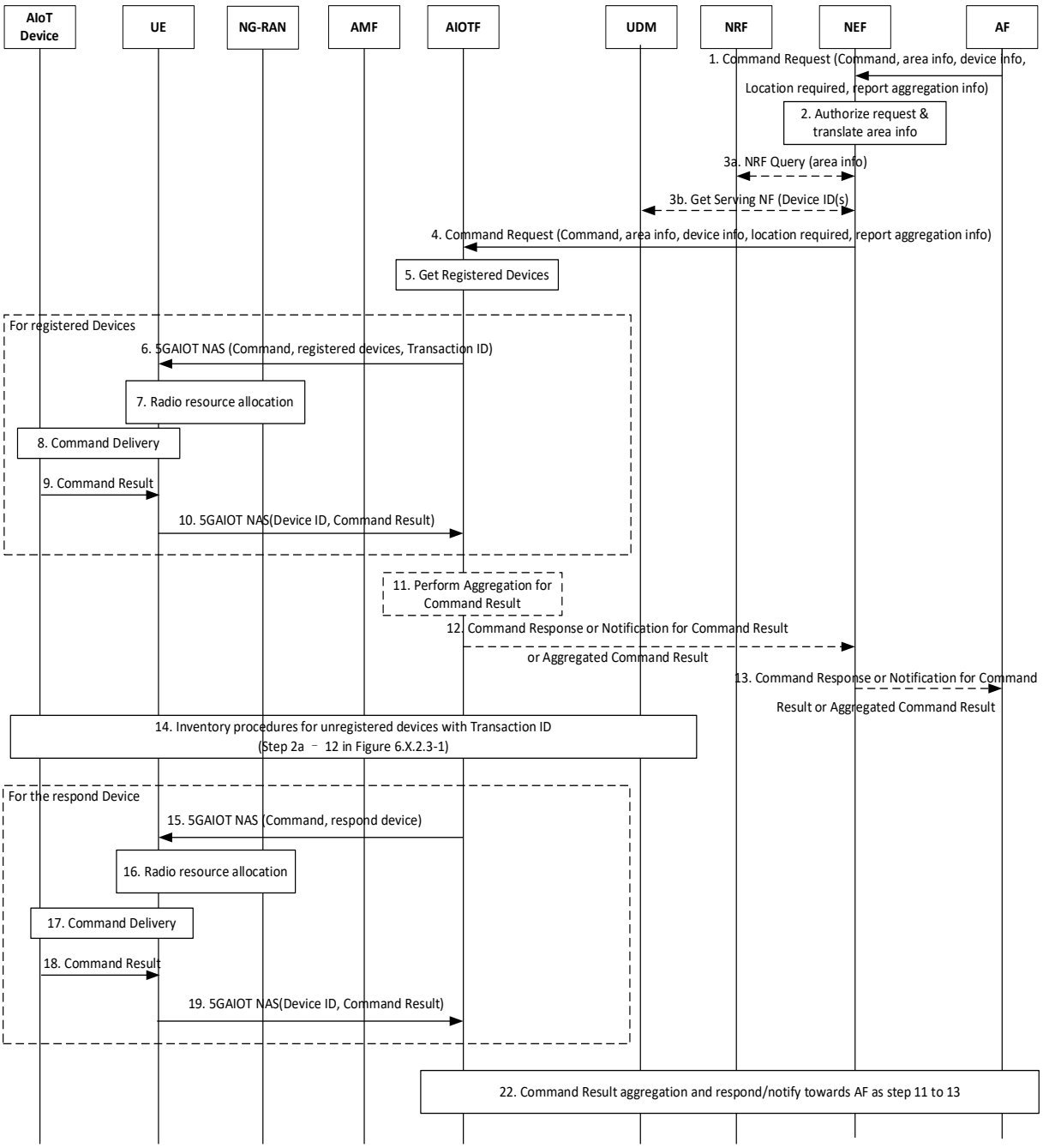


Figure 6.30.2.5-1: Command Procedure

1. The AF sends Command Request to the NEF, containing the command to be executed, area information, device information, optional inventory strategy information, and optional report aggregation info.

   - The command is to be executed in the AIoT devices. It contains the command (e.g. read, write, execute, etc.) together with the command parameters. The command specific parameters (e.g. what to be executed) are to be included in an app layer container between AF and AIOT devices.

   - The area information could be the external geographical area information.

   - The device information could be device ID, device group ID, and/or external device type.

   - The location required indicates whether the AF requests the location information of the AIoT devices provided.

   - The report aggregation info indicates whether the reports need to be aggregated or not for a specific aggregation period, and whether the reports are needed after the aggregation period.

2. The NEF authorizes the request from the AF and perform the area translation to translate external area information to the internal area information. Within the authorization, the NEF further check whether the AF is authorized to get the location information of the device. The NEF map the external device type to internal device type (i.e. device 1, device 2a, device 2b, etc.).

Editor's note: It is FFS and up to RAN to determine whether the device type is useful or not e.g. to optimize AS functionality such as spectrum usage, number of repetitions, etc.

Editor's note: It is FFS whether the external device type can be kept same as the internal device type.

Depending on whether area information is provided, step 3a or step 3b is performed:

3a. If area information is provided, the NEF sends NRF query with internal area information to query AIOTFs serving the area.

3b. If area information is not provided, and device ID(s) are provided, the NEF query the serving AIOTF from UDM.

4. The NEF sends the Command Request to the AIOTFs with the command, internal area information, device information, device type, and optional location required information.

5. The AIOTF allocates a Transaction ID for the command, which is unique per command. The AIOTF creates the AIOT Device NAS request message for the devices, based on the command information from the AF. The AIOTF get the registered devices, which match the device information. If no registered devices, step 6 - 13 are skipped.

6. The AIOTF creates the 5GAIOT NAS request message and sends to the Intermediate UE to deliver the Command. The 5GAIOT NAS request message contains the AIOT Device NAS message, CN allocated device IDs for the registered devices, the Transaction ID, and optional location required information. The 5GAIOT NAS message is delivered via the AMF and the NG-RAN.

7. The Intermediate UE determines radio resource or requests NG-RAN to determine radio resource, which is used between AIoT Devices and Intermediate UE.

8. The Intermediate UE initiates command delivery for the AIOT Device NAS request message based on CN allocated device IDs, using dedicated signalling. Transaction ID is also included.

NOTE 1: RAN signalling details, including how the command is sent, are to be further studied by RAN WGs.

9. The AIoT Device executes the command, creates an AIOT Device NAS response message containing the command result and delivers the AIOT Device NAS response message, if the result needs to be sent back. The application specific result can be included in an app layer container.

10. The Intermediate UE creates the 5GAIOT NAS response message, including the AIOT Device NAS response message, and sends to the AIOTF. The Intermediate UE may further provide location information (ULI) of the device, if requested from the AIOTF and allowed by local policy. The 5GAIOT NAS message is delivered via the NG-RAN and the AMF.

NOTE 2: It's up to RAN WG3 to determine the details and enhancements of the location information.

11. The AIOTF gets the command result from the AIOT Device NAS response message. The AIOTF may perform aggregation for the device ID, based on the report aggregation information provided by the AF. Within the aggregation period, the AIOTF will buffer the command results reported from the AIoT devices. The AIOTF may stop buffering and send report immediately, if it receives end indicator from the Intermediate UE in step 10. When the aggregation period expires, the AIOTF sends the report. For those command result after the aggregation period, if it is needed by the AF, the AIOTF sends the report. Otherwise, it will be dropped.

12. The AIOTF sends Command Response or Notification Request towards the NEF for the command result or the aggregated command result information.

13. The NEF sends Command Response or Notification Request towards the AF for the command result or the aggregated command result information.

If the AIOTF determines all required devices have executed the command (e.g. when AF provide the device ID(s) and those devices have executed the command in step 7 - 8), the following steps are skipped.

14. For those unregistered devices(s), the AIOTF initiate Inventory procedure as step 3 - 12 in Figure 6.30.2.3-1 (step 11 - 12 will be skipped, if the device is not able to persistent received information). In the inventory the Transaction ID is delivered, so that the registered devices which have executed command do not respond this inventory request from the Intermediate UE.

15. The AIOTF creates the 5GAIOT NAS request message and sends to the Intermediate UE to deliver the Command to the responded device. The 5GAIOT NAS request message contains the AIOT Device NAS request message, AIoT device ID and optional location required information. The 5GAIOT NAS request message is delivered via the AMF and the NG-RAN.

16. The Intermediate UE determines radio resource or requests NG-RAN to determine radio resource, which is used between AIoT Devices and Intermediate UE.

17. The Intermediate UE initiates command delivery for the AIOT Device NAS request message to the device.

18. The AIoT Device executes the command, creates an AIOT Device NAS response message containing the command result and delivers the AIOT Device NAS response message, if the result needs to be sent back. The application specific result can be included in an app layer container.

19. The Intermediate UE creates the 5GAIOT NAS response message, including the AIoT Device NAS response message, and sends to the AIOTF. The Intermediate UE may further provide location information (ULI) of the device, if requested from the AIOTF and allowed by local policy. The 5GAIOT NAS message is delivered via the NG-RAN and the AMF.

20. As step 11 - 13, the AIOTF get the command result from the AIOT Device NAS response message. It may perform the aggregation and send to NEF. The NEF delivers to the AF.

6.30.3 Impacts on services, entities and interfaces

Editor's note: The services, entities and interfaces are FFS.

6.31 Solution #31: fixed type of UE reader solution for Ambient IoT services

6.31.1 Description

This solution resolves Key Issue #3 about the Support of Ambient IoT Services. As we know, the following two connectivity topologies as defined in TR 38.848 [7] are to be studied:

- Topology 1: BS <--> Ambient IoT Device;

- Topology 2: BS <--> intermediate node <--> Ambient IoT Device: Only a UE can act as an intermediate node that is under the network control.

Due to UE's mobility feature, the UE reader can't be involved with Base Station readers together in some ambient IoT services, such as AF-initiated inventory service operation. The base Station reader is fixed and covers the designated service area. However, from the network perspective, the current solution doesn't know the UE position precisely, only the cell level. For example, we don't know whether it is in the drawer, stock, or other places. Besides, it isn't designed to cover a designated service area where service is always available.

Furthermore, the network side doesn't know its posture due to the UE reader's mobility. The UE reader's posture, such as the direction of the antenna, will significantly impact the result of the Ambient IoT service.

A reader UE can register into 5GC to be a fixed type of UE reader explicitly if it is mounted on the wall or rack to serve the designated serving area with a fixed posture; thus, the core network can treat the fixed UE reader as a mini Base Station reader to realize the same features as provided by Base Station reader to provide a cost-efficient and fast-deployed solution for these two following typical scenarios:

1) Hybrid fixed UE readers networking with Base Station Readers—Fixed UE readers are used to provide coverage for some blind spots not covered by Base Station Readers in scenarios with bigger service area requirements, such as Costco, Home Depot, supermarket, and so on.

2) Pure fixed UE readers networking—Treat fixed UE readers as mini Base Station Readers in scenarios with smaller service area requirements, such as convenience stores, 7-Eleven, and so on.

The principles related to fixed UE reader are depicted below:

- The UE reader can register as a fixed UE reader or mobile UE reader in 5GC.
- 5GC has the capability to involve the fixed UE readers in the AF-initiated service operations.
- 5GC can combine the response contents from both Base Station readers and fixed UE readers and get rid of duplicated records based on transaction ID and ambient IoT device ID
- the fixed UE reader is designed to serve a designated service area with fixed posture
- The fixed UE reader will re-initiate the registration message when it is moved to another place.

6.31.2 Procedures

6.31.2.1 Reader UE registration procedure with UE reader type info, Location, and AF ID

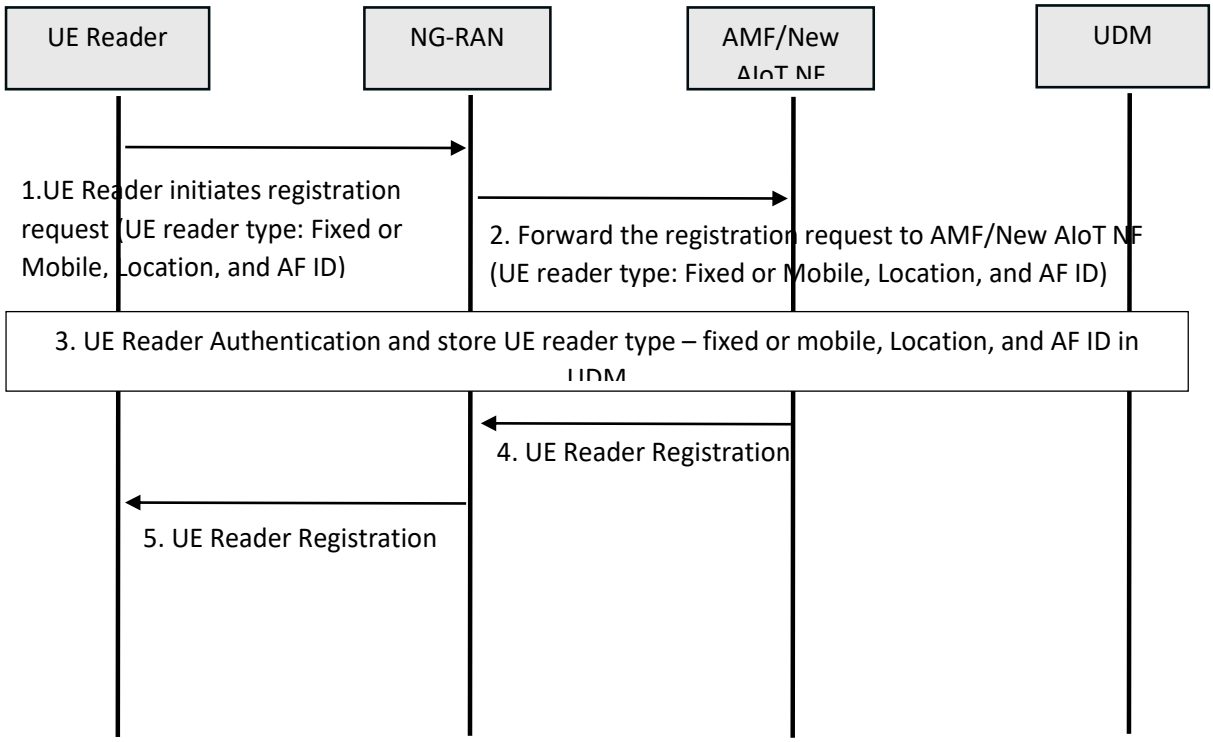


Figure 6.31.2.1-1: Reader UE registration procedure with UE reader type info, location, and AF ID

1. The UE reader initiates the registration procedure to 5GC:

   UE reader initiates registration request with UE reader type - fixed or mobile, Location, and AF ID. These parameters will be used by the core network to choose the correct targeted UE readers during the Ambient IoT service procedure.

Editor's note: The detailed information carried by the Location is FFS, e.g. coordinate, store ID, and so on.

Editor's note: It is FFS that UE reader type information (Fixed or Mobile) is trustable and accurate.

2. NG-RAN forwards the Registration Request message to AMF.
3. AMF and UDM do the authentication with the UE reader. Then, the UE reader profile is saved in UDM and AMF or New Ambient IoT NF, including UE reader type indication: Fixed or Mobile, Location, and AF ID.

4. AMF responds with a Registration Accept message to NG-RAN.
5. NG-RAN forward the Registration Accept message to the UE reader.
6. If the UE reader doesn't proactively include a location in the registration message, after the registration message, the core network can optionally launch the LCS procedure to obtain UE's coordinate info and save it into the UE reader's profile in UDM

6.31.2.2 Ambient IoT service - Network initiated Ambient IoT service procedure with fixed UE reader

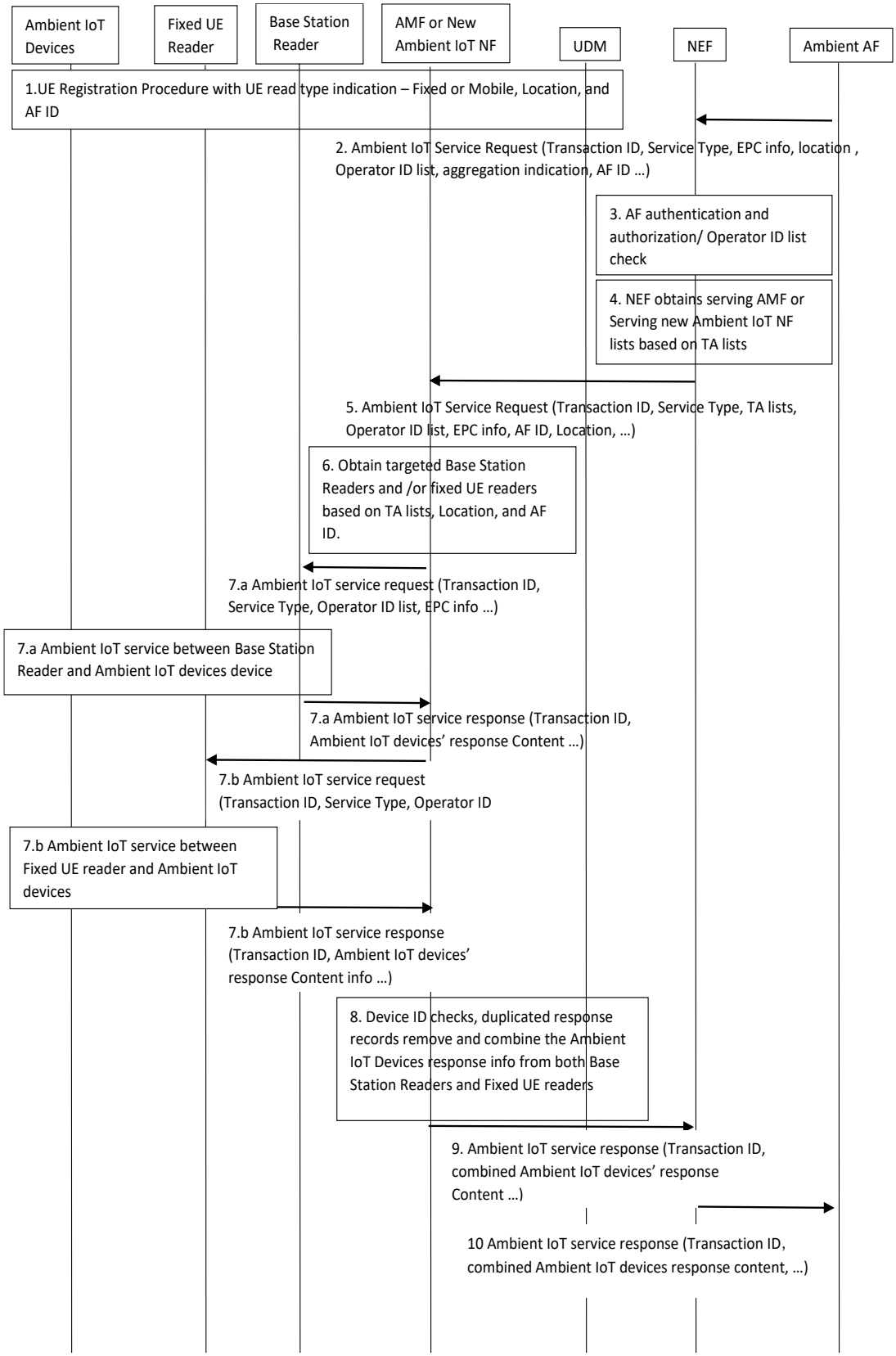


Figure 6.31.2.2-1: Network initiated ambient IoT service procedure with fixed UE reader

1. The UE Reader performs the registration procedure, including the fixed or mobile type of UE reader, Location, and AF ID.

2. The third Ambient AF has the store's location information and relevant AIoT serving operator info. The third Ambient AF launches an Ambient IoT service request towards NEF belonging to the AIoT serving operator. In the Ambient IoT service request, the third Ambient AF will provide Transaction ID, Service Type, EPC info, Location info, Operator ID list, Aggregation indication, and AF ID to NEF:

   - Service Type: This information is used to define Ambient IoT service types, such as Inventory, Read, Write, and so on.

   - EPC info: This information is used to define the targeted ambient IoT devices in the service from the products' perspective. it could be one specific ambient IoT device or a group of ambient IoT devices.

   - Operator ID list: This information is used to define the targeted ambient IoT devices in the service from the operator's perspective, supporting multiple operator ambient IoT device access. It could be one specific operator or a list of operators.

   - Location info: This information is used to define locations where service happens.

   - Aggregation indication: It is used to tell the 5GS how to handle the response messages -- aggregation response or not.

   - AF ID: which AF is initiating the AIoT service operation. NEF will do the authentication and authorization based on AF ID. This AF ID parameter will be transferred to AMF/New Ambient IoT NF to locate the targeted UE readers or Base Station Readers. In some cases, UE readers or base station readers can't be located only based on location.

3. NEF will perform the below actions on the third Ambient AF:

   - Perform authentication to the third Ambient AF to decide whether it is allowed to access 5GS or not.

   - Check the authorization to decide whether the third Ambient AF can perform certain Ambient IoT service operations.

   - Check the authorization to see whether multiple Operators' ambient IoT device access is allowed or not.

   - Converting location information to TA List information.

4. NEF will obtain serving AMF or serving New Ambient IoT NF based on TA lists.

5. NEF forwards Ambient IoT service requests to serving AMFs or Serving New Ambient IoT NFs. Both leveraging existing AMF and introducing new ambient IoT NF can be applied to this solution.

6. The AMF or New Ambient IoT NF determines the serving Base Station Readers and/or fixed UE readers based on TA lists, AF ID, and Location. There are following three scenarios:

   1) pure Base Station Readers are chosen; only step 7.a will be executed, and setp7.b will be skipped.

   2) pure fixed UE readers are chosen; only step 7.b will be executed, and step 7.a will be skipped.

   3) Mixed base Station Readers and fixed UE readers are chosen; both steps 7.a and 7.b will be executed.

NOTE 1: Only the fixed type of UE readers will be involved in AF-triggered AIoT service if there is no explicit UE GPSI parameter in AF-triggered ambient IoT service operations.

7.a The AMF or New Ambient IoT NF forwards the Ambient IOT service request to the Base Station Readers. Base Station Reader performs service operations toward the serving ambient IoT devices. The service operation will use both EPC info and the Operator ID list to identify the targeted ambient IoT devices. Base Station Reader will perform the response packet aggregation operation at the IP transportation level based on transaction ID in terms of aggregation indication to mitigate signal amounts between Base Station Reader and 5GC if Base Station Reader can't decode the response packets to look into the content. If Base Station Reader can decode the response packets to look into the content, Base Station Reader should perform the aggregation operation at the content level based on the transaction ID in terms of aggregation indication to further deduce signal amounts significantly between Base Station Reader and 5GC. Base Station Reader responds with an Ambient IoT service response message with parameters-transaction ID, Aggregated Inventory content, and so on-to AMF or the new Ambient IoT NF.

7.b The AMF or New Ambient IoT NF establishes connections with fixed UE readers and then forwards them the Ambient IOT service request. Fixed UE readers perform ambient IoT service operations toward the serving

ambient IoT devices. The service operation will use both EPC info and the Operator ID list to identify the targeted ambient IoT devices. Fixed UE readers will perform the response packet aggregation operation at the content level based on transaction ID in terms of aggregation indication to mitigate signal amounts between fixed UE readers and 5GC. Fixed UE readers respond with an Ambient IoT service response message with parameters-transaction ID, Aggregated Inventory content, and so on-to AMF or the new Ambient IoT NF.

NOTE 2: If the new Ambient IoT NF is used, communication between the UE reader and the NF is via AMF.

8. The AMF or new Ambient IoT NF will implement the Ambient IoT device ID check, remove duplicate Ambient IoT device records, and aggregate response packets from both Base Station Readers and/or fixed UE readers.
9. The AMF or new Ambient continues to forward the response to NEF IoT NF.
10. NEF will forward the response to the third ambient AF.

6.31.3 Impacts on services, entities, and interfaces

This solution impacts the following entities:

- **AMF or New Ambient IoT NF:**
  - Capability to combine response content from both Base Station Reader readers and fixed UE readers
  - Capability to choose fixed UE readers based on UE reader Type, TA, AF ID, and Location
- **UE:**
  - Capability to register UE reader type to 5GC
- **UDM:**
  - Capability to store UE reader type, AF ID, and Location in the profile.

6.32 Solution #32: AIoT communication for connectivity topology 2 and AIoT data transmission in the UP of PDU Session

6.32.1 Key Issue mapping

This solution addresses the following requirements:

- from Key Issue #1 the requirement:
  - *System architecture identified along with the solutions for KI#2 and KI#3.*
- from Key Issue #3 the requirements:
  - *Study how to support information transfer for Ambient IoT services and related system functionality, including the information transfer for an Ambient IoT device and for a group of Ambient IoT Devices.*

NOTE: Including whether there is a need to support session based transfer between Ambient IoT Device and the network considering the device types and capabilities.

  - *Study which of the enabled Ambient IoT services are exposed to AF and how, e.g. for the case AF requests Ambient IoT service for an Ambient IoT Device and for a group of Ambient IoT Devices.*

6.32.2 Functional Description

6.32.2.1 General

This solution focuses on the Topology 2. A UE, which is capable of AIoT transmissions towards AIoT devices, is called "UE AIoT Reader". The UE AIoT Reader indicates its AIoT capability during the registration with the network.

This solution proposes that the AIoT data/information transfer between the AIoT application server (AIoT AF) and the AIoT devices is executed via the User Plane (e.g. UP) of a PDU Session. A new NF called Ambient AIoT function (AIoTF) is introduced which is located the Control Plane (CP) and can be accesses via IP connectivity (e.g. via the UP and the N6 LAN). The assumed architecture is shown in Figure 6.32.2.1-1.

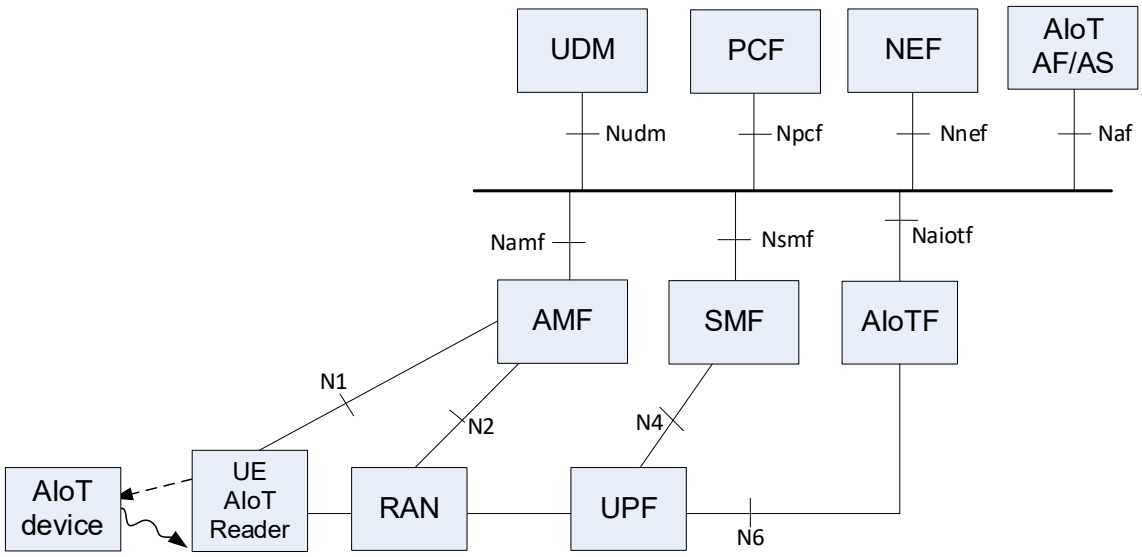


Figure 6.32.2.1-1: Assumed architecture for AIoT services using topology 2 and and AIoT data transmission in the UP of a PDU Session

The following principles are proposed:

- During the UE AIoT Reader registration procedure:
  - the UE AIoT Reader indicates its AIoT capabilities to the AMF and the AMF registers the UE AIoT Reader with the AIoTF.
  - The AMF or AIoTF may configure the UE AIoT Reader to use the U-plane for the AIoT data transmission and may configured the UE AIoT Reader with the allowed frequency resources to be used.
- The UE AIoT Reader establishes a PDU Session to transmit and receive AIoT data:
  - An AIoT application client resides at the UE AIoT Reader. The AIoT client performs 1) reception and transmission of AIoT data from/to AIoT-AF via the 5GS and 2) transmission and reception of AIoT data/signalling to/from the AIoT device.
  - The SMF may determine the AIoT configuration data for the UE Reader for the PDU Session either from the SM Subscription data or from the SM policy control information received from the PCF. The AIoT configuration data includes sending the AIoTF address to the UE, so that the UE Reader client can establish user plane connection to the AIoTF.
  - The SMF configures the UPF with a forwarding rule to receive and transmit AIoT data from/to the AIoTF on the N6 interface.
- An AIoTF connected in the C-plane and accessible via the U-plane is introduced. The AIoTF perform inspection of the AIoT data and/or perform verification of the AIoT devices.

6.32.2.2 Protocol Stack

The Figure 6.32.2.2-1 shows the protocol stack for AIoT communication between the AIoT device and AIoTF. The AIoT data is transmitted over the user plane of a PDU Session established by the UE AIoT Reader.

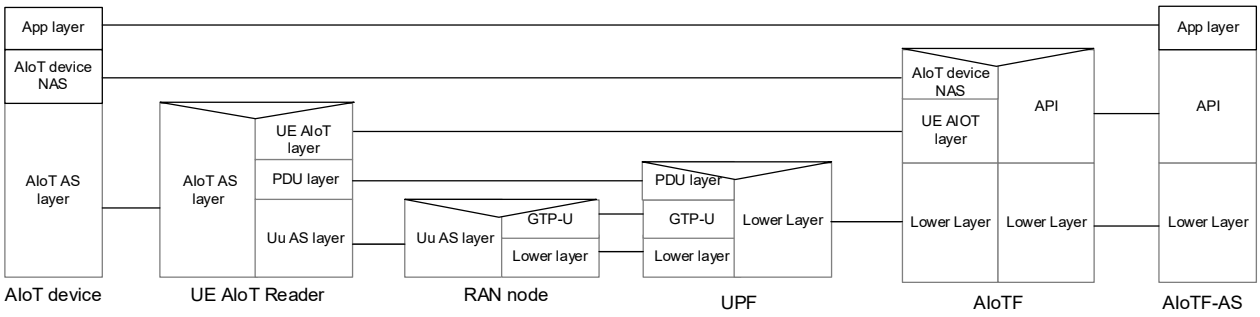


Figure 6.32.2-1: Protocol Stack for AF Based Solution (over user plane)

The following is applicable:

- App layer: the command layer protocol between AIoT devices and AIoT-AF/AS which includes the commands for "inventory" service and the "command" service (e.g. read, write, execute, etc.).

Editor's note: It is FFS whether the security for inventory service (to individual or group of AIoT devices) can be supported.

- UE AIoT layer: this the layer between the AIoTF and the UE Reader and it can be implemented as an IP protocol outside the scope of SA WG2. The UE AIoT layer encapsulates the AIoT device NAS PDUs.

- AIoT device NAS: the protocol between the AIoTF and AIoT device for authentication or verification of the AIoT device and also carrying the App layer data.

- Uu AS layer: the same as the current Uu interface.

NOTE: The AIoT device NAS protocol is assumed to be in the scope of SA WG2 (Stage 2 aspects) and CT WG1 (Stage 3 aspects).

6.32.3 Procedures

Two procedures are shown in Figure 6.32.3-1 to explain the concept of this solution:

- the Registration procedure of the UE AIoT Reader with the network and the AIoTF; and

- the PDU Session establishment procedure for the transmission of the AIoT data/signalling via the U-plane.

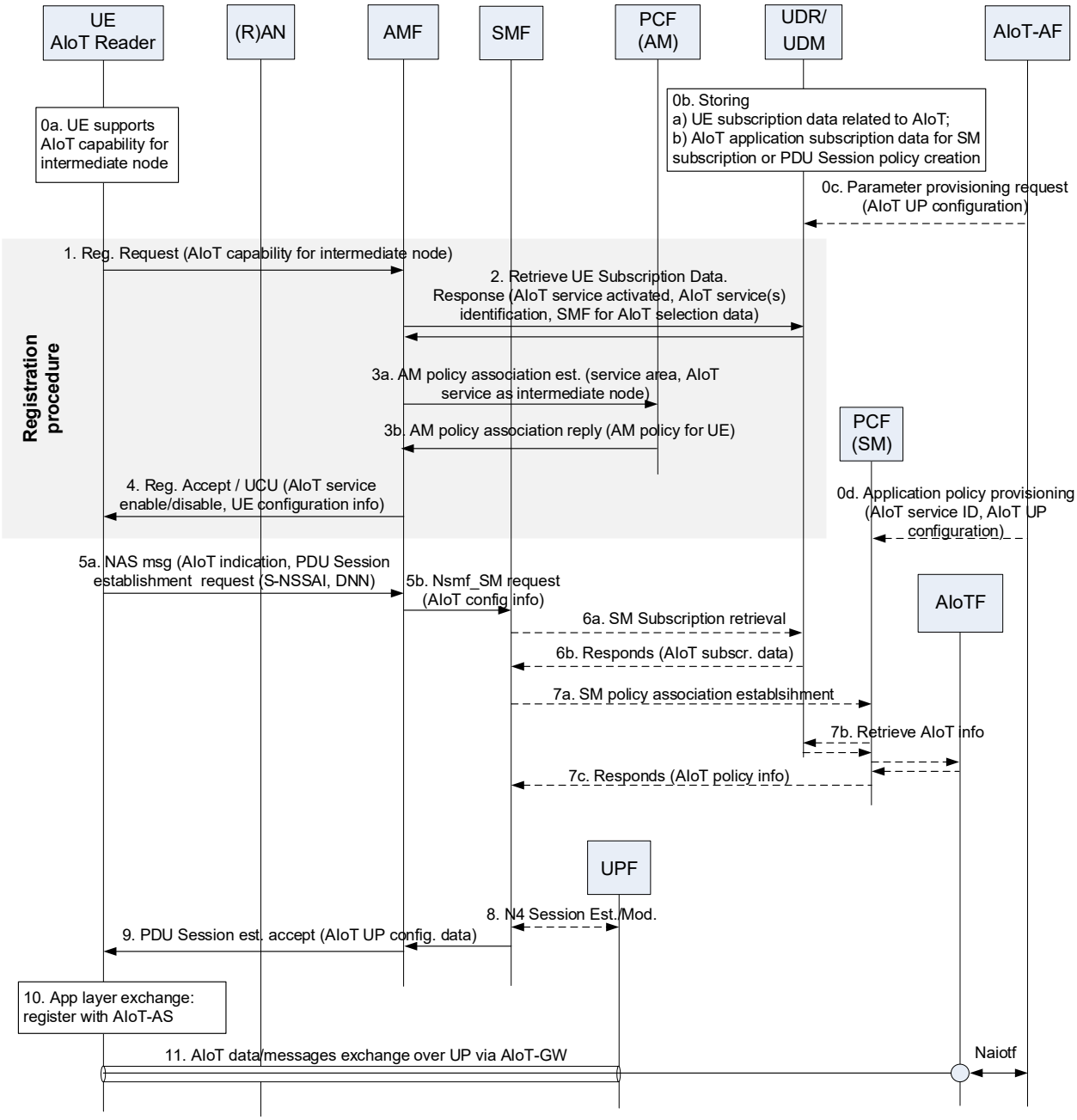


Figure 6.32.3-1: Signalling flow of the AIoT session establishment between the UE AIoT Reader and the AIoTF or AIoT-AS in the U-plane

The detailed description of the steps is provided as follows:

0a. The UE AIoT Reader may implement an AIoT client functionality which is responsible for: 1) reception and transmission of AIoT data/signalling via the U-plane (e.g. from AIoTF), and 2) transmission and reception of AIoT data/signalling to/from the AIoT devices.

0b. The UDM/UDR stores:

- UE AIoT Reader subscription data which may include the UE authorisation to act as AIoT intermediate node, a list of AIoT service(s) for which is allowed to transmit AIoT data/signalling to the AIoT devices, etc.

- AIoT application subscription data contains the configuration information for the AIoT application including credentials.

0c. The AIoT-AF may use parameter provisioning service to confiure or update the AIoT U-plane configuration for a specific AIoT service (idenitified by AIoT service ID).

1. The UE sends a Registration Request message which includes 1) indication that the UE is capable of AIoT communication serving as intermediate node, 2) whether the UE can receive and transmit AIoT data from/to the network via CP or UP communication path, 3) AIoT capabilities (e.g. supported frequency bands, range of communication or AIoT radio coverage area, one or more power modes.

2. The AMF can retrieve the UE's subscription data from the UDM/UDR. The UDM/UDR may indicate whether the UE is enabled/disabled to serve as AIoT intermediate node, a list of AIoT service(s) for which is allowed to transmit AIoT data/signalling to the AIoT devices, AIoTF selection information, etc.

3. The AMF requests AM policy association establishment with the PCF. The PCF may configure AM policy for the UE specific for the operation as UE AIoT Reader.

4. The AMF sends a Registration Accept message (or UE Configuration Update Command, UCU) which includes an indication whether the operation as UE AIoT Reader is allowed and which AIoT services are allowed to be served. The AMF may configure the frequency resources to be used by the UE Reader for the AIoT data transmission to/from the AIoT devices.

5a. The UE AIoT Reader's AIoT client functionality (please see step 0a) initiates a PDU Session establishment. The AMF may determine either based on the combination S-NSSAI/DNN or based on explicit indicaiton, that the PDU Session is for AIoT data.

5b. The AMF selects an appropriate SMF. If the AMF has AIoT configuration information, the AMF sends it to the SMF.

6. The SMF retrieves SM subscription data from the UDM, which may inlcude information about the user plane configuration (e.g. AIoTF address).

7. The SMF initiates the establishment of SM policy association to the PCF. The PCF may be preconfigured, or retrieve PDU Session subscription information from the UDR, or receive AF steering information in order to determine the AIoTF address which needs to be configured to the UE.

8. The SMF establishes N4 session with the selected UPF.

9. The SMF sends N2 SM configuration information to the RAN node and N1 SM container with configuration information to the UE. The N1 SM configuration information contains the AIoTF address.

10. The UE AIoT Reader may initiate application layer signalling to register with the AIoTF. During this step the UE and the AIoTF may exchange an AIoT session identifier (e.g. AIoT Service ID, UL port ID and DL port ID, etc.) which is used for the correct routing of the AIoT data/signalling in the UE AIoT Reader to one or more AIoT devices.

11. End-to-end communication:

    - In the DL direction: the AIoTF transmits AIoT data which is received from the AF via an API to the UE AIoT Reader (e.g. in an AIoT client functionality). The AIoT client functionality transmits the AIoT data to the AIoT device(s).

    - In the UL direction: the UE AIoT Reader (e.g. in an AIoT client functionality) receives AIoT data from the AIoT device. The UE AIoT Reader transmits the AIoT data the AIoT. The AIoTF transmits the AIoT data to the AF via the API.

6.32.4 Impacts on existing services, entities and interfaces

The following NFs and interfaces are impacted:

- AIoTF: optionally, a new function on the N6 interface can be introduced.

- UE AIoT Reader: 1) supporting AIoT capabilities to communication with the AIoT devices and 2) capabilities to receive configuration information from the SMF to initiate application layer communication with the AIoTF or AIoT-AS.

- PCF or UDM/UDR: store AIoT configuration information to configure PDU Session for the communicaiton between the UE AIoT Reader and AIoTF-UP.

6.33 Solution #33: Solution for different Ambient IoT Device Types

6.33.1 Description

6.33.1.1 Introduction

RAN has considered three types of AIoT devices (according to TR 38.848 [7]). The capabilities of these devices, vary quite significantly in terms of their baseband capabilities, storage, form factor from read-only devices to devices with writable memory, varying cost, complexity etc. As such it may not be possible to have the same solution for all AIoT device types and it may be prudent to classify AIoT devices (as an example) into two device types:

- the type A AIoT device that may not need to address privacy requirements and may also use a fixed permanent device identifier;
- the type B AIoT device that would use a temporary identifier, to address privacy concerns.

This solution describes for these two different types of Ambient IoT devices.

6.33.1.2 Functional Description

The solution for type A AIoT device is based on the following principles.

- The type A AIoT devices are ultra-low complexity devices with ultra-low power consumption for very-low end AIoT applications with limited or no energy storage capability, wherein UL transmission is backscattered, and the solution applies for both Topology 1 and 2.
- The AIoT device is pre-provisioned with a fixed permanent device identifier (device ID) that it uses for all communication.
- The AIoT device does not perform the 5GC Registration procedure with the network and the device does not authenticate with the network, though the network may perform certain checks to validate the AIoT device subscription and service operation based on the device ID. The AIoT device has no NAS or RRC states as defined in 5GS. The AIoT device can be paged.
- There is no PDU session created for session management and data transfer is done using NAS messages. The device communicates with the network directly by sending data (Ethernet, Unstructured frames) using its device ID.

The solution for type B AIoT devices is based on the following principles.

- The type B AIoT device are relatively low complexity devices with low power consumption and limited energy storage capability that can generate their own UL transmission and the solution applies for both Topology 1 and 2.
- The AIoT device registers with the network based on a network trigger and the network authenticates the device and assigns the device temporary identifiers. The AIoT device maintains some notion of NAS sates (registered, de-registered, etc.) and can be paged.
- The AIoT devices performs data transfer using NAS messages.
- The communication between the device and network is secure (may be encrypted and integrity protected).

Editor's note: Whether there is a need to differentiate Ambient IoT Type A and Type B devices is FFS.

6.33.1.3 Reference Architecture

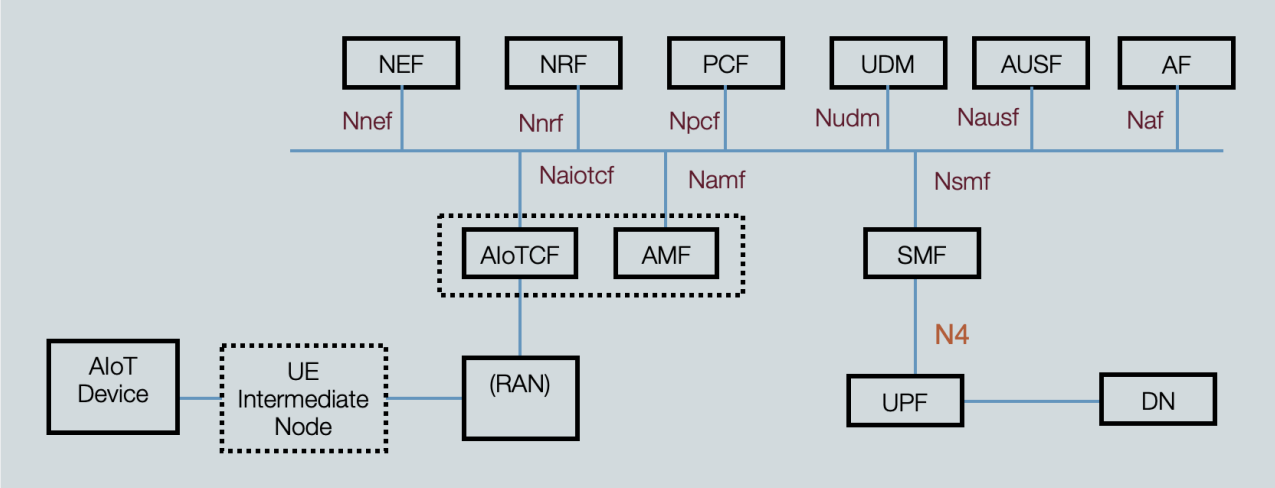


Figure 6.33.1.3-1: Reference Architecture to support Ambient IoT

The figure 6.33.1-3.1 shows the example reference architecture to support AIoT devices (both type A and type B) for Topology 2 wherein the UE (intermediate node) acts as the Reader. The same architecture also applies for Topology 1 (without the UE intermediate Node) and in that case the RAN acts as the Reader.

The 5GC has a new network function AIoTCF (Ambient IoT Controller Function) to support different AIoT service operations (command, inventory, etc.). The AIoTCF registers the Readers and authorizes the AFs for different AIoT service operations. The AIoTCF discovers and selects Readers for different service operations initiated by the AF. The AIoTCF can be integrated with AMF as well.

6.33.1.4 Protocol Stack

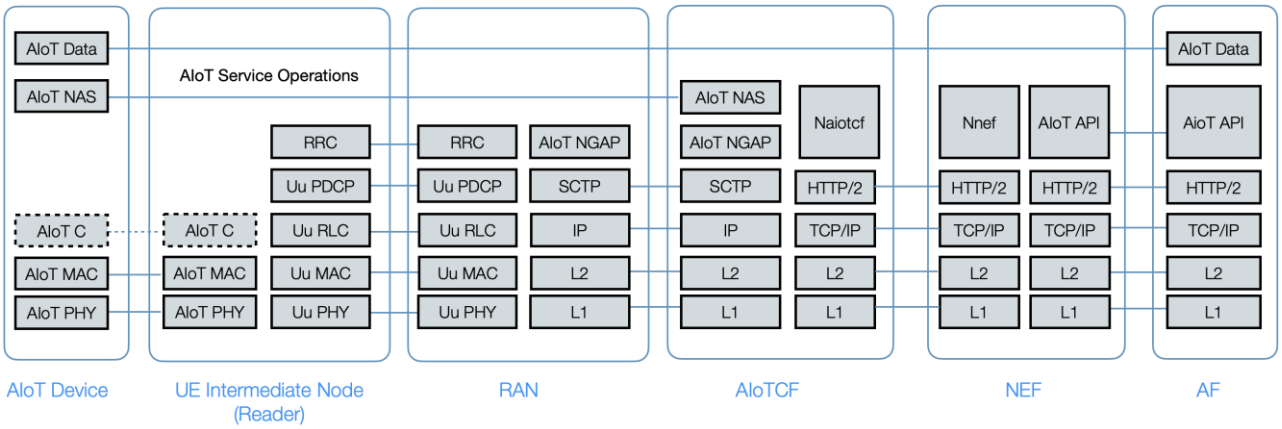


Figure 6.33.1.4-1: Protocol Stack to support Ambient IoT

The figure 6.33.1.4.1 shows the example protocol stack to support AIoT devices (both type A and type B) for Topology 2. The Reader supports the A-Uu air interface with AIoT device. The AIoT C is the optional control protocol between AIoT device and the Reader and will be defined by RAN. The AIoT NGAP protocol is the interface between the Reader and AIoTCF and can be a lightweight version of NGAP protocol tailored for AIoT or a new different protocol as well, which will be in scope of RAN WG3. The AIoTCF exposes the Naiotcf interface and in case the AIOTCF is integrated with AMF, the Namf interface can be enhanced. The AIoT APIs will be defined by CT WG3 and will expose the AIoT functionality to external entities. The AIoT NAS protocol enhancements will be defined by CT WG1. All stage-2 architectural aspects will be defined by SA WG2.

Editor's note: Whether AIoT C protocol is required and what is the content of AIoT C protocol will be defined by RAN.

6.33.2 Procedures

6.33.2.1 AIoT Service Operation Procedure for Type A AIoT Device

This clause describes the procedure for service operation (inventory, command, etc.) for a type A AIoT device as indicated in Figure 6.33.2.1-1. The procedure in general applies to both Topology 1 and Topology 2 configurations.

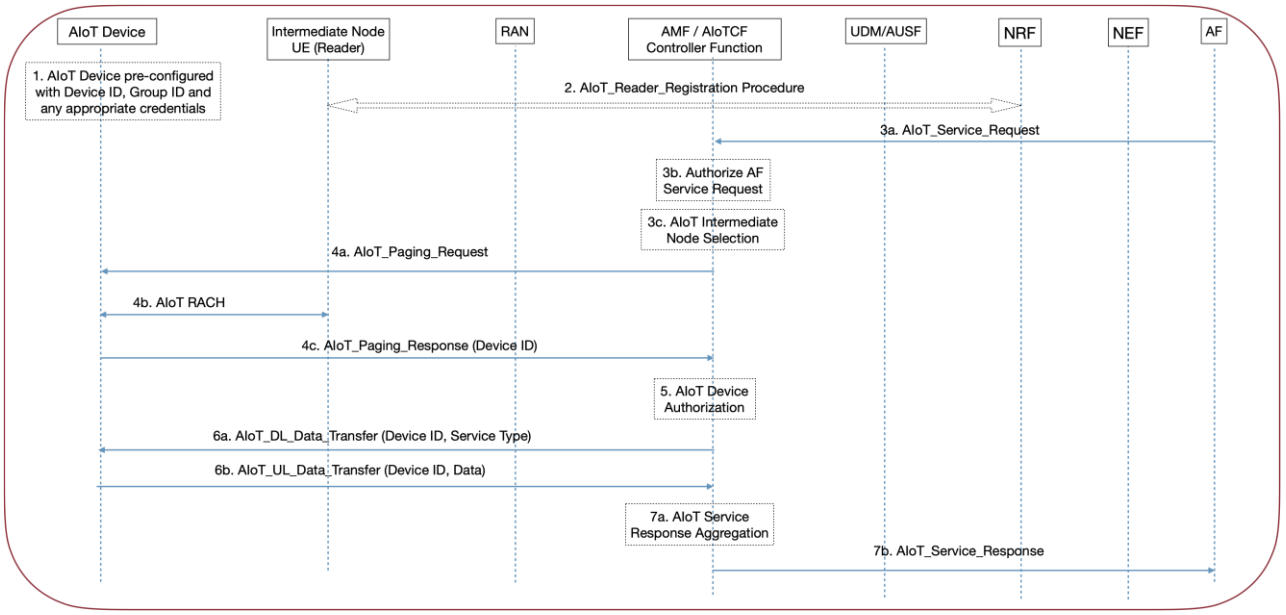


Figure 6.33.2.1-1: AIoT Service Operation Procedure for Type A AIoT Device

1. The type A AIoT device is pre-configured with Device ID, Group ID, and any other appropriate credentials. The type A AIoT device does not have UICC and may use the permanent device identifier as Device ID.

2. The Reader (UE intermediate node) registers with the AIoT Controller Function in the 5GC. The Reader reports (Reader ID, Capability to support AIoT, coverage area.) to AIoTCF. The AIoTCF aggregates the capability of all Readers and registers with NRF to provide AIoT service for AFs.

3. The AF sends a AIoT service operation (inventory, command, etc.) request to NEF which discovers and selects the appropriate AIoTCF registered with NRF, based on the included device information, and sends the request to the selected AIoTCF. The request includes device information (area of device operation, device identifier, reader information) and service information (type of service operation, periodicity, service specific control information, reporting information).

Editor's note: Whether reader information is included in the request to AIoT CF and the contents of reader information is FFS.

The AIoTCF authorizes the service request after verifying whether the AF is entitled to make the specified service operation request based on service agreements, operator policy, etc.

The AIoTCF selects the Ambient IoT capable Reader (or a set of Readers) based on the device information and information about the allowed service area. The selection of the intermediate node can be performed based on the geographical location of the target AIoT devices or the identity of the intermediate node from AF.

4. For each of the selected Reader, the AIoTCF sends the AIoT NGAP message to the RAN which results in the RAN triggering a paging like procedure and the Reader paging the type A AIoT device. The AIoT device performs random access channel (RACH) like procedure and sends a Paging response (including its Device ID) to the Reader which in turn relays this to the RAN which then sends this response in an AIoT NGAP message to the AIoTCF.

5. The AIoTCF retrieves the Device ID and performs validation on the type A AIoT device based on subscription information retrieved from UDM (device disabled or not, authorized for specified service operation, operating in permissible service area, etc.). If the deice is validated, the AIoTCF records the AIoT device location and performs the association with corresponding RAN and Reader. If the AIoT device cannot be validated the AIoTCF may invalidate the device and disable it, if possible.

6. The AIoTCF constructs the service operation request and sends it in an DL AIoT-NAS message to the AIoT Device. The AIoT Device performs the service operation and generates the results for specified operation and includes them in a UL AIoT-NAS message which is sent to the AIoTCF.

7. The AIoTCF can aggregate the results of service operation across Device IDs and report the results of service operation to NEF and the AF.

6.33.2.2 AIoT Service Operation Procedure for Type B AIoT Device

This clause describes the procedure for service operation (inventory, command, etc.) for a type B AIoT device as indicated in Figure 6.33.2.1-2.

1-2. These are same as in clause 6.33.2.1. The type B AIoT device may have a UICC.

3. The Reader triggers the paging like procedure for the type B AIoT Device. The type B AIoT device performs random access channel (RACH) like procedure to establish AS connection with Reader and thereafter initiates the AIoT Device 'Registration triggered by Network' procedure. The AIoTCF authenticates the AIoT Device and assigns the type B AIoT Device temporary identifier (TempDevID) as part of the authentication procedure. The AIoT Device uses the temporary identifier in subsequent communication. The AIoT device enters the registered state.

NOTE: It is expected that the temporary identifier (TempDevId) assigned to the type B AIoT device as part of Registration procedure can survive multiple "energy harvesting cycles". Otherwise, if the AIoT device loses its energy and cannot maintain results from the Registration procedure, it may not be feasible to conduct such a procedure, in its entirety.

Editor's note: The trigger for paging like procedure in step 3a and the identifiers used in this procedure are FFS.

4. Void.

5-9. These are same as steps 3-7 in clause 6.33.2.1.

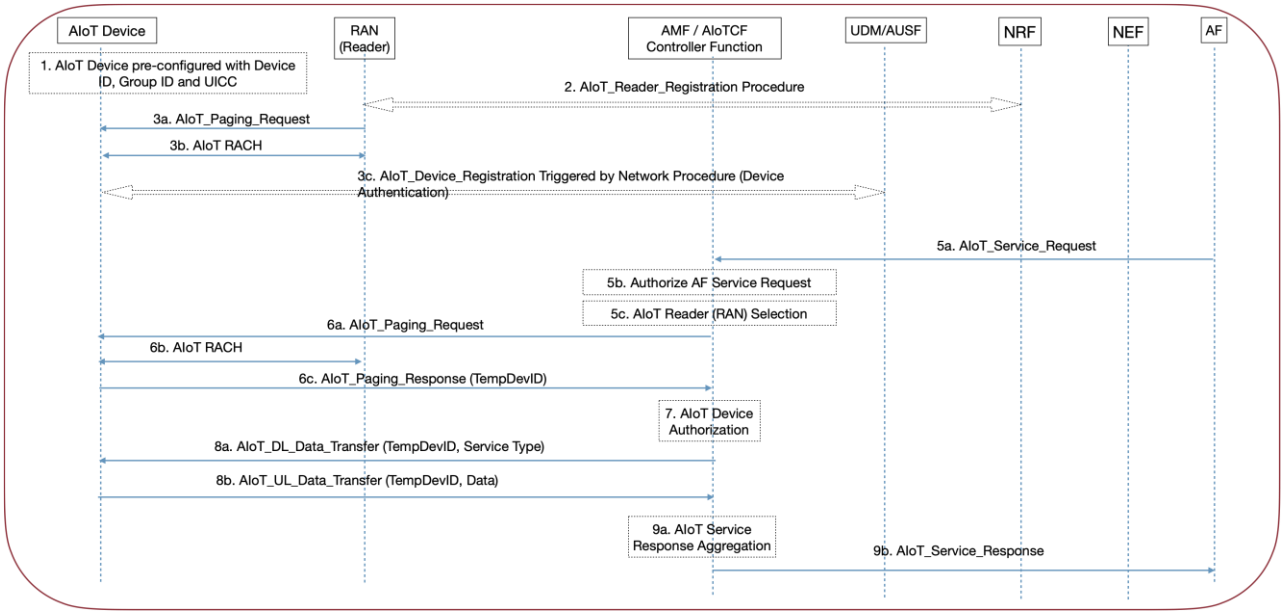


Figure 6.33.2.2-1: AIoT Service Operation Procedure for Type B AIoT Device

6.33.3 Impacts on services, entities and interfaces

Editor's note: This clause captures impacts on existing services, entities and interfaces.

6.34 Solution #34: UE assisted paging for AIoT Devices

6.34.1 Functional Description

This solution applies to Key Issue #2 "Identification, Subscription, Registration and Connection management".

This solution proposes that a UE Reader (i.e. intermediate node) handles paging and communication on behalf of AIoT devices using a network-assigned virtual TMSI (VTMSI). The UE receives the VTMSI when registering for AIoT devices paging and communication assistance. The UE then establishes a PDU Session used for the AIoT devices communication which the AMF and UE associates with the VTMSI. The UE monitors paging using the VTMSI to receive downlink data (e.g. command) for the AIoT devices. When paged with the VTMSI, the UE performs the Service Request including the PDU Session ID used for the AIoT service. The UE forwards the data received via the PDU Session to and from the AIoT devices (e.g. using AIoT devices supported RAT).

6.34.2 Procedures

The figures below describe the procedures (Registration, PDU Session establishment, network triggered Service Request) with enhancements to support AIoT devices paging and communication assistance by an UE.

Figure 6.34.2-1 shows the UE Registration and PDU Session establishment procedures for an UE.

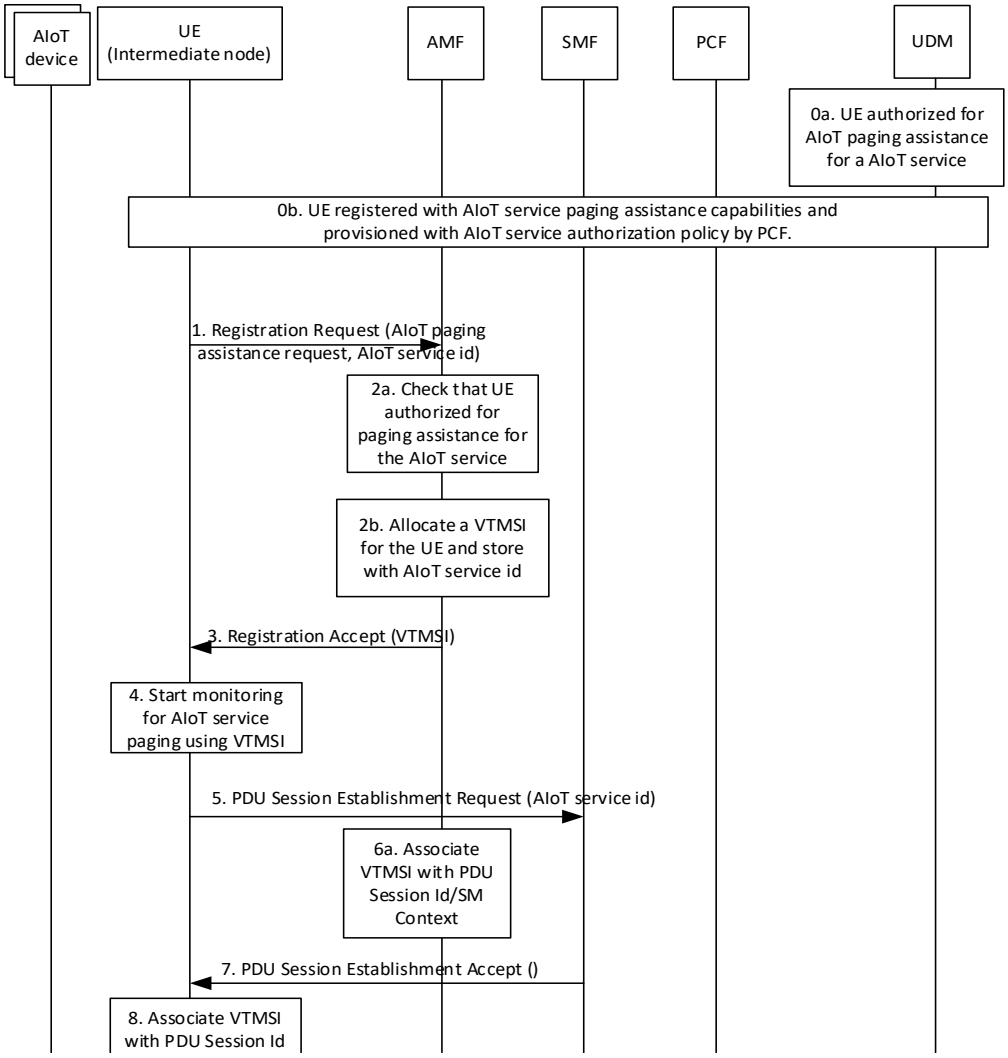


Figure 6.34.2-1: UE Registration and PDU Session establishment procedures

0. During Registration, the UE is authorized in the network (e.g. by the UDM/UDR) to act as a UE for an AIoT service. The authorization information included in the subscription data includes the AIoT Service ID. The UE

performs an initial Registration procedure providing its AIoT assistance capabilities. The UE is provisioned by a PCF with a policy authorizing the UE for paging and communication assistance for the AIoT Service ID(s).

1. Based on the AIoT authorization policy, the UE sends a Registration Update request message including an AIoT paging and communication assistance request indication and the AIoT Service ID and/or Service Provider ID (s) the UE wishes to provide assistance for.

2. The AMF checks that the UE is authorized as UE for the provided AIoT Service ID based on subscription information. If the device is authorized the AMF allocates a virtual temporary identifier (VTMSI) for each AIoT service and/or Service Provider ID that the UE wishes to provide assistance for. The VTMSI can be in the form of a 5G-S-TMSI to be used in Paging and Service Request procedures. The AMF stores the VTSMI in the UE Context along with the associated AIoT Service ID.

3. The UE receives from AMF a Registration Response message including the VTMSI associated with the AIoT Service ID provided by the UE in the Registration Request.

4. The UE stores the VTMSI along the AIoT Service ID and starts monitoring for AIoT device paging using the VTMSI.

NOTE: The UE performs regular Paging monitoring for its own paging using its 5G-S-TMSI (e.g. for non-AIoT data services). i.e. the PO calculation will still use 5G-S-TIMSI.

5. The UE establishes a PDU Session for the AIoT service. The triggering of this PDU Session establishment is not based on uplink data request. The UE sends the VTMSI to the AMF in the NAS message to the AMF that carries the PDU Session Establishment Request or provides a DNN/S-NSSAI combination used for the AioT service. If VTMSI is provided, the AMF can determine a DNN/S-NSSAI combination for the AIoT service (stored in 2b) based on UE's subscription information or location configuration and provide the DNN/S-NSSAI combination to the SMF when the PDU Session Establishment Request is sent to the SMF.

Editor's note: The details of the management (e.g. deactivation) of the PDU Session for AIoT service is FFS.

6. The AMF associates the VTMSI with the PDU Session ID (SM Context).

7. The UE receives a PDU Session Establishment Accept confirming that the UE is authorized to send and receive data on behalf of AIoT devices. The UE associates the PDU Session ID with the AIoT Service ID. The UE is ready to handle the paging and receipt of downlink data for the AIoT devices.

Figure 6.34.2-2 shows the network-triggered Service Request procedure for the UE.

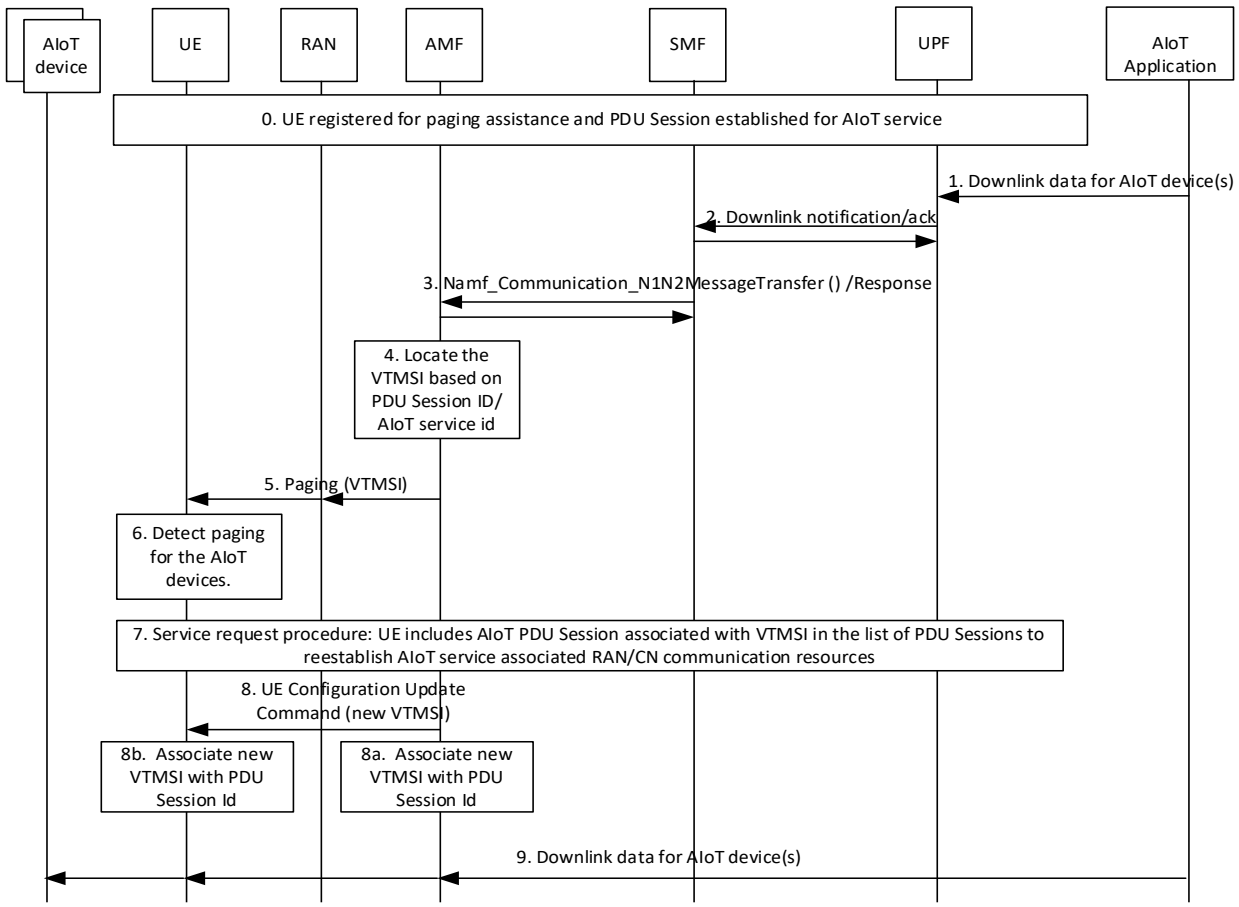


Figure 6.34.2-2: Network triggered service request for UE

0. The UE is authorized by the network to act as a UE for an AIoT service. The UE has established a PDU Session for the AIoT service communication. The AIoT application on the UE has established communication with the AIoT AS.

1. The AS sends some downlink data to the UE destined for the AIoT devices.

2. The UPF detects and notifies the SMF of the arrival of a downlink packet destined for the PDU Session of the IoT Assistant UE.

3. The SMF requests AMF transfer of downlink message to UE/RAN providing the PDU Session ID.

4. The AMF determines that the message is for a PDU Session used for AIoT device communication based on the PDU Session ID, associates stored AIoT Service ID, and retrieves the associated VTMSI.

5. The AMF sends a paging message to RAN to page the UE identified by VTMSI. RAN sends a paging message to the UE including the VTMSI. The paging cycle and the calculation of Paging Occasion uses the UE's original settings so the UE doesn't need a different set of Pos for Paging with VTMSI.

Editor's note: It is FFS in case there are simultaneous downlink data arrival for UE's own data or for multiple different AIoT services.

6. The UE monitors the paging and compares the UE identity in the received paging message with both VTMSI and its own 5G-S-TMSI. If the UE detects the paging for its VTMSI and determines that it needs to receive incoming downlink data for the AIoT devices on the PDU Session established for the AIoT service.

7. The UE sends a service request message and includes the PDU Session ID for the AIoT service in the list of PDU Sessions to request the re-establishment of RAN/CN communications resources.

8. Following the completion of the Service Request the AMF sends a UE Configuration Command message including optionally a new VTMSI to be used for a future AIoT service paging.

9. The UE receives downlink data for the AIoT devices. The UE transmits the data to the devices using the applicable AIoT RAT.

Editor's note: It is FFS how UE in RRC_INACTIVE state handles the paging.

6.34.3 Impacts on existing services, entities, and interfaces

UDM/UDR:

- Stores/Provides: AIoT Assistance Authorization information.

PCF:

- Provisions UE with AIoT Assistance Authorization policy.

AMF:

- Checks for AIoT Assistance Authorization from UDM.
- Allocates and Sends VTMSI to authorized UE (Registration Accept).
- Pages UE using VTMSI.

UE (AIoT Assistant):

- Receives from PCF and applies authorization policy to operate as AIoT Assistance UE.
- Sends to AMF:
  - AIoT Assistance Capabilities and AIoT Service ID (Registration Request).
  - VTMSI (NAS message for PDU Session Establishment request).
- Receives VTMSI from AMF.
- Monitors and receives paging messages for VTMSI.

6.35 Solution #35: An compatible format of AIoT device

6.35.1 Description

The solution addresses Key Issue 2.

In RFID domain, EPC has been well used. In AIoT scenario, the AIoT device may be owned by the 3rd party but can be operated by an operator network. So it is possible for a AIoT device without operator allocated device ID before it being operated by any reader in the operator network. Upon reception of inventory signalling, the AIoT device may provide EPC to the reader as the response.

In order to construct an open and wide-applicable business for AIoT service, the AIoT network should be compatible to the AIoT devices with either 3GPP defined ID or EPC. However, EPC format is not suitable to be used in the 3GPP interfaces and among NFs directly and it should be translated into 3GPP defined device ID format, i.e. the 3GPP defined device ID format should be compatible to EPC.

6.35.2 EPC format

The format of the EPC can be referred to GS1 TDS Release 2.1 [10]: "EPC Tag Data Standard". The structure of the EPC guarantees worldwide uniqueness of the EPC across all types of 920 physical objects and applications. The EPC URI is a string having the following form which is referred to clause 6 of GS1 TDS Release 2.1 [10]: "EPC Tag Data Standard":

urn:epc:id:scheme:component1.component2 . . .

An example of a specific EPC URI is the following, where the scheme is sgtin:

urn:epc:id:sgtin:9521141.012345.4711

Where Scheme names an EPC scheme. The components (i.e. component1, component2, and following parts are the 911 remainder) and the precise form of the EPC depends on which EPC scheme is used.

Table 6.35.2-1: Example of EPC Schemes defined in Figure 6.1 of GS1 TDS Release 2.1 [10]: "EPC Tag Data Standard"

EPC Scheme	Corresponding GS1 Key	Typical use
sgtin	GTIN (with added serial number)	Trade item
sscc	SSCC	Logistics unit
sgln	GLN (with or without additional extension)	Location
grai	GRAI (serial number mandatory)	Returnable asset
giai	GIAI	Fixed asset
…	…	…

It is observed that according EPC Scheme, the EPC format is determined.

Table 6.35.2-2

Scheme	The full name of Scheme	Component1	Component2	Component3
Sgtin	Serialised Global Trade Item Number	CompanyPrefix	ItemRefAndIndicator	SerialNumber
Sscc	Serial Shipping Container Code	CompanyPrefix	SerialReference	
Sgln	Global Location Number With or Without Extension	CompanyPrefix	LocationReference	Extension
Grai	Global Returnable Asset Identifier	CompanyPrefix	AssetType	SerialNumber
Giai	Global Individual Asset Identifier	CompanyPrefix	IndividulAssetReference	
Gsrn	Global Service Relation Number – Recipient	CompanyPrefix	ServiceReference	
Gsrnp	Global Service Relation Number – Provider	CompanyPrefix	ServiceReference	
Gdti	Global Document Type Identifier	CompanyPrefix	DocumentType	SerialNumber
Cpi	Component / Part Identifier	CompanyPrefix	ComponentPartReference	Serial
Sgcn	Serialised Global Coupon Number	CompanyPrefix	CouponReference	SerialComponent
Ginc	Global Identification Number for Consignment	CompanyPrefix	ConsignmentReference	
Gsin	Global Shipment Identification Number	CompanyPrefix	ShipperReference	
Itip	Individual Trade Item Piece	CompanyPrefix	ItemRefAndIndicator	Piece.Total.SerialNumber
Upui	Unit Pack Identifier	CompanyPrefix	ItemRefAndIndicator	TPX
Pgln	Global Location Number of Party	CompanyPrefix	PartyReference	
Gid	General Identifier	ManagerNumber	ObjectClass	SerialNumber
Usdod	US Department of Defense Identifier	CAGECodeOrDODAAC	SerialNumber	
Adi	Aerospace and Defense Identifier	CAGECodeOrDODAAC	OriginalPartNumber	Serial
Bic	BIC Container Code	BICContainerCode		
Imovn	IMO Vessel Number IMO (International Maritime Organization)	IMOvesselNumber		

And always, in the EPC URI, most of the component1 part is *CompanyPrefix*. The GS1 Company Prefix, assigned by GS1 to a managing entity or its delegates.

It is observed that a *CompanyPrefix* is common for most of EPC.

6.35.3 Design AIoT device ID

In this alternative, the AIoT device ID still reuse NAI, i.e. the form of username@realm.

For the realm part, it contains the following information:

- The Scheme present the scheme of the AIoT device ID:
  - If the AIoT device ID is mapped by EPC, it is the EPC scheme.
  - If the AIoT device ID is 3GPP operator-defined, it is 3GPP.
- The Assignment Mode present whether the Operating Entity ID is global unique assigned:
  - Option 1: the Operating Entity ID assigned as globally unique.
  - Option 2: the Operating Entity ID assigned as the combination of the Operating Entity ID and the Scheme is globally unique.
  - Option 3: the Operating Entity ID assigned as the combination of the Operating Entity ID and the PLMN ID is globally unique.
- The Operating Entity presents the owner/manager/company/credential holder of the AIoT device:
  - If the AIoT device ID is mapped by EPC, the value can be company prefix in EPC.
  - If the AIoT device ID is 3GPP self-defined, e.g. the value is the 3GPP allocated company value.

NOTE: If the AIoT device is owned by a network operator, the value could be the PLMN ID, since a network operator is also a company.

- Optional MCC+MNC, it is only needed when Operating Entity assigned such that the combination of the Operating Entity ID and the PLMN ID is globally unique.

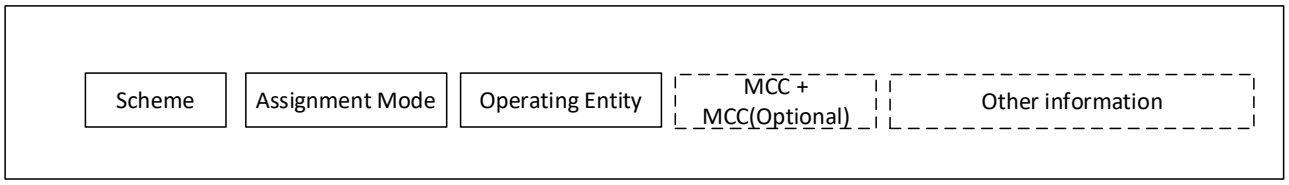


Figure 6.35.3-1: Format of AIoT device ID

For the username part, it contains the following information:

- It is EPC, if the AIoT device ID is mapped by EPC.
- It is 3GPP defined username part, if the AIoT device ID is 3GPP self-defined.

6.35.4 Procedure

The procedure is described below:

1. The reader sends signalling for Inventory to AIoT device.
2. When AIoT device response to the signalling for Inventory, it provides AIoT device ID, the AIoT device ID may be 3GPP allocated ID or the EPC mapped ID.
3. The other steps are not repeated here.

6.35.5 Impacts on services, entities and interfaces

Editor's note: This clause captures impacts on existing services, entities and interfaces.

6.36 Solution #36: Device ID(s) for AIoT Devices

6.36.1 Description

This solution addresses device ID aspect of Key Issue #2

In 5GS, there are the following identifiers which are described in TS 23.501 [4]:

- Subscription Permanent Identifier (SUPI).
- Subscription Concealed Identifier (SUCI).
- Permanent Equipment Identifier (PEI).
- 5G Globally Unique Temporary Identifier (5G-GUTI).
- Generic Public Subscription Identifier (GPSI).

Internal-Group Identifiers and External-Group Identifiers are used to identify a group of UEs.

The different identifiers in 5GS are used for different purposes. Depending on the use cases and required functionality for supporting AIoT, different AIoT device IDs should be introduced.

To be able to perform authentication and authorization procedures in 5GS, it is necessary to introduce SUPI like and SUCI like IDs for AIoT devices. The SUCI like AIoT device ID is a concealed format of SUPI like AIoT device ID, which is used to transmit the device ID from AIoT devices towards network, similar as SUCI in registration request.

The SUPI can be in different formats depending on the SUPI type. While a NAI based format with username@realm can be used a more condensed encoding can be beneficial, i.e. the encoding format can be left for stage 3 WGs to use a suitable encoding format.

5GS allows different SUPI types to be used, e.g. IMSI, NSI, GCI, GLI i.e. for Ambient IoT a new SUPI type allowing the content to be e.g. EPC ID could be used. However, also with non-EPC ID SUPI type the network operator could be provisioned with an EPC ID e.g. in UDM such that the AF input can be related to EPC ID while the 5G network internal handling is for a SUPI and also towards the AIoT Device the SUPI is used. If beneficial, the EPC ID can then be part of the GPSI, see GPSI aspects below.

The SUCI like AIoT device ID can be used to conceal part of the SUPI. The HNI part of the SUCI can include MCC and MNC for a PLMN and also a NID for an SNPN, and with a Routing Indicator the HNI and RID allows to route to the proper AUSF and UDM instances capable of handling authentication.

NOTE: How the SUCI is created in the AIOT Device is to be determined by other WGs, e.g. SA WG3.

For 5G-GUTI like AIoT device ID, similar as 5G-GUTI, could be allocated by the CN after authentication and authorization. The 5G-GUTI like AIoT device ID is passed from CN towards an AIoT device, and the AIoT device stores it and then uses it in the following communication towards network to identify the authenticated AIoT device. The 5G-GUTI like AIoT device ID can limit the possibility of device tracking.

Depending on AIoT device capabilities, the 5G-GUTI like AIoT device ID can be used.

The PEI is an identifier of the ME and is used e.g. to check whether the ME been stolen etc. 5GS allows different types of PEI to be used e.g. IMEI(-SV), MAC, EUI-64 i.e. in principle 5GS can be extended to allow TID as a PEI if seen useful over existing type of PEIs. Without any permanent device identifier, it would be hard to perform tracking of e.g. stolen devices.

To be able to communicate with external AF, a GPSI like AIoT device ID can be introduced. It is the NEF which perform the mapping between the SUPI like AIoT device ID and GPSI like AIoT device ID. The AF utilizes the GPSI like AIoT Device ID in requests towards the CN, and the NEF maps it to the SUPI like AIoT device ID, which is used inside CN. And when the CN responds or notifies the AF, the NEF maps the SUPI like AIoT device ID back to GPSI like AIoT device ID.

In RFID, RFID tags use EPC ID to communicate with RFID readers. It may make sense to introduce the EPC in GPSI like AIoT Device ID, to minimize the integration with RFID applications.

The GPSI can be defined as an External Identifier and have the form "<Local Identifier>@<Domain Identifier>". It could be an option to introduce EPC ID in the <Local Identifier>, while the <Domain Identifier> can be the domain name of the AIoT service provider.

Regarding the group of AIoT devices, Internal Group Identfier and External Group Identifier can be reused. And the NEF can perform the mapping between the Internal Group Identifiers and External Group Identifiers.

6.36.2 Procedures

The device IDs for AIoT devices are used in an equivalent way as current ones for UEs. No new procedures introduced due to AIoT device IDs. They are used in the inventory, command, (and also other) procedures for AIoT devices:

- SUCI like device ID is sent from AIoT devices to network to identify the device when unauthenticated e.g. response to inventory.
- SUPI like device ID is used in 5GC to uniquely identify the data stored in the network for enabling connectivity for the device. As to enable paging like functionality for AIoT devices that cannot store data like 5G-GUTI in non-volentary memory or before a capable AIoT Device has been allocated a 5G-GUTI, there may be a need to again open up for using SUPI in paging (i.e. IMSI based paging was removed in 5GS for security/privacy reasons).
- 5G-GUTI like device ID can be allocated by the core network and sent back to the device. If allocated it is used to page the device and identify the device.
- PEI like device ID: a permant identifier of the AIOT device, can be optionally provided by the AIOT device or is separately requested by the network.
- GPSI like device ID is used externally between AF and 5GC
- Internal Group Identifer is used to identify a group of AIoT devices internally within the 3GPP network
- External Group Identifier is used to identify a group of AIoT devices in AF.

Editor's note: Additional procedures e.g. security procedures is FFS.

NF Discovery Service via NRF:

- NFs that serves a specific set of AIOT devices stores the list of AIOT device IDs in the NF profile in NRF in the same way as current NF profiles can include, e.g. SUPI, SUPI ranges, or part of the SUPI, e.g. MCC/MNC, GPSI, GPSI ranges, etc.
- The NF discovery service operation discovers the set of NF Instances (and their associated NF Service Instances), represented by their NF Profile, that are currently registered in NRF and satisfy a number of other input query parameters as per current 5GC procedures.

6.36.3 Impacts on services, entities and interfaces

The logic of current 5GS usage of identifiers is re-used with the following differences or clarifications:

- AIOTF applies similar functionality as an AMF for handling, e.g. SUCI.
- Entities, e.g. AIoT Device, UDM that conceal and reveal SUCI to handle new SUPI type format.
- Entities handling SUPI may need to handle new SUPI format depending on stage 3 decision of format.
- Entities handling PEI, e.g. EIR may handle new PEI format.

6.37 Solution #37: Multiple Temporary Reusable Identities for AIoT Devices

6.37.1 Description

This solution addresses KI#2 and tackles the problem of efficiently and securely identifying an AIoT device while preserving the limited AIoT device computing resources. This is accomplished by providing the AIoT device with a set of temporary identifiers, which can be revoked and reallocated as explained below. Unique aspects of this solution compared to other solutions documented in this TR include the multiplicity of the temporary identities and their re-usability.

It is acknowledged that this solution would be advantageous in terms of power efficiency in scenarios where storing, accessing and removing multiple identifiers consumes less AIoT device power than separately transmitting multiple identifiers to the AIoT device.

NOTE 1: Additional considerations on non-volatile memory capabilities of the AIoT devices may be dependent on RAN WG1 discussions.

The proposed solution can be summarized as follows:

- The network should allocate and provide more than one temporary identity to the AIoT device, say N identities together with authentication information.
- The N allocated identities together with the authentication information can be sent to the AIoT device in one message.
- The AIoT device and the network will use one identity at time such that both entities know ahead of time which identity would be used next as will be described later.
- The ID combined to authentication information (e.g. tokens, encryption keys) are used to authenticate the AIoT device for each exchanged message.

NOTE 2: Mechanisms for authentication information generation, exchange and update are to be defined by SA WG3.

- The temporary IDs can be revoked from the AIoT device and re-allocated to another device e.g. in case of a pre-determined period of inactivity.
- When re-allocating the one or more ID(s) to another AIoT device, new authentication information is provided.

NOTE 3: The temporary identity may refer to any temporary identity which may be used in the 3GPP system or in the 5GS such as 5G-GUTI, or any identity that may be defined for AIoT devices.

In the following, it is described new capabilities that are required in the network and/or AIoT device to support the above functionality.

In terms of new functionality, the solution requires a new capability for multiple temporary identifiers to indicate its support for receiving, handling, and storing multiple temporary identifiers at a time. The AIoT device may also indicate how many identities it can receive at a given time, where this may be based on memory or resources in the device. The AIoT device may determine the number of identifiers which it can receive based on several factors, e.g. the total available memory in the AIoT device, the length or size of the identifier, a combination of available resources, etc. Furthermore, the network then may determine to allocate a number N of temporary identities to an AIoT device based on the capability indication provided by the AIoT device and other factors. The transmission of the set of identities should occur using a secured message.

6.37.2 Procedures

6.37.2.1 Temporary Reusable ID Allocation

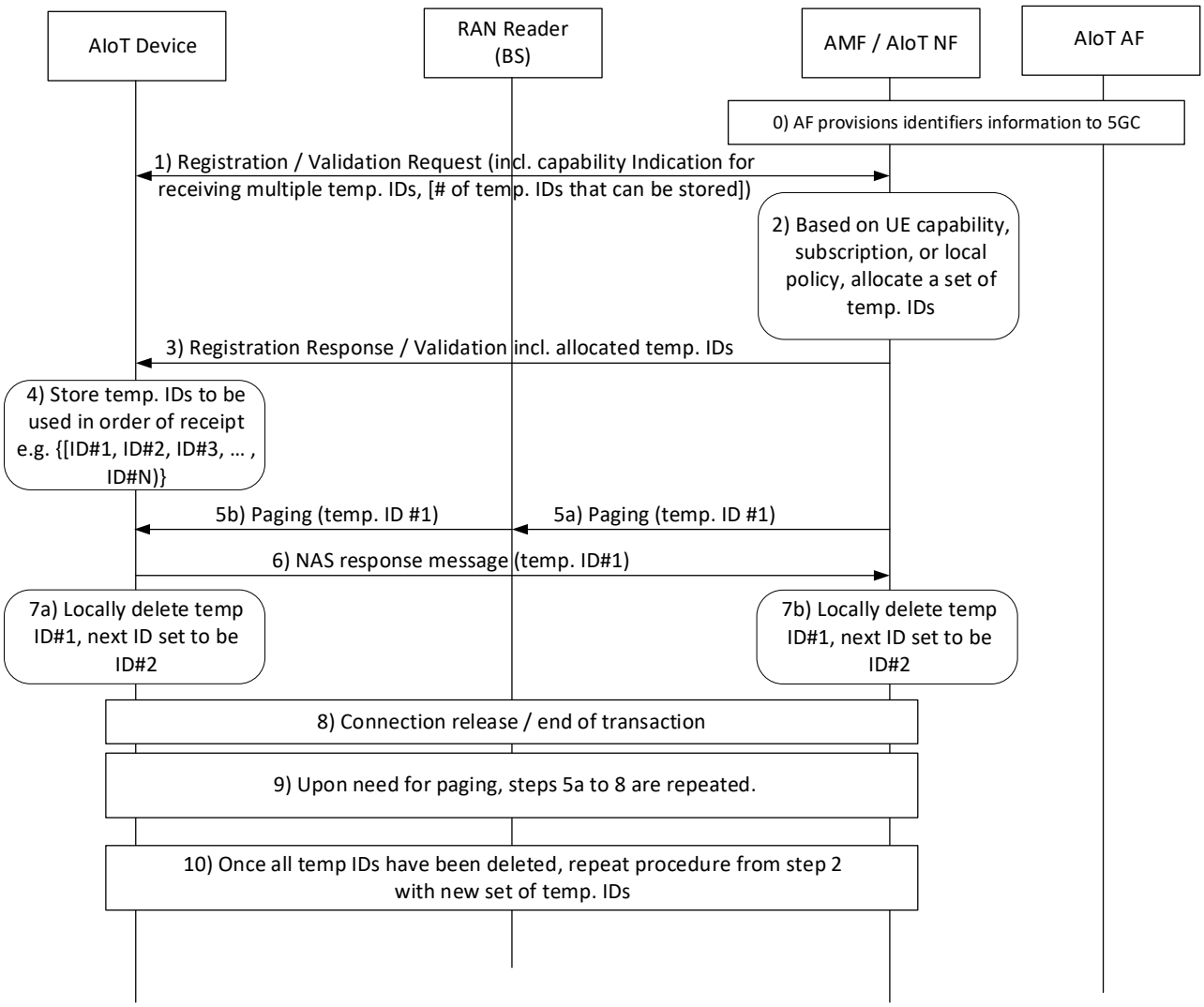


Figure 6.37.2.1-1: Procedure to support allocation of multiple temporary identities to AIoT devices

The procedure in Figure 6.37.2-1 to support allocation of multiple temporary identities to AIoT devices is described step by step below. The temporary ID allocation is assumed to be performed in a secure manner.

0. An AIoT AF may provision information related to the multiple and re-usable AIoT device identifiers to the 5GC (e.g. AIoT NF, AMF).

1. Upon a network trigger (not visible in Figure 6.37.2.1-1), the AIoT device sends a registration/validation request to the AMF (or AIoT NF). The request includes a capability indication for receiving multiple temporary identifiers. Optionally, the request may also include the maximum number of temporary identifiers that can be stored at the AIoT device.

2. The AMF (or AIoT NF) allocates a set of temporary identifiers and authentication information to the AIoT device with N entries. The allocation may be performed based on device capability, subscription, or local policy.

3. A registration response /validation notification is sent by the AMF (or AIoT NF) to the AIoT device including the set of allocated temporary identifiers with N entries together with any required authentication information.

4. The AIoT device stores the set of N temporary identifiers in the same order they were received in step 3. For example, if N identifiers were received, they should be stored so that the first identifier to be used is Temp ID#1, then Temp ID#2…, and lastly Temp ID#N.

5. A paging is sent from AMF (or AIoT NF) to the AIoT device via the RAN reader including the first temporary identifier, i.e. Temp ID#1.

6. A NAS response message is provided by the AIoT device including the temporary identifier contained in step 5, i.e. Temp ID#1 and the corresponding authentication information.

7. The temporary identifier in steps 5 and 6 is deleted locally both at the AIoT device (see step 7a) and AMF (or AIoT NF, see step 7b). In both cases, the next temporary identifier is set to be the next one in the set received in step 3, i.e. Temp ID#2.

NOTE: Whether the temporary ID is deleted here or at a later stage is FFS.

8. The connection with the AIoT device is released or the transaction/service is completed.

9. Steps 5 to 9 are repeated upon a new need for paging. Both the AIoT device and the AMF (or AIoT NF) locally delete any temporary identifier that has been used.

10. Once the full set of identifiers have been used and deleted, the procedure is repeated from step 2 with a new set of temporary identifiers being allocated by the AMF/AIoT NF to the AIoT device before the connection is released.

6.37.2.2 Temporary Reusable ID Revocation and Re-Allocation

The temporary reusable IDs may be revoked and re-allocated as described below:

- One or more temporary IDs can be revoked from the AIoT device and re-allocated to another device e.g. in case of a pre-determined period of inactivity after which the device is assumed to have left the network. A revocation notification may be sent to the device.

- When re-allocating the one or more ID(s) to another device, new authentication information is associated with the new allocation. The authentication information from a pre-defined number of previous allocations is also temporarily stored in the CN NF. The number of previous allocations that is stored per ID is assumed to be a configurable parameter.

- It is assumed that an AIoT device may not have received the revocation notification from the network and may communicate again at a later time by responding to a paging for the same ID that has been re-allocated to another device.

6.37.2.3 Temporary Reusable ID Conflict Detection and Resolution

If a device from which the ID had been previously revoked attempts to communicate again with the network by responding to the paging from the network, the following conflict detection and resolution mechanisms are performed:

- When the returning device responds to paging from the network using the temporary ID and the authentication information from a previous ID association, the network can determine that the device is using an ID which has been re-assigned, and identify it using the authentication information from the previous association.

- If the device is still authorized to communicate with the network, new temporary IDs and new authentication material are allocated to the device.

6.37.3 Impacts on services, entities and interfaces

Impacts on existing entities:

CN NF (AMF or AIoT NF):

- Support handling and allocation of multiple temporary reusable identifiers for AIoT devices.

- Support revocation and conflict detection and resolution for temporary reusable IDs for AIoT devices.

AIoT device:

- Support for multiple temporary identifiers capability indication.

- Handling the allocation of multiple temporary identifiers from the network.

6.38 Solution #38: AIoT device tracking for inventory

6.38.1 Description

This solution addresses KI#3.

The solution introduces a new role, "AIoT Device Companion UE", which is a normal UE and moves with the AIoT devices. A Companion UE is able to perform mobility/location update and support LCS procedures with the 5GC. Therefore, a Companion UE's accurate location may be known in the 5GC. When an AIoT application needs to reach one or a group of AIoT devices through the 5GC, e.g. for inventory purposes, and the target device's accurate location is not available, the 5GC may locate the Companion UE instead, and select the AIoT Function/Reader that serves the Companion UE's location, as the target devices are believed to be close to the Companion UE.

The association between the Companion UE and the AIoT devices that move with it can be provided by the AF, or reported by the Companion UE and stored in the 5GC. When a request, e.g. inventory request, is initiated from the AF for a target device, the 5GC may locate the Companion UE based on its mobility information or using LCS procedures. Based on the Companion UE's location, the 5GC can select the AIoT Function/Reader that serves the location and forward the request to the selected AIoT Function/Reader.

If the Companion UE is also AIoT capable, it can be directly selected as the Reader and the 5GC can forward the request directly to the Companion UE.

The solution can be used for both Topology 1 and 2. For the consideration of cost-effectiveness the solution is most suitable for use cases where large amount of AIoT devices are moving together (e.g. in a container ship) and share a companion UE.

6.38.2 Procedures

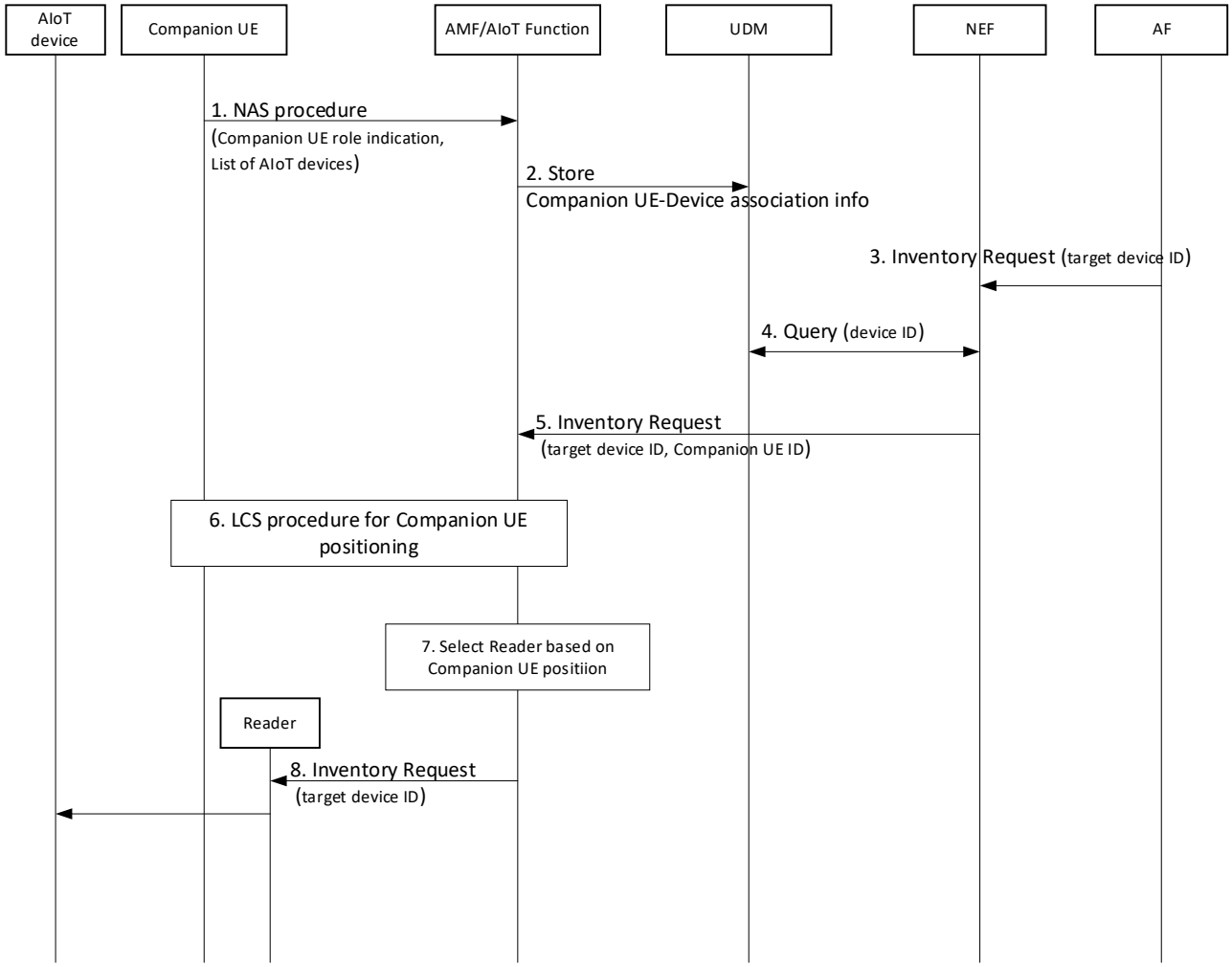


Figure 6.38.2-1: AIoT inventory procedure using Companion UE

1. Multiple AIoT devices move with a Companion UE. The Companion UE obtains a list of AIoT device IDs or group device IDs that move with it, either through preconfiguration or collecting via the radio interface. The UE reports its role of Companion UE and the list of AIoT device IDs it is serving to the AMF/AIoT Function using NAS procedure (e.g. Registration Update). If the Companion UE additionally supports UE Reader functionality, it may also initiate UE Reader authorization procedure with the network to serve as a UE Reader.

2. The AMF/AIoT Function stores the association between the Companion UE and AIoT device IDs or group IDs in UDM.

Editor's note: It's for FFS how the association is stored in the UDM/UDR.

3. The 5GC receives an inventory request from an AF. The target device ID(s) or group device ID(s) is indicated in the request. In case that no device ID or group ID is provided in the request, this Companion-UE-assisted inventory procedure is not used and the network may continue with normal inventory procedure as described in other solutions in the TR.

4. The NEF queries the UDM for information about the target device ID. It discovers that it has a Companion UE. The NEF obtains the serving AMF/AIoT Function address of the Companion UE.

5. The NEF forwards the inventory request to the AMF/AIoT Function and includes the Companion UE ID in the request.

6. The AMF/AIoT Function obtains the Companion UE location from the mobility information or using the LCS procedure.

7. Based on the Companion UE location, the AMF/AIoT Function selects a Reader (e.g. Base Station) that serves the Companion UE's location and forwards the request to the Reader. In case that the companion UE can serve as a UE Reader, the network may forward the request to the Companion UE.

8. The Reader delivers the inventory request to the AIoT devices and receives the response from the devices.

6.38.3 Impacts on services, entities and interfaces

UE:

- Support the role of Companion UE (obtaining AIoT device IDs, report their roles to the network).

AMF/AIoT Function:

- Receive and store Companion UE-AIoT device(s) ID association in the UDM.
- Select Reader based on Companion UE location.
- Forward inventory request to Reader.

UDM:

- Store Companion UE-AIoT device(s) ID association.
- Provide Companion UE information upon query.

NEF:

- Query Companion UE information from UDM.
- Forward inventory request to AMF/AIoT Function.

6.39 Solution #39: Configuration of Ambient IoT device frequency

6.39.1 Description

This solution addresses Key Issue #2 by providing simplified connection management for Ambient IoT devices. Based on the assumptions in clause 4.1 of the present document, there is no cell selection/re-selection like feature for Ambient IoT devices. Hence, it is unclear how the Ambient IoT devices is going to work with different operators as operators own spectrum in different frequency bands. Not being able to adjust an Ambient IoT device to work with different operators can cause limitations in deployments and supported use cases. For example, the use case would require coverage from one operator or operators working on the same frequency across borders. With limitations in deployments there is an increased risk Ambient IoT devices become one time use devices creating a large carbon footprint.

This solution proposes a mechanism to reconfigure Ambient IoT devices to listen to different frequencies with the ambition to support Ambient IoT services requiring coverage from multiple operators and creating a sustainable deployment of Ambient IoT devices.

Editor's note: Whether or not the Ambient IoT devices can backscatter data using a carrier wave over the complete frequency range or specific frequency band(s) is FFS.

Editor's note: This solution is dependent on RAN WG1 agreements.

This solution assumes a third party owns the Ambient IoT device(s), the Ambient IoT device(s) share an Ambient IoT group ID and a group key. The group key is used by Ambient IoT device(s) associated with the Ambient IoT group ID. It is up to deployment to determine how many Ambient IoT devices are associated with the Ambient IoT group ID, hence sharing the group key.

6.39.2 Procedures

This clause describes procedures to enable and disable communication for an Ambient IoT device in licensed spectrum.

6.39.2.1 Enable Ambient IoT device(s) communication in operator network

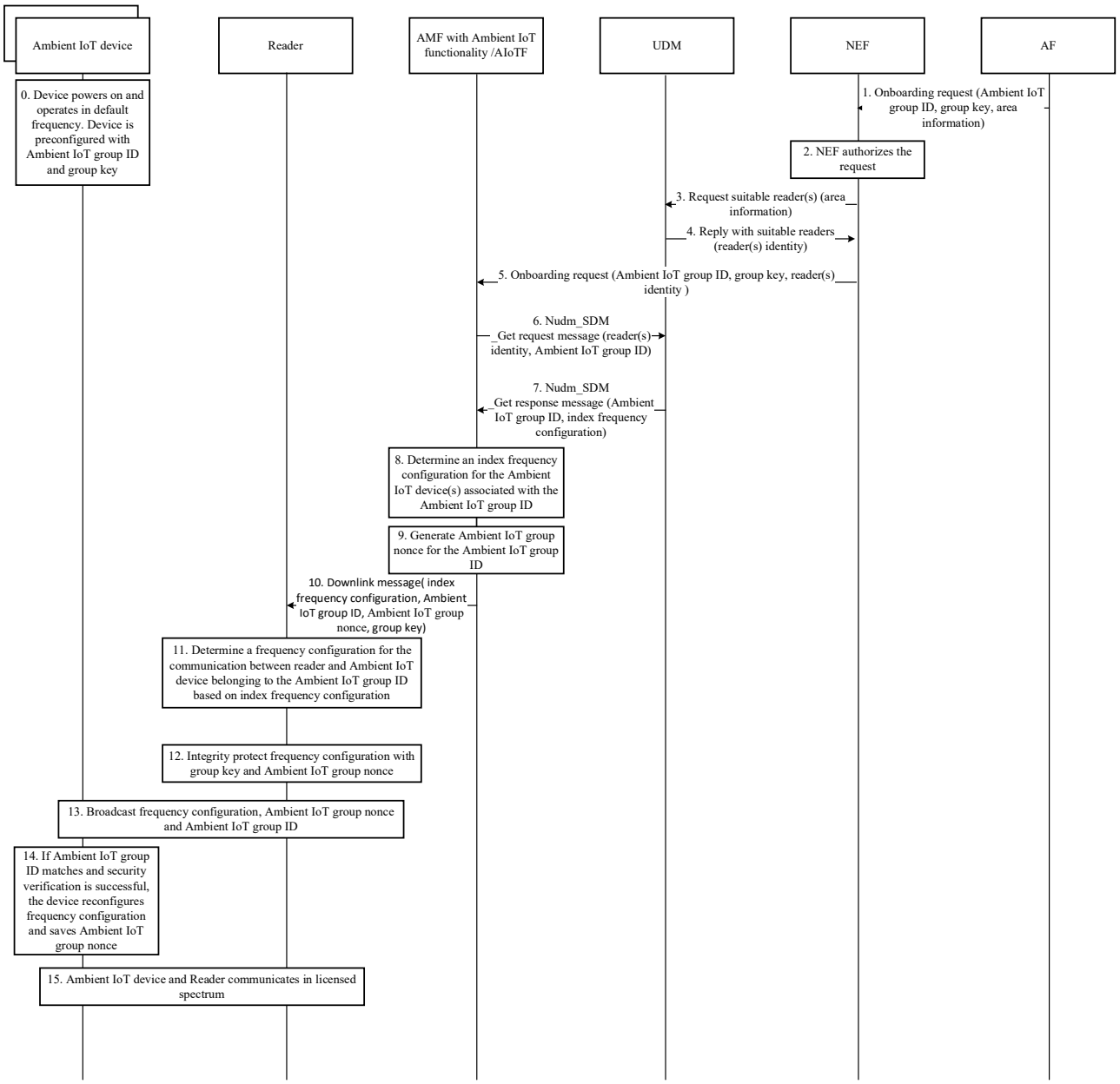


Figure 6.39.2.1-1: Enablement of communication in operator network

0. One or several Ambient IoT device(s) powers on and operates in a default frequency. The Ambient IoT device(s) operates in a default frequency. The default frequency is unlicensed. The Ambient IoT device(s) is preconfigured with Ambient IoT group ID and group key.

Editor's note: Whether or not the Ambient IoT devices can use a default frequency which is unlicensed is FFS.

1. An AF sends an Onboarding request to a serving network. The onboarding request contains the Ambient IoT group ID, the group key and area information.

NOTE 1: AF optionally includes the default frequency.

2. The NEF verifies the AF is authorized to make requests to the serving network.

3. The NEF sends a request to an UDM to retrieve suitable reader(s) to onboard the Ambient IoT device(s) and includes area information.

4. The UDM responds with suitable readers, serving AMF with IoT Functionality / AIoTF and includes the reader(s) identity. Reader identity can point to a UE reader or RAN reader.

5. The NEF sends the onboarding request to the AMF with IoT Functionality / AIoTF and includes the Ambient IoT group ID, the group key and the reader(s) identity.

6. The AMF with IoT Functionality / AIoTF sends a service based request message e.g. Nudm_SDM_Get request message to the UDM. The Nudm_SDM_Get message contains reader(s) identity and the Ambient IoT device ID. The purpose of the request is to obtain an index frequency configuration. If the AMF with IoT Functionality / AIoTF already has an index frequency configuration for Ambient IoT services this step may be omitted.

Editor's note: How the UDM receives the relevant information is FFS.

7. The UDM responds with a service based response message e.g. Nudm_SDM_Get response message. The UDM can have the index frequency configuration stored per reader, per Ambient IoT group ID, for Ambient IoT services e.g. based on an operator's preference. The Nudm_SDM_Get response message contains the Ambient IoT Group ID and the index frequency configuration. If the AMF with IoT Functionality / AIoTF already has an index frequency configuration for Ambient IoT services this step may be omitted.

8. The AMF with IoT Functionality / AIoTF determines the index frequency configuration for the Ambient IoT device(s) associated with the Ambient IoT group ID. The determination of index frequency configuration is based on the licensed spectrum the operator owns, an operator's preference of steering Ambient IoT services to a frequency band, local configuration or based on current load in the serving network.

9. The AMF with IoT Functionality / AIoTF generates a Ambient IoT group nonce for the broadcasting of a frequency configuration. The Ambient IoT group nonce is used to prevent replay attacks.

10. The AMF with IoT Functionality / AIoTF sends a downlink message towards the reader. The downlink message contains the index frequency configuration, group key, the Ambient IoT group nonce and Ambient IoT group ID.

11. The base station determines the frequency configuration based on the index frequency configuration. In Topology 1, the base station uses the frequency configuration for communication with Ambient IoT devices. In Topology 2, the base station forwards the frequency configuration and other parameters to the UE acting as the reader.

12. The reader integrity protects the frequency configuration with the group key and Ambient IoT group nonce. In Topology 2, the base station integrity protects the frequency configuration before sending the frequency configuration to the UE acting as a reader.

Editor's note: The level of necessary security and if security provided by the group key is for SA WG3 to decide, e.g. integrity and/or confidentiality protection with/without replay protection using NAS counters/nonce. Algorithm to use and support for key rotation.

13. The reader broadcasts the Ambient IoT group ID, frequency configuration and Ambient IoT group nonce.

14. The Ambient IoT device applies the frequency configuration if the broadcasted Ambient IoT group ID matches and frequency configuration is validated successfully with the group key and Ambient IoT group nonce. The Ambient IoT device stores the Ambient IoT group nonce to be able to detect replay attacks.

NOTE 2: The length of the Ambient IoT group nonce and how many Ambient IoT group nonces are saved and for how long is implementation specific.

15. The Ambient IoT device and reader can communicate in licensed spectrum owned by the serving network.

6.39.2.2 Disable Ambient IoT device(s) communication in operator network

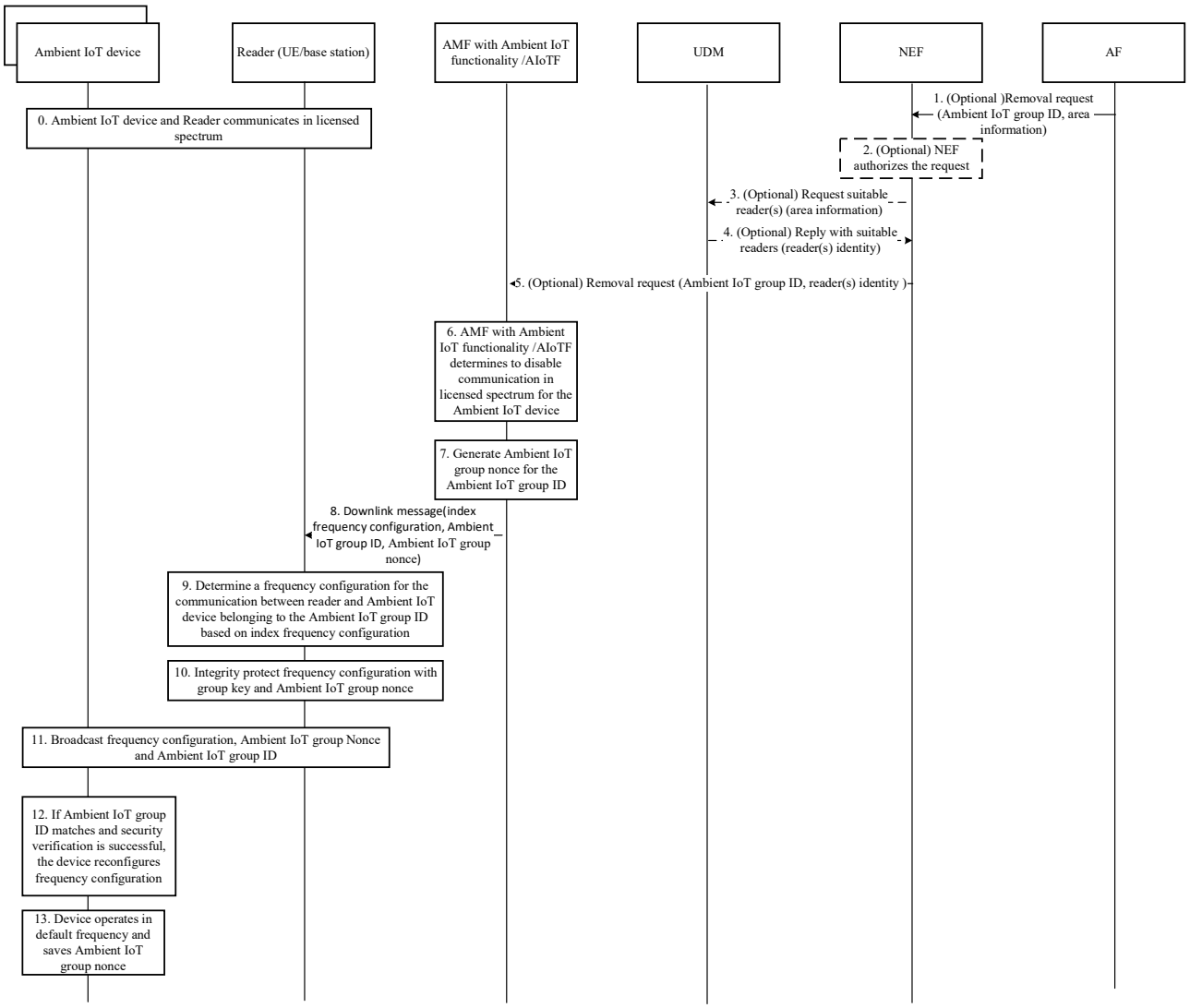


Figure 6.39.2.2-1: Disablement of communication in operator network

0. The Ambient IoT device and reader can communicate in licensed spectrum owned by the serving network.

1. Optional, the AF can send a Removal request to the NEF. The reason can be the AF is planning to move Ambient IoT devices from one serving network to another serving network, decommissioning of the service using the Ambient IoT device(s) or some other reason. The removal request contains the Ambient IoT group ID and area information.

Editor's note: Further details on how the AF can determine when to trigger the removal process of the Ambient IoT device(s) is FFS.

Editor's note: How to deal with a situation where the Ambient IoT device is no longer reachable in the configured frequency is FFS.

2. The NEF verifies the AF is authorized to make requests to the serving network.

3. The NEF sends a request to an UDM to retrieve suitable reader(s) to disable communication for the Ambient IoT device(s) in the operator network and includes area information.

4. The UDM responds with suitable readers, serving AMF with IoT Functionality / AIoTF and includes the reader(s) identity. Reader identity can point to a UE reader or RAN reader.

5. The NEF sends the removal request to the AMF with IoT Functionality / AIoTF and includes the Ambient IoT group ID and the reader(s) identity.

6. The AMF with IoT Functionality / AIoTF determines to disable communication in licensed spectrum for the Ambient IoT device(s). Disabling communication can be based on receiving the removal request from the AF via the NEF, local configuration, reconfiguration of resources, the Ambient IoT device(s) being taken out of service, the Ambient IoT device(s) being moved from one serving network to another serving network etc. The AMF with IoT Functionality / AIoTF produces a index frequency configuration indicating preference for unlicensed spectrum.

7. The AMF with IoT Functionality / AIoTF sends generates a Ambient IoT group nonce for the broadcasting of a frequency configuration. The Ambient IoT group nonce is used to prevent replay attacks.

8. The AMF with IoT Functionality / AIoTF sends a downlink message towards the reader. The downlink message contains the index frequency configuration, group key, the Ambient IoT group nonce and Ambient IoT group ID.

9. The base station determines the frequency configuration based on the index frequency configuration. In Topology 1, the base station uses the frequency configuration for communication with Ambient IoT devices. In Topology 2, the base station configures the UE acting as a reader with the frequency configuration and forwards parameters to the UE acting as the reader.

10. The reader integrity protects the frequency configuration with the group key and Ambient IoT group nonce.

11. The reader broadcasts the Ambient IoT group ID, frequency configuration and Ambient IoT group nonce.

12. The Ambient IoT device applies the frequency configuration if the broadcasted Ambient IoT group ID matches and frequency configuration is validated successfully with the group key and Ambient IoT group nonce. The Ambient IoT device stores the Ambient IoT group nonce to be able to detect replay attacks.

13. The Ambient IoT device is disabled and can no longer communicate in licensed spectrum of the serving network.

6.39.3 Impacts on services, entities and interfaces

AF:

- Support requesting onboarding and removal of Ambient IoT devices from a serving network.

NEF:

- Supports signalling for onboarding and removal of Ambient IoT devices in the serving network.

AMF with AIoT functionality / AIoTF:

- Support index frequency configuration for Ambient IoT e.g. steer to licensed spectrum or unlicensed spectrum.
- Generate Ambient IoT group nonce to protect against replay attacks.

UDM:

- Store index frequency configuration for a group of Ambient IoT devices.

Ambient IoT device:

- Support at least two frequency bands.
- Storage of Ambient IoT group nonce, group key and Ambient IoT group ID.

6.40 Solution #40: Reader dependent Inventory Response

6.40.1 Description

This solution address aspects of Key Issue #3 and specifically relates to inventory and "how to support information transfer for Ambient IoT services and related system functionality, including the information transfer for an Ambient IoT device and for a group of Ambient IoT Devices".

In some scenarios, it may not be permitted for certain readers to receive application payloads from the AIoT Device. The owner of the AIoT Device may not trust certain readers to inappropriately read or save the application payloads that

are transmitted by the AIoT Devices. However, the owner of the AIoT Devices may allow the devices to transmit just the AIoT Device ID via some of these "less-trusted" readers. These "less-trusted" readers might be useful in scenarios where the location of the AIoT device needs to be determined even if application payloads cannot be obtained.

The principle of this solution is that the AIoT Device can be configured with information that it can use to determine whether to respond to an inventory request with both an AIoT Device ID and an Application Payload, or only an AIoT Device ID, or with neither of them.

6.40.2 Procedures

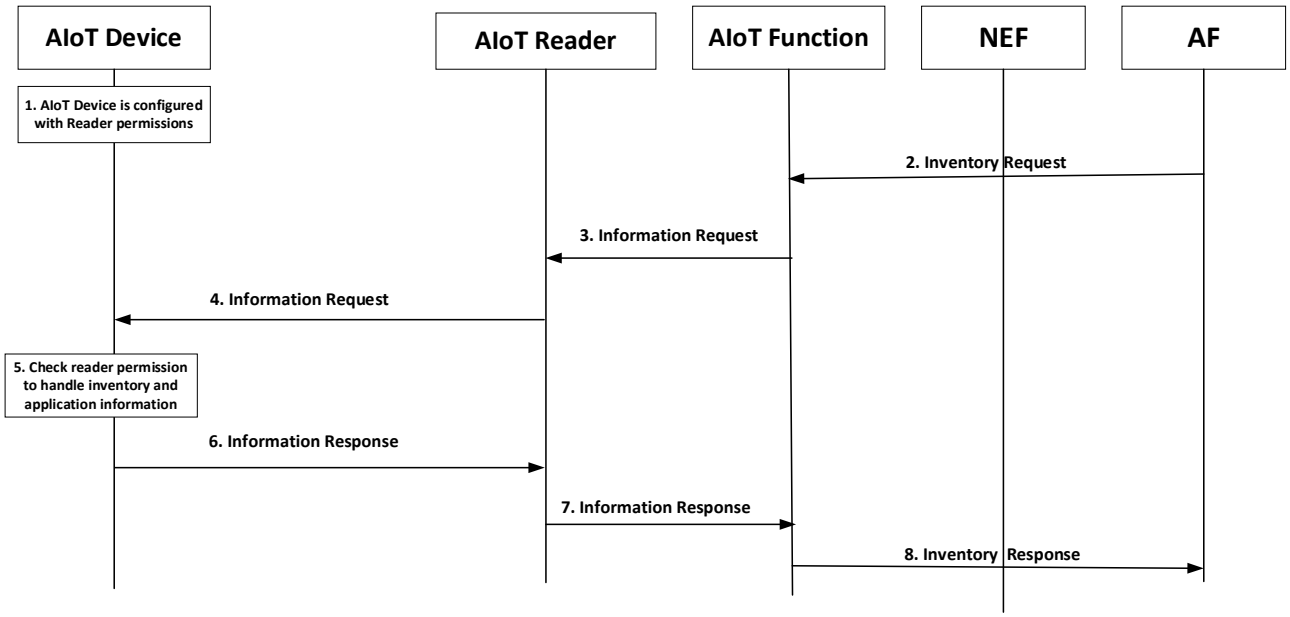


Figure 6.40.2-1: Inventory Procedure where the AIoT Device Checks Reader Permissions before Responding

1. The AIoT Device is configured with reader(s) identities and permissions.

   The reader(s) identities and permissions can be either pre-configured in the AIoT Device or sent to the AIoT Device by the network. The configuration information can include the following information for each reader:

   - Reader's identity:

     - Whether the AIoT Device is allowed to respond to the reader, and if the AIoT device is allowed to respond to the reader:

     - Whether the AIoT Device is allowed to respond to the reader with only it's AIoT Device ID or

     - Whether AIoT device is allowed to respond to the reader with both its AIoT Device ID and an Application Payload. If the AIoT Device is allowed to respond to the reader with an Application Payload, then the configuration information can include the service/application identifier(s) associated with the application payloads that are allowed.

NOTE 1: Whether the AIoT Device is able to store the Reader identities and other information depends on the device capability.

2. The AF initiates an inventory request toward the AIoT Device. The request includes the AIoT Device ID and can include the service/application identifier of the application on the AIoT Device that the AF wants to obtain data from.

3. The AIoT function sends an information request to a AIoT reader. The request has the AIoT Device ID and can include the service/application identifier.

4. The reader sends the information request to the AIoT Device. This request has the reader identity, AIoT Device ID, and can include the service/application identifier.

NOTE 2: The solutions assumes that the air interface Paging message can accommodate the reader identity, AIoT Device ID, and can include the service/application identifier.

5. The AIoT Device checks the configuration information that was received in step 1 to determine how to respond to the reader.

6. The AIoT Device uses the configuration information from step 1 to determine how to respond to the request. For example, the AIoT Device responds by sending the AIoT Device identifier and application data or only the AIoT Device Identifier. If no Reader identity is included in the request, the device may respond with only the device identifier or ignore the request, based on the configuration.

7. The Reader sends the AIoT Device response to the AIoT function.

8. The AIoT Function sends the AIoT Device response to the AF.

NOTE 3: Potential security threats, e.g. a malicious player posing as a Reader, require further study in SA WG3.

6.40.3 Impacts on services, entities and interfaces

Reader:

- Supports sending the request to AIoT devices with Reader Identity, AIoT Device ID and Service/Application ID.

AIoT Device:

- Supports per-reader configuration information.
- Uses the per-reader configuration to determine how to respond to an inventory request (e.g. whether to include application data or to only send the AIoT Device Identifier).

6.41 Solution #41: Simple AIoT device and AIoT solution in the certain deployment

6.41.1 Description

One typical AIoT use case is inventory in the warehouse, where the environment is closed and controllable by the warehouse owner (as the 3rd party/AF) and the AIoT network operator.

For the AIoT device with very limited capabilities and power can access to the network, the AIoT device operation including security and privacy can be compromised if it is agreed by the AIoT network operator and the 3rd party.

In addition to the architectures in Figure 6.3.1.4-1 and Figure 6.3.1.4-2, the architecture is enhanced with a core network function AIoT device Management, which can manage the device including the indication that this AIoT device can be operated without security and privacy requirements on the device and the related Reader(s).

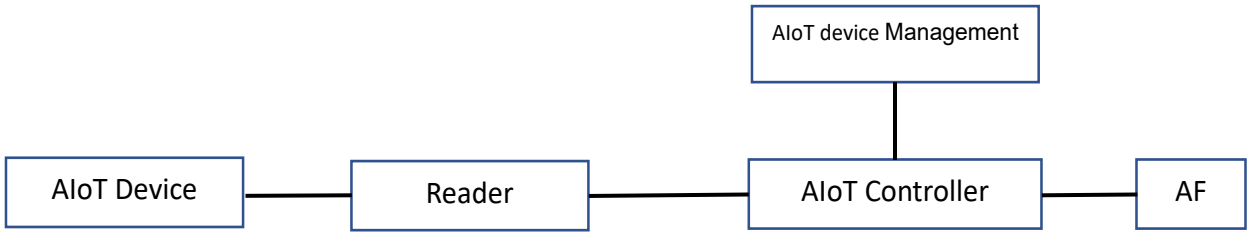


Figure 6.41.1-1: Ambient IoT system architecture for Topology 1

6.41.2 Procedures

In this solution, the AIoT device does not need to have security or privacy capabilities internally. For the AIoT device having security or privacy capabilities, either the warehouse owner or the network operator configures the AIoT device without security or privacy enabled in this case. From the AIoT network and 3rd party's perspectives, the device only needs to store its device ID, and responds to a reader with its device ID when the reader requests for inventory. Based on the agreement between AF and AIoT device management, AIoT controller authorizes the simple AIoT operation without security and privacy.

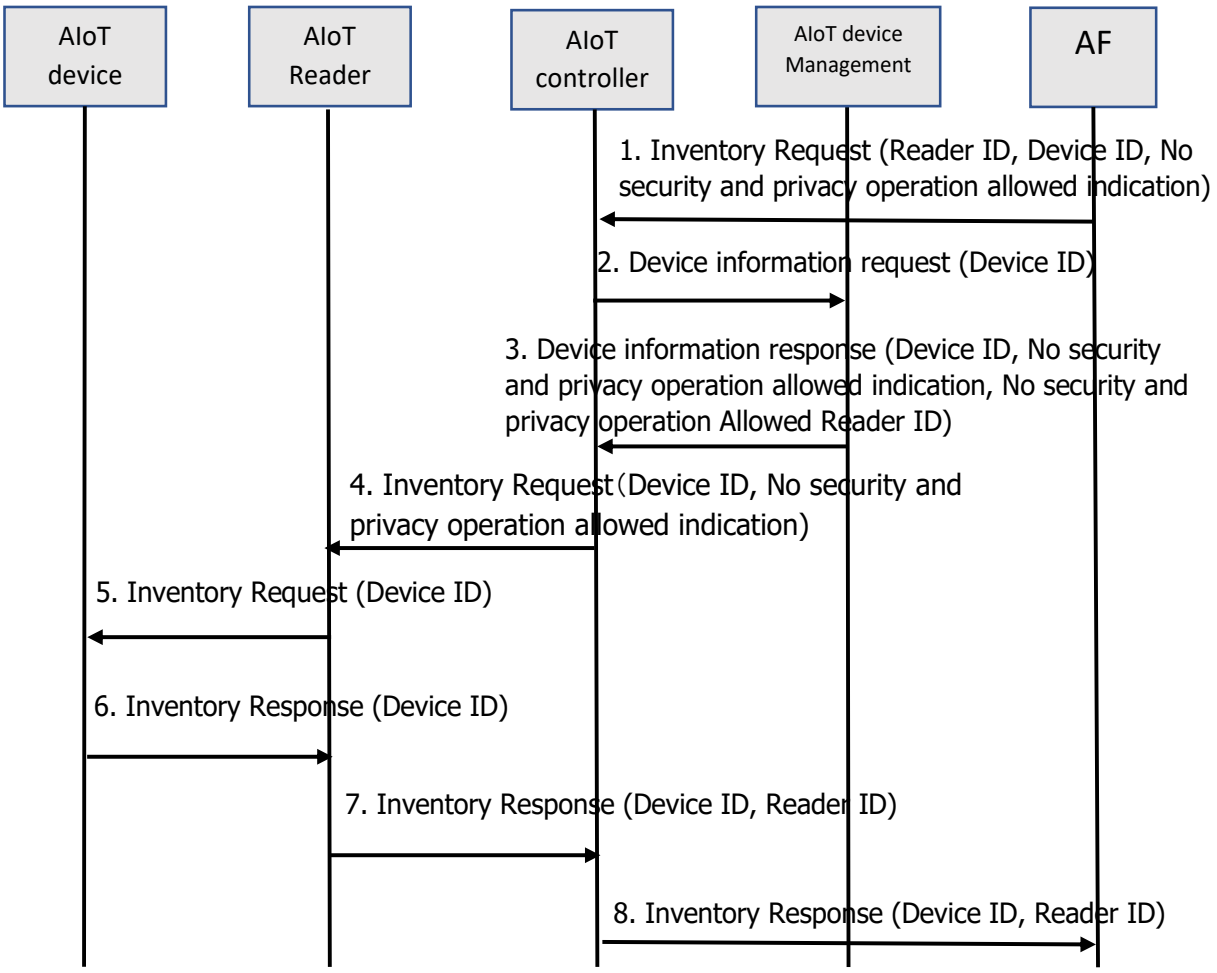


Figure 6.41.2-1: Inventory procedure

1. AF sends Inventory request to AIoT controller, which includes Reader ID, Device ID and No security and privacy operation allowed indication. The Reader ID indicates the reader requested for Inventory. Multiple Reader IDs can be included. The Device ID indicates the AIoT device for Inventory. Multiple Device IDs can be included.

2. AIoT controller sends Device information request to AIoT device management, which includes Device ID received in step 1. If multiple Device IDs are received in step 1, AIoT controller can repeat this step per Device ID for multiple times, or AIoT controller can send the Device ID list to AIoT device management in the same message.

3. AIoT device management sends Device information response to AIoT controller, which includes the Device ID, No security and privacy operation allowed indication, No security and privacy operation allowed Reader ID(s). If Multiple Device IDs are received in step 2, this message may include all device information corresponding to the Device IDs.

4. Based on the received information in step 3 and the received request in step 1, if the AF and the AIoT device management both indicate No security and privacy operation for the AIoT device, the AIoT controller authorizes the No security and privacy operation for the AIoT device under the Reader(s). The reader ID(s) shall be the

common parts both from the step 1 and step 3, the differences between them shall be excluded. The AIoT controller sends the inventory Request to the reader(s), which includes Device ID, No security and privacy operation allowed indication.

5. The Reader(s) send Inventory Request to AIoT device, which includes the Device ID.
6. If the received Device ID is matched with the Device ID stored in the AIoT device, the AIoT device sends Inventory Response to the Reader with its Device ID.
7. The Reader sends Inventory Response to the AIoT controller with the Device ID and Reader ID.
8. AIoT controller stores the Device ID and the Reader ID, if multiple Inventory Responses are received from different Readers, the AIoT controller stores the Device ID and multiple Reader IDs. AIoT controller sends the Inventory Response to AF with the Device ID and Reader ID(s).

6.41.3 Impacts on services, entities and interfaces

AF:

- It indicates the 3rd party's preference with No security and privacy operation allowed indication to AIoT network.

AIoT device Management:

- A core network function manages the AIoT device including the indication that this AIoT device can be operated without security and privacy requirements.

AIoT controller:

- A core network function authorizes No security and privacy operation and indicates to the selected Reader.

Reader and AIoT device:

- No security and privacy operation.

NOTE: Whether security and privacy can be disabled is to be determined by SA WG3.

6.42 Solution #42: AIoT data forwarding over UE reader in topology 2

6.42.1 Description

This solution address aspects of key issue #3 on Support of Ambient IoT Services.

This solution can be applied for only Topology 2.

There are multiple methods that a UE reader forward data received from the AIoT devices to the network.

1. Over user plane.
2. Over control plane where the UE reader in RRC CONNECT state (UL NAS).
3. Over control plane where the UE in RRC IDLE state (SDT with CIoT optimisation).

Each mechanism has its own advantages and disadvantages depending on specific factors as described below:

1. Latency Requirements:
   a. User Plane: Suitable for applications requiring low latency and high data rates, as it allows for direct data transfer.
   b. UL NAS: Typically has higher latency compared to the User Plane but might be adequate for less time-sensitive data.

c. CIoT optimisation: Designed for small data bursts and can provide lower latency without full RRC connection setup, making it efficient for small, infrequent data transmissions.

2. Data Volume:

   a. User Plane: Ideal for large data volumes since it supports higher throughput.

   b. UL NAS: More appropriate for moderate data volumes, where full control signalling is acceptable.

   c. CIoT optimisation: Best for small data payloads, as it avoids the overhead of establishing and maintaining an RRC connection.

3. Energy Efficiency:

   a. User Plane: Generally, consumes more power due to continuous connection and data transfer.

   b. UL NAS: Might offer a balance between power consumption and connectivity requirements.

   c. CIoT optimisation: Optimized for low power consumption, making it suitable for battery-operated devices with small data needs.

4. QoS Requirements:

   a. User Plane: Offers better support for QoS management, ensuring data prioritization and reliability.

   b. UL NAS: May not provide the same level of QoS granularity as the User Plane but can still ensure reliable delivery.

   c. CIoT optimisation: Limited QoS capabilities but can be sufficient for non-critical small data transmissions.

Considering the diversity of AIoT deployment scenarios, it is envisaged that all three mechanisms mentioned above would need to be supported and selectively used, depending on several factors including the application requirements and/or the network conditions.

This solution proposes a mechanism to select and indicate which forwarding mechanism should be used by the UE reader.

6.42.2 Procedures

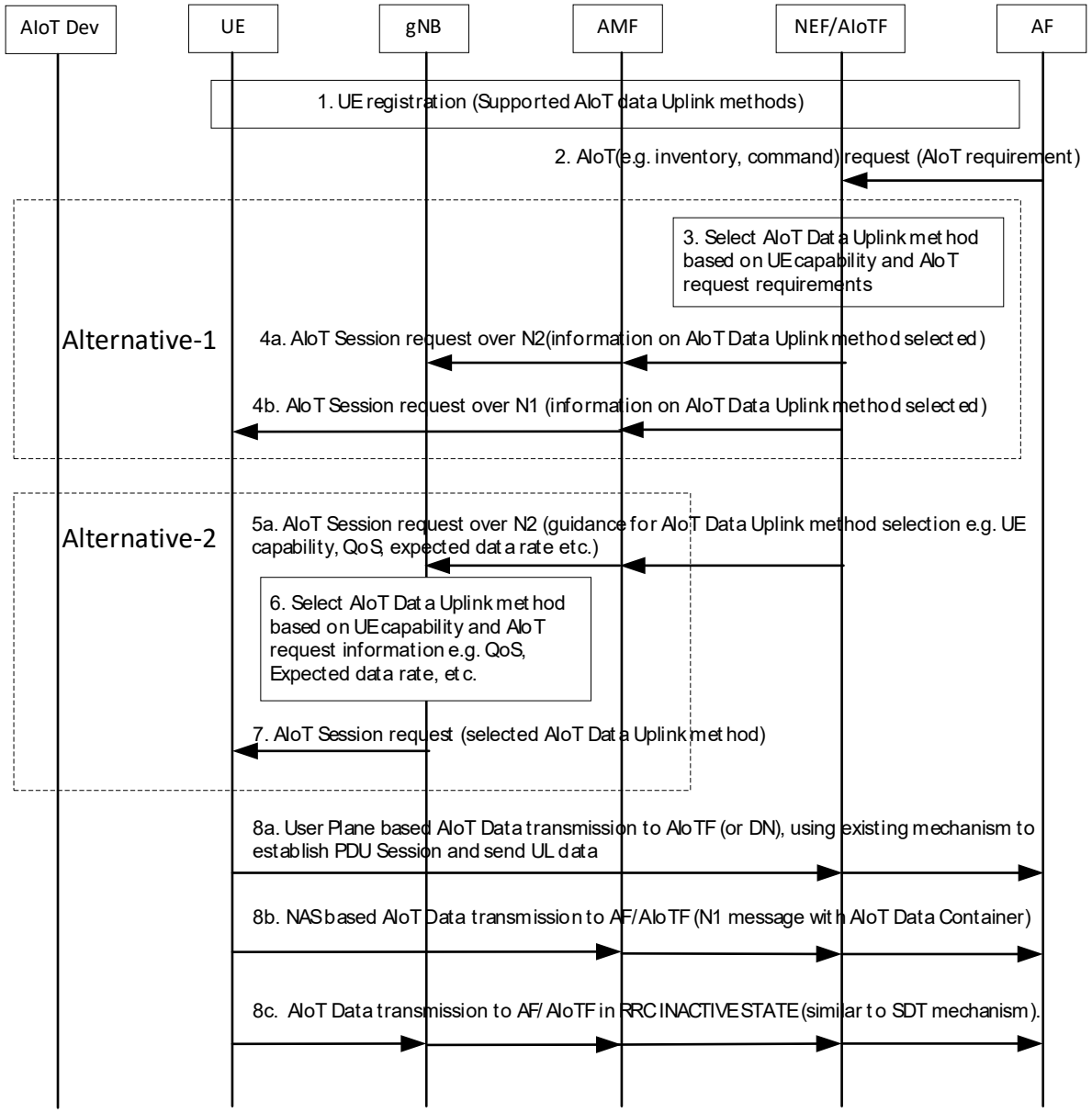


Figure 6.42.2-1: Information Flow for AIoT communication optimization

1. When a UE registers with the 5GC, it provides its capability, indicating which AIoT data forwarding (UL) methods (e.g. UP, UL NAS and/or CIoT optimisation) it supports.

2. An AF sends an AIoT service (e.g. inventory or command) request message to the 5GC. This message may contain the AIoT service requirements such as maximum latency, reliability level, energy level etc.

3. The AIoTF may select the AIoT data forwarding method to be used be the UE reader based on the AIoT service requirements given in step 2 as well as the UE capability indicated in step 1.

NOTE: How the AIoTF determines which UL data forwarding mechanism to select is up to the AIoTF implementation, taking into account the information provided by the AF, the UE capability and/or the local policy.

4a. The AIoTF may send a AIoT session request message to the gNB over N2 interface, indicating the selected AIoT data forwarding method.

4b. The AIoTF may send a AIoT session request message to the UE over N1 interface, indicating the selected AIoT data forwarding method.

5. As an alternative to step 3, the AIoTF may not select the AIoT data forwarding method, but only provide the necessary information, including the UE capability and the AIoT service requirements, to the gNB.

6. The gNB selects the AIoT data forwarding method to be used be the UE reader based on the information given in step 5.

7. The gNB may send a AIoT session request message to the UE, indicating the selected AIoT data forwarding method.

8a. When the UP method is selected, the UE establishes a PDU session targeting to the AIoTF (or directly DN), using the existing mechanisms and forwards the AIoT data.

8b. When the UL NAS method is selected, the UE forwards the AIoT data to the AMF, which then deliver the message to the AIoTF.

8c. When the SDT with CIoT optimisation method is selected, the UE forwards the AIoT data to the gNB in the RRC RESUME request message. The gNB will then further forward the message to the AIoTF over AMF.

6.42.3 Impacts on services, entities and interfaces

Entity in 5GC, responsible for AIoT service handling (e.g. AIoTF):

- Supports selecting the AIoT data forwarding methods or providing the relevant information to the gNB.

AIoT UE reader:

- Supports indicating its capability for AIoT data forwarding methods

RAN node:

- Supports selecting the AIoT data forwarding methods and indicating the result to the AIoT UE reader

6.43 Solution #43: Harmonized RAN-CN solution using AIOTF for Topology 2

6.43.1 Description

This solution proposes a complementary solution to Solution #8. It enables the support of Ambient IoT devices for topology 2 with harmonized RAN-CN architecture, which addresses KI#1, KI#2 and KI#3. Together with Solution #8 which focus on Topology 1, the solutions can support AIoT devices for both Topologies.

In this solution, NG-RAN performs NG Setup towards AIOTF besides AMF. AIOTF is responsible for generating the device NAS message and delivering as non-UE associated signalling towards NG-RAN. It is NG-RAN who is in charge of the intermediate UE for the Ambient IoT operations, including selecting intermediate UEs (optionally based on the candidate UE list form the AIOTF), sending operation commands towards intermediate UEs and forwards the results from intermediate UEs towards the AIOTF. As the licensed spectrum is owned by MNO, it is proposed to let NG-RAN provide the radio resource information towards the intermediate UEs about the spectrum information for the over-the-air interface between Intermediate UEs and AIoT devices.

6.43.1.1 Reference Architecture

The Figure 6.43.1.1-1 illustrates the architecture for harmonized RAN-CN solution using AIOTF for topology 2:

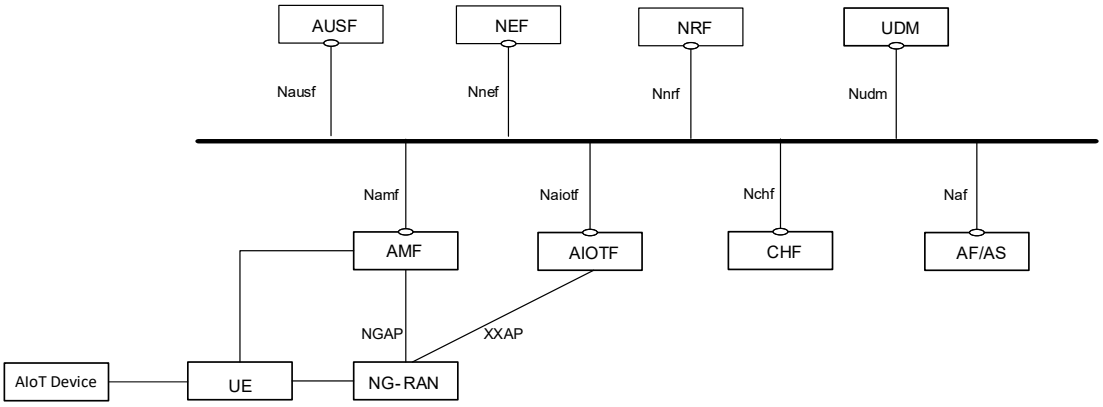


Figure 6.43.1.1-1: System Architecture of harmonized RAN-CN solution using AIOTF for Topology 2

This solution focuses on Topology 2, which works together with Solution #8 addressing Topology 1.

The functional entities defined in Solution #8 and TS 23.501 [4] are reused with the differences:

- AIOTF: Distribute AIoT device NAS messages to NG-RAN.
- NG-RAN: Determine Intermediate UEs, allocate radio resources for communication between Intermediate UEs and AIoT devices, forwards device NAS messages to Intermediate UEs, and forwards the response to AIOTF.

NOTE: The XXAP is to be determined by RAN WG3.

6.43.1.2 Protocol Stack

The Figure 6.43.1.2-1 illustrates the protocol stack:

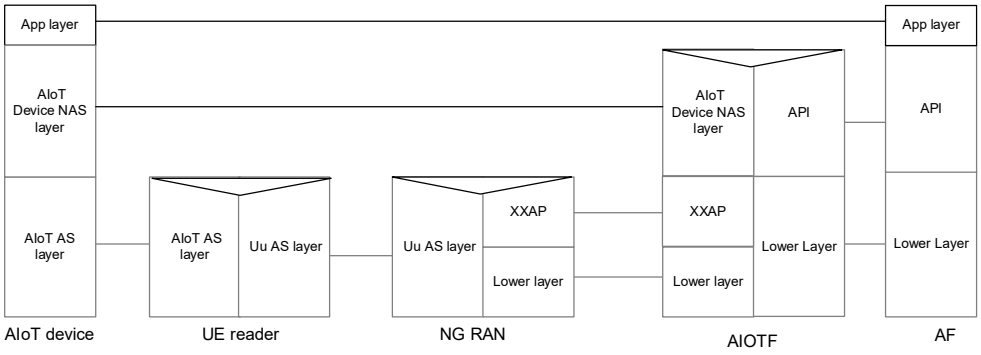


Figure 6.43.1.2-1: Protocol Stack for harmonized RAN-CN solution using AIOTF for Topology 2

As solution #8, within the protocol stack:

- AIOT device NAS layer: The NAS protocol between AIOTF and AIoT devices.
- AIOT AS layer: The AS protocol layers between UE reader and AIoT devices, including physical layer, MAC layer, etc.
- App layer: The application layer protocol between AIoT devices and AF.

In NG-RAN, the AIoT service operation information received in NGAP will be mirrored to Uu AS layer.

NOTE: The XXAP is to be determined by RAN WG3.

6.43.2 Procedures

NOTE: The message names in the procedures below are descriptive. It is assumed that the names are updated with corresponding SBI based names where applicable during the normative phase.

6.43.2.1 NG Setup

The NG-RAN triggers NG Setup procedure not only towards the AMF, but also towards the AIOTF. Via the NG connection, the AIOTF delivers AIoT device NAS messages towards the NG-RAN, as well as receives AIoT device NAS messages from the NG-RAN.

6.43.2.2 AIoT Service Authorization for Intermediate UE

The procedures for AIoT service authorization for Intermediate UEs are the same as described in clause 6.30.2.1 in solution #30.

The Registration procedure for UE is performed as defined in clause 4.2.2.2 of TS 23.502 [5] with the following additions:

- UE includes the AIoT Intermediate node capability as part of "5GMM capability" in Registration Request message.
- The AMF obtains the AIoT Subscription data as part of the user subscription data from UDM using Nudm_SDM service.
- The AMF determines whether the UE is authorized to work as Intermediate UE for AIoT based on UE's AIoT Intermediate node capability and the AIoT Subscription data. The AMF includes the authorization information as part of UE context in NGAP message sent to NG-RAN.

In Service Request procedure, N2 Handover procedure, Xn Handover procedure, and when receiving Subscriber Data Update to AMF, the AMF includes the authorization information in NGAP message sent to NG-RAN.

To ensure the Intermediate UE is reachable when performing AIOT service operations:

- The NG-RAN does not release the authorized Intermediate UEs to RRC_IDLE state (UEs can be released to RRC_INACTIVE when needed); or
- The NG-RAN may release the authorized Intermediate UEs to RRC_IDLE state. And in this case, the Intermediate UEs need to be paged back by CN before the AIoT service operation.

Editor's note: It is FFS whether there are requirement to charge I-UE for performing AIOT procedures and how the charging can be perform since the CN doesn't know the 1-UE is selected by the NG RAN.

6.43.2.3 AIOTF Keep Track of Intermediate UE

The procedures to enable AIOTF to keep track of Intermediate UEs are the same as described in clause 6.30.2.2 in solution #30.

For authorized Intermediate UEs (Those Ambient IoT Capable UEs which have been authorized by the AMF as described in clause 6.43.2.2), in Initial Registration procedure as defined in clause 4.2.2.2 of TS 23.502 [5] with the following additions:

- The AMF sends Create UE Context Request to the AIOTF for the Intermediate UE.

In Deregistration procedure as defined in clause 4.2.2.3 of TS 23.502 [5], the following additions are performed:

- The AMF sends Release UE Context Request to the AIOTF for the Intermediate UE.

The AIOTF can subscribe the UE mobility event of the Intermediate UE towards the AMF to be updated for the location information.

If there is AMF relocation, the AIOTF information can be part of UE context transferred from the old AMF towards the new AMF.

When AMF determines UE moves to another AIOTF service area, it performs Release UE Context towards the old AIOTF and Create UE Context towards new AIOTF

NOTE: The service area of AIOTF is on TA granularity.

6.43.2.4 Application Inventory

The Inventory procedure can be initiated by the AF to discover one or more AIoT devices in a specific area via Intermediate UEs.

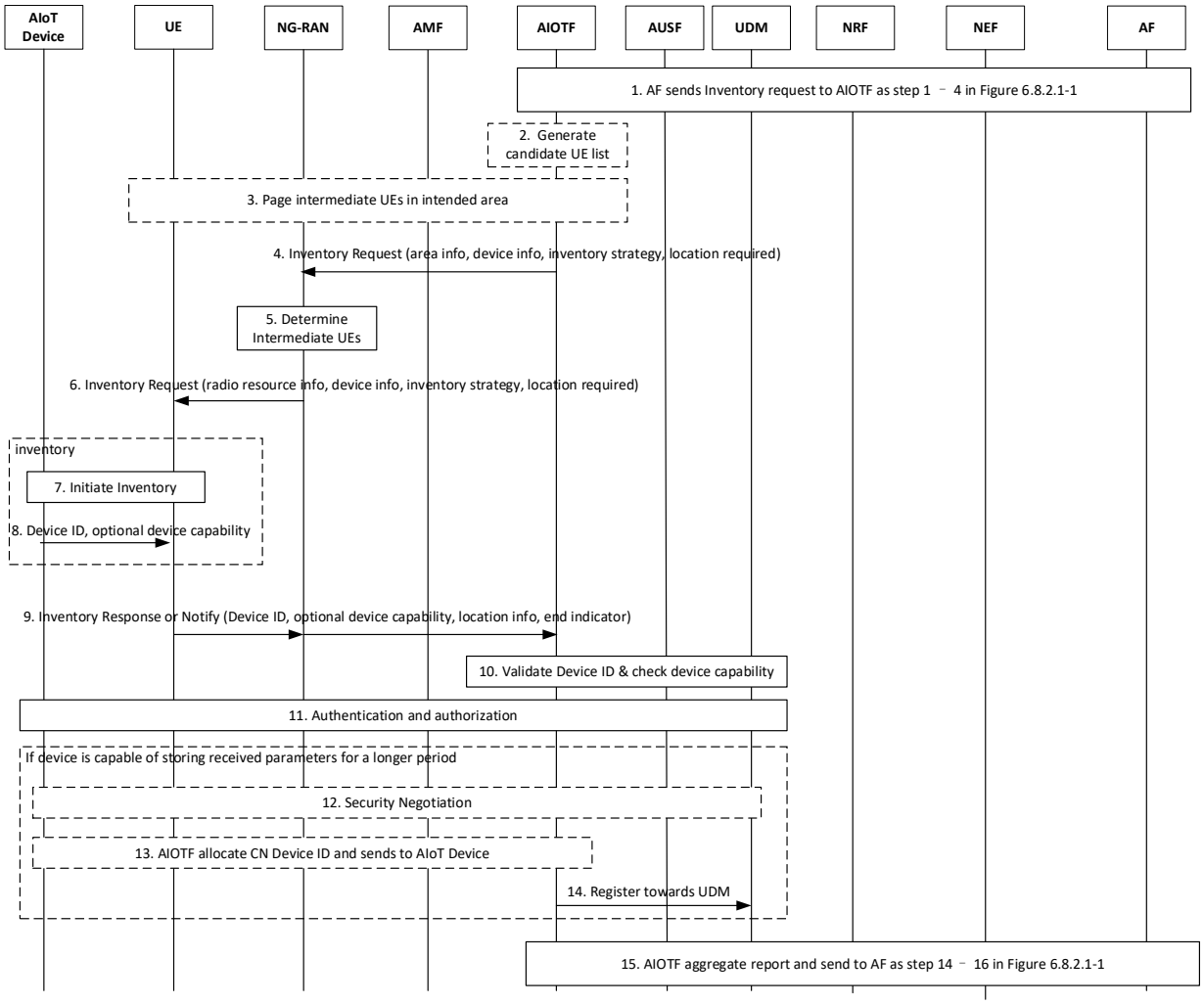


Figure 6.43.2.4-1: Inventory Procedure

1. AF sends Inventory Request to the AIOTF, which is the same as step 1 - 4 in Figure 6.8.2.1-1 in Solution #8.
2. The AIOTF may generate the candidate UE list inside the intended area, if the NG-RAN may release the intermediate UEs to RRC_IDLE state.
3. The AIOTF sends enable UE reachability request with the UE list towards the AMF to page the intermediate UEs inside the intended area. The AMF checks the CM states of those intermediate UEs, and triggers paging towards the NG-RAN(s) for those UEs in CM-IDLE state.
4. The AIOTF discovers NG-RANs based on the intended area and sends Inventory Request towards the NG-RAN with the area info, device info, inventory strategy and location required. The AIOTF uses non-UE associated signalling towards NG-RAN.

Editor's note: The enhancement of the non-UE associated signalling to support inventory in case of specific UE readers are used (e.g. indicated by the AF) is FFS.

5. The NG-RAN checks whether the inventory request can be performed. The NG-RAN triggers RAN paging for the intermediate UEs in RRC_INACTIVE state and then selects intermediate UEs. The NG-RAN should select intermediate UEs among the authorized intermediate UEs.

NOTE 1: RAN WGs will need to determine how to further select the Intermediate UE(s). Whether and how the UE posture information can be considered, it is up to RAN WGs to determine.

6. The NG-RAN determines radio resource for the AIoT service operation between the intermediate UEs and AIoT devices, and sends inventory request towards determined intermediate UEs, together with the determined radio resources.

NOTE 2: It is up to RAN WG2 to determine the Uu AS layer impacts for inventory request delivery.

NOTE 3: SA WG3 can further evaluate any security aspects of RAN sending the AIOT information to Intermediate UE using AS security established for normal PLMN functionality.

7. The Intermediate UE initiates inventory based on device information as well as the inventory strategy information provided by the AF. The Intermediate UE may provide reader identity information to enable the AIoT devices to understand they are read by which Intermediate UE. Considering the mobility of the Intermediate UE, the reader identity information can be a combination of a UE ID and the location information or the RAN ID that serves the Intermediate UE.

NOTE 4: The use of reader identity information can enable the device to understand whether it is read by the same reader or not.

8. The AIoT Device reports the device ID in an AIOT device NAS message to the Intermediate UE. If the Inventory procedure indicates delta inventory only, and the AIoT Device has performed the inventory procedure towards this UE reader in the same location or served by the same RAN node, it should skip the reporting. The device capability index can be provided by AIoT device optionally.

9. The Intermediate UE forwards AIOT device NAS message towards the NG-RAN, including the AIoT Device ID and optional device capability index. The NG-RAN forwards the AIoT device NAS message towards the AIOTF.

   The Intermediate UE may further provide location information (ULI) of the device, if requested from the AIOTF and allowed by local policy. The Intermediate UE may further provide an end indicator to inform the NG-RAN that will inform AIOTF whether it is the last inventory response for the inventory round, e.g. based on timeout in Intermediate UE.

NOTE 5: It is up to RAN WG3 to determine the details and enhancements of the location information, and the ability to provide an end indicator from a reader is assumed to be aligned between UE as reader and RAN as reader.

10. The AIOTF validates the concealed device ID via interacting with AUSF and UDM. The AIOTF may further get the device capability information from the device capability profile stored in the UDM based on the device capability index provided by the device.

11. The AIOTF together with AUSF and UDM, triggers authentication and authorization procedures (i.e. like the authentication request/response between UE and network) towards the AIoT device. In Topology 2, the message between the AIOTF and the AIOT device is transmitted via the NG-RAN and the Intermediate UE.

Based on the device capability information from UDM, if the AIoT device is capable of storing received parameters for a longer period, step 12 - step 14 are executed i.e. the AIOT device is then considered registered:

12. The security mode negotiation and security parameter exchanges are performed (i.e. like the security mode command/complete between UE and network). In Topology 2, the message between the AIOTF and the AIOT device is transmitted via the NG-RAN and the Intermediate UE.

NOTE 6: Further details of the security procedures are to be defined by SA WG3.

13. The AIOTF may further allocate CN device ID and sends it to the AIoT device.

14. The AIOTF registers with the UDM (UECM) for the AIoT device access.

15. The AIOTF may further aggregate the reported device IDs and send them to AF via the NEF, as for steps 14-16 in Figure 6.8.2.1-1.

Compared with solution#30, this solution is different in steps 2-6 and step 9: the AIOTF pages candidate intermediate UEs in the intended area and delivers the device NAS message over NGAP (e.g. inventory request) towards NG-RANs in the intended area. NG-RAN determines intermediate UEs inside the area and delivers the device NAS message over Uu AS layer (e.g. inventory request) towards intermediate UEs. The device response is sent from intermediate UEs to NG-RAN and forwarded to AIOTF. In steps 11-13, the device NAS message delivery follows the same principle.

6.43.2.5 Periodic Inventory Procedure

The periodic inventory may be requested by the AF, so that network can trigger the inventory periodically without the request from the AF directly.

Considering the movement of the Intermediate UEs which are not fixed, and the inventory is based on area information, the periodic inventory should be initiated by the CN (i.e. the AIOTF), which follows the instructions from the AF. When the period expires in the AIOTF, the AIOTF pages the Intermediate UEs based on the area information from the AF and the location information for the Intermediate UEs, and then triggers inventory procedure, as steps 2-15 in Figure 6.43.2.4-1.

6.43.2.6 Command Procedure

The command procedure is initiated by the AF to request one or more specific AIoT devices or a group of AIoT devices in an area to execute the command. The command can be read, write, enable, disable, or execution request from the AF. The command result from the devices may or may not be required.

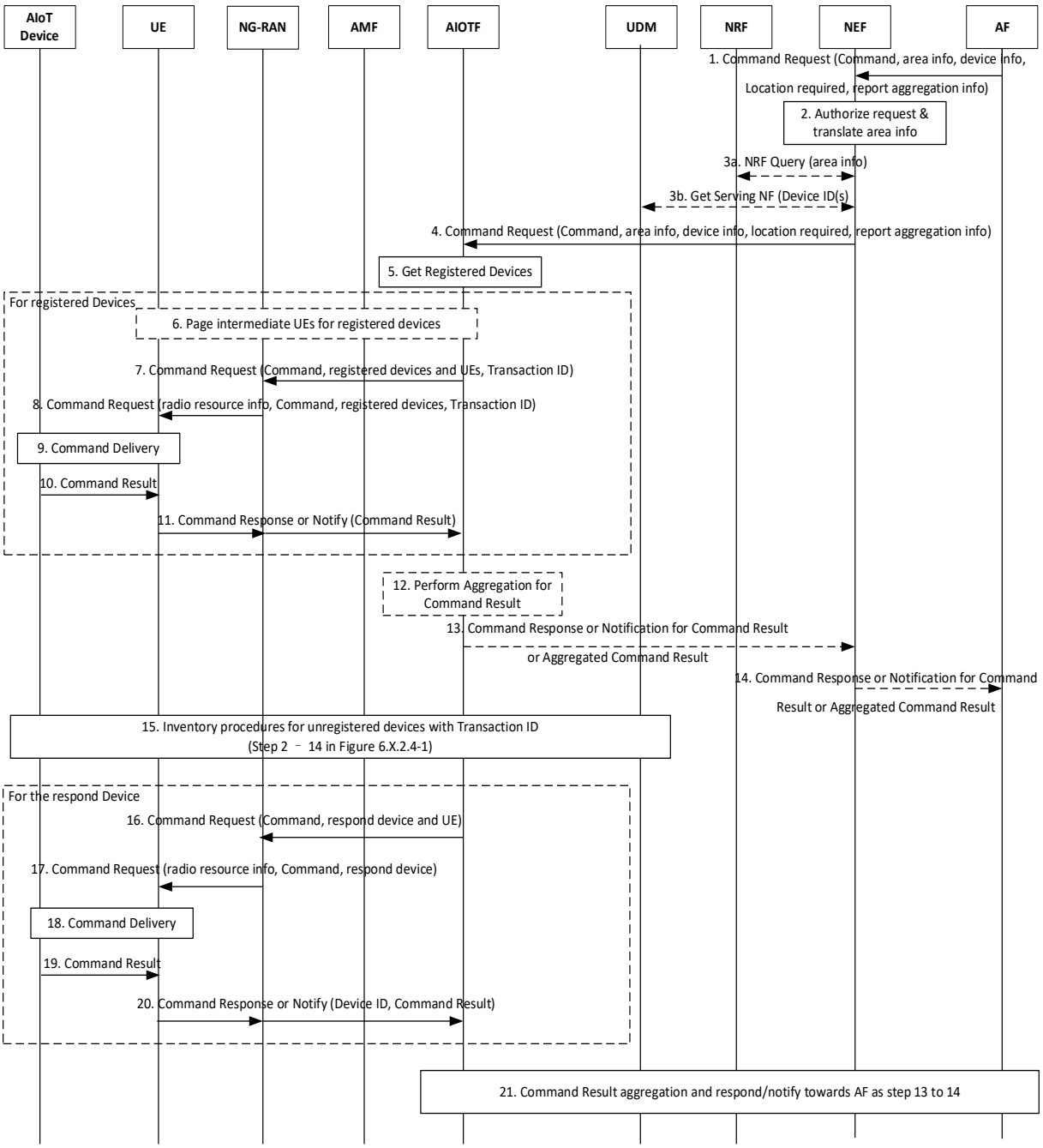


Figure 6.43.2.6-1: Command Procedure

1. The AF sends Command Request to the NEF, containing the command to be executed, area information, device information, optional inventory strategy information, and optional report aggregation info.

   - The command is to be executed in the AIoT devices. It contains the command (e.g. read, write, execute, etc.) together with the command parameters. The command specific parameters (e.g. what to be executed) are to be included in an app layer container between AF and AIOT devices.

   - The area information could be the external geographical area information.

   - The device information could be device ID, device group ID, and/or external device type.

   - The location required indicates whether the AF requests the location information of the AIoT devices provided.

   - The report aggregation info indicates whether the reports need to be aggregated or not for a specific aggregation period, and whether the reports are needed after the aggregation period.

2. The NEF authorizes the request from the AF and perform the area translation to translate external area information to the internal area information. Within the authorization, the NEF further check whether the AF is authorized to get the location information of the device. The NEF map the external device type to internal device type (i.e. device 1, device 2a, device 2b, etc.).

NOTE 1: It is up to RAN to determine whether the device type is useful or not e.g. to optimize AS functionality such as spectrum usage, number of repetitions, etc.

Depending on whether area information is provided, step 3a or step 3b is performed:

3a. If area information is provided, the NEF sends NRF query with internal area information to query AIOTFs serving the area.

3b. If area information is not provided, and device ID(s) are provided, the NEF query the serving AIOTF from UDM.

4. The NEF sends the Command Request to the AIOTFs with the command, internal area information, device information, device type and optional location required information.

5. The AIOTF allocates a Transaction ID for the command, which is unique per command. The AIOTF creates the AIOT Device NAS request message for the devices, based on the command information from the AF. The AIOTF get the registered devices, which match the device information. If no registered devices, steps 6-14 are skipped.

6. The AIOTF sends enable UE reachability request towards the AMF to page the intermediate UEs for the registered devices, if the NG-RAN may release the intermediate UEs to RRC_IDLE state. The AMF check the CM states of those intermediate UEs, and triggers paging towards the NG-RAN(s) for those UEs in CM-IDLE state.

7. The AIOTF creates an AIOT device NAS request message containing the Command. The AIOTF sends the NG-RAN with the AIOT device NAS request message, registered devices together with UEs, and the Transaction ID.

8. The NG-RAN checks whether the command request can be performed. The NG-RAN performs RAN paging for the intermediate UEs in RRC_INACTIVE state. The NG-RAN determines radio resource for the AIoT service operation between Intermediate UEs and the AIoT devices, and sends the Command Request towards the Intermediate UEs for the registered devices with the AIOT device NAS message, radio resource information, registered devices, and the Transaction ID.

NOTE 2: It is up to RAN WG2 to determine the Uu AS layer impacts for command request delivery.

9. The Intermediate UE initiates command delivery to transfer the AIOT device NAS request messages based on CN allocated device IDs of the registered devices, using dedicated signalling. Transaction ID is also included.

NOTE 3: RAN signalling details, including how the command is sent, are to be further studied by RAN WGs.

10. The AIoT Device executes the command, creates an AIOT device NAS response message containing the command result and delivers the AIOT device NAS response message, if the result needs to be sent back. The application specific result can be included in an app layer container.

11. The Intermediate UE forwards the AIOT device NAS response message to the NG-RAN, and the NG-RAN forwards to the AIOTF. The Intermediate UE may further provide location information (ULI) of the device, if requested from the AIOTF and allowed by local policy.

NOTE 4: It is up to RAN WG3 to determine the details and enhancements of the location information.

12. The AIOTF gets the command result from the AIOT Device NAS response message. The AIOTF may perform aggregation, based on the report aggregation information provided by the AF. Within the aggregation period, the AIOTF will buffer the command results reported from the AIoT devices. The AIOTF may stop buffering and send report immediately, if it receives end indicator from the Intermediate UE in step 11. When the aggregation period expires, the AIOTF sends the report. For those command result after the aggregation period, if it is needed by the AF, the AIOTF sends the report. Otherwise, it will be dropped.

13. The AIOTF sends Command Response or Notification Request towards the NEF for the command result or the aggregated command result information.

14. The NEF sends Command Response or Notification Request towards the AF for the command result or the aggregated command result information.

If the AIOTF determines all required devices have executed the command (e.g. when AF provide the device ID(s) and those devices have executed the command in step 9 - 10), the following steps are skipped.

15. For those unregistered devices(s), the AIOTF initiate Inventory procedure as steps 2-14 in Figure 6.43.2.4-1 (steps 12-13 will be skipped, if the device is not able to persistent received information). In the inventory the Transaction ID is delivered, so that the registered devices which have executed command do not respond this inventory request from the Intermediate UE.

16. The AIOTF creates an AIOT device NAS request message containing the Command. The AIOTF sends the NG-RAN with the AIOT device NAS request message, the respond device together with the serving UE.

17. The NG-RAN checks whether the command request can be performed. The NG-RAN determines radio resource which is used between AIoT Devices and Intermediate UE. The NG-RAN sends the AIOT device NAS request message and the respond device.

18. The Intermediate UE initiates command delivery for the AIOT Device NAS request message to the device.

19. The AIoT Device executes the command, creates an AIOT Device NAS response message containing the command result and delivers the AIOT Device NAS response message to the intermediate UE, if the result needs to be sent back. The application specific result can be included in an app layer container.

20. The Intermediate UE forwards the AIOT device NAS response message to the NG-RAN, and the NG-RAN forwards to the AIOTF. The Intermediate UE may further provide location information (ULI) of the device, if requested from the AIOTF and allowed by local policy.

21. As steps 13-14, the AIOTF get the command result from the AIOT Device NAS response message. It may perform the aggregation and send to NEF. The NEF delivers to the AF.

Compared with solution#30, this solution is different in steps 6-8 and step 11: the AIOTF pages intermediate UEs and delivers the device NAS message over NGAP (i.e. command request) towards NG-RANs. NG-RAN delivers the device NAS message over Uu AS layer (e.g. command request) towards intermediate UEs. The device response is sent from intermediate UEs to NG-RAN and forwarded to AIOTF. In steps 16-17 and step 20, the device NAS message delivery follows the same principle.

6.43.3 Impacts on services, entities and interfaces

The following NFs are impacted on top of the impacts of solution #8:

- NG-RAN:
  - NGAP association with AIOTF.
  - Determine radio resources between Intermediate UE and AIoT devices.
  - Receive request from AIOTF, (optional) determine Intermediate UEs, forward to Intermediate UEs.
  - Receive response from Intermediate UEs and forward to AIOTF.
  - Keep Intermediate UEs in RRC_CONNECTED or RRC_INACTIVE state (optional).
- Ambient IoT Function (AIOTF):
  - NG-RAN connectivity.
  - Routing the request from AF (via NEF) to NG-RAN, for DO-DTT/DT traffic types.
  - Routing the response from NG-RAN to AF (via NEF) for DO-DTT traffic type.
  - Page Intermediate UEs via AMF (optional, if NG-RAN does not release the authorized Intermediate UEs to RRC_IDLE state, it is not needed).
- AMF: Authorization of Intermediate UEs and create UE context towards AIOTF.
- UDM: Enhance authorization information for Intermediate UEs.
- Intermediate UE:

- Authorization of Intermediate UEs.
- Receive the request and radio resource information from NG-RAN and perform Ambient IoT operations (e.g. inventory, command, etc.) on the proper spectrum.
- Receive responses from AIoT devices and send to NG-RAN.

7 Overall Evaluation

Editor's note: This clause will provide a general evaluation and comparison of the solutions per Key Issue #<X>.

8 Conclusions

NOTE: The cconclusions in this clause regarding Topology 2 (i.e. mainly in clause 8.1.3) have not been updated since v1.2.0 and will not be further updated considering that the Topology 2 is not pursued in Rel-19 normative phase as per RAN WI in RP-243326.

8.1 Conclusion on Key Issue #1

8.1.1 General

Key issue #1 includes the following aspects:

- System architecture identified along with the solutions for KI#2 and KI#3.

Key issue#2 aspect on "Ambient IoT Device subscription management" and key issue#3 aspect on "Ambient IoT service exposure" is considered in this clause.

In this Release, normative work will take place for Topology 1 and no normative work will take place for Topology 2.

The following aspects common for Topology 1 and Topology 2 are concluded as principles for normative work:

1. A new core network function is introduced to support Ambient IoT (e.g. AIOTF) service for both the topology 1 and topology 2. The AIoTF performs the following functionality.

   a. The AIOTF registers itself in the NRF with its NF profile, this is to enable the discovery of AIOTF instances e.g. by an NEF. The NF profile at least includes the AIOTF ID/address, NF type, information used to allow the NEF to discover AIOT instances, e.g. based on the target area information in AF request. Other details of the NF profile, if needed, will be completed in normative phase.

   b. For topology 1, the AIOTF selects AIOT RAN nodes and optionally a list of the BS readers. For topology 2, the AIOTF selects the UE readers (e.g. candidate or final UE readers) and provides the selected UE Reader list to the RAN.

NOTE 1: Providing the UE Reader list to the RAN only applies to the RRC-based option.

For topology 1, NGAP is terminated at an AIOT RAN node. An AIOT RAN node that supports one or more BS readers, may report its supported BS reader information e.g. BS reader IDs and their corresponding service areas. The OAM may configure the AIOTF with BS Reader information including e.g. their corresponding service area when the BS Readers are integrated in the network.

NOTE 2: Whether RAN3 will specify BS reader information (e.g. BS reader ID, service area) reporting to AIOTF over NGAP depends on RAN3 decision and if so SA WG2 may include this feature in SA WG2 specifications.

NOTE 3: If the request from the AIOTF to the AIOT RAN does not include a BS reader list, then AIOT RAN can use all available BS readers.

NOTE 4: The BS reader’s service area needs to coordinate with RAN WG(s).

c. The AIOTF receives an AIoT service request from the AF and triggers the BS/UE Readers to perform AIoT service operations towards the AIoT Devices(s).

d. The AIOTF aggregates the service operation results (including the removal of the duplicated devices records) from BS Readers and UE Readers and sends to AF.

e. The AIOTF may provide the following assistance information to AIoT RAN/UE Reader:

- AIoT service type (e.g. Inventory, Command);
- approximate number of AIoT devices based on AF request;
- approximate D2R message size based on AF request.

NOTE 5: If there are multiple Readers selected for the AIoT Service, the AIOTF may provide the approximate number of AIoT devices to each Reader based on implementation.

NOTE 6: The approximate D2R message size considering the overhead of AIoT Device NAS layer will be determined later in cooperation with CT WG1 and SA WG3.

NOTE 7: Further assistance information can be added during the normative phase and in cooperation with other WGs if necessary.

Editor's note: For RRC based solution of topology 2, whether the aggregation can be performed by the RAN is FFS and coordination with RAN WGs is needed.

f. The AIOTF may provide the AIoT aggregation assistance information to the AIoT RAN. Aggregation assistance information is used to assist in the aggregation of AIoT responses received from multiple AIoT devices. The AIOTF may determine the aggregation assistance information based on operator policy, considering information provided from the AF, which consists of:

- Time interval: The fixed time interval for which AIoT RAN collects AIoT Device responses before AIoT RAN reports the aggregated AIoT responses to AIOTF. This may potentially happen multiple times until the AIoT RAN completes the request operation. If the AF has provided a time interval, then the AIOTF should signal a time interval to AIOT RAN that is equal or shorter than the time interval received from the AF.

NOTE 8: Further details how to use the AIoT aggregation assistance information are subject to RAN3 decision.

NOTE 9: The AIoT responses aggregated from the AIoT devices share the same correlation identifier.

NOTE 10: AIOTF is expected to receive an end indication from AIoT RAN, signaling the completion of reporting for a specific correlation ID.

NOTE 11: If the AIOTF does not provide aggregation assistance information, the aggregation process in the AIoT RAN is determined by implementation. As a result, the interpretation of aggregated data by the AIOTF or AF also depends on individual implementation choices.

g. When the AIOTF sends an operation request to a UE Reader or BS Reader (via AIOT RAN), a response and one or more reports with the results of the AIoT service operation is returned to the AIOTF with the results of the AIoT service operation, and the AIOTF needs to correlate the results to a given operation request. The AMF (if used to route the requests) additionally provides an AIOTF identifier with the request from the AIOTF which is returned with the response(s) related to the request, so the AMF can be routed back the requesting AIOTF.

2. A Permanent AIoT Device ID is stored in the AIoT Device and the network or a Credential Holder's AAA server. The AIOTF checks whether the AIoT Device ID from AIoT Device has a subscription and retrieves.

Editor's note: How the AIoT Device ID privacy protection and including ID authentication is done will be concluded by SA WG3. SA2 alignment with SA WG3 TR conclusions for AIoT Device security including AIoT Device ID privacy protection is FFS..

3. The AIoT Device does not distinguish whether the connectivity topology is Topology 1 or Topology 2, nor the transport used by the AIoT Reader.

NOTE 12: The AIoT device is also agnostic to the potential different architectures if more than one architecture is concluded for both the topology 1 and topology 2.

4. AIoT Device NAS protocol is supported between the AIoT Device and the AIOTF. The AIoT Device NAS layer supports Inventory Response and Command (e.g. Read and Write) Request and Response.

5. The AIOTF may store and manage the AIoT device related information (also known as device context information) locally that includes e.g. the AIOT device permanent ID, the last known reader information of the AIoT device. The last known reader information can be used to support the AIOTF to select the serving reader to forward the message towards the specific AIoT device(s).

NOTE 13: The storage of security related information at the AIOTF is up to the coordination with SA WG3 later.

8.1.2 Architecture to Support Topology 1

8.1.2.1 General

The principles and aspects in this clause are agreed to support Topology 1:

- The new core network function (AIOTF) is introduced to support Ambient IoT functionality, described in clause 8.1.1, with the following features for topology 1:

  - Send requests that trigger BS Reader(s) (via AIoT RAN) to perform AIoT operations, either directly or via an AMF.

NOTE 1: It is assumed AIoT RAN supports Ambient IoT specific functionalities. There is no assumption about whether the AIoT RAN (i.e. BS Reader) also has any gNB functionality for NR-Uu or not.

NOTE 2: In direct communication, NGAP is used between the AIOTF and the AIOT RAN (i.e. over the Nx reference point). In indirect communication, NGAP is used between the AMF and AIOT RAN (i.e. over the N2 reference point).

NOTE 3: It is assumed Ambient IoT services can be deployed independently from existing deployments.

NOTE 4: It is not expected a deployment will use both direct communication between AIOT RAN and an AIOTF and indirect communication between AIOT RAN and an AIOTF via an AMF. The deployment choice can be based on, for example, using a direct communication path for a local deployments.

  - The signalling transport for NGAP at the A-IoT RAN node is SCTP/IP.

8.1.2.2 AIOT RAN and the AIOTF communicate directly

When a AIOT RAN and the AIOTF communicate directly:

- The AIOTF communicates with a AIOT RAN via a direct interface reference point Nx.

- Figure 8.1.2.2-1 below shows the aspects related to Topology 1 (direct path) architecture in reference point representation architecture with other NFs removed.

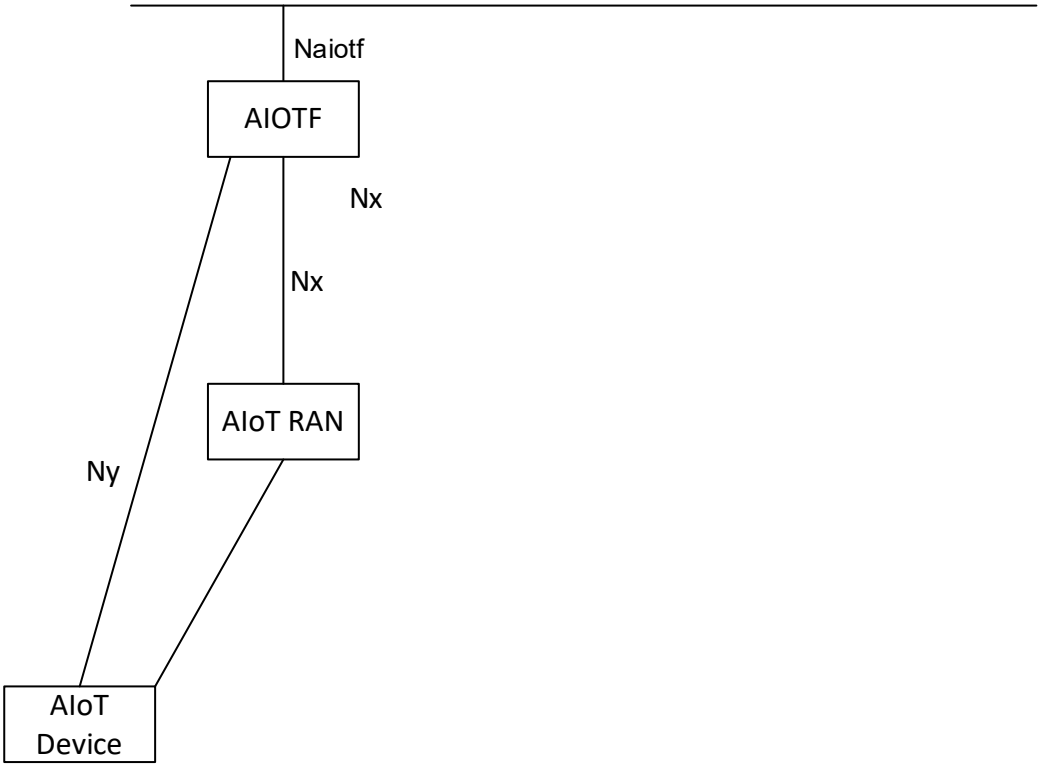


Figure 8.1.2.2-1: Non-Roaming 5G System Architecture (Direct Path)

- Figure 8.1.2.2-2 below shows the aspects related to Topology 1 (direct path) architecture in reference point representation with other NFs removed.

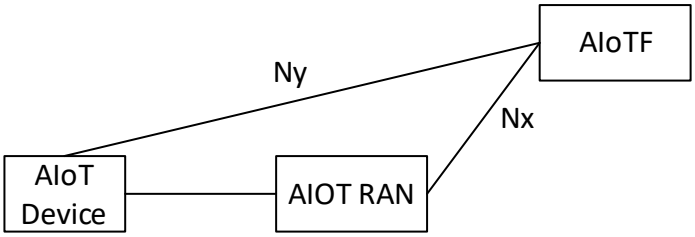


Figure 8.1.2.2-2: Non-Roaming 5G System Architecture in reference point representation (Direct Path)

NOTE 1: NGAP used over Nx reference point will support procedures and information to be exchanged as specified by RAN WG2, RAN WG3 and SA WG2.

NOTE 2: The protocol stack used between the AIoTF and the AIOT RAN will be concluded by RAN WG3.

- Figure 8.1.2.2-3 below shows the aspects related to Topology 1 (direct path) protocol stack between the AIOT RAN and AIoTF.

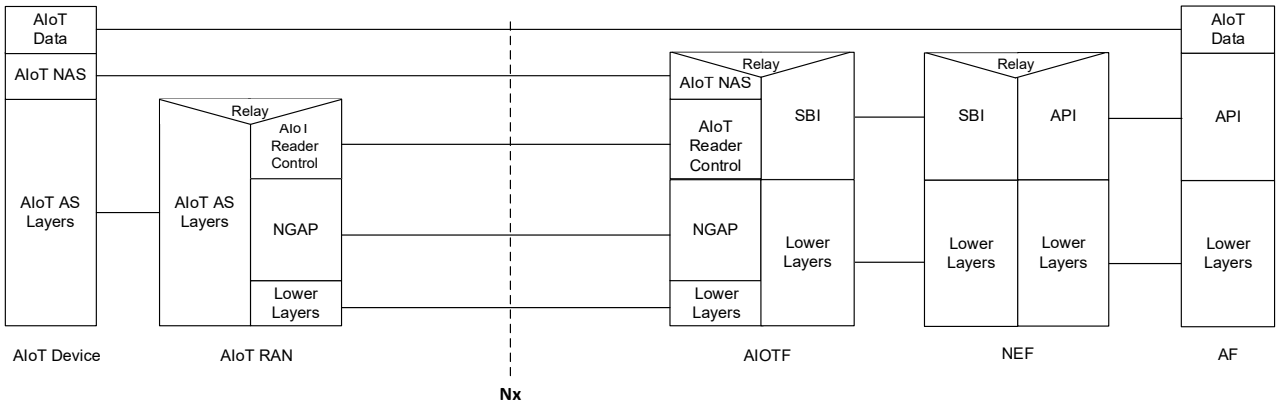


Figure 8.1.2.2-3: Example Protocol Stack between AIOTF and AIoT Device for Topology 1 (Direct Path)

NOTE 3: Whether AIoT Reader Control is transported by NGAP or is part of the NGAP protocol will be determined by RAN WG3.

8.1.2.3 AIOT RAN and the AIOTF communicate indirectly via an AMF

When AIOT RAN and the AIoTF communicate indirectly via an AMF:

- The AIoTF connects with AIOT RAN via an AMF. NGAP over the N2 reference point between the AIOT RAN and AMF supports Ambient IoT services including delivery of inventory/command messages.

NOTE 1: If network isolation is required an AMF instance is deployed for supporting the AIOTF communication with AIOT RAN.

- NGAP between AIOT RAN and the AMF is enhanced to support Ambient IoT Services.

NOTE 2: The details of the enhancements will be concluded by RAN WG3.

- The AMF shall be enhanced to support:
  - On the Nz reference point, the AMF supports Services which are used by an AIoTF for Ambient IoT Operations e.g. to send AIoT requests towards AIOT RAN and to receive AIoT responses from the AIOT RAN.
  - On the N2 reference point, the AMF supports sending AIOT information to the AIOT RAN (e.g. operation requests) and receiving responses from the AIOT RAN.
  - The AMF routes the AIoT messages between the AIOT RAN (over N2 reference point) and the AIOTF (over Nz reference point).

NOTE 3: Whether to enhance an existing service or define a new service will be determined in the normative phase.

- Figure 8.1.2.3-1 below shows the aspects related to Topology 1 (indirect path via AMF) architecture in reference point representation with other NFs removed.

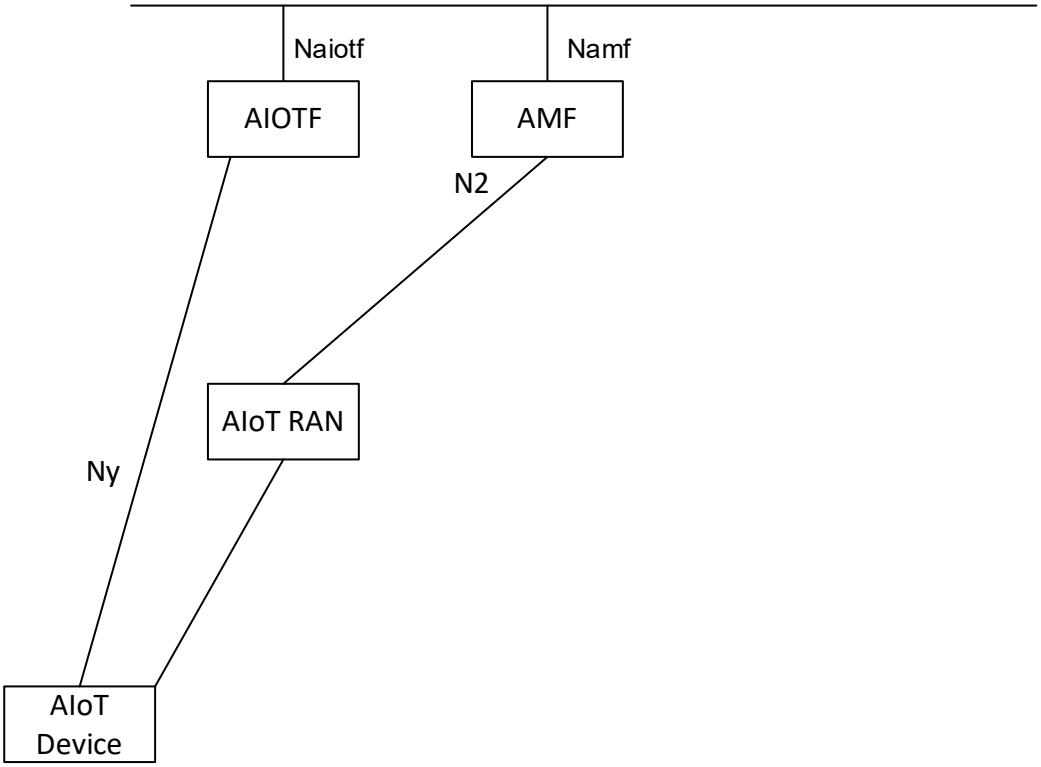


Figure 8.1.2.3-1: Non-Roaming 5G System Architecture (indirect Path via AMF)

- Figure 8.1.2.3-2 below shows the aspects related to Topology 1 (indirect Path via AMF) architecture in reference point representation with other NFs removed.

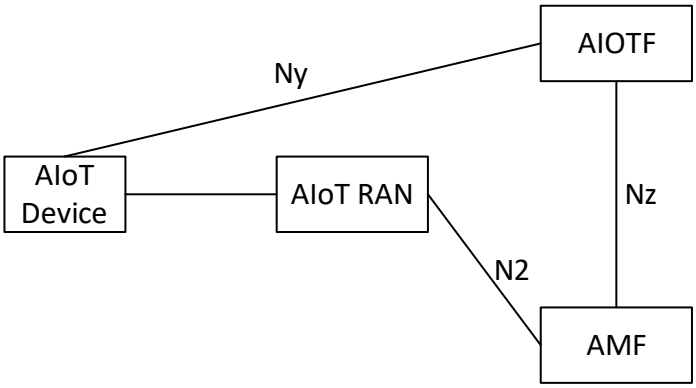


Figure 8.1.2.3-2: Non-Roaming 5G System Architecture in reference point representation (indirect Path via AMF)

- Figure 8.1.2.3-3 below shows the aspects related to Topology 1 (indirect path via AMF) protocol stack between AIOT RAN and AIOTF.

  - AIoT Data represents information exchanged between the AIoT Device and AF (application specific content) and AIoT Reader Control represents the requests and responses between the AIOTF and AIOT RAN.

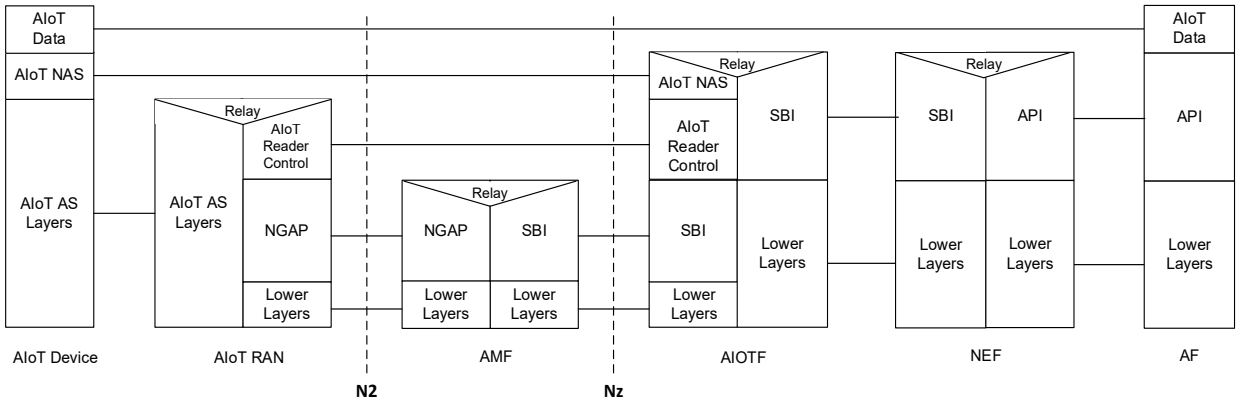


Figure 8.1.2.3-3: Example Protocol Between AIOTF and AIoT Device for Topology 1 (indirect Path via AMF)

NOTE 4: Whether AIoT Reader Control is transported by NGAP or is part of the NGAP protocol will be determined by RAN WG3.

NOTE 5: UL traffic is routed based on an AIoT correlation identifier (i.e. AIOT-RAN identifies the appropriate AMF to which to forward UL messages base on the association between AIoT correction identifier and the AMF); details will be determined during normative work.

8.1.3 Architecture to Support Topology 2

8.1.3.1 General

The following principles apply:

- Architecture: Two architecture options will be specified for Topology 2:
  - User-plane option as defined in clause 8.1.3.2.
  - RRC-based option as defined in clause 8.1.3.3.
- Subscription aspects:
  - The UE subscription in the UDM will be extended with UE Reader subscription information, which consists of the following:
    - information indicating whether the UE is allowed to operate as a UE Reader.

Editor's note: Additional subscription information, e.g. validity information, for the UE Reader is FFS.

  - UE Reader subscription information is available to AMF and AIoTF.
  - If AMF receives, as part of the subscription information, the indication that the UE is authorized to operate as a UE Reader, then AMF informs NG-RAN that the UE is authorized to operate as a UE Reader.

Editor's note: Whether and how to enable authorization to the UE is FFS.

- Radio resource management for UE Reader operation:
  - If the gNB has received the indication that a UE is authorized to operate as a UE Reader, then the gNB may assign radio resources to the UE for UE Reader operation.
  - The UE Reader is assumed to request radio resources for Reader operation (e.g. taking into account assistance information received from the AIoTF) from the gNB.

NOTE 1: Assistance information is further clarified in other clauses.

NOTE 2: The details of UE Reader radio resource management will be defined by RAN WG2.

8.1.3.2 User-plane option

The following additional principles apply:

- As depicted in Figure 8.1.3.2-1, the UE Reader connects to the AIoTF based on the AIoT Application Protocol (AIoT-AP) using an IP PDU Session between the UE and the UPF as transport. The related protocol stack is shown in Figure 8.1.3.2-2. The AIoT AP protocol will support procedures and information to be exchanged as specified by RAN WG2, RAN WG3 and SA WG2.

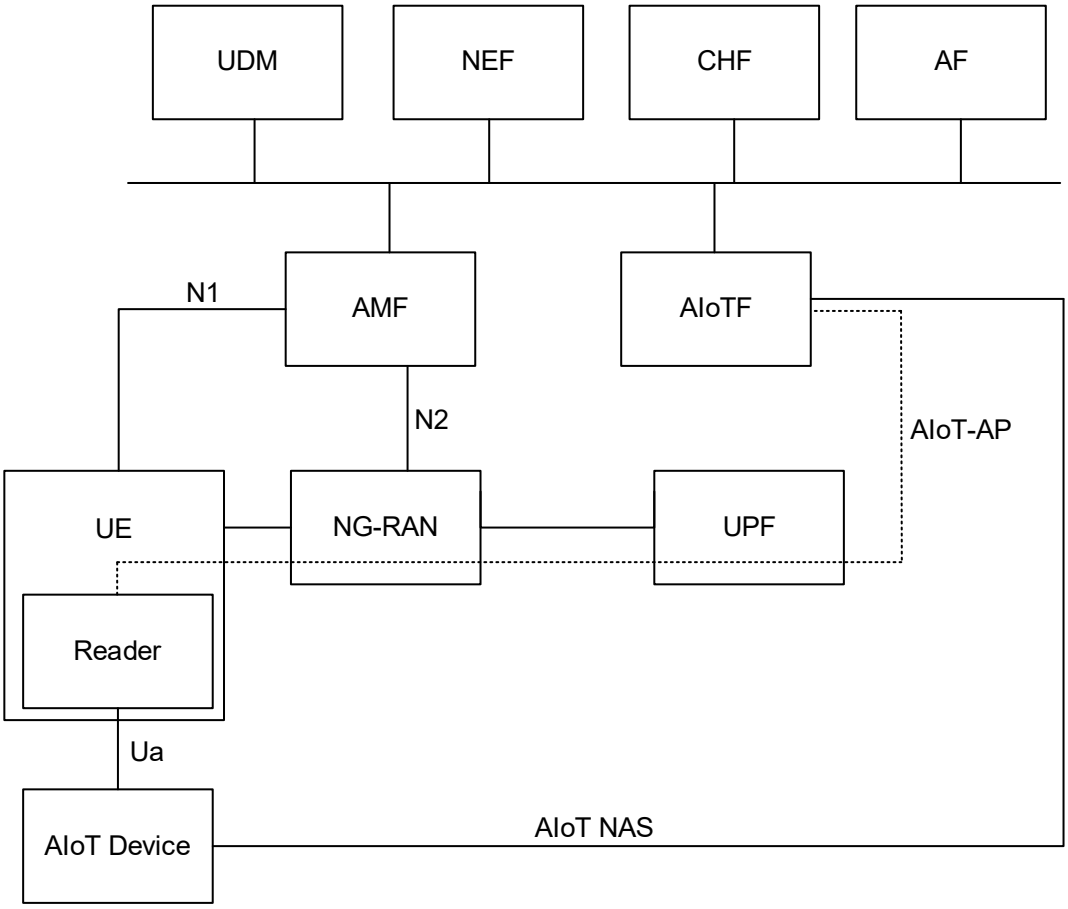


Figure 8.1.3.2-1: User-plane architecture for Topology 2

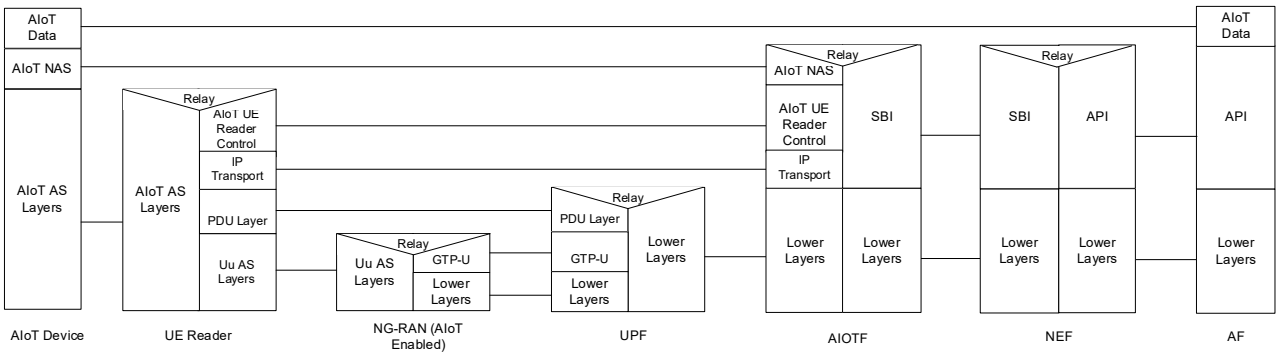


Figure 8.1.3.2-2: Protocol stack for the user-plane architecture for Topology 2

NOTE 1: Which transport protocol to use for AIoT-AP can be decided by CT WG1.

NOTE 2: Security for AIoT-AP is assumed to be defined by SA WG3.

- To operate as a UE Reader, the UE establishes a PDU Session to a specific DNN/S-NSSAI and establishes an association with the AIoTF identified by an FQDN using the AIoT AP protocol. In this release, the UE is preconfigured with the specific DNN/S-NSSAI and the AIoTF FQDN.

Editor's note: Whether and which additional methods to dynamically provision the UE with this information can be specified and in which release is FFS.

- If the AIoTF determines that the UE is not authorized to operate as a UE Reader according to the UE's subscription, then the AIoTF rejects the UE's association request.

- If the AIoTF detects that the Reader function in the UE does not respond to a request by the AIoTF (e.g. an Inventory Request or Command Request), then the AIoTF considers the UE Reader unreachable and locally deletes the association for the UE Reader.

8.1.3.3 RRC option

The following principles apply:

- Messages between the UE Reader and the AIoTF are delivered using RRC between UE and RRC and NGAP between gNB and AMF, and using an SBI interface between AMF and AIoTF. The related protocol stack is shown in Figure 8.1.3.3-1.

Editor's note: How addressing works for UL for Option B is FFS.

Editor's note: Further details how the RRC option works are FFS.

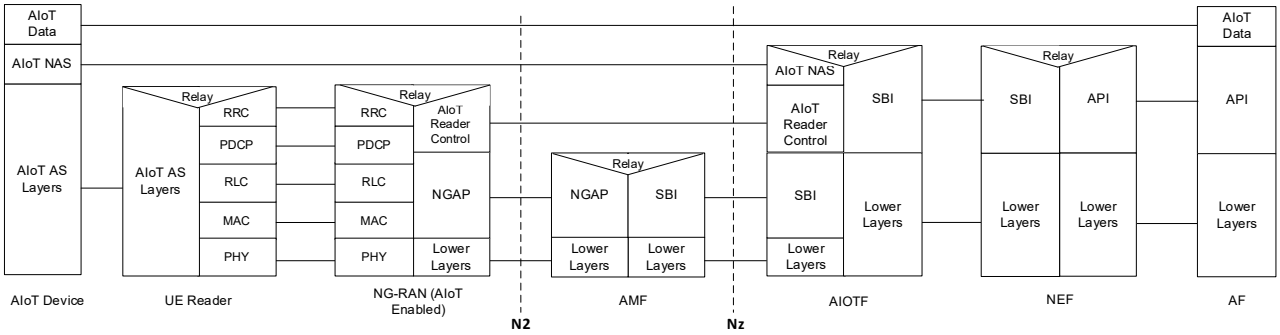


Figure 8.1.3.3-1: Protocol Stack for the RRC option

8.1.3.4 Interim Conclusion for authorization

The following interim conclusion are agreed for the UE Reader authorization in Topology 2:

- The subscription data of UE Reader includes the indication that whether the UE is allowed to act as an AIoT Reader.

- The AMF provides the UE Reader authorization indication to the NG-RAN serving the UE Reader.

Editor's note: How to authorize the UE Reader will be determined later.

8.2 Conclusion on Key Issue #2

8.2.1 Identifier and Identification Management

The following aspects and principles are considered and agreed for the interim conclusion on Identifier and Identification Management.

The following principles are agreed for Identifier and Identification Management:

- An Ambient IoT Device is configured with a permanent Ambient IoT Device Identifier which can be assigned by an operator or by a third party. The Identifier is used to identify Ambient IoT Device and locate the corresponding authentication server.

NOTE 1: How to configure Ambient IoT Device with the permanent Ambient IoT Device Identifier is out of 3GPP scope.

The permanent Ambient IoT Device Identifier includes the following information:

1) Part1information:
   - The ID type, including
     - information indicating whether the network identifier is included or not.
     - information indicating whether information used to identify a 3rd party is included or not.
     - the Part2 type indicating EPC or the unstructured format.
   - A network identifier (i.e. MCC+MNC and/or NID), when the ID includes the network identifier.
   - Information used to identify a 3rd party when the ID includes the information used to identify a 3rd party.
2) Part2 information:
   - The information (i.e. EPC or unstructured format) used to distinguish different Ambient IoT Devices within the scope identified by the Part1 information.

NOTE 3: The coding for the above information and how to code the length for Part2 are left to stage 3.

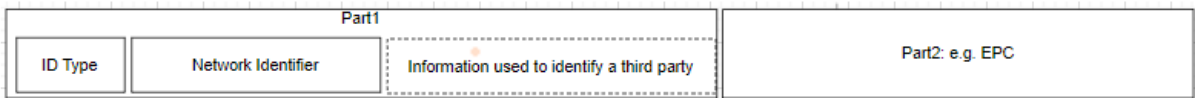


Figure 8.2.1-1: The example of Operator allocated ID

For operator allocated Part 1 information, the network identifier is mandatory and can be used to index the authentication server or not. If it is not used to index the authentication server, the ID may further include the information used to identify a 3rd party. The third party may be the credential holder or not. If it is not the credential holder, the network should be provided with third party related context including the information used to locate the authentication server.



Figure 8.2.1-2: The example third party allocated ID

For third party allocated Part 1 information, the network identifier is not needed. The third party may be the credential holder or not. If it is not the credential holder, the network should be provided with third party related context including the information used to locate the authentication server.

NOTE 4: Within each type of ID, the length of Part 1 is fixed (i.e. the length of ID type, the Network Identifier (if present) and the Information used to identify a third party (if present) are fixed) and the length of Part2 is dynamic. The detail is left to stage 3.

With the above information, the AIoT device ID is globally unique.

8.2.2 Subscription Management

The AIoT device may or may not have operator's subscription data in the network. If the AIoT device doesn't have the operator's subscription data in the network, the corresponding related data (e.g. Device ID or credentials) is stored in the AAA server external to the network.

Subscription data for an Ambient IoT Device is required in case the Ambient IoT Device is managed by the network:

- It is used by the AIOTF to check whether the device is subscribed.

NOTE 1: Checking whether the device is subscribed needs coordination with SA WG3 study outcomes on validation of the ambient IoT Device identifier.

- It is dedicated subscription data for Ambient IoT Devices, i.e. different with UE subscription data.

- It contains at least a permanent Ambient IoT Device ID, etc.

NOTE 2: SA WG2 will align security related materials in subscription data with SA WG3 during the normative phase.

The Ambient IoT Device ID is used by the AIOTF together with local configuration, 3rd party related context to locate the entity storing the AIoT device profile data of an Ambient IoT Device. The AIoT device profile data includes the AIoT Device ID. When the AIoT device profile data is managed by the 5GC, it is stored in the ADM (Ambient IoT Data Management), possibly together with UDR, which exclusively supports AIoT devices.

NOTE 3: Whether the AIoT device profile data can include credentials for AIoT device will be determined by SA WG3.

NOTE 4: Storage of AIoT device profile data (i.e. AIoT device subscription like data) in the network does not impact the UDM.

Authorization data for 3rd party AF is required:

- It is used by the AIOTF together with SLA and operator policy at NEF to perform authorization of the AF request targeting for Ambient IoT service. This includes: 1) checking whether the AF is authorized for the requested service operation; 2) checking whether the AF is authorized for the requested target area; 3) checking whether the AF is authorized for the requested target AIoT device(s), including individual devices and a group of devices.

- it is stored at ADM within 5G network or configured in the NEF.

NOTE 5: Items within the authorization data for 3rd party AF will be determined during normative phase.

NOTE 6: The NEF performs AIoT specific AF authorization, as specified in TS 33.501 [15], based on operator policy and local configuration. The AIOTF may perform AIoT specific AF authorization based on the authorization data stored in the ADM. Further clarifications on the roles of NEF and AIOTF will be defined during normative phase.

UE Subscription data is enhanced to include an indication that whether the UE is allowed to operate as a UE reader.

8.3 Conclusion on Key Issue #3

8.3.1 General

The clause concludes the following aspects:

- Study how to support information transfer for Ambient IoT services and related system functionality, including the information transfer for an Ambient IoT device and for a group of Ambient IoT Devices.

NOTE: The above aspect includes studying whether there is a need to support session based transfer between Ambient IoT Device and the network considering the device types and capabilities.

- Study which of the enabled Ambient IoT services are exposed to AF and how, e.g. for the case AF requests Ambient IoT service for an Ambient IoT Device and for a group of Ambient IoT Devices.

8.3.2 AIoT services supported by the 5GC

The following Ambient IoT services are agreed to be supported by the 5GC, which apply to both topology 1 and topology 2:

- Inventory: Request to perform an inventory operation.

- Read: Request to read information from an AIoT Device.

- Write: Request to write information to an AIoT Device.

- Disable: Request that an AIoT Device has its capability to transmit RF permanently disabled.

NOTE: The temporarily disabling and enabling of AIoT devices did not reach conclusion and will therefore not be supported in Release 19.

8.3.3 NEF exposure

NEF supports to expose Ambient IoT service towards the AF.

- New NEF services (e.g. Nnef_AIoT) to support the AIoT services including Service Operations of Inventory, Read, Write, and Permanent Disable will be defined.

8.3.4 Principles on the procedures to support AIoT services

To support the services provided by 5GC and the NEF exposure of those AIoT services, the following procedures are supported:

- Inventory Procedure.
- Command Procedure, to e.g. transfer AF AIoT Data to/from AIoT Device(s) as AIoT specific NAS messages.

There are requests that are used from the AIOTF towards the Reader and responses from the Reader to the AIOTF. The routing of the request and response messages and their encoding depends on the topology and transport to the Reader (see KI#1).

All the procedures follow have the following steps:

1. The AF makes a service request to the NEF, including parameters as defined in clause 8.3.5.
2. The NEF determines an AIOTF for the requested operation, and invokes a new service operation on the AIOTF.
3. For the requested operation the AIOTF:

   1). Performs initial reader selection by either:

   - The AF provide information to identify readers to include in the initial reader selection network. The information can identify multiple or an individual Reader.

NOTE 1: Which readers are identified by the information from the AF is up to the network deployment, configuration or implementation.

   - If a single UE Reader ID is provided by the AF via the NEF for the operation, then that is used as the selected Reader.
   - If reader selection information or UE Reader ID is not provided, then how the AIOTF determines which readers to use is based on implementation. The AIOTF may be e.g. preconfigured with which readers to use, or take the requested target AIoT Devices last known location into account, etc.

     If no readers can be selected then the request is rejected.

   2). Determines AIoT Device Identification information based on the information from the AF, to be included in the paging message on the AIoT radio interface to find the AIoT Devices. AIoT Devices compare the AIoT Device Identification information with their own AIoT Device Identifier (part of or full AIoT Device Identifier) to determine whether respond to the paging message.

   3). Determines Reader Assistance information required for the operation used to the Reader, taking into account assistance information from the AF.

   4). Constructs a request for an Inventory operation using the determined A-IoT Device Identification information page the AIoT Devices, and a correlation identifier for the AIOTF to correlate the inventory responses to the request. The Inventory request is routed to the Readers determined by the initial reader selection.

     See clause 8.1 for how to provide the request to a Reader.

   5). The Reader executes the inventory request, reporting AIoT specific NAS message responses from the AIoT Device to the AIOTF, including its Reader ID and correlation identifier from the AIOTF. The Reader may aggregate results from multiple AIoT Devices in the responding messages. The AIOTF can determine which request the results are for using the correlation identifier.

   6). The AIOTF may, depending on the information within the AIoT Device identifier, obtain subscription-like information from either:

- the serving network performing the operation (either as identified by a AIoT Device Identifier or operator policy to check it been provided with information for a specific AIoT Device), or
- another network as identified by a AIoT Device Identifier, or
- A third party as identified by a AIoT Device Identifier.

7). Checks if the AIoT Device is subscribed.

NOTE 2: Whether and how AIOT Device Identifiers are verified depends on SA WG3.

8). If the operation is a command operation, the AIOT generates a request, including an AIoT specific NAS message for the command, along with any additional information required by the Reader to execute the command, and a correlation identifier to determine the destination of command responses. The request is then routed to the Reader. The Reader executes the command, passing the AIoTF specific NAS message to the AIoT Device and collecting any AIoT specific NAS responses. The AIoT specific NAS responses are then routed back to the AIOTF.

NOTE 3: Whether and how security protection is applied to the AIoT specific NAS message send to the AIoT Device and the response AIoT specific NAS message from the AIoT Device depends on SA WG3.

9). Provide the results of the operation to the NEF. Results from multiple AIoT Devices may be included/aggregated in the service response(s).

4. Provide the results of the operation from the NEF to the AF.

8.3.5 Information provided by the AF

The AF provides one or some of the following information:

- Information about AIoT services:
  - Service type indicating e.g. inventory or command (e.g. read or write).
- Information to be used for AIoT reader selection:
  - At least one of the UE reader ID or the target area information.
- Optionally, Information about the target AIoT device(s):
  - At least one of the AIoT device ID(s) or the filtering information which can be used to associate with multiple AIoT devices.
- Optionally, Information to be used for resource allocation that is detailed in TR 38.769 [8]:
  - Approximate number of AIoT devices.
  - Approximate D2R message size.

NOTE: Information to be used for resource allocation will be aligned with RAN WGs during the normative phase.

Annex A:
Overview of the Electronic Product Code

The EPC Tag Data Standard (GS1 TDS Release 2.1 [10]), issued and maintained by the GS1 organisation [11], defines the Electronic Product Code as follows:

> *The EPC is a universal identifier that provides a unique identity for any physical object. The EPC is designed to be unique across all physical objects in the world, over all time, and across all categories of physical objects. It is expressly intended for use by business applications that need to track all categories of physical objects, whatever they may be.*

In line with this definition, it is important to emphasize that the EPC Tag Data Standard (GS1 TDS Release 2.1 [10]) defines EPC as a means for unique identification of physical objects independently of any technology for storing and transmitting such EPC information. In other words, EPCs can be used independently of existing RFID tags.

Despite the word "product" that is an integral part of the term Electronic Product Code, EPCs are not only used for instance for retail items. Instead, the EPC Tag Data Standard (GS1 TDS Release 2.1 [10]) defined identification schemes for various business domains to enable identification of:

- individual trade items based on the Serialised Global Trade Item Number (SGTIN) EPC scheme;
- logistics handling units (e.g. a pallet load) based on the Serial Shipping Container Code (SSCC) EPC scheme;
- returnable assets such as boxes, pallets and casks based on the Global Returnable Asset Identifier (GRAI);
- aircraft parts based on the Aerospace and Defence Identifier (ADI) EPC scheme;
- patients and the services provided to them based on the Global Service Relation Number (GSRN) EPC scheme (see also [12] for more information on the use of EPC in the healthcare supply chain);
- etc.

The reason that EPCs are unique despite being used by independently operating organizations in many different domains is the typical structure of the underlying EPC schemes, which typically include an organizational identifier.

For example, as illustrated in Table A-1, the Serial Shipping Container Code (SSCC) as defined in clause 6.3.2 of the EPC Tag Data Standard (GS1 TDS Release 2.1 [10]), consists of a GS1 Company Prefix and a Serial Reference. The GS1 Company Prefix number space is managed by the GS1 organization, which assigns GS1 Company Prefix numbers to individual organizations. The serial reference is then assigned by the organization itself. Together this yields a unique EPC.

Table A-1: Serial Shipping Container Code EPC scheme

GS1 Company Prefix	Serial Reference

The EPC Tag Data Standard (GS1 TDS Release 2.1 [10]) follows the same approach also for other EPC schemes, e.g. the Serialised Global Trade Item Number (SGTIN), which is used for individual trade items such as an instance of a specific product (e.g. an individual TV). As depicted in Table A-2, the SGTIN also contains a unique GS1 Company Prefix. In addition, the SGTN contains the Item Reference (to differentiate different object classes) and a Serial number, which are assigned by the organization identified by the company prefix.

Table A-2: Serialised Global Trade Item Number (SGTIN) EPC scheme

GS1 Company Prefix	Item Reference	Serial Reference

In clause 14, the EPC Tag Data Standard (GS1 TDS Release 2.1 [10]) also defines a binary encoding of the EPC schemes, i.e. the standard also defines how EPCs are serialized as a string of bits, e.g. for storage on tags.

As shown in Table A-3, the key idea is that a binary representation of an EPC starts with a header value that indicates the EPC scheme followed by the bitwise representation of a particular encoding scheme. For instance, SSCC-96 defines how the GS1 Company Prefix and the Serial Reference of the Serial Shipping Container Code are represented as a string of bits. The typical size of EPCs, e.g. for the Serialised Global Trade Item Number using the SGTIN-96 coding scheme is 96 bits (including the Header value).

Table A-3: The EPC binary representation consists of the header value that identifies the EPC scheme and coding scheme specific bits, e.g. SSCC-96 for the Serial Shipping Container Code as defined in GS1 TDS Release 2.1 [10]. (Note: This table shows a subset only)

EPC scheme	Header value	Coding scheme specific bits
Serial Shipping Container Code (SSCC)	0011 0001	SSCC-96 as defined in clause 14.6.2.1 of GS1 TDS Release 2.1 [10]
Serialised Global Trade Item Number (SGTIN)	0011 0000	SGTIN-96 as defined in clause 14.6.1.1 of GS1 TDS Release 2.1 [10]

Annex B:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-01	SA2#160 AH-e	S2-2400509	-	-	-	Proposed skeleton agreed at SA2#160AH-e	0.0.0
2024-09	SA#105	SP-241274	-	-	-	MCC editorial update for presentation to TSG SA for information	1.0.0
2024-11	SA#106	SP-241506	-	-	-	MCC editorial update for presentation to TSG SA for information (second presentation)	1.2.1
2025-02	SA#107	SP-250070	-	-	-	MCC editorial update for presentation to TSG SA for approval	2.0.0
2025-03	SA#107	-	-	-	-	MCC editorial update for publication after TSG SA approval	19.0.0